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DATA PROCESSING SYSTEMS INCLUDING OPTICAL COMMUNICATION MODULES

Abstract

An apparatus includes: at least one of a circuit board or a substrate; and a first structure attached to the at least one of a circuit board or a substrate. The first structure is configured to enable an optical module with connector to be removably coupled to the first structure, and the optical module with connector is configured to enable an optical fiber connector to be removably coupled to the optical module with connector. For example, the circuit board or the substrate includes first electrical contacts, the first structure includes walls that define a first opening, the walls also define one or more retaining mechanisms such that when the optical module with connector is inserted into the first opening, the one or more retaining mechanisms on the walls of the first structure engage one or more latch mechanisms on the optical module with connector to secure the optical module with connector to the first structure, and second electrical contacts on the optical module with connector are electrically coupled to the first electrical contacts on the circuit board or the substrate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/617,137, filed on Mar. 26, 2024, which is a continuation of U.S. patent application Ser. No. 17/495,338, now U.S. Pat. No. 11,988,874, filed on Oct. 6, 2021, which claims priority to U.S. Provisional Patent Application 63/210,437, filed on Jun. 14, 2021, U.S. Provisional Patent Application 63/088,914, filed on Oct. 7, 2020, U.S. Provisional Patent Application 63/245,005, filed on Sep. 16, 2021, U.S. Provisional Patent Application 63/116,660, filed on Nov. 20, 2020, U.S. Provisional Patent Application 63/146,421, filed on Feb. 5, 2021, U.S. Provisional Patent Application 63/145,368, filed on Feb. 3, 2021, U.S. Provisional Patent Application 63/159,768, filed on Mar. 11, 2021, U.S. Provisional Patent Application 63/225,779, filed on Jul. 26, 2021, U.S. Provisional Patent Application 63/175,021, filed on Apr. 14, 2021, U.S. Provisional Patent Application 63/208,759, filed on Jun. 9, 2021 U.S. Provisional Patent Application 63/173,253, filed on Apr. 9, 2021, U.S. Provisional Patent Application 63/245,011, filed on Sep. 16, 2021, U.S. Provisional Patent Application 63/178,501, filed on Apr. 22, 2021, U.S. Provisional Patent Application 63/192,852, filed on May 25, 2021, and U.S. Provisional Patent Application 63/223,685, filed on Jul. 20, 2021. The entire disclosures of the above applications are hereby incorporated by reference.

TECHNICAL FIELD

[0002] This document describes data processing systems that include optical communication modules.

BACKGROUND

[0003] This section introduces aspects that can help facilitate a better understanding of the

disclosure. Accordingly, the statements of this section are to be read in this light and are not to be understood as admissions about what is in the prior art or what is not in the prior art.

[0004] As the input/output (I/O) capacities of electronic processing chips increase, electrical signals may not provide sufficient input/output capacity across the limited size of a practically viable electronic chip package. For example, some data centers include racks of data processing servers (e.g., switch servers) and use optical fibers to transmit optical signals between the data processing servers. Each data processing server receives first optical signals from optical fiber cables, converts the first optical signals to first electrical signals, perform operations (e.g., switching operations) on the first electrical signals to generate second electrical signals, convert the second electrical signals to second optical signals, and outputs the second optical signals through the optical fiber cables.

SUMMARY OF THE INVENTION

[0005] In a general aspect, an apparatus includes: at least one of a circuit board or a substrate; a data processor unit mounted on a first side of the circuit board or the substrate; and a first structure attached to a second side of the circuit board or the substrate. The second side of the circuit board or the substrate includes a two-dimensional arrangement of second electrical contacts. The first structure is configured to enable an optical module with a two-dimensional arrangement of first electrical contacts to be removably coupled to the first structure and enable the optical module to be electrically coupled to the circuit board or the substrate through the two-dimensional arrangement of first electrical contacts and the two-dimensional arrangement of second electrical contacts.

[0006] Implementations can include one or more of the following features. The optical module can include a connector configured to enable an optical fiber connector to be removably coupled to the optical module.

[0007] The data processor unit can include at least one of a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, a storage device, or an application specific integrated circuit (ASIC) that is configured to process electrical signals received from or transmitted to the optical module.

[0008] The data processor unit can include at least one of (i) a bare die processor that is mounted on the first side of the circuit board or the substrate, or (ii) a packaged processor that includes a package substrate that is mounted on the first side of the circuit board or the substrate.

[0009] The first structure can include walls that define a first opening. The walls can also define one or more retaining mechanisms such that when the optical module is inserted into the first opening, the one or more retaining mechanisms on the walls of the first structure engage one or more latch mechanisms on the optical module to secure the optical module to the first structure, and the first electrical contacts on the optical module are electrically coupled to the second electrical contacts on the circuit board or the substrate.

[0010] The apparatus can include at least one optical module. The optical module can include a mechanical connector structure configured to removably couple the optical module to the circuit board or the substrate to enable electrical signals to be transmitted between the optical module and the second electrical contacts of the circuit board or the substrate.

[0011] The optical module can include a connector configured to enable an optical fiber connector to be removably coupled to the optical module. The mechanical connector structure can be configured to receive the optical fiber connector to enable light signals from the optical fiber connector to be transmitted to the optical module.

[0012] The apparatus can include the optical fiber connector. The optical fiber connector can be optically coupled to a fiber cable that includes a plurality of optical fibers. The optical fiber connector can be configured to transmit optical signals carried in the optical fibers to the optical module.

[0013] The optical module can include a photonic integrated circuit that includes a two-dimensional array of optical coupling elements. The optical fiber connector can be configured to

couple a two-dimensional array of optical fibers to the two-dimensional array of optical coupling elements on the photonic integrated circuit.

[0014] The two-dimension array of optical fibers can include at least a 3×12 array of optical fibers.

[0015] The arrangement of second electrical contacts of the circuit board or the substrate can include at least four rows and four columns of second electrical contacts. The arrangement of first electrical contacts of the optical module can include at least four rows and four columns of first electrical contacts. The first structure can be configured to enable the at least four rows and four columns of first electrical contacts of the optical module to be electrically coupled to the at least four rows and four columns of second electrical contacts of the circuit board or the substrate when the optical module is coupled to the first structure.

[0016] The first structure can include a grid structure that defines multiple openings. Each opening can be configured to receive a corresponding optical module.

[0017] The grid structure can be configured such that when the optical modules are inserted into the openings of the grid structure, the optical modules are oriented such that each optical module is rotated 90 degrees relative to at least one adjacent optical module, and the rotational axis is perpendicular to the circuit board or the substrate.

[0018] The first structure can be configured to enable the optical modules to be mounted on the first structure in a 90-degree rotate checkerboard fashion.

[0019] The first structure can be configured to function as a heat spreader.

[0020] The first structure can have an opening located opposite the data processor unit relative to the circuit board or the substrate, discrete circuit components can be mounted on the second side of the circuit board or the substrate, and the discrete circuit components can extend from the circuit board or the substrate into the opening in the first structure.

[0021] The apparatus can include the circuit board and the substrate. The first structure can be attached to the second side of the circuit board, the data processor unit can be mounted on the first side of the substrate, the opening in the first structure can be located opposite the data processor unit relative to the substrate, discrete circuit components can be mounted on the second side of the substrate, and the discrete circuit components can extend from the substrate into the opening in the structure.

[0022] The discrete circuit components can include at least one of one or more voltage regulators, one or more filters, or one or more capacitors.

[0023] A second structure can be attached to a first side of the circuit board or the substrate, and the first structure can be mechanically and thermally attached to the second structure.

[0024] The first structure can be attached to the second structure by screws that pass through the printed circuit board or the substrate.

[0025] The first structure can be attached to the second structure by thermal vias.

[0026] The apparatus can include a heat sink attached to at least one of the first structure or the second structure.

[0027] The apparatus can include a snap-in mechanism that is configured to secure the optical module when the optical module is inserted into the first structure.

[0028] The snap-in mechanism can be configured to enable the optical module to be pulled away from the first structure when a force above a threshold is applied to the optical module.

[0029] The snap-in mechanism can include one or more grooves formed on walls of the first structure, the optical module can include one or more elastic wings, and each elastic wing can include a tongue that is configured to engage a corresponding groove when the optical module is inserted into the first structure.

[0030] The snap-in mechanism can include a lever-based latch mechanism, the latch mechanism can be movable between a first position and a second position, the latch mechanism can engage a support structure on the first structure when in the first position and disengage from the support structure when in the second position, and the optical module can be secured to the first structure

when the latch mechanism is in the first position and released from the first structure when the latch mechanism is in the second position.

[0031] A lever can be provided as part of the optical module. The lever can be movable between a first position and a second position. The lever can be configured such that moving the lever to the first position causes the latch mechanism to move to the first position, and moving the lever to the second position causes the latch mechanism to move to the second position.

[0032] A lever can be provided as part of a tool used to insert or remove the optical module into or from the first structure.

[0033] The optical module can include a co-packaged optical module.

[0034] The co-packaged optical module can include: a photonic integrated circuit configured to perform at least one of converting first optical signals to first electrical signals or converting second electrical signals to second optical signals; and the two-dimensional arrangement of first electrical contacts. The two-dimensional arrangement of first electrical contacts can be configured to at least one of (i) enable the first electrical signals to be transmitted from the photonic integrated circuit to the circuit board or the substrate, or (ii) enable the second electrical signals to be transmitted from the circuit board or the substrate to the photonic integrated circuit.

[0035] The co-packaged optical module can include an optical fiber connector part configured to be optically coupled to an array of optical fibers. The optical fiber connector part can be configured to at least one of (i) receive the first optical signals from the array of optical fibers and transmit the first optical signals to the photonic integrated circuit, or (ii) receive the second optical signals from the photonic integrated circuit and transmit the second optical signals to the array of optical fibers.

[0036] The co-packaged optical module can include at least one of: micro-optics assembly facilitating fiber-to-photonic integrated circuit coupling, one or more driver amplifiers, one or more trans-impedance amplifiers, one or more SerDes integrated circuits, one or more digital signal processors, one or more lasers or light emitting diodes, or one or more microcontrollers.

[0037] The apparatus can include the optical module. The optical module can include a snap-in mechanism configured such that the optical fiber connector locks in place the snap-in mechanism when the optical fiber connector is coupled to the optical module.

[0038] The optical module can include a mechanical connector structure. The optical fiber connector can snap into a part of the mechanical connector structure to hold the optical fiber connector in place when the optical fiber connector is coupled to the optical module.

[0039] The apparatus can include the optical fiber connector. The optical fiber connector and the optical module can include a ball-detent mechanism configured to hold the optical fiber connector in place when the optical fiber connector is coupled to the optical module.

[0040] In another general aspect, an apparatus includes: an optical module configured to be removably coupled to a first structure that is attached to a circuit board or a substrate. The optical module includes a two-dimensional arrangement of at least 16 electrical contacts distributed in a two-dimensional surface area of the optical module. The optical module includes a photonic integrated circuit, the optical module is configured to hold the photonic integrated circuit in place when the optical module is coupled to the first structure and to enable electronic signals from the photonic integrated circuit to be transmitted to the circuit board or the substrate through the two-dimensional arrangement of at least 16 electrical contacts. The optical module includes a connector part that is configured to enable an optical fiber connector to be removably coupled to the optical module, in which the optical module is configured to enable optical signals from the optical fiber connector to be transmitted to the photonic integrated circuit.

[0041] Implementations can include one or more of the following features. The optical fiber connector can be configured to couple a two-dimensional array of optical fibers to a two-dimensional array of optical coupling elements on the photonic integrated circuit.

[0042] The two-dimension array of optical fibers can include at least a 3×12 array of optical fibers.

[0043] The apparatus can include at least one of a network switch, a central processor unit, a

graphics processor unit, a tensor network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, a storage device, or an application specific integrated circuit (ASIC) that is coupled to the circuit board or the substrate and configured to process electrical signals received from or transmitted to the photonic integrated circuit.

[0044] The apparatus can include the first structure and at least one of the circuit board or the substrate. The circuit board or the substrate can be attached to a first side of the first structure, and the optical module can be configured to be removably coupled to the first structure from the second side of the first structure, and the second side of the first structure is opposite to the first side of the first structure.

[0045] The apparatus can include two or more optical modules that are removably coupled to the first structure in a 90-degree rotate checkerboard fashion.

[0046] The apparatus can include the circuit board and the substrate. The circuit board can be attached to the first side of the first structure, the substrate can be attached to the circuit board, and the optical module can be configured to hold the photonic integrated circuit in place when the optical module is coupled to the first structure and to enable electronic signals from the photonic integrated circuit to be transmitted to the substrate.

[0047] The first structure can include a grid structure that enables two or more optical modules to be removably coupled to the first structure in an array defined by the grid structure.

[0048] In another general aspect, a system includes: a housing; and any of the apparatuses described above, in which the optical module is configured to be removably coupled to, partially through, or through a front panel of the housing.

[0049] Implementations can include one or more of the following features. The first structure can be part of the front panel of the housing.

[0050] The circuit board can be part of the front panel of the housing.

[0051] The first structure can be positioned near and spaced apart from the front panel of the housing.

[0052] The first structure can have an overall structure that extends along a plane that is substantially parallel to the front panel of the housing.

[0053] The circuit board can be substantially parallel to the front panel of the housing.

[0054] In another general aspect, a method includes: providing at least one of a circuit board or a substrate, in which a data processor unit is mounted on a first side of the circuit board or the substrate, a first structure is attached to a second side of the circuit board or the substrate, and the second side of the circuit board or the substrate includes a two-dimensional arrangement of second electrical contacts. The method includes removably coupling an optical module to the first structure, in which the optical module includes a two-dimensional arrangement of first electrical contacts. The method includes electrically coupling the optical module to the second side of the circuit board or the substrate through the two-dimensional arrangement of first electrical contacts and the two-dimensional arrangement of second electrical contacts.

[0055] Implementations can include one or more of the following features. The method can include removably coupling an optical fiber connector to the optical module.

[0056] The method can include: inserting the optical module through a first opening defined by walls of the first structure; using one or more retaining mechanisms on the walls of the first structure to engage one or more latch mechanisms on the optical module to secure the optical module to the first structure; and electrically coupling the second electrical contacts on the circuit board or the substrate to the first electrical contacts on the optical module.

[0057] The method can include: transmitting optical signals from an optical fiber connector to the optical module; at the optical module, generating electrical signals based on the optical signals; and transmitting the electrical signals from the optical module to the circuit board or the substrate.

[0058] The optical module can include a photonic integrated circuit that includes a two-

dimensional array of optical coupling elements. The method can include using the optical fiber connector to couple a two-dimensional array of optical fibers to the two-dimensional array of optical coupling elements on the photonic integrated circuit.

[0059] The two-dimension array of optical fibers can include at least a 3×12 array of optical fibers.

[0060] The arrangement of second electrical contacts on the circuit board or the substrate can include an array of at least four rows and four columns of electrical contacts, and the arrangement of first electrical contacts of the optical module can include an array of at least four rows and four columns of electrical contacts. Electrically coupling the optical module to the second side of the circuit board or the substrate can include electrically coupling the array of at least four rows and four columns of electrical contacts of the optical module to the array of at least four rows and four columns of electrical contacts of the circuit board or the substrate.

[0061] The first structure can include a grid structure that defines multiple first openings. The method can include: removably coupling a plurality of optical modules to the grid structure, including inserting each optical module through a corresponding first opening defined by the walls of the grid structure. The method can include for each of the plurality of optical modules, using one or more retaining mechanisms on the walls of the grid structure to engage one or more latch mechanisms on the optical module to secure the optical module to the grid structure. The method can include for each of the plurality of optical modules, electrically coupling the first electrical contacts on the optical module to corresponding second electrical contacts on the circuit board or the substrate.

[0062] The method can include orienting the optical modules such that each optical module is rotated 90 degrees relative to at least one adjacent optical module, and the rotational axis is perpendicular to the circuit board or the substrate.

[0063] The method can include removably coupling a plurality of optical modules to the first structure; and orienting the optical modules coupled to the first structure in a 90-degree rotate checkerboard fashion.

[0064] Removably coupling the optical module to the first structure can include securing, using a snap-in mechanism, the optical module to the first structure.

[0065] The method can include applying a force above a threshold to the optical module to disengage the snap-in mechanism, and pulling the optical module away from the first structure.

[0066] The snap-in mechanism can include one or more grooves formed on walls of the first structure, the optical module includes one or more elastic wings, and each elastic wing includes a tongue. The method can include engaging the tongue of the elastic wing with a corresponding groove formed on walls of the first structure when inserting the optical module into an opening defined by the walls of the first structure.

[0067] The snap-in mechanism can include a lever-based latch mechanism, the latch mechanism can be movable between a first position and a second position, and the latch mechanism can engage a support structure on the first structure when in the first position and disengages from the support structure when in the second position. The method can include placing the latch mechanism in the first position to secure the optical module to the first structure; and placing the latch mechanism in the second position to release the optical module from the first structure.

[0068] A lever can be provided as part of the optical module. The method can include: moving the lever to a first position to cause the latch mechanism to move to the first position, and moving the lever to a second position causes the latch mechanism to move to the second position.

[0069] The optical module can include a co-packaged optical module.

[0070] The co-packaged optical module can include: a photonic integrated circuit configured to perform at least one of converting first optical signals to first electrical signals or converting second electrical signals to second optical signals; and the two-dimensional arrangement of first electrical contacts. The two-dimensional arrangement of first electrical contacts can be configured to at least one of (i) enable the first electrical signals to be transmitted from the photonic integrated circuit to

the circuit board or the substrate, or (ii) enable the second electrical signals to be transmitted from the circuit board or the substrate to the photonic integrated circuit.

[0071] The co-packaged optical module can include an optical fiber connector part configured to be optically coupled to an array of optical fibers. The optical fiber connector part can be configured to at least one of (i) receive the first optical signals from the array of optical fibers and transmit the first optical signals to the photonic integrated circuit, or (ii) receive the second optical signals from the photonic integrated circuit and transmit the second optical signals to the array of optical fibers.

[0072] The co-packaged optical module can include at least one of: micro-optics assembly facilitating fiber-to-photonic integrated circuit coupling, one or more driver amplifiers, one or more trans-impedance amplifiers, one or more SerDes integrated circuits, one or more digital signal processors, one or more lasers or light emitting diodes, or one or more microcontrollers.

[0073] In another general aspect, a method includes: attaching a first structure to a first side of a circuit board or a substrate, in which the circuit board or the substrate includes a plurality of sets of two-dimensional arrangements of electrical contacts on the first side. The method includes mounting an application specific integrated circuit on a second side of the circuit board or the substrate. The first structure has a first opening located opposite the application specific integrated circuit relative to the circuit board or the substrate, the first structure defines second openings that are configured to receive optical modules, each second opening corresponding to a set of two-dimensional arrangement of electrical contacts in the plurality of sets of two-dimensional arrangements of electrical contacts. The first structure is configured to enable each optical module to be electrically coupled to the circuit board or the substrate through the corresponding set of two-dimensional arrangement of electrical contacts when the optical module is inserted through the corresponding second opening defined by the first structure. The method includes mounting discrete circuit components on the first side of the circuit board or the substrate, in which the discrete circuit components extend from the circuit board or the substrate into the first opening in the first structure.

[0074] Implementations can include one or more of the following features. The method can include: attaching the first structure to the first side of the circuit board; mounting the application specific integrated circuit on the second side of the substrate; and mounting discrete circuit components on the first side of the substrate. The discrete circuit components can extend from the substrate into the opening in the structure.

[0075] The discrete circuit components can include at least one of one or more voltage regulators, one or more filters, or one or more capacitors.

[0076] The method can include attaching a second structure to a second side of the circuit board or the substrate, and mechanically and thermally coupling the first structure to the second structure.

[0077] The method can include attaching the first structure to the second structure using screws that pass through the circuit board or the substrate.

[0078] The method can include attaching the first structure to the second structure using thermal vias.

[0079] The method can include attaching a heat sink to at least one of the first structure or the second structure.

[0080] In another general aspect, an apparatus includes: at least one of a circuit board or a substrate; and a first structure attached to the at least one of a circuit board or a substrate, in which the first structure is configured to enable an optical module with connector to be removably coupled to the first structure, and the optical module with connector is configured to enable an optical fiber connector to be removably coupled to the optical module with connector.

[0081] Implementations can include one or more of the following features. The circuit board or the substrate can include first electrical contacts, the first structure can include walls that define a first opening, the walls also define one or more retaining mechanisms such that when the optical module with connector is inserted into the first opening, the one or more retaining mechanisms on the walls

of the first structure engage one or more latch mechanisms on the optical module with connector to secure the optical module with connector to the first structure, and second electrical contacts on the optical module with connector are electrically coupled to the first electrical contacts on the circuit board or the substrate.

[0082] The apparatus can include at least one optical module with connector. The optical module with connector can include an optical module and a mechanical connector structure, the mechanical connector structure can be configured to removably couple the optical module to the circuit board or the substrate to enable electrical signals output from the optical module to be transmitted to the first electrical contacts of the circuit board or the substrate.

[0083] The mechanical connector structure can be configured to receive the optical fiber connector to enable light signals from the optical fiber connector to be transmitted to the optical module.

[0084] The apparatus can include the optical fiber connector, in which the optical fiber connector can be optically coupled to a fiber cable that includes a plurality of optical fibers, and the optical fiber connector can be configured to transmit optical signals carried in the optical fibers to the optical module.

[0085] The circuit board or the substrate can include an array of first electrical contacts including at least four rows and four columns of electrical contacts, the optical module can include an array of second electrical contacts including at least four rows and four columns of electrical contacts, and the first structure can be configured to enable the array of second electrical contacts of the optical module to be electrically coupled to the array of first electrical contacts of the circuit board or the substrate when the optical module with connector is coupled to the first structure.

[0086] The first structure can include a grid structure that defines multiple openings, and each opening can be configured to receive a corresponding optical module with connector.

[0087] The grid structure can be configured such that when the optical module with connectors are inserted into the openings of the grid structure, the optical module with connectors are oriented such that each optical module with connector is rotated 90 degrees relative to at least one adjacent optical module with connector, and the rotational axis is perpendicular to the circuit board or the substrate.

[0088] The first structure can be configured to enable the optical module with connectors to be mounted on the first structure in a 90-degree rotate checkerboard fashion.

[0089] The first structure can be configured to function as a heat spreader.

[0090] Each of the at least one of the circuit board or substrate can include a first side and a second side, the first structure can be attached to the first side of the circuit board or the substrate, an application specific integrated circuit can be mounted on the second side of the circuit board or substrate, the first structure can have an opening, the opening can be located opposite the application specific integrated circuit relative to the circuit board or substrate, discrete circuit components can be mounted on the first side of the circuit board or substrate, and the discrete circuit components can extend from the circuit board or substrate into the opening in the structure.

[0091] The apparatus can include the circuit board and the substrate. The first structure can be attached to the first side of the circuit board, an application specific integrated circuit can be mounted on the second side of the substrate, the first structure can have an opening, the opening can be located opposite the application specific integrated circuit relative to the substrate, discrete circuit components can be mounted on the first side of the substrate, and the discrete circuit components can extend from the substrate into the opening in the structure.

[0092] The discrete circuit components can include at least one of one or more voltage regulators, one or more filters, or one or more capacitors.

[0093] The circuit board or substrate can include a first side and a second side, the first structure can be attached to the first side of the circuit board or substrate, a second structure can be attached to the second side of the circuit board or substrate, and the first structure can be mechanically and thermally attached to the second structure.

[0094] The first structure can be attached to the second structure by screws that pass through the printed circuit board or substrate.

[0095] The first structure can be attached to the second structure by thermal vias.

[0096] The apparatus can include a heat sink attached to at least one of the first structure or the second structure.

[0097] The apparatus can include a snap-in mechanism that is configured to secure the optical module with connector when the optical module with connector is inserted into the first structure.

[0098] The snap-in mechanism can be configured to enable the optical module with connector to be pulled away from the first structure when a force above a threshold is applied to the optical module with connector.

[0099] The snap-in mechanism can include one or more grooves formed on walls of the first structure, the optical module with connector can include one or more elastic wings, and each elastic wing can include a tongue that is configured to engage a corresponding groove when the optical module with connector is inserted into the first structure.

[0100] The snap-in mechanism can include a lever-based latch mechanism, the latch mechanism can be movable between a first position and a second position, the latch mechanism can engage a support structure on the first structure when in the first position and disengage from the support structure when in the second position, the optical module with connector can be secured to the first structure when the latch mechanism is in the first position and released from the first structure when the latch mechanism is in the second position.

[0101] A lever can be provided as part of the optical module with connector, the lever can be movable between a first position and a second position, the lever can be configured such that moving the lever to the first position causes the latch mechanism to move to the first position, and moving the lever to the second position can cause the latch mechanism to move to the second position.

[0102] A lever can be provided as part of a tool used to insert or remove the optical module with connector into or from the first structure.

[0103] In any of the apparatuses described above, the optical module with connector can include a co-packaged optical module.

[0104] The apparatus can include the optical module with connector, in which the optical module with connector can include a snap-in mechanism configured such that the optical fiber connector locks in place the snap-in mechanism when the optical fiber connector is coupled to the optical module with connector.

[0105] The optical module with connector can include a mechanical connector structure, and the optical fiber connector can snap into a part of the mechanical connector structure to hold the optical fiber connector in place when the optical fiber connector is coupled to the optical module with connector.

[0106] The apparatus can include the optical fiber connector, in which the optical fiber connector and the optical module with connector can include a ball-detent mechanism configured to hold the optical fiber connector in place when the optical fiber connector is coupled to the optical module with connector.

[0107] In another general aspect, an apparatus includes: an optical module with connector configured to be removably coupled to a first structure that is attached to a circuit board or a substrate. The optical module includes a photonic integrated circuit, the optical module with connector is configured to hold the photonic integrated circuit in place when the optical module with connector is coupled to the first structure and to enable electronic signals from the photonic integrated circuit to be transmitted to the circuit board or the substrate. The optical module with connector is configured to enable an optical fiber connector to be removably coupled to the optical module with connector, in which the optical module with connector is configured to enable optical signals from the optical fiber connector to be transmitted to the photonic integrated circuit.

[0108] Implementations can include one or more of the following features. The apparatus can include at least one of a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, a storage device, or an application specific integrated circuit (ASIC) that is coupled to the circuit board or the substrate and configured to process electrical signal received from or transmitted to the photonic integrated circuit.

[0109] The apparatus can include the first structure and at least one of the circuit board or the substrate. The circuit board or the substrate can be attached to a first side of the first structure, and the optical module with connector can be configured to be removably coupled to the first structure from the second side of the first structure, and the second side of the first structure is opposite to the first side of the first structure.

[0110] The apparatus can include two or more optical module with connectors that are removably coupled to the first structure in a 90-degree rotate checkerboard fashion.

[0111] The apparatus can include the circuit board and the substrate. The circuit board can be attached to a first side of the first structure, the substrate can be attached to the circuit board, and the optical module with connector can be configured to hold the photonic integrated circuit in place when the optical module with connector is coupled to the first structure and to enable electronic signals from the photonic integrated circuit to be transmitted to the substrate.

[0112] The first structure can include a grid structure that enables two or more optical module with connectors to be removably coupled to the first structure in an array defined by the grid structure.

[0113] In another general aspect, a system includes: a housing; and any of the apparatuses described above, in which the optical module with connector is configured to be removably coupled to, partially through, or through a front panel of the housing.

[0114] Implementations can include one or more of the following features. The first structure can be part of the front panel of the housing.

[0115] The circuit board can be part of the front panel of the housing.

[0116] The first structure can be positioned near and spaced apart from the front panel of the housing.

[0117] The first structure can have an overall structure that extends along a plane that is substantially parallel to the front panel of the housing.

[0118] The circuit board can be substantially parallel to the front panel of the housing.

[0119] In another general aspect, a method includes: removably coupling an optical fiber connector to an optical module with connector; removably coupling the optical module with connector to a first structure attached to at least one of a circuit board or a substrate; and electrically coupling the optical module with connector to the circuit board or the substrate.

[0120] Implementations can include one or more of the following features. The method can include: inserting the optical module with connector through a first opening defined by walls of the first structure; using one or more retaining mechanisms on the walls of the first structure to engage one or more latch mechanisms on the optical module with connector to secure the optical module with connector to the first structure; and electrically coupling first electrical contacts on the circuit board or the substrate to second electrical contacts on the optical module with connector.

[0121] The method can include: transmitting optical signals from the optical fiber connector to the optical module with connector; at the optical module with connector, generating electrical signals based on the optical signals; and transmitting the electrical signals from the optical module with connector to the circuit board or the substrate.

[0122] The circuit board or the substrate can include an array of first electrical contacts including at least four rows and four columns of electrical contacts, the optical module with connector can include an array of second electrical contacts including at least four rows and four columns of electrical contacts. Electrically coupling the optical module with connector to the circuit board or the substrate can include electrically coupling the array of second electrical contacts of the optical

module with connector to the array of first electrical contacts of the circuit board or the substrate.

[0123] The first structure can include a grid structure that defines multiple first openings. The method can include: removably coupling a plurality of optical modules with connectors to the grid structure, including inserting each optical module with connector through a corresponding first opening defined by the walls of the grid structure. The method can include: for each of the plurality of optical modules with connectors, using one or more retaining mechanisms on the walls of the grid structure to engage one or more latch mechanisms on the optical module with connector to secure the optical module with connector to the grid structure. The method can include: for each of the plurality of optical modules with connectors, electrically coupling second electrical contacts on the optical module with connector to corresponding first electrical contacts on the circuit board or the substrate.

[0124] The method can include orienting the optical modules with connectors such that each optical module with connector is rotated 90 degrees relative to at least one adjacent optical module with connector, and the rotational axis is perpendicular to the circuit board or the substrate.

[0125] The method can include removably coupling a plurality of optical modules with connectors to the first structure; and orienting the optical modules with connectors coupled to the first structure in a 90-degree rotate checkerboard fashion.

[0126] Removably coupling the optical module with connector to the first structure can include securing, using a snap-in mechanism, the optical module with connector to the first structure.

[0127] The method can include applying a force above a threshold to the optical module with connector to disengage the snap-in mechanism, and pulling the optical module with connector away from the first structure.

[0128] The snap-in mechanism can include one or more grooves formed on walls of the first structure, the optical module with connector can include one or more elastic wings, and each elastic wing can include a tongue. The method can include engaging the tongue of the elastic wing with a corresponding groove formed on walls of the first structure when inserting the optical module with connector into an opening defined by the walls of the first structure.

[0129] The snap-in mechanism can include a lever-based latch mechanism, the latch mechanism can be movable between a first position and a second position, the latch mechanism can engage a support structure on the first structure when in the first position and disengage from the support structure when in the second position. The method can include placing the latch mechanism in the first position to secure the optical module with connector to the first structure; and placing the latch mechanism in the second position to release the optical module with connector from the first structure.

[0130] A lever can be provided as part of the optical module with connector. The method can include: moving the lever to a first position to cause the latch mechanism to move to the first position, and moving the lever to a second position causes the latch mechanism to move to the second position.

[0131] The optical module with connector can include a co-packaged optical module.

[0132] In another general aspect, a method includes: attaching a first structure to a first side of a circuit board or a substrate; and mounting an application specific integrated circuit on a second side of the circuit board or the substrate. The first structure has an opening located opposite the application specific integrated circuit relative to the circuit board or the substrate. The method includes: mounting discrete circuit components on the first side of the circuit board or the substrate. The discrete circuit components extend from the circuit board or the substrate into the opening in the structure.

[0133] Implementations can include one or more of the following features. The method can include: attaching the first structure to the first side of the circuit board; mounting the application specific integrated circuit on the second side of the substrate; and mounting discrete circuit components on the first side of the substrate. The discrete circuit components can extend from the

substrate into the opening in the structure.

[0134] The discrete circuit components can include at least one of one or more voltage regulators, one or more filters, or one or more capacitors.

[0135] The method can include attaching a second structure to a second side of the circuit board or the substrate, and mechanically and thermally coupling the first structure to the second structure.

[0136] The method can include attaching the first structure to the second structure using screws that pass through the circuit board or the substrate.

[0137] The method can include attaching the first structure to the second structure using thermal vias.

[0138] The method can include attaching a heat sink to at least one of the first structure or the second structure.

[0139] Other aspects include other combinations of the features recited above and other features, expressed as methods, apparatus, systems, program products, and in other ways.

[0140] Interconnecting electronic chip packages using optical signals can have the advantage that the optical signals can be delivered with a higher input/output capacity per unit area compared to electrical input/outputs.

[0141] Particular embodiments of the subject matter described in this specification can be implemented to realize one or more of the following advantages. The data processing system has a high power efficiency, a low construction cost, a low operation cost, and high flexibility in reconfiguring optical network connections.

[0142] The details of one or more embodiments of the subject matter described in this specification are set forth in the accompanying drawings and the description below. Other features, aspects, and advantages of the invention will become apparent from the description, the drawings, and the claims.

[0143] Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. In case of conflict with patent applications or patent application publications incorporated herein by reference, the present specification, including definitions, will control.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0144] The disclosure is best understood from the following detailed description when read in conjunction with the accompanying drawings. It is emphasized that, according to common practice, the various features of the drawings are not to-scale. The dimensions of the various features can be arbitrarily expanded or reduced for clarity.

[0145] FIG. 1 is a block diagram of an example optical communication system.

[0146] FIG. 2 is a schematic side view of an example data processing system.

[0147] FIG. 3 is a schematic side view of an example integrated optical device.

[0148] FIG. 4 is a schematic side view of an example data processing system.

[0149] FIG. 5 is a schematic side view of an example integrated optical device.

[0150] FIGS. 6 and 7 are schematic side views of examples of data processing systems.

[0151] FIG. 8 is an exploded perspective view of an integrated optical communication device.

[0152] FIGS. 9 and 10 are diagrams of example layout patterns of optical and electrical terminals of integrated optical devices.

[0153] FIGS. 11, 12, 13, and 14 are schematic side views of examples of data processing systems.

[0154] FIGS. 15 and 16 are bottom views of examples of integrated optical devices.

[0155] FIG. 17 is a diagram showing various types of integrated optical communication devices that can be used in a data processing system.

[0156] FIG. **18** is a diagram of an example octal serializers/deserializers block.

[0157] FIG. **19** is a diagram of an example electronic communication integrated circuit.

[0158] FIG. **20** is a functional block diagram of an example data processing system.

[0159] FIG. **21** is a diagram of an example rackmount data processing system.

[0160] FIGS. **22, 23, 24, 25, 26A, 26B, 26C, 27, 28A, and 28B** are top view diagrams of examples of rackmount data processing systems incorporating optical interconnect modules.

[0161] FIGS. **29A and 29B** are diagrams of an example rackmount data processing system incorporating multiple optical interconnect modules.

[0162] FIGS. **30 and 31** are block diagrams of example data processing systems.

[0163] FIG. **32** is a schematic side view of an example data processing system.

[0164] FIG. **33** is a diagram of an example electronic communication integrated circuit that includes octal serializers/deserializers blocks.

[0165] FIG. **34** is a flow diagram of an example process for processing optical and electrical signals using a data processing system.

[0166] FIG. **35A** is a diagram an optical communications system.

[0167] FIGS. **35B and 35C** are diagrams of co-packaged optical interconnect modules.

[0168] FIGS. **36 and 37** are diagrams of examples of optical communications systems.

[0169] FIGS. **38 and 39** are diagrams of examples of serializers/deserializers blocks.

[0170] FIGS. **40A, 40B, 41A, 41B, and 42** are diagrams of examples of bus processing units.

[0171] FIG. **43** is an exploded view of an example of a front-mounted module of a data processing system.

[0172] FIG. **44** is an exploded view of an example of the internals of an optical module.

[0173] FIG. **45** is an assembled view of the internals of an optical module.

[0174] FIG. **46** is an exploded view of an optical module.

[0175] FIG. **47** is an assembled view of an optical module.

[0176] FIG. **48** is a diagram of a portion of a grid structure and a circuit board.

[0177] FIG. **49** is a diagram showing a lower mechanical part prior to insertion into the grid structure.

[0178] FIG. **50** is a diagram of an example of a partially populated front-view of an assembled system.

[0179] FIG. **51A** is a front view of an example of the mounting of the module.

[0180] FIG. **51B** is a side view of an example of the mounting of the module.

[0181] FIG. **52A** is a front view of an example of the mechanical connector structure and an optical module mounted within a grid structure.

[0182] FIG. **52B** is a side view of an example of the mechanical connector structure and an optical module mounted within a grid structure.

[0183] FIGS. **53 and 54** are diagrams of an example of an assembly that includes a fiber cable, an optical fiber connector, a mechanical connector module, and a grid structure.

[0184] FIGS. **55A and 55B** are perspective views of the mechanisms shown in FIGS. **53 and 54** before the optical fiber connector is inserted into the mechanical connector structure.

[0185] FIG. **56** is a perspective view showing that the optical module and the mechanical connector structure are inserted into the grid structure.

[0186] FIG. **57** is a perspective view showing that the optical fiber connector is mated with the mechanical connector structure.

[0187] FIGS. **58A to 58D** are diagrams of an example an optical module that includes a latch mechanism.

[0188] FIG. **59** is a diagram of an alternative example of the optical module.

[0189] FIGS. **60A and 60B** are diagrams of an example implementation of the lever and the latch mechanism in the optical module with connector.

[0190] FIG. **61** is a diagram of cross section of the module viewed from the front mounted in the

assembly with the connector.

[0191] FIGS. **62** to **65** are diagrams showing cross-sectional views of an example of a fiber cable connection design.

[0192] FIG. **66** is a map of electrical contact pads.

[0193] FIG. **67** is a top view of an example of a rackmount server.

[0194] FIG. **68A** is a top view of an example of a rackmount server.

[0195] FIG. **68B** is a diagram of an example of a front panel of the rackmount server.

[0196] FIG. **68C** is a perspective view of an example of a heat sink.

[0197] FIG. **69A** is a top view of an example of a rackmount server.

[0198] FIG. **69B** is a diagram of an example of a front panel of the rackmount server.

[0199] FIG. **70** is a top view of an example of a rackmount server.

[0200] FIG. **71A** is a top view of an example of a rackmount server.

[0201] FIG. **71B** is a front view of the rackmount server.

[0202] FIG. **72** is a top view of an example of a rackmount server.

[0203] FIG. **73A** is a top view of an example of a rackmount server.

[0204] FIG. **73B** is a front view of the rackmount server.

[0205] FIG. **74A** is a top view of an example of a rackmount server.

[0206] FIG. **74B** is a front view of the rackmount server.

[0207] FIG. **75A** is a top view of an example of a rackmount server.

[0208] FIG. **75B** is a front view of the rackmount server.

[0209] FIG. **75C** is a diagram of the air flow in the rackmount server.

[0210] FIG. **76** is a diagram of a network rack that includes a plurality of rackmount servers.

[0211] FIG. **77A** is a side view of an example of a rackmount server.

[0212] FIG. **77B** is a top view of the rackmount server.

[0213] FIG. **78** is a top view of an example of a rackmount server.

[0214] FIG. **79** is a block diagram of an example of an optical communication system.

[0215] FIG. **80A** is a diagram of an example of an optical communication system.

[0216] FIG. **80B** is a diagram of an example of an optical cable assembly used in the optical communication system of FIG. **80A**.

[0217] FIG. **80C** is an enlarged diagram of the optical cable assembly of FIG. **80B**.

[0218] FIG. **80D** is an enlarged diagram of the upper portion of the optical cable assembly of FIG. **80B**.

[0219] FIG. **80E** is an enlarged diagram of the lower portion of the optical cable assembly of FIG. **80B**.

[0220] FIG. **81** is a block diagram of an example of an optical communication system.

[0221] FIG. **82A** is a diagram of an example of an optical communication system.

[0222] FIG. **82B** is a diagram of an example of an optical cable assembly.

[0223] FIG. **82C** is an enlarged diagram of the optical cable assembly of FIG. **82B**.

[0224] FIG. **82D** is an enlarged diagram of the upper portion of the optical cable assembly of FIG. **82B**.

[0225] FIG. **82E** is an enlarged diagram of the lower portion of the optical cable assembly of FIG. **82B**.

[0226] FIG. **83** is a block diagram of an example of an optical communication system.

[0227] FIG. **84A** is a diagram of an example of an optical communication system.

[0228] FIG. **84B** is a diagram of an example of an optical cable assembly.

[0229] FIG. **84C** is an enlarged diagram of the optical cable assembly of FIG. **84B**.

[0230] FIGS. **85** to **87B** are diagrams of examples of data processing systems.

[0231] FIG. **88** is a diagram of an example of connector port mapping for an optical fiber interconnection cable.

[0232] FIGS. **89** and **90** are diagrams of examples of fiber port mapping for optical fiber

interconnection cables.

[0233] FIGS. **91** and **92** are diagrams of examples of viable port mapping for optical fiber connectors of universal optical fiber interconnection cables.

[0234] FIG. **93** is a diagram of an example of a port mapping for an optical fiber connector that is not appropriate for a universal optical fiber interconnection cable.

[0235] FIGS. **94** and **95** are diagrams of examples of viable port mapping for optical fiber connectors of universal optical fiber interconnection cables.

[0236] FIG. **96** is a top view of an example of a rackmount server.

[0237] FIG. **97A** is a perspective view of the rackmount server of FIG. **96**.

[0238] FIG. **97B** is a perspective view of the rackmount server of FIG. **96** with the top panel removed.

[0239] FIG. **98** is a diagram of the front portion of the rackmount server of FIG. **96**.

[0240] FIG. **99** includes perspective front and rear views of the front panel of the rackmount server of FIG. **96**.

[0241] FIG. **100** is a top view of an example of a rackmount server.

[0242] FIGS. **101**, **102**, **103A**, and **103B** are diagrams of examples of optical fiber connectors.

[0243] FIGS. **104** and **105** are a top view and a front view, respectively, of an example of a rackmount device that includes a vertical printed circuit board on which co-packaged optical modules are mounted.

[0244] FIG. **106** is a diagram of an example of an optical cable assembly.

[0245] FIG. **107** is a front view diagram of the rackmount device with the optical cable assembly.

[0246] FIG. **108** is a top view diagram of an example of a rackmount device that includes a vertical printed circuit board on which co-packaged optical modules are mounted.

[0247] FIG. **109** is a front view diagram of the rackmount device with the optical cable assembly.

[0248] FIGS. **110** and **111** are a top view and a front view, respectively, of an example of a rackmount device.

[0249] FIG. **112** is diagram of an example of a rackmount device with example parameter values.

[0250] FIGS. **113** and **114** show another example of a rackmount device with example parameter values.

[0251] FIGS. **115** and **116** are a top view and a front view, respectively, of an example of a rackmount device.

[0252] FIGS. **117** to **122** are diagrams of examples of systems that include co-packaged optical modules.

[0253] FIG. **123** is a diagram of an example of a vertically mounted processor blade.

[0254] FIG. **124** is a top view of an example of a rack system that includes several vertically mounted processor blades.

[0255] FIG. **125A** is a side view of an example of a rackmount server that has a hinged front panel.

[0256] FIG. **125B** is a diagram of an example of a rackmount server that has pluggable modules.

[0257] FIGS. **126A** to **127** are diagrams of examples of rackmount servers that have pluggable modules.

[0258] FIG. **128** is a diagram of an example of a fiber guide that includes one or more photon supplies.

[0259] FIG. **129** is a diagram of an example of a rackmount server that includes guide rails/cage to assist the insertion of fiber guides.

[0260] FIG. **130** is a diagram of an example of a CPO module with a compression plate.

[0261] FIG. **131** is a diagram of an example of a compression plate.

[0262] FIG. **132** is a diagram of an example of a U-shaped bolt.

[0263] FIG. **133** is a diagram of an example of a wave spring.

[0264] FIGS. **134** and **135A** to **135C** are diagrams of an example of compression plates secured to a front lattice structure.

[0265] FIG. **136** is an exploded front perspective view of an example of an assembly in a rackmount system that includes a substrate, a printed circuit board, a front lattice structure, a rear lattice structure, and a heat dissipating device.

[0266] FIG. **137** is an exploded rear perspective view of an example of the assembly shown in FIG. **136**.

[0267] FIG. **138** is an exploded top view of an example of the assembly shown in FIG. **136**.

[0268] FIG. **139** is an exploded side view of an example of the assembly shown in FIG. **136**.

[0269] FIG. **140** is a front perspective view of an example of the assembly that has been fastened together.

[0270] FIG. **141** is a front perspective view of an example of the assembled assembly without the front lattice structure.

[0271] FIG. **142** is a front perspective view of an example of the substrate, the rear lattice structure, and the heat dissipating device that have been fastened together.

[0272] FIG. **143** is a front perspective view of an example of the rear lattice structure and the heat dissipating device that have been fastened together.

[0273] FIG. **144** is a front perspective view of an example of the heat dissipating device and the screws.

[0274] FIG. **145** is a rear perspective view of an example of the assembly that has been fastened together.

[0275] FIG. **146** is a rear perspective view of an example of the assembly without the rear lattice structure.

[0276] FIG. **147** is a rear perspective view of an example of the front lattice structure, the printed circuit board, and the substrate that have been fastened together.

[0277] FIG. **148** is a rear perspective view of an example of the front lattice structure and the printed circuit board that have been fastened together.

[0278] FIG. **149** is a rear perspective view of an example of the front lattice structure.

[0279] FIG. **150** is a diagram of an example of a configuration for connecting a data processing chip to CPO modules.

[0280] FIGS. **151** to **153** are diagrams of examples of an assembly in a rackmount system that includes a substrate, a printed circuit board, a front lattice structure, a rear lattice structure, and a heat dissipating device.

[0281] FIG. **154** is a diagram of an example of a CPO module.

[0282] FIGS. **155A** and **155B** are perspective views of examples of LGA sockets, optical modules, and compression plates.

[0283] FIG. **156** is a front view of an example of an array of compression plates.

[0284] FIG. **157** is a front perspective view of an example of an assembly that includes a substrate, optical modules, and compression plates.

[0285] FIG. **158** is a top view of an example of an assembly that includes a substrate, a data processing integrated circuit, optical modules, and compression plates.

[0286] FIG. **159** is a side view of an example of a rackmount server that has a hinge-mounted front panel.

[0287] FIG. **160** is a top view of an example of a rackmount server that has a hinge-mounted front panel.

[0288] FIG. **161** is a diagram of an example of an optical cable.

[0289] FIGS. **162** to **166** are diagrams of examples of a system that can provide a large memory bank or memory pool.

[0290] FIGS. **167** to **171D** are diagrams of examples of packaging configurations for compact co-packaged optical modules.

DETAILED DESCRIPTION

[0291] This document describes a novel system for high bandwidth data processing, including

novel input/output interface modules for coupling bundles of optical fibers to data processing integrated circuits (e.g., network switches, central processing units, graphics processor units, tensor processing units, digital signal processors, and/or other application specific integrated circuits (ASICs)) that process the data transmitted through the optical fibers. In some implementations, the data processing integrated circuit is mounted on a circuit board positioned near the input/output interface module through a relatively short electrical signal path on the circuit board. The input/output interface module includes a first connector that allows a user to conveniently connect or disconnect the input/output interface module to or from the circuit board. The input/output interface module includes a second connector that allows the user to conveniently connect or disconnect the bundle of optical fibers to or from the input/output interface module. In some implementations, a rack mount system having a front panel is provided in which the circuit board (which supports the input/output interface modules and the data processing integrated circuits) is vertically mounted in an orientation substantially parallel to, and positioned near, the front panel. In some examples, the circuit board functions as the front panel or part of the front panel. The second connectors of the input/output interface modules face the front side of the rack mount system to allow the user to conveniently connect or disconnect bundles of optical fibers to or from the system.

[0292] In some implementations, a feature of the high bandwidth data processing system is that, by vertically mounting the circuit board that supports the input/output interface modules and the data processing integrated circuits to be near the front panel, or configuring the circuit board as the front panel or part of the front panel, the optical signals can be routed from the optical fibers through the input/output interface modules to the data processing integrated circuits through relatively short electrical signal paths. This allows the signals transmitted to the data processing integrated circuits to have a high bit rate (e.g., over 50 Gbps) while maintaining low crosstalk, distortion, and noise, hence reducing power consumption and footprint of the data processing system.

[0293] In some implementations, a feature of the high bandwidth data processing system is that the cost of maintenance and repair can be lower compared to traditional systems. For example, the input/output interface modules and the fiber optic cables are configured to be detachable, a defective input/output interface module can be replaced without taking apart the data processing system and without having to re-route any optical fiber. Another feature of the high bandwidth data processing system is that, because the user can easily connect or disconnect the bundles of the optical fibers to or from the input/output interface modules through the front panel of the rack mount system, the configurations for routing of high bit rate signals through the optical fibers to the various data processing integrated circuits is flexible and can easily be modified. For example, connecting a bundle of hundreds of strands of optical fibers to the optical connector of the rack mount system can be almost as simple as plugging a universal serial bus (USB) cable into a USB port. A further feature of the high bandwidth data processing system is that the input/output interface module can be made using relatively standard, low cost, and energy efficient components so that the initial hardware costs and subsequent operational costs of the input/output interface modules can be relatively low, compared to conventional systems.

[0294] In some implementations, optical interconnects can co-package and/or co-integrate optical transponders with electronic processing chips. It is useful to have transponder solutions that consume relatively low power and that are sufficiently robust against significant temperature variations as may be found within an electronic processing chip package. In some implementations, high speed and/or high bandwidth data processing systems can include massively spatially parallel optical interconnect solutions that multiplex information onto relatively few wavelengths and use a relatively large number of parallel spatial paths for chip-to-chip interconnection. For example, the relatively large number of parallel spatial paths can be arranged in two-dimensional arrays using connector structures such as those disclosed in U.S. patent application Ser. No. 16/816,171, filed on Mar. 11, 2020, now U.S. Pat. No. 11,287,585, and incorporated herein by reference in its entirety.

[0295] FIG. 1 shows a block diagram of a communication system **100** that incorporates one or more novel features described in this document. In some implementations, the system **100** includes nodes **101_1** to **101_6** (collectively referenced as **101**), which in some embodiments can each include one or more of: optical communication devices, electronic and/or optical switching devices, electronic and/or optical routing devices, network control devices, traffic control devices, synchronization devices, computing devices, and data storage devices. The nodes **101_1** to **101_6** can be suitably interconnected by optical fiber links **102_1** to **102_12** (collectively referenced as **102**) establishing communication paths between the communication devices within the nodes. The optical fiber links **102** can include the fiber-optic cables described in U.S. patent application Ser. No. 16/822,103 filed on Mar. 18, 2020, now U.S. Pat. No. 11,194,109, and incorporated herein by reference in its entirety. The system **100** can also include one or more optical power supply modules **103** producing one or more light outputs, each light output comprising one or more continuous-wave (CW) optical fields and/or one or more trains of optical pulses for use in one or more of the optical communication devices of the nodes **101_1** to **101_6**. For illustration purposes, only one such optical power supply module **103** is shown in FIG. 1. A person of ordinary skill in the art will understand that some embodiments can have more than one optical power supply module **103** appropriately distributed over the system **100** and that such multiple power supply modules can be synchronized, e.g., using some of the techniques disclosed in U.S. patent application Ser. No. 16/847,705, filed on Apr. 14, 2020, now U.S. Pat. No. 11,153,670, and incorporated herein by reference in its entirety.

[0296] Some end-to-end communication paths can pass through an optical power supply module **103** (e.g., see the communication path between the nodes **101_2** and **101_6**). For example, the communication path between the nodes **101_2** and **101_6** can be jointly established by the optical fiber links **102_7** and **102_8**, whereby light from the optical power supply module **103** is multiplexed onto the optical fiber links **102_7** and **102_8**.

[0297] Some end-to-end communication paths can pass through one or more optical multiplexing units **104** (e.g., see the communication path between the nodes **101_2** and **101_6**). For example, the communication path between the nodes **101_2** and **101_6** can be jointly established by the optical fiber links **102_10** and **102_11**. Multiplexing unit **104** is also connected, through the link **102_9**, to receive light from the optical power supply module **103** and, as such, can be operated to multiplex said received light onto the optical fiber links **102_10** and **102_11**.

[0298] Some end-to-end communication paths can pass through one or more optical switching units **105** (e.g., see the communication path between the nodes **101_1** and **101_4**). For example, the communication path between the nodes **101_1** and **101_4** can be jointly established by the optical fiber links **102_3** and **102_12**, whereby light from the optical fiber links **102_3** and **102_4** is either statically or dynamically directed to the optical fiber link **102_12**.

[0299] As used herein, the term “network element” refers to any element that generates, modulates, processes, or receives light within the system **100** for the purpose of communication. Example network elements include the node **101**, the optical power supply module **103**, the optical multiplexing unit **104**, and the optical switching unit **105**.

[0300] Some light distribution paths can pass through one or more network elements. For example, optical power supply module **103** can supply light to the node **101_4** through the optical fiber links **102_7**, **102_4**, and **102_12**, letting the light pass through the network elements **101_2** and **105**.

[0301] Various elements of the communication system **100** can benefit from the use of optical interconnects, which can use photonic integrated circuits comprising optoelectronic devices, co-packaged and/or co-integrated with electronic chips comprising integrated circuits.

[0302] As used herein, the term “photonic integrated circuit” (or PIC) should be construed to cover planar lightwave circuits (PLCs), integrated optoelectronic devices, wafer-scale products on substrates, individual photonic chips and dies, and hybrid devices. A substrate can be made of, e.g., one or more ceramic materials, or organic “high density build-up” (HDBU). Example material

systems that can be used for manufacturing various photonic integrated circuits can include but are not limited to III-V semiconductor materials, silicon photonics, silica-on-silicon products, silica-glass-based planar lightwave circuits, polymer integration platforms, lithium niobate and derivatives, nonlinear optical materials, etc. Both packaged devices (e.g., wired-up and/or encapsulated chips) and unpackaged devices (e.g., dies) can be referred to as planar lightwave circuits.

[0303] Photonic integrated circuits are used for various applications in telecommunications, instrumentation, and signal-processing fields. In some implementations, a photonic integrated circuit uses optical waveguides to implement and/or interconnect various circuit components, such as for example, optical switches, couplers, routers, splitters, multiplexers/demultiplexers, filters, modulators, phase shifters, lasers, amplifiers, wavelength converters, optical-to-electrical (O/E) and electrical-to-optical (E/O) signal converters, etc. For example, a waveguide in a photonic integrated circuit can be an on-chip solid light conductor that guides light due to an index-of-refraction contrast between the waveguide's core and cladding. A photonic integrated circuit can include a planar substrate onto which optoelectronic devices are grown by an additive manufacturing process and/or into which optoelectronic devices are etched by a subtractive manufacturing processes, e.g., using a multi-step sequence of photolithographic and chemical processing steps.

[0304] In some implementations, an “optoelectronic device” can operate on both light and electrical currents (or voltages) and can include one or more of: (i) an electrically driven light source, such as a laser diode; (ii) an optical amplifier; (iii) an optical-to-electrical converter, such as a photodiode; and (iv) an optoelectronic component that can control the propagation and/or certain properties (e.g., amplitude, phase, polarization) of light, such as an optical modulator or a switch. The corresponding optoelectronic circuit can additionally include one or more optical elements and/or one or more electronic components that enable the use of the circuit's optoelectronic devices in a manner consistent with the circuit's intended function. Some optoelectronic devices can be implemented using one or more photonic integrated circuits.

[0305] As used herein, the term “integrated circuit” (IC) should be construed to encompass both a non-packaged die and a packaged die. In a typical integrated circuit-fabrication process, dies (chips) are produced in relatively large batches using wafers of silicon or other suitable material(s). Electrical and optical circuits can be gradually created on a wafer using a multi-step sequence of photolithographic and chemical processing steps. Each wafer is then cut (“diced”) into many pieces (chips, dies), each containing a respective copy of the circuit that is being fabricated. Each individual die can be appropriately packaged prior to being incorporated into a larger circuit or be left non-packaged.

[0306] The term “hybrid circuit” can refer to a multi-component circuit constructed of multiple monolithic integrated circuits, and possibly some discrete circuit components, all attached to each other to be mountable on and electrically connectable to a common base, carrier, or substrate. A representative hybrid circuit can include (i) one or more packaged or non-packaged dies, with some or all of the dies including optical, optoelectronic, and/or semiconductor devices, and (ii) one or more optional discrete components, such as connectors, resistors, capacitors, and inductors. Electrical connections between the integrated circuits, dies, and discrete components can be formed, e.g., using patterned conducting (such as metal) layers, ball-grid arrays, solder bumps, wire bonds, etc. Electrical connections can also be removable, e.g., by using land-grid arrays and/or compression interposers. The individual integrated circuits can include any combination of one or more respective substrates, one or more redistribution layers (RDLs), one or more interposers, one or more laminate plates, etc.

[0307] In some embodiments, individual chips can be stacked. As used herein, the term “stack” refers to an orderly arrangement of packaged or non-packaged dies in which the main planes of the stacked dies are substantially parallel to each other. A stack can typically be mounted on a carrier in an orientation in which the main planes of the stacked dies are parallel to each other and/or to the

main plane of the carrier.

[0308] A “main plane” of an object, such as a die, a photonic integrated circuit, a substrate, or an integrated circuit, is a plane parallel to a substantially planar surface thereof that has the largest sizes, e.g., length and width, among all exterior surfaces of the object. This substantially planar surface can be referred to as a main surface. The exterior surfaces of the object that have one relatively large size, e.g., length, and one relatively small size, e.g., height, are typically referred to as the edges of the object.

[0309] FIG. 2 is a schematic cross-sectional diagram of a data processing system **200** that includes an integrated optical communication device **210** (also referred to as an optical interconnect module), a fiber-optic connector assembly **220**, a package substrate **230**, and an electronic processor integrated circuit **240**. The data processing system **200** can be used to implement, e.g., one or more of devices **101_1** to **101_6** of FIG. 1. FIG. 3 shows an enlarged cross-sectional diagram of the integrated optical communication device **210**.

[0310] Referring to FIGS. 2 and 3, the integrated optical communication device **210** includes a substrate **211** having a first main surface **211_1** and a second main surface **211_2**. The main surfaces **211_1** and **211_2**, respectively, include arrays of electrical contacts **212_1** and **212_2**. In some embodiments, the minimum spacing $d_{sub.1}$ between any two contacts within the array of contacts **212_1** is larger than the minimum spacing $d_{sub.2}$ between any two contacts within the array of contacts **212_2**. In some embodiments the minimum spacing between any two contacts within the array of contacts **212_2** is between 40 and 200 micrometers. In some embodiments, the minimum spacing between any two contacts within the array of contacts **212_1** is between 200 micrometers and 1 millimeter. At least some of the contacts **212_1** are electrically connected through the substrate **211** with at least some of the contacts **212_2**. In some embodiments, the contacts **212_1** can be permanently attached to a corresponding array of electrical contacts **232_1** on the package substrate **230**. In some embodiments, the contacts **212_1** can include mechanisms to allow the device **210** to be removably connected to the package substrate **230**, as indicated by a double arrow **233**. For example, the system can include mechanical mechanisms (e.g., one or more snap-on or screw-on mechanisms) to hold the various modules in place. In some embodiments, the contacts **212_1**, **212_2**, and/or **232_1** can include one or more of solder balls, metal pillars, and/or metal pads, etc. In some embodiments, the contacts **212_1**, and/or **232_1** can include one or more of spring-loaded elements, compression interposers, and/or land-grid arrays.

[0311] In some embodiments, the integrated optical communication device **210** can be connected to the electronic processor integrated circuit **240** using traces **231** embedded in one or more layers of the package substrate **230**. In some embodiments, the processor integrated circuit **240** can include monolithically embedded therein an array of serializers/deserializers (SerDes) **247** electrically coupled to the traces **231**. In some embodiments, the processor integrated circuit **240** can include electronic switching circuitry, electronic routing circuitry, network control circuitry, traffic control circuitry, computing circuitry, synchronization circuitry, time stamping circuitry, and data storage circuitry. In some implementations, the processor integrated circuit **240** can be a network switch, a central processing unit, a graphics processor unit, a tensor processing unit, a digital signal processor, or an application specific integrated circuit (ASIC).

[0312] Because the electronic processor integrated circuit **240** and the integrated communication device **210** are both mounted on the package substrate **230**, the electrical connectors or traces **231** can be made shorter, as compared to mounting the electronic processor integrated circuit **240** and the integrated communication device **210** on separate circuit boards. Shorter electrical connectors or traces **231** can transmit signals that have a higher data rate with lower noise, lower distortion, and/or lower crosstalk.

[0313] In some implementations, the electrical connectors or traces can be configured as differential pairs of transmission lines, e.g., in a ground-signal-ground-signal-ground configuration. In some examples, the speed of such signal links can be 10 Gbps or more; 56 Gbps or more; 112

Gbps or more; or 224 Gbps or more.

[0314] In some implementations, the integrated optical communication device **210** further includes a first optical connector part **213** having a first surface **213_1** and a second surface **213_2**. The connector part **213** is configured to receive a second optical connector part **223** of the fiber-optic connector assembly **220**, optically coupled to the connector part **213** through the surfaces **213_1** and **213_2**. In some embodiments the connector part **213** can be removably attached to the connector part **223**, as indicated by a double-arrow **234**, e.g., through a hole **235** in the package substrate **230**. In some embodiments the connector part **213** can be permanently attached to the connector part **223**. In some embodiments, the connector parts **213** and **223** can be implemented as a single connector element combining the functions of both the connector parts **213** and **223**.

[0315] In some implementations, the optical connector part **223** is attached to an array of optical fibers **226**. In some embodiments, the array of optical fibers **226** can include one or more of: single-mode optical fiber, multi-mode optical fiber, multi-core optical fiber, polarization-maintaining optical fiber, dispersion-compensating optical fiber, hollow-core optical fiber, or photonic crystal fiber. In some embodiments, the array of optical fibers **226** can be a linear (1D) array. In some other embodiments, the array of optical fibers **226** can be a two-dimensional (2D) array. For example, the array of optical fibers **226** can include 2 or more optical fibers, 4 or more optical fibers, 10 or more optical fibers, 100 or more optical fibers, 500 or more optical fibers, or 1000 or more optical fibers. Each optical fiber can include, e.g., 2 or more cores, or 10 or more cores, in which each core provides a distinct light path. Each light path can include a multiplex of, e.g., 2 or more, 4 or more, 8 or more, or 16 or more serial optical signals, e.g., by use of wavelength division multiplexing channels, polarization-multiplexed channels, coherent quadrature-multiplexed channels. The connector parts **213** and **223** are configured to establish light paths through the first main surface **211_1** of the substrate **211**. For example, the array of optical fibers **226** can include n_1 optical fibers, each optical fiber can include n_2 cores, and the connector parts **213** and **223** can establish $n_1 \times n_2$ light paths through the first main surface **211_1** of the substrate **211**. Each light path can include a multiplex of n_3 serial optical signals, resulting in a total of $n_1 \times n_2 \times n_3$ serial optical signals passing through the connector parts **213** and **223**. In some embodiments, the connector parts **213** and **223** can be implemented, e.g., as disclosed in U.S. patent application Ser. No. 16/816,171, now U.S. Pat. No. 11,287,585.

[0316] In some implementations, the integrated optical communication device **210** further includes a photonic integrated circuit **214** having a first main surface **214_1** and a second main surface **214_2**. The photonic integrated circuit **214** is optically coupled to the connector part **213** through its first main surface **214_1**, e.g., as disclosed in U.S. patent application Ser. No. 16/816,171, now U.S. Pat. No. 11,287,585. For example, the connector part **213** can be configured to optically couple light to the photonic integrated circuit **214** using optical coupling interfaces, e.g., vertical grating couplers or turning mirrors. In the example above, a total of $n_1 \times n_2 \times n_3$ serial optical signals can be coupled through the connector parts **213** and **223** to the photonic integrated circuit **214**. Each serial optical signal is converted to a serial electrical signal by the photonic integrated circuit **214**, and each serial electrical signal is transmitted from the photonic integrated circuit **214** to a deserializer unit, or a serializer/deserializer unit, described below.

[0317] In some embodiments, the connector part **213** can be mechanically connected (e.g., glued) to the photonic integrated circuit **214**. The photonic integrated circuit **214** can contain active and/or passive optical and/or opto-electronic components including optical modulators, optical detectors, optical phase shifters, optical power splitters, optical wavelength splitters, optical polarization splitters, optical filters, optical waveguides, or lasers. In some embodiments, the photonic integrated circuit **214** can further include monolithically integrated active or passive electronic elements such as resistors, capacitors, inductors, heaters, or transistors.

[0318] In some implementations, the integrated optical communication device **210** further includes an electronic communication integrated circuit **215** configured to facilitate communication between

the array of optical fibers **226** and the electronic processor integrated circuit **240**. A first main surface **215_1** of the electronic communication integrated circuit **215** is electrically coupled to the second main surface **214_2** of the photonic integrated circuit **214**, e.g., through solder bumps, copper pillars, etc. The first main surface **215_1** of the electronic communication integrated circuit **215** is further electrically connected to the second main surface **211_2** of the substrate **211** through the array of electrical contacts **212_2**. In some embodiments, the electronic communication integrated circuit **215** can include electrical pre-amplifiers and/or electrical driver amplifiers electrically coupled, respectively, to photodetectors and modulators within the photonic integrated circuit **214** (see also FIG. **14**). In some embodiments, the electronic communication integrated circuit **215** can include a first array of serializers/deserializers (SerDes) **216** (also referred to as a serializers/deserializers module) whose serial inputs/outputs are electrically connected to the photodetectors and the modulators of the photonic integrated circuit **214** and a second array of serializers/deserializers **217**, whose serial inputs/outputs are electrically coupled to the contacts **212_1** through the substrate **211**. Parallel inputs of the array of serializers/deserializers **216** can be connected to parallel outputs of the array of serializers/deserializers **217** and vice versa through a bus processing unit **218**, which can be, e.g., a parallel bus of electrical lanes, a cross-connect device, or a re-mapping device (gearbox). For example, the bus processing unit **218** can be configured to enable switching of the signals, allowing the routing of signals to be re-mapped. For example, $N \times 50$ Gbps electrical lanes can be remapped into $N/2 \times 100$ Gbps electrical lanes, N being a positive even integer. An example of a bus processing unit **218** is shown in FIG. **40A**.

[0319] For example, the electronic communication integrated circuit **215** includes a first serializers/deserializers module that includes multiple serializer units and multiple deserializer units, and a second serializers/deserializers module that includes multiple serializer units and multiple deserializer units. The first serializers/deserializers module includes the first array of serializers/deserializers **216**. The second serializers/deserializers module includes the second array of serializers/deserializers **217**.

[0320] In some implementations, the first and second serializers/deserializers modules have hardwired functional units so that which units function as serializers and which units function as deserializers are fixed. In some implementations, the functional units can be configurable. For example, the first serializers/deserializers module is capable of operating as serializer units upon receipt of a first control signal, and operating as deserializer units upon receipt of a second control signal. Likewise, the second serializers/deserializers module is capable of operating as serializer units upon receipt of a first control signal, and operating as deserializer units upon receipt of a second control signal.

[0321] Signals can be transmitted between the optical fibers **226** and the electronic processor integrated circuit **240**. For example, signals can be transmitted from the optical fibers **226** to the photonic integrated circuit **214**, to the first array of serializers/deserializers **216**, to the second array of serializers/deserializers **217**, and to the electronic processor integrated circuit **240**. Similarly, signals can be transmitted from the electronic processor integrated circuit **240** to the second array of serializers/deserializers **217**, to the first array of serializers/deserializers **216**, to the photonic integrated circuit **214**, and to the optical fibers **226**.

[0322] In some implementations, the electronic communication integrated circuit **215** is implemented as a first integrated circuit and a second integrated circuit that are electrically coupled each other. For example, the first integrated circuit includes the array of serializers/deserializers **216**, and the second integrated circuit includes the array of serializers/deserializers **217**.

[0323] In some implementations, the integrated optical communication device **210** is configured to receive optical signals from the array of optical fibers **226**, generate electrical signals based on the optical signals, and transmit the electrical signals to the electronic processor integrated circuit **240** for processing. In some examples, the signals can also flow from the electronic processor integrated circuit **240** to the integrated optical communication device **210**. For example, the electronic

processor integrated circuit **240** can transmit electronic signals to the integrated optical communication device **210**, which generates optical signals based on the received electronic signals, and transmits the optical signals to the array of optical fibers **226**.

[0324] In some implementations, the photodetectors of the photonic integrated circuit **214** convert the optical signals transmitted in the optical fibers **226** to electrical signals. In some examples, the photonic integrated circuit **214** can include transimpedance amplifiers for amplifying the currents generated by the photodetectors, and drivers for driving output circuits (e.g., driving optical modulators). In some examples, the transimpedance amplifiers and drivers are integrated with the electronic communication integrated circuit **215**. For example, the optical signal in each optical fiber **226** can be converted to one or more serial electrical signals. For example, one optical fiber can carry multiple signals by use of wavelength division multiplexing. The optical signals (and the serial electrical signals) can have a high data rate, such as 50 Gbps, 100 Gbps, or more. The first serializers/deserializers module **216** converts the serial electrical signals to sets of parallel electrical signals. For example, each serial electrical signal can be converted to a set of N parallel electrical signals, in which N can be, e.g., 2, 4, 8, 16, or more. The first serializers/deserializers module **216** conditions the serial electrical signals upon conversion into sets of parallel electrical signals, in which the signal conditioning can include, e.g., one or more of clock and data recovery, and signal equalization. The first serializers/deserializers module **216** sends the sets of parallel electrical signals to the second serializers/deserializers module **217** through the bus processing unit **218**. The second serializers/deserializers module **217** converts the sets of parallel electrical signals to high speed serial electrical signals that are output to the electrical contacts **212_2** and **212_1**.

[0325] The serializers/deserializers module (e.g., **216**, **217**) can perform functions such as fixed or adaptive signal pre-distortion on the serialized signal. Also, the parallel-to-serial mapping can use a serialization factor M different from N, e.g., 50 Gbps at the input to the first serializers/deserializers module **216** can become 50×1 Gbps on a parallel bus, and two such parallel buses from two serializers/deserializers modules **216** having a total of 100×1 Gbps can then be mapped to a single 100 Gbps serial signal by the serializers/deserializers module **217**. An example of the bus processing unit **218** for performing such mapping is shown in FIG. 40B. Also, the high-speed modulation on the serial side can be different, e.g., the serializers/deserializers module **216** can use 50 Gbps Non-Return-to-Zero (NRZ) modulation whereas the serializers/deserializers module **217** can use 100 Gbps Pulse-Amplitude Modulation 4-Level (PAM4) modulation. In some implementations, coding (line coding or error-correction coding) can be performed at the bus processing unit **218**. The first and second serializers/deserializers modules **216** and **217** can be commercially available high quality, low power serializers/deserializers that can be purchased in bulk at a low cost.

[0326] In some implementations, the package substrate **230** can include connectors on the bottom side that connects the package substrate **230** to another circuit board, such as a motherboard. The connection can use, e.g., fixed (e.g., by use of solder connection) or removable (e.g., by use of one or more snap-on or screw-on mechanisms). In some examples, another substrate can be provided between the electronic processor integrated circuit **240** and the package substrate **230**.

[0327] Referring to FIG. 4, in some implementations, a data processing system **250** includes an integrated optical communication device **252** (also referred to as an optical interconnect module), a fiber-optic connector assembly **220**, a package substrate **230**, and an electronic processor integrated circuit **240**. The data processing system **250** can be used, e.g., to implement one or more of devices **101_1** to **101_6** of FIG. 1. The integrated optical communication device **252** is configured to receive optical signals, generate electrical signals based on the optical signals, and transmit the electrical signals to the electronic processor integrated circuit **240** for processing. In some examples, the signals can also flow from the electronic processor integrated circuit **240** to the integrated optical communication device **252**. For example, the electronic processor integrated circuit **240** can transmit electronic signals to the integrated optical communication device **252**,

which generates optical signals based on the received electronic signals, and transmits the optical signals to the array of optical fibers **226**.

[0328] The system **250** is similar to the data processing system **200** of FIG. **2** except that in the system **250**, in the direction of the cross section of the figure, a portion **254** of the top surface of the photonic integrated circuit **214** is not covered by the first serializers/deserializers module **216** and the second serializers/deserializers module **217**. For example, the portion **254** can be used to couple to other electronic components, optical components, or electro-optical components, either from the bottom (as shown in FIG. **4**) or from the top (as shown in FIG. **6**). In some examples, the first serializers/deserializers module **216** can have a high temperature during operation. The portion **254** is not covered by the first serializers/deserializers module **216** and can be less thermally coupled to the first serializers/deserializers module **216**. In some examples, the photonic integrated circuit **214** can include modulators that modulate the phases of optical signals by modifying the temperature of waveguides and thereby modifying the refractive indices of the waveguides. In such devices, using the design shown in the example of FIG. **4** can allow the modulators to operate in a more thermally stable environment.

[0329] FIG. **5** shows an enlarged cross-sectional diagram of the integrated optical communication device **252**. In some implementations, the substrate **211** includes a first slab **256** and a second slab **258**. The first slab **256** provides electrical connectors to fan out the electrical contacts, and the second slab **258** provides a removable connection to the package substrate **230**. The first slab **256** includes a first set of contacts arranged on the top surface and a second set of contacts arranged on the bottom surface, in which the first set of contacts has a fine pitch and the second set of contacts has a coarse pitch. The minimum distance between contacts in the second set of contacts is greater than the minimum distance between contacts in the first set of contacts. The second slab **258** can include, e.g., spring-loaded contacts **259**.

[0330] Referring to FIG. **6**, in some implementations, a data processing system **260** includes an integrated optical communication device **262** (also referred to as an optical interconnect module), a fiber-optic connector assembly **270**, a package substrate **230**, and an electronic processor integrated circuit **240**. The data processing system **260** can be used, e.g., to implement one or more of devices **101_1** to **101_6** of FIG. **1**. The integrated optical communication device **262** includes a photonic integrated circuit **264**. The photonic integrated circuit **264** can include components that perform functions similar to those of the photonic integrated circuit **214** of FIGS. **2-5**. The integrated optical communication device **262** further includes a first optical connector part **266** that is configured to receive a second optical connector part **268** of the fiber-optic connector assembly **270**. For example, snap-on or screw-on mechanisms can be used to hold the first and second optical connector parts **266** and **268** together.

[0331] The connector parts **266** and **268** can be similar to the connector parts **213** and **223**, respectively, of FIG. **4**. In some examples, the optical connector part **268** is attached to an array of optical fibers **272**, which can be similar to the fibers **226** of FIG. **4**.

[0332] The photonic integrated circuit **264** has a top main surface and bottom main surface. The terms “top” and “bottom” refer to the orientations shown in the figure. It is understood that the devices described in this document can be positioned in any orientation, so for example the “top surface” of a device can be oriented facing downwards or sideways, and the “bottom surface” of the device can be oriented facing upwards or sideways. A difference between the photonic integrated circuit **264** and the photonic integrated circuit **214** (FIG. **4**) is that the photonic integrated circuit **264** is optically coupled to the connector part **268** through the top main surface, whereas the photonic integrated circuit **214** is optically coupled to the connector part **213** through the bottom main surface. For example, the connector part **266** can be configured to optically couple light to the photonic integrated circuit **214** using optical coupling interfaces, e.g., vertical grating couplers or turning mirrors, similar to the way that the connector part **213** optically couples light to the photonic integrated circuit **214**.

[0333] The integrated optical communication devices **252** (FIG. 4) and **262** (FIG. 6) provide flexibility in the design of the data processing systems, allowing the fiber-optic connector assembly **220** or **270** to be positioned on either side of the package substrate **230**.

[0334] Referring to FIG. 7, in some implementations, a data processing system **280** includes an integrated optical communication device **282** (also referred to as an optical interconnect module), a fiber-optic connector assembly **270**, a package substrate **230**, and an electronic processor integrated circuit **240**. The data processing system **280** can be used, e.g., to implement one or more of devices **101_1** to **101_6** of FIG. 1.

[0335] The integrated optical communication device **282** includes a photonic integrated circuit **284**, a circuit board **286**, a first serializers/deserializers module **216**, a second serializers/deserializers module **217**, and a control circuit **287**. The photonic integrated circuit **284** can include components that perform functions similar to those of the photonic integrated circuit **214** (FIGS. 2-5) and **264** (FIG. 6). The control circuit **287** controls the operation of the photonic integrated circuit **284**. For example, the control circuit **287** can control one or more photodetector and/or modulator bias voltages, heater voltages, etc., either statically or adaptively based on one or more sensor voltages that the control circuit **287** can receive from the photonic integrated circuit **284**. The integrated optical communication device **282** further includes a first optical connector part **288** that is configured to receive a second optical connector part **268** of the fiber-optic connector assembly **270**. The optical connector part **268** is attached to an array of optical fibers **272**.

[0336] The circuit board **286** has a top main surface **290** and a bottom main surface **292**. The photonic integrated circuit **284** has a top main surface **294** and bottom main surface **296**. The first and second serializers/deserializers modules **216**, **217** are mounted on the top main surface **290** of the circuit board **286**. The top main surface **294** of the photonic integrated circuit **284** has electrical terminals that are electrically coupled to corresponding electrical terminals on the bottom main surface **292** of the circuit board **286**. In this example, the photonic integrated circuit **284** is mounted on a side of the circuit board **286** that is opposite to the side of the circuit board **286** on which the first and second serializers/deserializers modules **216**, **217** are mounted. The photonic integrated circuit **284** is electrically coupled to the first serializers/deserializers **216** by electrical connectors **300** that pass through the circuit board **286** in the thickness direction. In some embodiments, the electrical connectors **300** can be implemented as vias.

[0337] The connector part **288** has dimensions that are configured such that the fiber-optic connector assembly **270** can be coupled to the connector part **288** without bumping into other components of the integrated optical communication device **282**. The connector part **288** can be configured to optically couple light to the photonic integrated circuit **284** using optical coupling interfaces, e.g., vertical grating couplers or turning mirrors, similar to the way that the connector part **213** or **266** optically couples light to the photonic integrated circuit **214** or **264**, respectively.

[0338] When the integrated optical communication device **282** is coupled to the package substrate **230**, the photonic integrated circuit **284** and the control circuit **287** are positioned between the circuit board **286** and the package substrate **230**. The integrated optical communication device **282** includes an array of contacts **298** arranged on the bottom main surface **292** of the circuit board **286**. The array of contacts **298** is configured such that after the circuit board **286** is coupled to the package substrate **230**, the array of contacts **298** maintains a thickness d_3 between the circuit board **286** and the package substrate **230**, in which the thickness d_3 is slightly larger than the thicknesses of the photonic integrated circuit **284** and the control circuit **287**.

[0339] FIG. 8 is an exploded perspective view of the integrated optical communication device **282** of FIG. 7. The photonic integrated circuit **284** includes an array of optical coupling components **310**, e.g., vertical grating couplers or turning mirrors, as disclosed in U.S. patent application Ser. No. 16/816,171, now U.S. Pat. No. 11,287,585, that are configured to optically couple light from the optical connector part **288** to the photonic integrated circuit **214**. The optical coupling components **310** are densely packed and have a fine pitch so that optical signals from many optical

fibers can be coupled to the photonic integrated circuit **284**. For example, the minimum distance between adjacent optical coupling components **310** can be as small as, e.g., 5 μm , 10 μm , 50 μm , or 100 μm .

[0340] An array of electrical terminals **312** arranged on the top main surface **294** of the photonic integrated circuit **284** are electrically coupled to an array of electrical terminals **314** arranged on the bottom main surface **292** of the circuit board **286**. The array of electrical terminals **312** and the array of electrical terminals **314** have a fine pitch, in which the minimum distance between two adjacent electrical terminals can be as small as, e.g., 10 μm , 40 μm , or 100 μm . An array of electrical terminals **316** arranged on the bottom main surface of the first serializers/deserializers **216** are electrically coupled to an array of electrical terminals **318** arranged on the top main surface **290** of the circuit board **286**. An array of electrical terminals **320** arranged on the bottom main surface of the second serializers/deserializers module **217** are electrically coupled an array of electrical terminals **322** arranged on the top main surface **290** of the circuit board **286**.

[0341] For example, the arrays of electrical terminals **312**, **314**, **316**, **318**, **320**, and **322** have a fine pitch (or fine pitches). For simplicity of description, in the example of FIG. 8, for each of the arrays of electrical terminals **312**, **314**, **316**, **318**, **320**, and **322**, the minimum distance between adjacent terminals is d_2 , which can be in the range of, e.g., 10 μm to 200 μm . In some examples, the minimum distance between adjacent terminals for different arrays of electrical terminals can be different. For example, the minimum distance between adjacent terminals for the arrays of electrical terminals **314** (which are arranged on the bottom surface of the circuit board **286**) can be different from the minimum distance between adjacent terminals for the arrays of electrical terminals **318** arranged on the top surface of the circuit board **286**. The minimum distance between adjacent terminals for the arrays of electrical terminals **316** of the first serializers/deserializers **216** can be different from the minimum distance between adjacent terminals for the arrays of electrical terminals **320** of the second serializers/deserializers module **217**.

[0342] An array of electrical terminals **324** arranged on the bottom main surface of the circuit board **286** are electrically coupled to the array of contacts **298**. The array of electrical terminals **324** can have a coarse pitch. For example, the minimum distance between adjacent electrical terminals is d_1 , which can be in the range of, e.g., 200 μm to 1 mm. The array of contacts **298** can be configured as a module that maintains a distance that is slightly larger than the thicknesses of the photonic integrated circuit **284** and the control circuit **287** (which is not shown in FIG. 8) between the integrated optical communication device **282** and the package substrate **230** after the integrated optical communication device **282** is coupled to the package substrate **230**. The array of contacts **298** can include, e.g., a substrate that has embedded spring loaded connectors.

[0343] FIG. 9 is a diagram of an example layout design for optical and electrical terminals of the integrated optical communication device **282** of FIGS. 7 and 8. FIG. 9 shows the layout of the optical and electrical terminals when viewed from the top or bottom side of the device **282**. In this example, the photonic integrated circuit **284** has a width of about 5 mm and a length of about 2.2 mm to 18 mm. For the example in which the length of the photonic integrated circuit **284** is about 2.2 mm, the optical signals provided to the photonic integrated circuit **284** can have a total bandwidth of about 1.6 Tbps. For the example in which the length of the photonic integrated circuit is about 18 mm, the optical signals provided to the photonic integrated circuit can have a total bandwidth of about 12.8 Tbps. The width of the integrated optical communication device **282** can be about 8 mm.

[0344] An array **330** of optical coupling components **310** is provided to allow optical signals to be provided to the photonic integrated circuit **284** in parallel. The first serializers/deserializers **216** include an array **332** of electrical terminals **316** arranged on the bottom surface of the first serializers/deserializers **216**. The second serializers/deserializers module **217** include an array **334** of electrical terminals **320** arranged on the bottom surface of the second serializers/deserializers module **217**. The arrays **332** and **334** of electrical terminals **316**, **320** have a fine pitch, and the

minimum distance between adjacent terminals can be in the range of, e.g., 40 μm to 200 μm . An array **336** of electrical terminals **324** is arranged on the bottom main surface of the circuit board **286**. The array **336** of electrical terminals **324** has a coarse pitch, and the minimum distance between adjacent terminals can be in the range of, e.g., 200 μm to 1 mm. For example, the array **336** of electrical terminals **324** can be part of a compression interposer that has a pitch of about 400 μm between terminals.

[0345] FIG. **10** is a diagram of an example layout design for optical and electrical terminals of the integrated optical communication device **210** of FIG. **2**. FIG. **10** shows the layout of the optical and electrical terminals when viewed from the top or bottom side of the device **210**. In this embodiment, the photonic integrated circuit **214** is implemented as a single chip. In some embodiments, the photonic integrated circuit **214** can be tiled across multiple chips. Likewise, the electronic communication integrated circuit **215** is implemented as a single chip in this embodiment. In some embodiments, the electronic communication integrated circuit **215** can be tiled across multiple chips. In this embodiment, the electronic communication integrated circuit **215** is implemented using 16 serializers/deserializers blocks **216_1** to **216_16** that are electrically connected to the photonic integrated circuit **214** and 16 serializers/deserializers blocks **217_1** to **217_16**, which are electrically connected to an array of contacts **2121** by electrical connectors that pass through the substrate **211** in the thickness direction. The 16 serializers/deserializers blocks **216_1** to **216_16** are electrically coupled to the 16 serializers/deserializers blocks **217_1** to **217_16** by bus processing units **218_1** to **218_16**, respectively. In this embodiment, each serializers/deserializers block (**216** or **217**) is implemented using 8 serial differential transmitters (TX) and 8 serial differential receivers (RX). In order to transfer the electrical signals from the serializers/deserializers blocks **217** to ASIC **240**, a total of $8 \times 16 \times 2 = 256$ electrical differential signal contacts **212_1** in addition to $8 \times 17 \times 2 = 272$ ground (GND) contacts **212_1** can be used. Other contact arrangements that beneficially reduce crosstalk, e.g., placing a ground contact between every pair of TX and RX contacts, can also be used as will be appreciated by a person skilled in the art. The transmitter contacts are collectively referenced as **340**, the receiver contacts are collectively referenced as **342**, and the ground contacts are collectively referenced as **344**.

[0346] The electrical contacts of the serializers/deserializers blocks **216_1** to **216_12** and **217_1** to **217_12** have a fine pitch, and the minimum distance between adjacent terminals can be in the range of, e.g., 40 μm to 200 μm . The electrical contacts **212_1** have a coarse pitch, and the minimum distance between adjacent terminals can be in the range of, e.g., 200 μm to 1 mm.

[0347] FIG. **11** is a schematic side view of an example data processing system **350**, which includes an integrated optical communication device **374**, a package substrate **230**, and a host application specific integrated circuit **240**. The integrated optical communication device **374** and the host application specific integrated circuit **240** are mounted on the top side of the package substrate **230**. The integrated optical communication device **374** includes a first optical connector **356** that allows optical signals transmitted in optical fibers to be coupled to the integrated optical communication device **374**, in which a portion of the optical fibers connected to the first optical connector **356** are positioned at a region facing the bottom side of the package substrate **230**.

[0348] The integrated optical communication device **374** includes a photonic integrated circuit **352**, a combination of drivers and transimpedance amplifiers (D/T) **354**, a first serializers/deserializers module **216**, a second serializers/deserializers module **217**, the first optical connector **356**, a control module **358**, and a substrate **360**. The host application specific integrated circuit **240** includes an embedded third serializers/deserializers module **247**.

[0349] In this example, the photonic integrated circuit **352**, the drivers and transimpedance amplifiers **354**, the first serializers/deserializers module **216**, and the second serializers/deserializers module **217** are mounted on the top side of the substrate **360**. In some embodiments, the drivers and transimpedance amplifiers **354**, the first serializers/deserializers module **216**, and the second serializers/deserializers module **217** can be monolithically integrated

into a single electrical chip. The first optical connector **356** is optically coupled to the bottom side of the photonic integrated circuit **352**. The control module **358** is electrically coupled to electrical terminals arranged on the bottom side of the substrate **360**, whereas the photonic integrated circuit **352** is connected to electrical terminals arranged on the top side of the substrate **360**. The control module **358** is electrically coupled to the photonic integrated circuit **352** through electrical connectors **362** that pass through the substrate **360** in the thickness direction. In some embodiments, the substrate **360** can be removably connected to the package substrate **230**, e.g., using a compression interposer or a land grid array.

[0350] The photonic integrated circuit **352** is electrically coupled to the drivers and transimpedance amplifiers **354** through electrical connectors **364** on or in the substrate **360**. The drivers and transimpedance amplifiers **354** are electrically coupled to the first serializers/deserializers module **216** by electrical connectors **366** on or in the substrate **360**. The second serializers/deserializers module **216** has electrical terminals **370** on the bottom side that are electrically coupled to electrical terminals **366** arranged on the bottom side of the substrate **360** through electrical connectors **368** that pass through the substrate **360** in the thickness direction. The electrical terminals **370** have a fine pitch, whereas the electrical terminals **366** have a coarse pitch. The electrical terminals **366** are electrically coupled to the third serializers/deserializers module **247** through electrical connectors or traces **372** on or in the package substrate **230**.

[0351] In some implementations, optical signals are converted by the photonic integrated circuit **352** to electrical signals, which are conditioned by the first serializers/deserializers module **216** (or the second serializers/deserializers module **217**), and processed by the host application specific integrated circuit **240**. The host application specific integrated circuit **240** generates electrical signals that are converted by the photonic integrated circuit **352** into optical signals.

[0352] FIG. **12** is a schematic side view of an example data processing system **380**, which includes an integrated optical communication device **382**, a package substrate **230**, and a host application specific integrated circuit **240**. The integrated optical communication device **382** is similar to the integrated optical communication device **374** (FIG. **11**), except that the transimpedance amplifiers and drivers are implemented in a separate chip **384** from the chip housing the serializers/deserializers modules **216** and **217**.

[0353] FIG. **13** is a schematic side view of an example data processing system **390** that includes an integrated optical communication device **402**, a package substrate **230**, and a host application specific integrated circuit (not shown in the figure). The integrated optical communication device **402** includes photonic integrated circuit **392**, a first serializers/deserializers module **394**, a second serializers/deserializers module **396**, a third serializers/deserializers module **398**, and a fourth serializers/deserializers module **400** that are mounted on a substrate **410**. The photonic integrated circuit **392** can include transimpedance amplifiers and drivers, or such amplifiers and/or drivers can be included in the serializers/deserializers modules **394** and **398**. The first serializers/deserializers module **394** and the second serializers/deserializers module **396** are positioned on the right side of the photonic integrated circuit **392**. The third serializers/deserializers module **398** and the fourth serializers/deserializers module **400** are positioned on the left side of the photonic integrated circuit **392**. Here, the term “left” and “right” refer to the relative positions shown in the figure. It is understood that the system **390** can be positioned in any orientation so that the first serializers/deserializers module **394** and the second serializers/deserializers module **396** are not necessarily at the right side of the photonic integrated circuit **392**, and the third serializers/deserializers module **398** and the fourth serializers/deserializers module **400** are not necessarily at the left side of the photonic integrated circuit **392**.

[0354] The photonic integrated circuit **392** receives optical signals from a first optical connector **404**, generates serial electrical signals based on the optical signals, sends the serial electrical signals to the first and second serializers/deserializers modules **394** and **398**. The first and second serializers/deserializers modules **394** and **398** generate parallel electrical signals based on the

received serial electrical signals, and send the parallel electrical signals to the third and fourth serializers/deserializers modules **396** and **400**, respectively. The third and fourth serializers/deserializers modules **396** and **400** generate serial electrical signals based on the received parallel electrical signals, and send the serial electrical signals to electrical terminals **406** and **408**, respectively, arranged on the bottom side of the substrate **410**.

[0355] The first optical connector **404** is optically coupled to the bottom side of the photonic integrated circuit **392**. In some embodiments, the optical connector **404** can also be placed on the top of the photonic integrated circuit **392** and couple light to the top side of the photonic integrated circuit **392** (not shown in the figure). The first optical connector **404** is optically coupled to a second optical connector, which in turn is optically coupled to a plurality of optical fibers. In the configuration shown in FIG. **13**, the first optical connector **404**, the second optical connector, and/or the optical fibers pass through an opening **412** in the package substrate **230**. The electrical terminals **406** are arranged on the right side of the first optical connector **404**, and the electrical terminals **408** are arranged on the left side of the first optical connector **404**. The electrical terminals **406** and **408** are configured such that the substrate **410** can be removably coupled to the package substrate **230**.

[0356] FIG. **14** is a schematic side view of an example data processing system **420** that includes an integrated optical communication device **428**, a package substrate **230**, and a host application specific integrated circuit (not shown in the figure). The integrated optical communication device **428** includes a photonic integrated circuit **422** (which does not include a transimpedance amplifier and driver), a first serializers/deserializers module **394**, a second serializers/deserializers module **396**, a third serializers/deserializers module **398**, and a fourth serializers/deserializers module **400** that are mounted on a substrate **410**. The integrated optical communication device **428** includes a first set of transimpedance amplifiers and driver circuits **424** positioned at the right of the photonic integrated circuit **422**, and a second set of transimpedance amplifiers and driver circuits **426** positioned at the left of the photonic integrated circuit **422**. The first set of transimpedance amplifiers and driver circuits **424** is positioned between the photonic integrated circuit **422** and a first serializers/deserializers module **394**. The second set of transimpedance amplifiers and driver circuits **424** is positioned between the photonic integrated circuit **422** and a third serializers/deserializers module **398**.

[0357] In some implementations, the integrated optical communication device **402** (or **408**) can be modified such that the first optical connector **404** couples optical signals to the top side of the photonic integrated circuit **392** (or **422**).

[0358] FIG. **32** is a schematic side view of an example data processing system **510** that includes an integrated optical communication device **512**, a package substrate **230**, and a host application specific integrated circuit (not shown in the figure). The integrated optical communication device **512** includes a substrate **514** that includes a first slab **516** and a second slab **518**. The first slab **516** provides electrical connectors to fan out the electrical contacts. The first slab **516** includes a first set of contacts arranged on the top surface and a second set of contacts arranged on the bottom surface, in which the first set of contacts has a fine pitch and the second set of contacts has a coarse pitch. The second slab **518** provides a removable connection to the package substrate **230**. A photonic integrated circuit **524** is mounted on the bottom side of the first slab **516**. A first optical connector **520** passes through an opening in the substrate **514** and couples optical signals to the top side of the photonic integrated circuit **524**.

[0359] A first serializers/deserializers module **394**, a second serializers/deserializers module **396**, a third serializers/deserializers module **398**, and a fourth serializers/deserializers module **400** are mounted on the top side of the first slab **516**. The photonic integrated circuit **524** is electrically coupled to the first and third serializers/deserializers modules **394** and **398** by electrical connectors **522** that pass through the substrate **514** in the thickness direction. For example, the electrical connectors **522** can be implemented as vias. In some examples, drivers and transimpedance

amplifiers can be integrated in the photonic integrated circuit **524**, or integrated in the serializers/deserializers modules **394** and **398**. In some examples, the drivers and transimpedance amplifiers can be implemented in a separate chip (not shown in the figure) positioned between the photonic integrated circuit **524** and the serializers/deserializers modules **394** and **398**, similar to the example in FIG. **14**. A control chip (not shown in the figure) can be provided to control the operation of the photonic integrated circuit **512**.

[0360] FIG. **15** is a bottom view of an example of the integrated optical communication device **428** of FIG. **14**. The photonic integrated circuit **422** includes modulator and photodetector blocks on both sides of a center line **432** in the longitudinal direction. The photonic integrated circuit **422** includes a fiber coupling region **430** arranged either at the bottom side of the photonic integrated circuit **392** or at the top side of the photonic integrated circuit (see FIG. **32**), in which the fiber coupling region **430** includes multiple optical coupling elements **310**, e.g., receiver optical coupling elements (RX), transmitter optical coupling elements (TX), and remote optical power supply (e.g., **103** in FIG. **1**) optical coupling elements (PS).

[0361] Complementary metal oxide semiconductor (CMOS) transimpedance amplifier and driver blocks **424** are arranged on the right side of the photonic integrated circuit **422**, and CMOS transimpedance amplifier and driver blocks **426** are arranged on the left side of the photonic integrated circuit **422**. A first serializers/deserializers module **394** and a second serializers/deserializers module **396** are arranged on the right side of the CMOS transimpedance amplifier and driver blocks **424**. A third serializers/deserializers module **398** and a fourth serializers/deserializers module **400** are arranged on the left side of the CMOS transimpedance amplifier and driver blocks **426**.

[0362] In this example, each of the first, second, third, and fourth serializers/deserializers module **394**, **396**, **398**, **400** includes 8 serial differential transmitter blocks and 8 serial differential receiver blocks. The integrated optical communication device **428** has a width of about 3.5 mm and a length of slightly more than about 3.6 mm.

[0363] FIG. **16** is a bottom view of an example of the integrated optical communication device **428** of FIG. **14**, in which the electrical terminals **406** and **408** are also shown. As shown in the figure, the electrical terminals **406** and **408** have a coarse pitch, the minimum distance between terminals in the array of electrical terminals **406** or **408** is much larger than the minimum distance between terminals in the array of electrical terminals of the first, second, third, and fourth serializers/deserializers modules **394**, **396**, **398**, and **400**. For example, the array of electrical terminals **406** and **408** can be part of a compression interposer that has a pitch of about 400 μm between terminals.

[0364] In some implementations, the electrical terminals (e.g., **406** and **408**) can be arranged in a configuration as shown in FIG. **66**. FIG. **66** shows a pad map **1020** that shows the locations of various contact pads as viewed from the bottom of the package. The contact pads occupy an area that is about 9.8 mm $\times$ 9.8 mm, in which 400 μm pitch pads are used.

[0365] The middle rectangle **1022** is a cutout that connects the photonic integrated circuit to the optics that leave from the top of the module. The bigger rectangle **1024** represents the photonic integrated circuit. The two gray rectangles **1026a**, **1026b** represent circuitry in a serializers/deserializers chip **1028a**. The two gray rectangles **1026c**, **1026d** represent circuitry in another serializers/deserializers chip **1028b**. The serializers/deserializers chips are positioned on the top of the package, and the photonic integrated circuit is positioned on the bottom of the package. The overlap between the photonic integrated circuit and the serializers/deserializers chips **1028a**, **1028b** is designed so that vias (not shown in the figure) can directly connect these integrated circuits through the package. In some implementations, the serializers/deserializers chips **1028a**, **1028b** and/or other electronic integrated circuits can be placed around three or four sides of the optical connector (represented by the rectangle **1022**), similar to the examples shown in FIGS. **168** to **170**.

[0366] In the examples of the data processing systems shown in FIGS. 2-8, 11-14, and 32, the integrated optical communication device (e.g., 210, 252, 262, 282, 374, 382, 402, 428, 512, which includes the photonic integrated circuit and the serializers/deserializers modules) is mounted on the package substrate 230 on the same side (top side in the examples shown in the figures) as the electronic processor integrated circuit (or host application specific integrated circuit) 240. The data processing systems can also be modified such that the integrated optical communication device is mounted on the package substrate 230 on the opposite side as the electronic processor integrated circuit (or host application specific integrated circuit) 240. For example, the electronic processor integrated circuit 240 can be mounted on the top side of the package substrate 230 and one or more integrated optical communication devices of the form disclosed in FIGS. 2-8, 11-14, and 32 can be mounted on the bottom side of the package substrate 230.

[0367] FIG. 17 is a diagram showing four types of integrated optical communication devices that can be used in a data processing system 440. In these examples, the integrated optical communication device does not include serializers/deserializers modules. At least some of the signal conditioning is performed by the serializers/deserializers module(s) in the digital application specific integrated circuit. The integrated optical communication device is mounted on the side of the printed circuit board that is opposite to the side on which the digital application specific integrated circuit is mounted, allowing the connectors to be short.

[0368] In a first example, the data processing system includes a digital application specific integrated circuit 444 mounted on the top side of a substrate 442, and an integrated optical communication device 448 mounted on the bottom side of the first circuit board. In some implementations, the integrated optical communication device 448 includes a photonic integrated circuit 450 and a set of transimpedance amplifiers and drivers 452 that are mounted on the bottom side of a substrate 454 (e.g., a second circuit board). The top side of the photonic integrated circuit 450 is electrically coupled to the bottom side of the substrate 454. A first optical connector part 456 is optically coupled to the bottom side of the photonic integrated circuit 450. The first optical connector part 456 is configured to be optically coupled to a second optical connector part 458 that is optically coupled to a plurality of optical fibers (not shown in the figure). An array of electrical terminals 460 is arranged on the top side of the substrate 454 and configured to enable the integrated optical communication device 448 to be removably coupled to the substrate 442.

[0369] The optical signals from the optical fibers are processed by the photonic integrated circuit 450, which generates serial electrical signals based on the optical signals. The serial electrical signals are amplified by the set of transimpedance amplifiers and drivers 452, which drives the output signals that are transmitted to a serializers/deserializers module 446 embedded in the digital application specific integrated circuit 444.

[0370] In a second example, an integrated optical communication device 462 can be mounted on the bottom side of the substrate 442 to provide an optical/electrical communications interface between the optical fibers and the digital application specific integrated circuit 444. The integrated optical communication device 462 includes a photonic integrated circuit 464 that is mounted on the bottom side of a substrate 454 (e.g., a second circuit board). The top side of the photonic integrated circuit 464 is electrically coupled to the bottom side of the substrate 454. A first optical connector part 456 is optically coupled to the bottom side of the photonic integrated circuit 450. An array of electrical terminals 460 is arranged on the top side of the substrate 454 and configured to enable the integrated optical communication device 462 to be removably coupled to the substrate 442. The integrated optical communication device 462 is similar to the integrated optical communication device 448, except that either the photonic integrated circuit 464 or the serializers/deserializers module 446 includes the set of transimpedance amplifiers and driver circuitry. In some examples, the serializers/deserializers module 446 is configured to directly accept electrical signals emerging from photonic integrated circuit 464, e.g., by having a high enough receiver input impedance that converts the photocurrent generated within the photonic integrated circuit 464 to a voltage swing

suitable for further electrical processing. For example, the serializers/deserializers module **446** is configured to have a low transmitter output impedance, and provide an output voltage swing that allows direct driving of optical modulators embedded within the photonic integrated circuit **464**. [0371] In a third example, an integrated optical communication device **466** can be mounted on the bottom side of the substrate **442** to provide an optical/electrical communications interface between the optical fibers and the digital application specific integrated circuit **444**. The integrated optical communication device **466** includes a photonic integrated circuit **468** that is mounted on the top side of a substrate **470** (e.g., a second circuit board). The bottom side of the photonic integrated circuit **468** is electrically coupled to the top side of the substrate **470**. A first optical connector part **456** is optically coupled to the bottom side of the photonic integrated circuit **468**. An array of electrical terminals **460** is arranged on the top side of the substrate **470** and configured to enable the integrated optical communication device **466** to be removably coupled to the substrate **442**. In some examples, either the photonic integrated circuit **468** or the serializers/deserializers module **446** includes the set of transimpedance amplifiers and driver circuitry. In some examples, the serializers/deserializers module **446** is configured to directly accept electrical signals emerging from the photonic integrated circuit **464**.

[0372] In a fourth example, an integrated optical communication device **472** can be mounted on the bottom side of the substrate **442** to provide an optical/electrical communications interface between the optical fibers and the digital application specific integrated circuit **444**. The integrated optical communication device **472** includes a photonic integrated circuit **474** and a set of transimpedance amplifiers and drivers **476** that are mounted on the top side of a substrate **470** (e.g., a second circuit board). The bottom side of the photonic integrated circuit **474** is electrically coupled to the top side of the substrate **470**. A first optical connector part **456** is optically coupled to the bottom side of the photonic integrated circuit **468**. An array of electrical terminals **460** is arranged on the top side of the substrate **470** and configured to enable the integrated optical communication device **466** to be removably coupled to the substrate **442**. The integrated optical communication device **472** is similar to the integrated optical communication device **466**, except that neither the photonic integrated circuit **464** nor the serializers/deserializers module **446** include a set of transimpedance amplifiers and driver circuitry, and the set of transimpedance amplifiers and drivers **476** is implemented as a separate integrated circuit.

[0373] FIG. **18** is a diagram of an example octal serializers/deserializers block **480** that includes 8 serial differential transmitters (TX) **482** and 8 serial differential receivers (RX) **484**. Each serial differential receiver **484** receives a serial differential signal, generates parallel signals based on the serial differential signal, and provides the parallel signals on the parallel bus **488**. Each serial differential transmitter **482** receives parallel signals from the parallel bus **488**, generates a serial differential signal based on the parallel signals, and provides the serial differential signal on an output electrical terminal **490**. The serializers/deserializers block **480** outputs and/or receives parallel signals through a parallel bus interface **492**.

[0374] In the examples described above, such as those shown in FIGS. **2-14**, the integrated optical communication device (e.g., **210**, **252**, **262**, **282**, **374**, **382**, **402**, **428**) includes a first serializers/deserializers module (e.g., **216**, **394**, **398**) and a second serializers/deserializers module (e.g., **217**, **396**, **400**). The first serializers/deserializers module serially interfaces with the photonic integrated circuit, and the second serializers/deserializers module serially interfaces with the electronic processor integrated circuit or host application specific integrated circuit (e.g., **240**). In some implementations, the electronic communication integrated circuit **215** includes an array of serializers/deserializers that can be logically partitioned into a first sub-array of serializers/deserializers and a second sub-array of serializers/deserializers. The first sub-array of serializers/deserializers corresponds to the serializers/deserializers module (e.g., **216**, **394**, **398**), and the second sub-array of serializers/deserializers corresponds to the second serializers/deserializers module (e.g., **217**, **396**, **400**).

[0375] FIG. 38 is a diagram of an example octal serializers/deserializers block **480** coupled to a bus processing unit **218**. The octal serializers/deserializers block **480** includes 8 serial differential transmitters (TX1 to TX8) **482** and 8 serial differential receivers (RX1 to RX4) **484**. In some implementations, the transmitters and receivers are partitioned such that the transmitters TX1, TX2, TX3, TX4 and receivers RX1, RX2, RX3, RX4 form a first serializers/deserializers module **840**, and the transmitters TX5, TX6, TX7, TX8 and receivers RX5, RX6, RX7, RX8 form a second serializers/deserializers module **842**. Serial electrical signals received at the receivers RX1, RX2, RX3, RX4 are converted to parallel electrical signals and routed by the bus processing unit **218** to the transmitters TX5, TX6, TX7, TX8, which convert the parallel electrical signals to serial electrical signals. For example, the photonic integrated circuit can send serial electrical signals to the receivers RX1, RX2, RX3, RX4, and the transmitters TX5, TX6, TX7, TX8 can transmit serial electrical signals to the electronic processor integrated circuit or host application specific integrated circuit.

[0376] For example, the bus processing unit **218** can re-map the lanes of signals and perform coding on the signals, such that the bit rate and/or modulation format of the serial signals output from the transmitters TX5, TX6, TX7, TX8 can be different from the bit rate and/or modulation format of the serial signals received at the receivers RX1, RX2, RX3, RX4. For example, 4 lanes of T Gbps NRZ serial signals received at the receivers RX1, RX2, RX3, RX4 can be re-encoded and routed to transmitters TX5, TX6 to output 2 lanes of $2 \times T$ Gbps PAM4 serial signals.

[0377] Similarly, serial electrical signals received at the receivers RX5, RX6, RX7, RX8 are converted to parallel electrical signals and routed by the bus processing unit **218** to the transmitters TX1, TX2, TX3, TX4, which convert the parallel electrical signals to serial electrical signals. For example, the electronic processor integrated circuit or host application specific integrated circuit can send serial electrical signals to the receivers RX5, RX6, RX7, RX8, and the transmitters TX1, TX2, TX3, TX4 can transmit serial electrical signals to the photonic integrated circuit.

[0378] For example, the bus processing unit **218** can re-map the lanes of signals and perform coding on the signals, such that the bit rate and/or modulation format of the serial signals output from the transmitters TX1, TX2, TX3, TX4 can be different from the bit rate and/or modulation format of the serial signals received at the receivers RX5, RX6, RX7, RX8. For example, 2 lanes of $2 \times T$ Gbps PAM4 serial signals received at receivers RX5, RX6 can be re-encoded and routed to the transmitters TX5, TX6, TX7, TX8 to output 4 lanes of T Gbps NRZ serial signals.

[0379] FIG. 39 is a diagram of another example octal serializers/deserializers block **480** coupled to a bus processing unit **218**, in which the transmitters and receivers are partitioned such that the transmitters TX1, TX2, TX5, TX6 and receivers RX1, RX2, RX5, RX6 form a first serializers/deserializers module **850**, and the transmitters TX3, TX4, TX7, TX8 and receivers RX3, RX4, RX7, RX8 form a second serializers/deserializers module **852**. Serial electrical signals received at the receivers RX1, RX2, RX5, RX6 are converted to parallel electrical signals and routed by the bus processing unit **218** to the transmitters TX3, TX4, TX7, TX8, which convert the parallel electrical signals to serial electrical signals. For example, the photonic integrated circuit can send serial electrical signals to the receivers RX1, RX2, RX5, RX6, and the transmitters TX3, TX4, TX7, TX8 can transmit serial electrical signals to the electronic processor integrated circuit or host application specific integrated circuit.

[0380] Similarly, serial electrical signals received at the receivers RX3, RX4, RX7, RX8 are converted to parallel electrical signals and routed by the bus processing unit **218** to the transmitters TX1, TX2, TX5, TX6, which convert the parallel electrical signals to serial electrical signals. For example, the electronic processor integrated circuit or host application specific integrated circuit can send serial electrical signals to the receivers RX3, RX4, RX7, RX8, and the transmitters TX1, TX2, TX5, TX6 can transmit serial electrical signals to the photonic integrated circuit.

[0381] In some implementations, the bus processing unit **218** can re-map the lanes of signals and perform coding on the signals, such that the bit rate and/or modulation format of the serial signals

output from the transmitters TX3, TX4, TX7, TX8 can be different from the bit rate and/or modulation format of the serial signals received at the receivers RX1, RX2, RX5, RX6. Similarly, the bus processing unit **218** can re-map the lanes of signals and perform coding on the signals such that the bit rate and/or modulation format of the serial signals output from the transmitters TX1, TX2, TX5, TX6 can be different from the bit rate and/or modulation format of the serial signals received at the receivers RX4, RX4, RX7, RX8.

[0382] FIGS. **38** and **39** show two examples of how the receivers and transmitters can be partitioned to form the first serializers/deserializers module and the second serializers/deserializers module. The partitioning can be arbitrarily determined based on application, and is not limited to the examples shown in FIGS. **38** and **39**. The partitioning can be programmable and dynamically changed by the system.

[0383] FIG. **19** is a diagram of an example electronic communication integrated circuit **480** that includes a first octal serializers/deserializers block **482** electrically coupled to a second octal serializers/deserializers block **484**. For example, the electronic communication integrated circuit **480** can be used as the electronic communication integrated circuit **215** of FIGS. **2** and **3**. The first octal serializers/deserializers block **482** can be used as the first serializers/deserializers module **216**, and the second octal serializers/deserializers block **484** can be used as the second serializers/deserializers module **217**. For example, the first octal serializers/deserializers block **482** can receive 8 serial differential signals, e.g., through electrical terminals arranged at the bottom side of the block, and generate 8 sets of parallel signals based on the 8 serial differential signals, in which each set of parallel signals is generated based on the corresponding serial differential signal. The first octal serializers/deserializers block **482** can condition serial electrical signals upon conversion into the 8 sets of parallel signals, such as performing clock and data recovery, and/or signal equalization. The first octal serializers/deserializers block **482** transmits the 8 sets of parallel signals to the second octal serializers/deserializers block **484** through a parallel bus **485** and a parallel bus **486**. The second octal serializers/deserializers block **484** can generate 8 serial differential signals based on the 8 sets of parallel signals, in which each serial differential signal is generated based on the corresponding set of parallel signals. The second octal serializers/deserializers block **484** can output the 8 serial differential signals through, e.g., electrical terminals arranged at the bottom side of the block.

[0384] Multiple serializers/deserializers blocks can be electrically coupled to multiple serializers/deserializers blocks through a bus processing unit that can be, e.g., a parallel bus of electrical lanes, a static or a dynamically reconfigurable cross-connect device, or a re-mapping device (gearbox). FIG. **33** is a diagram of an example electronic communication integrated circuit **530** that includes a first octal serializers/deserializers block **532** and a second octal serializers/deserializers block **534** electrically coupled to a third octal serializers/deserializers block **536** through a bus processing unit **538**. In this example, the bus processing unit **538** is configured to enable switching of the signals, allowing the routing of signals to be re-mapped, in which 8×50 Gbps serial electrical signals using NRZ modulation that are serially interfaced to the first and second octal serializers/deserializers blocks **532** and **534** are re-routed or combined into 8×100 Gbps serial electrical signals using PAM4 modulation that are serially interfaced to the third octal serializers/deserializers block **536**. An example of the bus processing unit **538** is shown in FIG. **41A**. In some examples, the bus processing unit **538** enables N lanes of T Gbps serial electrical signals to be remapped into N/M lanes of M×T Gbps serial electrical signals, N and M being positive integers, T being a real value, in which the N serially interfacing electrical signals can be modulated using a first modulation format and the M serially interfacing electrical signals can be modulated using a second modulation format.

[0385] In some other examples, the bus processing unit **538** can allow for redundancy to increase reliability. For example, the first and the second serializers/deserializers blocks **532** and **534** can be jointly configured to serially interface to a total of N lanes of $T \times N / (N - k)$ Gbps electrical signals,

while the third serializers/deserializers block **536** can be configured to serially interface to N lanes of T Gbps electrical signals. The bus processing unit **538** can then be configured to remap the data from only N-k out of the N lanes serially interfacing to the first and the second serializers/deserializers blocks **532** and **534** (carrying an aggregate bit rate of $(N-k) \times T \times N / (N-k) = T \times N$) to the third serializers/deserializers block **536**. This way, the bus processing unit **538** allows for k out of N serially interfacing electrical links to the first and the second serializers/deserializers blocks **532** and **534** to fail while still maintaining an aggregate of $T \times N$ Gbps of data serially interfacing to the third serializers/deserializers block **536**. The number k is a positive integer. In some embodiments, k can be approximately 1% of N. In some other embodiments, k can be approximately 10% of N. In some embodiment, the selection of which N-k of the N serially interfacing electrical links to the first and the second serializers/deserializers blocks **532** and **534** to remap to the third serializers/deserializers block **536** using bus processing unit **538** can be dynamically selected, e.g., based on signal integrity and signal performance information extracted from the serially interfacing signals by the serializers/deserializers blocks **532** and **534**. An example of the bus processing unit **538** is shown in FIG. **41B**, in which $N=16$, $k=2$, $T=50$ Gbps.

[0386] In some examples, using the redundancy technique discussed above, the bus processing unit **538** enables N lanes of $T \times N / (N-k)$ Gbps serial electrical signals to be remapped into N/M lanes of $M \times T$ Gbps serial electrical signals. The bus processing unit **538** enables k out of N serially interfacing electrical links to fail while still maintaining an aggregate of $T \times N$ Gbps of data serially interfacing to the third serializers/deserializers block **536**.

[0387] FIG. **20** is a functional block diagram of an example data processing system **200**, which can be used to implement, e.g., one or more of devices **101_1** to **101_6** of FIG. **1**. Without implied limitation, the data processing system **200** is shown as part of the node **101_1** for illustration purposes. The data processing system **200** can be part of any other network element of the system **100**. The data processing system **200** includes an integrated communication device **210**, a fiber-optic connector assembly **220**, a package substrate **230**, and an electronic processor integrated circuit **240**.

[0388] The connector assembly **220** includes a connector **223** and a fiber array **226**. The connector **223** can include multiple individual fiber-optic connectors **423_i** ($i \in \{R1 \dots RM; S1 \dots SK; T1 \dots TN\}$ with K, M, and N being positive integers). In some embodiments, some or all of the individual connectors **423_i** can form a single physical entity. In some embodiments some or all of the individual connectors **423_i** can be separate physical entities. When operating as part of the network element **101_1** of the system **100**, (i) the connectors **423_S1** through **423_SK** can be connected to optical power supply **103**, e.g., through link **102_6**, to receive supply light; (ii) the connectors **423_R1** through **423_RM** can be connected to the transmitters of the node **101_2**, e.g., through the link **102_1**, to receive from the node **101_2** optical communication signals; and (iii) the connectors **423_T1** through **423_TN** can be connected to the receivers of the node **101_2**, e.g., through the link **102_1**, to transmit to the node **101_2** optical communication signals.

[0389] In some implementations, the communication device **210** includes an electronic communication integrated circuit **215**, a photonic integrated circuit **214**, a connector part **213**, and a substrate **211**. The connector part **213** can include multiple individual optical connectors **413_i** to photonic integrated circuit **214** ($i \in \{R1 \dots RM; S1 \dots SK; T1 \dots TN\}$ with K, M, and N being positive integers). In some embodiments, some or all of the individual connectors **413_i** can form a single physical entity. In some embodiments some or all of the individual connectors **413_i** can be separate physical entities. The optical connectors **413_i** are configured to optically couple light to the photonic integrated circuit **214** using optical coupling interfaces **414**, e.g., vertical grating couplers, turning mirrors, etc., as disclosed in U.S. patent application Ser. No. 16/816,171, now U.S. Pat. No. 11,287,585.

[0390] In operation, light entering the photonic integrated circuit **214** from the link **102_6** through

coupling interfaces **414_S1** through **414_SK** can be split using an optical splitter **415**. The optical splitter **415** can be an optical power splitter, an optical polarization splitter, an optical wavelength demultiplexer, or any combination or cascade thereof, e.g., as disclosed in U.S. patent application Ser. No. 16/847,705, now U.S. Pat. No. 11,153,670, and in U.S. patent application Ser. No. 16/888,890, filed on Jun. 1, 2020, now U.S. Pat. No. 11,621,795, which is incorporated herein by reference in its entirety. In some embodiments, one or more splitting functions of the splitter **415** can be integrated into the optical coupling interfaces **414** and/or into optical connectors **413**. For example, in some embodiments, a polarization-diversity vertical grating coupler can be configured to simultaneously act as a polarization splitter **415** and as a part of optical coupling interface **414**. In some other embodiments, an optical connector that includes a polarization-diversity arrangement can simultaneously act as an optical connector **413** and as a polarization splitter **415**.

[0391] In some embodiments, light at one or more outputs of the splitter **415** can be detected using a receiver **416**, e.g., to extract synchronization information as disclosed in U.S. patent application Ser. No. 16/847,705, now U.S. Pat. No. 11,153,670. In various embodiments, the receiver **416** can include one or more p-i-n photodiodes, one or more avalanche photodiodes, one or more self-coherent receivers, or one or more analog (heterodyne/homodyne) or digital (intradyn) coherent receivers. In some embodiments, one or more opto-electronic modulators **417** can be used to modulate onto light at one or more outputs of the splitter **415** data for communication to other network elements.

[0392] Modulated light at the output of the modulators **417** can be multiplexed in polarization or wavelength using a multiplexer **418** before leaving the photonic integrated circuit **214** through optical coupling interfaces **414_T1** through **414_TN**. In some embodiments, the multiplexer **418** is not provided, i.e., the output of each modulator **417** can be directly coupled to a corresponding optical coupling interface **414**.

[0393] On the receiver side, light entering the photonic integrated circuit **214** through a coupling interfaces **414_R1** through **414_RM** from, e.g., the link **101_2**, can first be demultiplexed in polarization and/or in wavelength using an optical demultiplexer **419**. The outputs of the demultiplexer **419** are then individually detected using receivers **421**. In some embodiments, the demultiplexer **419** is not provided, i.e., the output of each coupling interface **414_R1** through **414_RM** can be directly coupled to a corresponding receiver **421**. In various embodiments, the receiver **421** can include one or more p-i-n photodiodes, one or more avalanche photodiodes, one or more self-coherent receivers, or one or more analog (heterodyne/homodyne) or digital (intradyn) coherent receivers.

[0394] The photonic integrated circuit **214** is electrically coupled to the integrated circuit **215**. In some implementations, the photonic integrated circuit **214** provides a plurality of serial electrical signals to the first serializers/deserializers module **216**, which generates sets of parallel electrical signals based on the serial electrical signals, in which each set of parallel electrical signal is generated based on a corresponding serial electrical signal. The first serializers/deserializers module **216** conditions the serial electrical signals, demultiplexes them into the sets of parallel electrical signals and sends the sets of parallel electrical signals to the second serializers/deserializers module **217** through a bus processing unit **218**. In some implementations, the bus processing unit **218** enables switching of signals and performs line coding and/or error-correcting coding functions. An example of the bus processing unit **218** is shown in FIG. 42.

[0395] The second serializers/deserializers module **217** generates a plurality of serial electrical signals based on the sets of parallel electrical signals, in which each serial electrical signal is generated based on a corresponding set of parallel electrical signal. The second serializers/deserializers module **217** sends the serial electrical signals through electrical connectors that pass through the substrate **211** in the thickness direction to an array of electrical terminals **500** that are arranged on the bottom surface of the substrate **211**. For example, the array of electrical terminals **500** configured to enable the integrated communication device **210** to be easily coupled

to, or removed from, the package substrate **230**.

[0396] In some implementations, the electronic processor integrated circuit **240** includes a data processor **502** and an embedded third serializers/deserializers module **504**. The third serializers/deserializers module **504** receives the serial electrical signals from the second serializers/deserializers module **217**, and generates sets of parallel electrical signals based on the serial electrical signals, in which each set of parallel electrical signal is generated based on a corresponding serial electrical signal. The data processor **502** processes the sets of parallel signals generated by the third serializers/deserializers module **504**.

[0397] In some implementations, the data processor **502** generates sets of parallel electrical signals, and the third serializers/deserializers module **504** generates serial electrical signals based on the sets of parallel electrical signals, in which each serial electrical signal is generated based on a corresponding set of parallel electrical signal. The serial electrical signals are sent to the second serializers/deserializers module **217**, which generates sets of parallel electrical signals based on the serial electrical signals, in which each set of parallel electrical signal is generated based on a corresponding serial electrical signal. The second serializers/deserializers module **217** sends the sets of parallel electrical signals to the first serializers/deserializers module **216** through the bus processing unit **218**. The first serializers/deserializers module **216** generates serial electrical signals based on the sets of parallel electrical signals, in which each serial electrical signal is generated based on a corresponding set of parallel electrical signals. The first serializers/deserializers module **216** sends the serial electrical signals to the photonic integrated circuit **214**. The opto-electronic modulators **417** modulate optical signals based on the serial electrical signals, and the modulated optical signals are output from the photonic integrated circuit **214** through optical coupling interfaces **414_T1** through **414_TN**.

[0398] In some embodiments, supply light from the optical power supply **103** includes an optical pulse train, and synchronization information extracted by the receiver **416** can be used by the serializers/deserializers module **216** to align the electrical output signals of the serializers/deserializers module **216** with respective copies of the optical pulse trains at the outputs of the splitter **415** at the modulators **417**. For example, the optical pulse train can be used as an optical power supply at the optical modulator. In some such implementations, the first serializers/deserializers module **216** can include interpolators or other electrical phase adjustment elements.

[0399] Referring to FIG. **21**, in some implementations, a data processing system **540** includes an enclosure or housing **542** that has a front panel **544**, a bottom panel **546**, side panels **548** and **550**, a rear panel **552**, and a top panel (not shown in the figure). The system **540** includes a printed circuit board **558** that extends substantially parallel to the bottom panel **546**. A data processing chip **554** is mounted on the printed circuit board **558**, in which the chip **554** can be, e.g., a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, or an application specific integrated circuit (ASIC).

[0400] At the front panel **544** are pluggable input/output interfaces **556** that allow the data processing chip **554** to communicate with other systems and devices. For example, the input/output interfaces **556** can receive optical signals from outside of the system **540** and convert the optical signals to electrical signals for processing by the data processing chip **554**. The input/output interfaces **556** can receive electrical signals from the data processing chip **554** and convert the electrical signals to optical signals that are transmitted to other systems or devices. For example, the input/output interfaces **556** can include one or more of small form-factor pluggable (SFP), SFP+, SFP28, QSFP, QSFP28, or QSFP56 transceivers. The electrical signals from the transceiver outputs are routed to the data processing chip **554** through electrical connectors on or in the printed circuit board **558**.

[0401] In the examples shown in FIGS. **21** to **29B**, **69A**, **70**, **71A**, **72**, **72A**, **74A**, **75A**, **75C**, **76**,

77A, 77B, 78, 96 to 98, 100, 110, 112, 113, 115, 117 to 122, 125A to 127, 129, 136 to 149, 159, and 160, various embodiments can have various form factors, e.g., in some embodiments the top panel and the bottom panel 546 can have the largest area, in other embodiments the side panels 548 and 550 can have the largest area, and in yet other embodiments the front panel 544 and the rear panel 552 can have the largest area. In various embodiments, the printed circuit board 558 can be substantially parallel to the two side panels, e.g., the data processing system 540 as shown in FIG. 21 can stand on one of its side panels during normal operation (such that the side panel 550 is positioned at the bottom, and the bottom panel 546 is positioned at the side). In various embodiments, the data processing system 540 can comprise two or more printed circuit boards some of which can be substantially parallel to the bottom panel and some of which can be substantially parallel to the side panels. For example, in some computer systems for machine learning/artificial intelligence applications have vertical circuit boards that are plugged into the systems. As used herein, the distinction between “front” and “back” is made based on where the majority of input/output interfaces 556 are located, irrespective of what a user may consider the front or back of data processing system 540.

[0402] FIG. 22 is a diagram of a top view of an example data processing system 560 that includes a housing 562 having side panels 564 and 566, and a rear panel 568. The system 560 includes a vertically mounted printed circuit board 570 that can also function as the front panel. The surface of the printed circuit board 570 is substantially perpendicular to the bottom panel of the housing 562. The term “substantially perpendicular” is meant to take into account of manufacturing and assembly tolerances, so that if a first surface is substantially perpendicular to a second surface, the first surface is at an angle in a range from 85° to 95° relative to the second surface. On the printed circuit board 570 are mounted a data processing chip 572 and an integrated communication device 574. In some examples, the data processing chip 572 and the integrated communication device 574 are mounted on a substrate (e.g., a ceramic substrate), and the substrate is attached (e.g., electrically coupled) to the printed circuit board 570. The data processing chip 572 can be, e.g., a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, or an application specific integrated circuit (ASIC). A heat sink 576 is provided on the data processing chip 572.

[0403] In some implementations, the integrated communication device 574 includes a photonic integrated circuit 586 and an electronic communication integrated circuit 588 mounted on a substrate 594. The electronic communication integrated circuit 588 includes a first serializers/deserializers module 590 and a second serializers/deserializers module 592. The printed circuit board 570 can be similar to the package substrate 230 (FIGS. 2, 4, 11-14), the data processing chip 572 can be similar to the electronic processor integrated circuit or application specific integrated circuit 240, and the integrated communication device 574 can be similar to the integrated communication device 210, 252, 374, 382, 402, 428. In some embodiments, the integrated communication device 574 is soldered to the printed circuit board 570. In some other embodiments, the integrated communication device 574 is removably connected to the printed circuit board 570, e.g., via a land grid array or a compression interposer. Related holding fixtures including snap-on or screw-on mechanisms are not shown in the figure.

[0404] In some examples, the integrated communication device 574 includes a photonic integrated circuit without serializers/deserializers modules, and drivers/transimpedance amplifiers (TIA) are provided separately. In some examples, the integrated communication device 574 includes a photonic integrated circuit and drivers/transimpedance amplifiers but without serializers/deserializers modules.

[0405] The integrated communication device 574 includes a first optical connector 578 that is configured to receive a second optical connector 580 that is coupled to a bundle of optical fibers 582. The integrated communication device 574 is electrically coupled to the data processing chip

572 through electrical connectors or traces 584 on or in the printed circuit board 570. Because the data processing chip 572 and the integrated communication device 574 are both mounted on the printed circuit board 570, the electrical connectors or traces 584 can be made shorter, compared to the electrical connectors that electrically couple the transceivers 556 to the data processing chip 554 of FIG. 21. Using shorter electrical connectors or traces 584 allows the signals to have a higher data rate with lower noise, lower distortion, and/or lower crosstalk. Mounting the printed circuit board 570 perpendicular to the bottom panel of the housing allows for more easily accessible connections to the integrated communication device 574 that may be removed and re-connected without, e.g., removing the housing from a rack.

[0406] In some examples, the bundle of optical fibers 582 can be firmly attached to the photonic integrated circuit 586 without the use of the first and second optical connectors 578, 580.

[0407] The printed circuit board 570 can be secured to the side panels 564 and 566, and the bottom and top panels of the housing using, e.g., brackets, screws, clips, and/or other types of fastening mechanisms. The surface of the printed circuit board 570 can be oriented perpendicular to bottom panel of the housing, or at an angle (e.g., between -60° to 60°) relative to the vertical direction (the vertical direction being perpendicular to the bottom panel). The printed circuit board 570 can have multiple layers, in which the outermost layer (i.e., the layer facing the user) has an exterior surface that is configured to be aesthetically pleasing.

[0408] The first optical connector 578, the second optical connector 580, and the bundle of optical fibers 582 can be similar to those shown in FIGS. 2, 4, and 11-16. As described above, the bundle of fibers 582 can include 10 or more optical fibers, 100 or more optical fibers, 500 or more optical fibers, or 1000 or more optical fibers. The optical signals provided to the photonic integrated circuit 586 can have a high total bandwidth, e.g., about 1.6 Tbps, or about 12.8 Tbps, or more.

[0409] Although FIG. 22 shows one integrated communication device 574, there can be additional integrated communication devices 574 that are electrically coupled to the data processing chip 572. The data processing system 560 can include a second printed circuit board (not shown in the figure) oriented parallel to the bottom panel of the housing 562. The second printed circuit board can support other optical and/or electronic devices, such as storage devices, memory chips, controllers, power supply modules, fans, and other cooling devices.

[0410] In some examples of the data processing system 540 (FIG. 21), the transceiver 556 can include circuitry (e.g., integrated circuits) that perform some type of processing of the signals and/or the data contained in the signals. The signals output from the transceiver 556 need to be routed to the data processing chip 554 through longer signal paths that place a limit on the data rate. In some data processing systems, the data processing chip 554 outputs processed data that are routed to one of the transceivers and transmitted to another system or device. Again, the signals output from the data processing chip 554 need to be routed to the transceiver 556 through longer signal paths that place a limit on the data rate. By comparison, in the data processing system 560 (FIG. 22), the electrical signals that are transmitted between the integrated communication devices 574 and the data processing chip 572 pass through shorter signal paths and thus support a higher data rate.

[0411] FIG. 23 is a diagram of a top view of an example data processing system 600 that includes a housing 602 having side panels 604 and 606, and a rear panel 608. The system 600 includes a vertically mounted printed circuit board 610 that functions as the front panel. The surface of the printed circuit board 610 is substantially perpendicular to the bottom panel of the housing 602. A data processing chip 572 is mounted on an interior side of the printed circuit board 610, and an integrated communication device 612 is mounted on an exterior side of the printed circuit board 610. In some examples, the data processing chip 572 is mounted on a substrate (e.g., a ceramic substrate), and the substrate is attached to the printed circuit board 610. In some embodiments, the integrated communication device 612 is soldered to the printed circuit board 610. In some other embodiments, the integrated communication device 612 is removably connected to the printed

circuit board **610**, e.g., via a land grid array or a compression interposer. Related holding fixtures including snap-on or screw-on mechanisms are not shown in the figure. A heat sink **576** is provided on the data processing chip **572**.

[0412] In some implementations, the integrated communication device **612** includes a photonic integrated circuit **614** and an electronic communication integrated circuit **588** mounted on a substrate **618**. The electronic communication integrated circuit **588** includes a first serializers/deserializers module **590** and a second serializers/deserializers module **592**. The integrated communication device **612** includes a first optical connector **578** that is configured to receive a second optical connector **580** that is coupled to a bundle of optical fibers **582**. The integrated communication device **612** is electrically coupled to the data processing chip **572** through electrical connectors or traces **616** that pass through the printed circuit board **610** in the thickness direction. Because the data processing chip **572** and the integrated communication device **612** are both mounted on the printed circuit board **610**, the electrical connectors or traces **616** can be made shorter, thereby allowing the signals to have a higher data rate with lower noise, lower distortion, and/or lower crosstalk. Mounting the integrated communication device **612** on the outside of the printed circuit board **610** perpendicular to the bottom panel of the housing and accessible from outside the housing allows for more easily accessible connections to the integrated communication device **612** that may be removed and re-connected without, e.g., removing the housing from a rack.

[0413] In some examples, the integrated communication device **612** includes a photonic integrated circuit without serializers/deserializers modules, and drivers and transimpedance amplifiers (TIA) are provided separately. In some examples, the integrated communication device **612** includes a photonic integrated circuit and drivers/transimpedance amplifiers but without serializers/deserializers modules. In some examples, the bundle of optical fibers **582** can be firmly attached to the photonic integrated circuit **614** without the use of the first and second optical connectors **578**, **580**.

[0414] In some examples, the data processing chip **572** is mounted on the rear side of the substrate, and the integrated communication device **612** are removably attached to the front side of the substrate, in which the substrate provides high speed connections between the data processing chip **572** and the integrated communication device **612**. For example, the substrate can be attached to a front side of a printed circuit board, in which the printed circuit board includes an opening that allows the data processing chip **572** to be mounted on the rear side of the substrate. The printed circuit board can provide from a motherboard electrical power to the substrate (and hence to the data processing chip **572** and the integrated communication device **612**, and allow the data processing chip **572** and the integrated communication device **612** to connect to the motherboard using low-speed electrical links.

[0415] The printed circuit board **610** can be secured to the side panels **604** and **606**, and the bottom and top panels of the housing using, e.g., brackets, screws, clips, and/or other types of fastening mechanisms. The surface of the printed circuit board **610** can be oriented perpendicular to bottom panel of the housing, or at an angle (e.g., between -60° to 60°) relative to the vertical direction (the vertical direction being perpendicular to the bottom panel). The printed circuit board **610** can have multiple layers, in which the portion of the outermost layer (i.e., the layer facing the user) not covered by the integrated communication device **612** has an exterior surface that is configured to be aesthetically pleasing.

[0416] FIGS. **24-27** below illustrate four general designs in which the data processing chips are positioned near the input/output communication interfaces. FIG. **24** is a top view of an example data processing system **630** in which a data processing chip **640** is mounted near an optical/electrical communication interface **644** to enable high bandwidth data paths (e.g., one, ten, or more Gigabits per second per data path) between the data processing chip **640** and the optical/electrical communication interface **644**. In this example, the data processing chip **640** and

the optical/electrical communication interface **644** are mounted on a circuit board **642** that functions as the front panel of an enclosure **632** of the system **630**, thus allowing optical fibers to be easily coupled to the optical/electrical communication interface **644**. In some examples, the data processing chip **640** is mounted on a substrate (e.g., a ceramic substrate), and the substrate is attached to the circuit board **642**.

[0417] The enclosure **632** has side panels **634** and **636**, a rear panel **638**, a top panel, and a bottom panel. In some examples, the circuit board **642** is perpendicular to the bottom panel. In some examples, the circuit board **642** is oriented at an angle in a range -60° to 60° relative to a vertical direction of the bottom panel. The side of the circuit board **642** facing the user is configured to be aesthetically pleasing.

[0418] The optical/electrical communication interface **644** is electrically coupled to the data processing chip **640** by electrical connectors or traces **646** on or in the circuit board **642**. The circuit board **642** can be a printed circuit board that has one or more layers. The electrical connectors or traces **646** can be signal lines printed on the one or more layers of the printed circuit board **642** and provide high bandwidth data paths (e.g., one or more Gigabits per second per data path) between the data processing chip **640** and the optical/electrical communication interface **644**.

[0419] In a first example, the data processing chip **640** receives electrical signals from the optical/electrical communication interface **644** and does not send electrical signals to the optical/electrical communication interface **644**. In a second example, the data processing chip **640** receives electrical signals from, and sends electrical signals to, the optical/electrical communication interface **644**. In the first example, the optical/electrical communication interface **644** receives optical signals from optical fibers, generates electrical signals based on the optical signals, and sends the electrical signals to the data processing chip **640**. In the second example, the optical/electrical communication interface **644** also receives electrical signals from the data processing chip, generates optical signals based on the electrical signals, and sends the optical signals to the optical fibers.

[0420] An optical connector **648** is provided to couple optical signals from the optical fibers to the optical/electrical communication interface **644**. In this example, the optical connector **648** passes through an opening in the circuit board **642**. In some examples, the optical connector **648** is securely fixed to the optical/electrical communication interface **644**. In some examples, the optical connector **648** is configured to be removably coupled to the optical/electrical communication interface **644**, e.g., by using a pluggable and releasable mechanism, which can include one or more snap-on or screw-on mechanisms. In some other examples, an array of 10 or more fibers is securely or fixedly attached to the optical connector **648**.

[0421] The optical/electrical communication interface **644** can be similar to, e.g., the integrated communication device **210** (FIG. 2), **252** (FIG. 4), **374** (FIG. 11), **382** (FIG. 12), **402** (FIG. 13), and **428** (FIG. 14). In some examples, the optical/electrical communication interface **644** can be similar to the integrated optical communication device **448**, **462**, **466**, **472** (FIG. 17), except that the optical/electrical communication interface **644** is mounted on the same side of the circuit board **642** as the data processing chip **640**. The optical connector **648** can be similar to, e.g., the first optical connector part **213** (FIGS. 2, 4), the first optical connector **356** (FIGS. 11, 12), the first optical connector **404** (FIGS. 13, 14), and the first optical connector part **456** (FIG. 17). In some examples, a portion of the optical connector **648** can be part of the optical/electrical communication interface **644**. In some examples, the optical connector **648** can also include the second optical connector part **223** (FIGS. 2, 4), **458** (FIG. 17) that is optically coupled to the optical fibers. FIG. 24 shows that the optical connector **648** passes through the circuit board **642**. In some examples, the optical connector **648** can be short so that the optical fibers pass through, or partly through, the circuit board **642**. In some examples, the optical connector is not attached vertically to a photonic integrated circuit that is part of the optical/electrical communication interface **644** but rather can be attached in-plane to the photonic integrated circuit using, e.g., V-groove fiber attachments, tapered

or un-tapered fiber edge coupling, etc., followed by a mechanism to direct the light interfacing to the photonic integrated circuit to a direction that is substantially perpendicular to the photonic integrated circuit, such as one or more substantially 90-degree turning mirrors, one or more substantially 90-degree bent optical fibers, etc. Any such solution is conceptually included in the vertical optical coupling attachment schematically visualized in FIGS. 24-27.

[0422] FIG. 25 is a top view of an example data processing system 650 in which a data processing chip 670 is mounted near an optical/electrical communication interface 652 to enable high bandwidth data paths (e.g., one, ten, or more Gigabits per second per data path) between the data processing chip 670 and the optical/electrical communication interface 652. In this example, the data processing chip 670 and the optical/electrical communication interface 652 are mounted on a circuit board 654 that is positioned near a front panel 656 of an enclosure 658 of the system 630, thus allowing optical fibers to be easily coupled to the optical/electrical communication interface 652. In some examples, the data processing chip 670 is mounted on a substrate (e.g., a ceramic substrate), and the substrate is attached to the circuit board 654.

[0423] The enclosure 658 has side panels 660 and 662, a rear panel 664, a top panel, and a bottom panel. In some examples, the circuit board 654 and the front panel 656 are perpendicular to the bottom panel. In some examples, the circuit board 654 and the front panel 656 are oriented at an angle in a range -60° to 60° relative to a vertical direction of the bottom panel. In some examples, the circuit board 654 is substantially parallel to the front panel 656, e.g., the angle between the surface of the circuit board 654 and the surface of the front panel 656 can be in a range of -5° to 5° . In some examples, the circuit board 654 is at an angle relative to the front panel 656, in which the angle is in a range of -45° to 45° .

[0424] The optical/electrical communication interface 652 is electrically coupled to the data processing chip 670 by electrical connectors or traces 666 on or in the circuit board 654, similar to those of the system 630. The signal path between the data processing chip 670 and the optical/electrical communication interface 652 can be unidirectional or bidirectional, similar to that of the system 630.

[0425] An optical connector 668 is provided to couple optical signals from the optical fibers to the optical/electrical communication interface 652. In this example, the optical connector 668 passes through an opening in the front panel 656 and an opening in the circuit board 654. The optical connector 668 can be securely fixed, or releasably connected, to the optical/electrical communication interface 652, similar to that of the system 630.

[0426] The optical/electrical communication interface 652 can be similar to, e.g., the integrated communication device 210 (FIG. 2), 252 (FIG. 4), 374 (FIG. 11), 382 (FIG. 12), 402 (FIG. 13), and 428 (FIG. 14). In some examples, the optical/electrical communication interface 652 can be similar to the integrated optical communication device 448, 462, 466, 472 (FIG. 17), except that the optical/electrical communication interface 652 is mounted on the same side of the circuit board 654 as the data processing chip 640. The optical connector 668 can be similar to, e.g., the first optical connector part 213 (FIGS. 2, 4), the first optical connector 356 (FIGS. 11, 12), the first optical connector 404 (FIGS. 13, 14), and the first optical connector part 456 (FIG. 17). In some examples, the optical connector is not attached vertically to a photonic integrated circuit that is part of the optical/electrical communication interface 652 but rather can be attached in-plane to the photonic integrated circuit using, e.g., V-groove fiber attachments, tapered or un-tapered fiber edge coupling, etc., followed by a mechanism to direct the light interfacing to the photonic integrated circuit to a direction that is substantially perpendicular to the photonic integrated circuit, such as one or more substantially 90-degree turning mirrors, one or more substantially 90-degree bent optical fibers, etc. In some examples, a portion of the optical connector 668 can be part of the optical/electrical communication interface 652. In some examples, the optical connector 668 can also include the second optical connector part 223 (FIGS. 2, 4), 458 (FIG. 17) that is optically coupled to the optical fibers. FIG. 25 shows that the optical connector 668 passes through the front panel 656 and the

circuit board **654**. In some examples, the optical connector **668** can be short so that the optical fibers pass through, or partly through, the front panel **656**. The optical fibers can also pass through, or partly through, the circuit board **654**.

[0427] In the examples of FIGS. **24** and **25**, only one optical/electrical communication interface (**544**, **652**) is shown in the figures. It is understood that the systems **630**, **650** can include multiple optical/electrical communication interfaces that are mounted on the same circuit board as the data processing chip to enable high bandwidth data paths (e.g., one, ten, or more Gigabits per second per data path) between the data processing chip and each of the optical/electrical communication interfaces.

[0428] FIG. **26A** is a top view of an example data processing system **680** in which a data processing chip **681** is mounted near optical/electrical communication interfaces **682a**, **682b**, **682c** (collectively referenced as **682**) to enable high bandwidth data paths (e.g., one, ten, or more Gigabits per second per data path) between the data processing chip **681** and each of the optical/electrical communication interfaces **682**. The data processing chip **681** is mounted on a first side of a circuit board **683** that functions as a front panel of an enclosure **684** of the system **680**. In some examples, the data processing chip **681** is mounted on a substrate (e.g., a ceramic substrate), and the substrate is attached to the circuit board **683**. The optical/electrical communication interfaces **682** are mounted on a second side of the circuit board **683**, in which the second side faces the exterior of the enclosure **684**. In this example, the optical/electrical communication interfaces **682** are mounted on an exterior side of the enclosure **684**, allowing optical fibers to be easily coupled to the optical/electrical communication interfaces **682**.

[0429] The enclosure **684** has side panels **685** and **686**, a rear panel **687**, a top panel, and a bottom panel. In some examples, the circuit board **683** is perpendicular to the bottom panel. In some examples, the circuit board **683** is oriented at an angle in a range -60° to 60° (or -30° to 30° , or -10° to 10° , or -1° to 1°) relative to a vertical direction of the bottom panel.

[0430] Each of the optical/electrical communication interfaces **682** is electrically coupled to the data processing chip **681** by electrical connectors or traces **688** that pass through the circuit board **683** in the thickness direction. For example, the electrical connectors or traces **688** can be configured as vias of the circuit board **683**. The signal paths between the data processing chip **681** and each of the optical/electrical communication interfaces **682** can be unidirectional or bidirectional, similar to those of the systems **630** and **650**.

[0431] For example, the system **680** can be configured such that signals are transmitted unidirectionally between the data processing chip **681** and one of the optical/electrical communication interfaces **682**, and bidirectionally between the data processing chip **681** and another one of the optical/electrical communication interfaces **682**. For example, the system **680** can be configured such that signals are transmitted unidirectionally from the optical/electrical communication interface **682a** to the data processing chip **681**, and unidirectionally from the data processing chip to the optical/electrical communication interface **682b** and/or optical/electrical communication interface **682c**.

[0432] Optical connectors **689a**, **689b**, **689c** (collectively referenced as **689**) are provided to couple optical signals from the optical fibers to the optical/electrical communication interfaces **682a**, **682b**, **682c**, respectively. The optical connectors **689** can be securely fixed, or releasably connected, to the optical/electrical communication interfaces **682**, similar to those of the systems **630** and **650**.

[0433] The optical/electrical communication interface **682** can be similar to, e.g., the integrated communication device **210** (FIG. **2**), **252** (FIG. **4**), **374** (FIG. **11**), **382** (FIG. **12**), **402** (FIG. **13**), **428** (FIG. **14**), and **512** (FIG. **32**), except that the optical/electrical communication interface **682** is mounted on the side of the circuit board **683** opposite to the side of the data processing chip **681**. In some examples, the optical/electrical communication interface **682** can be similar to the integrated optical communication device **448**, **462**, **466**, **472** (FIG. **17**). The optical connector **689** can be similar to, e.g., the first optical connector part **213** (FIGS. **2**, **4**), the first optical connector **356**

(FIGS. 11, 12), the first optical connector **404** (FIGS. 13, 14), the first optical connector part **456** (FIG. 17), and the first optical connector part **520** (FIG. 32). In some examples, the optical connector is not attached vertically to a photonic integrated circuit that is part of the optical/electrical communication interface **682** but rather can be attached in-plane to the photonic integrated circuit using, e.g., V-groove fiber attachments, tapered or un-tapered fiber edge coupling, etc., followed by a mechanism to direct the light interfacing to the photonic integrated circuit to a direction that is substantially perpendicular to the photonic integrated circuit, such as one or more substantially 90-degree turning mirrors, one or more substantially 90-degree bent optical fibers, etc. In some examples, a portion of the optical connector **689** can be part of the optical/electrical communication interface **682**. In some examples, the optical connector **689** can also include the second optical connector part **223** (FIGS. 2, 4), **458** (FIG. 17) that is optically coupled to the optical fibers.

[0434] In some examples, the optical/electrical communication interfaces **682** are securely fixed (e.g., by soldering) to the circuit board **683**. In some examples, the optical/electrical communication interfaces **682** are removably connected to the circuit board **683**, e.g., by use of mechanical mechanisms such as one or more snap-on or screw-on mechanisms. An advantage of the system **680** is that in case of a malfunction at one of the optical/electrical communication interfaces **682**, the faulty optical/electrical communication interface **682** can be replaced without opening the enclosure **684**.

[0435] FIG. 26B is a top view of an example data processing system **690b** in which a data processing chip **691b** is mounted near optical/electrical communication interfaces **692a**, **692b**, **692c** (collectively referenced as **692**) to enable high bandwidth data paths (e.g., one, ten, or more Gigabits per second per data path) between the data processing chip **691b** and each of the optical/electrical communication interfaces **692**. The data processing chip **691** is mounted on a first side of a circuit board **693b** that functions as a front panel of an enclosure **694b** of the system **690b**. In this example, the optical/electrical communication interface **692a** is mounted on the first side of the circuit board **693b** and the optical/electrical communication interfaces **692b** and **692c** are mounted on a second side of the circuit board **693b**, in which the second side faces the exterior of the enclosure **694b**. In this example, the optical/electrical communication interfaces **692b** and **692c** are mounted on an exterior side of the enclosure **694b**, allowing connection to optical fiber from the front of the enclosure **694b** while the optical/electrical communication interface **692a** is located internal to the enclosure **694b**, for example, to allow connection to optical fiber at the rear of the enclosure **694b**. In some examples, two or more of the optical/electrical communication interfaces **692** can be located internal to the enclosure **694b** and connect to optical fibers at the rear of the enclosure **694b**.

[0436] The enclosure **694b** has side panels **695b** and **696b**, a rear panel **697b**, a top panel, and a bottom panel. In some examples, the circuit board **693b** is perpendicular to the bottom panel. In some examples, the circuit board **693b** is oriented at an angle in a range -60° to 60° (or -30° to 30° , or -10° to 10° , or -1° to 1°) relative to a vertical direction of the bottom panel.

[0437] Each of the optical/electrical communication interfaces **692** is electrically coupled to the data processing chip **691b** by electrical connectors or traces **698b** that pass through the circuit board **693b** in the thickness direction. For example, the electrical connectors or traces **698b** can be configured as vias of the circuit board **693b**. In this example, the electrical connectors or traces **698b** extend to both sides of the circuit board **693b** (e.g., for connecting to optical/electrical communication interfaces **692** located internal to and external of the enclosure **694b**). The signal paths between the data processing chip **691b** and each of the optical/electrical communication interfaces **692** can be unidirectional or bidirectional, similar to those of the systems **630**, **650** and **680**.

[0438] For example, the system **690b** can be configured such that signals are transmitted unidirectionally between the data processing chip **691b** and one of the optical/electrical

communication interfaces **692**, and bidirectionally between the data processing chip **691b** and another one of the optical/electrical communication interfaces **692**. For example, the system **690b** can be configured such that signals are transmitted unidirectionally from the optical/electrical communication interface **692a** to the data processing chip **691b**, and unidirectionally from the data processing chip **691b** to the optical/electrical communication interface **692b** and/or optical/electrical communication interface **692c**.

[0439] Optical connectors **699a**, **699b**, **699c** (collectively referenced as **699**) are provided to couple optical signals from the optical fibers to the optical/electrical communication interfaces **692a**, **692b**, **692c**, respectively. The optical connectors **699** can be securely fixed, or releasably connected, to the optical/electrical communication interfaces **692**, similar to those of the systems **630**, **650**, and **680**. In this example, optical connector **699b** and optical connector **699c** can connect to optical fibers at the front of the enclosure **694b** and the optical connector **699a** can connect to optical fibers at the rear of the enclosure **694b**. In the illustrated example, the optical connector **699a** connects to an optical fiber at the rear of the enclosure **694b** by being connected to a fiber **1000b** that connects to a rear panel interface **1001b** (e.g., a backplane, etc.) that is mounted to the rear panel **697b**. In some examples, the optical connectors **699** can be securely or fixedly attached to communication interfaces **692**. In some examples, the optical connectors **699** can be securely or fixedly attached to an array of optical fibers.

[0440] The optical/electrical communication interface **692** can be similar to, e.g., the integrated communication device **210** (FIG. 2), **252** (FIG. 4), **374** (FIG. 11), **382** (FIG. 12), **402** (FIG. 13), **428** (FIG. 14), and **512** (FIG. 32), except that the optical/electrical communication interfaces **692b** and **692c** are mounted on the side of the circuit board **693b** opposite to the side of the data processing chip **691b**. In some examples, the optical/electrical communication interface **692** can be similar to the integrated optical communication device **448**, **462**, **466**, **472** (FIG. 17). The optical connector **699** can be similar to, e.g., the first optical connector part **213** (FIGS. 2, 4), the first optical connector **356** (FIGS. 11, 12), the first optical connector **404** (FIGS. 13, 14), the first optical connector part **456** (FIG. 17), and the first optical connector part **520** (FIG. 32). In some examples, the optical connector is not attached vertically to a photonic integrated circuit that is part of the optical/electrical communication interface **692** but rather can be attached in-plane to the photonic integrated circuit using, e.g., V-groove fiber attachments, tapered or un-tapered fiber edge coupling, etc., followed by a mechanism to direct the light interfacing to the photonic integrated circuit to a direction that is substantially perpendicular to the photonic integrated circuit, such as one or more substantially 90-degree turning mirrors, one or more substantially 90-degree bent optical fibers, etc. In some examples, a portion of the optical connector **699** can be part of the optical/electrical communication interface **692**. In some examples, the optical connector **699** can also include the second optical connector part **223** (FIGS. 2, 4), **458** (FIG. 17) that is optically coupled to the optical fibers.

[0441] In some examples, the optical/electrical communication interfaces **692** are securely fixed (e.g., by soldering) to the circuit board **693b**. In some examples, the optical/electrical communication interfaces **692** are removably connected to the circuit board **693b**, e.g., by use of mechanical mechanisms such as one or more snap-on or screw-on mechanisms. An advantage of the system **690b** is that in case of a malfunction at one of the optical/electrical communication interfaces **692**, the faulty optical/electrical communication interface **692** can be replaced without opening the enclosure **694b**.

[0442] FIG. 26C is a top view of an example data processing system **690c** in which a data processing chip **691c** is mounted near optical/electrical communication interfaces **692d**, **692e**, **692f** (collectively referenced as **692**) to enable high bandwidth data paths (e.g., one, ten, or more Gigabits per second per data path) between the data processing chip **691c** and each of the optical/electrical communication interfaces **692**. The data processing chip **691c** is mounted on a first side of a circuit board **693c** that functions as a front panel of an enclosure **694c** of the system

690c. In this example, the optical/electrical communication interface **692d** is mounted on the first side of the circuit board **693c** and the optical/electrical communication interfaces **692e** and **692f** are mounted on a second side of the circuit board **693c**, in which the second side faces the exterior of the enclosure **694c**. In this example, the optical/electrical communication interfaces **692e** and **692f** are mounted on an exterior side of the enclosure **694c**, allowing connection to optical fibers from the front of the enclosure **694c** while the optical/electrical communication interface **692d** is located internal to the enclosure **694c**, for example, to allow connection to optical fiber at the rear of the enclosure **694c**. In some examples, two or more of the optical/electrical communication interfaces **692** can be located internal to the enclosure **694c** and connect to optical fibers at the rear of the enclosure **694c**.

[0443] The enclosure **694c** has side panels **695c** and **696c**, a rear panel **697c**, a top panel, and a bottom panel. In some examples, the circuit board **693c** is perpendicular to the bottom panel. In some examples, the circuit board **693c** is oriented at an angle in a range -60° to 60° (or -30° to 30° , or -10° to 10° , or -1° to 1°) relative to a vertical direction of the bottom panel.

[0444] Each of the optical/electrical communication interfaces **692** is electrically coupled to the data processing chip **691c** by electrical connectors or traces **698c** that pass through the circuit board **693c** in the thickness direction. For example, the electrical connectors or traces **698c** can be configured as vias of the circuit board **693c**. In this example, the electrical connectors or traces **698c** extend to both sides of the circuit board **693b** (e.g., for connecting to optical/electrical communication interfaces **692** located internal to and external of the enclosure **694b**). The signal paths between the data processing chip **691c** and each of the optical/electrical communication interfaces **692** can be unidirectional or bidirectional, similar to those of the systems **630**, **650** and **680**.

[0445] For example, the system **690c** can be configured such that signals are transmitted unidirectionally between the data processing chip **691c** and one of the optical/electrical communication interfaces **692**, and bidirectionally between the data processing chip **691c** and another one of the optical/electrical communication interfaces **692**. For example, the system **690c** can be configured such that signals are transmitted unidirectionally from the optical/electrical communication interface **692d** to the data processing chip **691c**, and unidirectionally from the data processing chip **691c** to the optical/electrical communication interface **692e** and/or optical/electrical communication interface **692f**.

[0446] Optical connectors **699d**, **699e**, **699f** (collectively referenced as **699**) are provided to couple optical signals from the optical fibers to the optical/electrical communication interfaces **692d**, **692e**, **692f**, respectively. The optical connectors **699** can be securely fixed, or releasably connected, to the optical/electrical communication interfaces **692**, similar to those of the systems **630**, **650**, and **680**. In the illustrated example, the optical/electrical communication interfaces **692d** and optical connector **699d** are oriented differently compared to the optical/electrical communication interfaces **692a** and optical connector **699a** of FIG. 26B. Here the orientation change is a counter clockwise rotation of 90 degrees. Other types of orientation changes (e.g., rotations, pitches, tipping, etc.) may be implemented. Position changes (e.g., translations) and other types of location changes may also be employed. In this example, optical connector **699e** and optical connector **699f** can connect to optical fibers at the front of the enclosure **694c** and the optical connector **699d** can connect to optical fibers the rear of the enclosure **694c**. In the illustrated example, the optical connector **699d** connects to an optical fiber at the rear of the enclosure **694c** by being connected to a fiber **1000c** that connects to a rear panel interface **1001c** (e.g., a backplane, etc.) that is mounted to the rear panel **697c**.

[0447] The optical/electrical communication interface **692** can be similar to, e.g., the integrated communication device **210** (FIG. 2), **252** (FIG. 4), **374** (FIG. 11), **382** (FIG. 12), **402** (FIG. 13), **428** (FIG. 14), and **512** (FIG. 32), except that the optical/electrical communication interface **692e** and **692f** are mounted on the side of the circuit board **693c** opposite to the side of the data processing

chip **691c**. In some examples, the optical/electrical communication interface **692** can be similar to the integrated optical communication device **448, 462, 466, 472** (FIG. **17**). The optical connector **699** can be similar to, e.g., the first optical connector part **213** (FIGS. **2, 4**), the first optical connector **356** (FIGS. **11, 12**), the first optical connector **404** (FIGS. **13, 14**), the first optical connector part **456** (FIG. **17**), and the first optical connector part **520** (FIG. **32**). In some examples, the optical connector is not attached vertically to a photonic integrated circuit that is part of the optical/electrical communication interface **692** but rather can be attached in-plane to the photonic integrated circuit using, e.g., V-groove fiber attachments, tapered or un-tapered fiber edge coupling, etc., followed by a mechanism to direct the light interfacing to the photonic integrated circuit to a direction that is substantially perpendicular to the photonic integrated circuit, such as one or more substantially 90-degree turning mirrors, one or more substantially 90-degree bent optical fibers, etc. In some examples, a portion of the optical connector **699** can be part of the optical/electrical communication interface **692**. In some examples, the optical connector **699** can also include the second optical connector part **223** (FIGS. **2, 4**), **458** (FIG. **17**) that is optically coupled to the optical fibers.

[0448] In some examples, the optical/electrical communication interfaces **692** are securely fixed (e.g., by soldering) to the circuit board **693c**. In some examples, the optical/electrical communication interfaces **692** are removably connected to the circuit board **693c**, e.g., by use of mechanical mechanisms such as one or more snap-on or screw-on mechanisms. An advantage of the system **690c** is that in case of a malfunction at one of the optical/electrical communication interfaces **692**, the faulty optical/electrical communication interface **692** can be replaced without opening the enclosure **694c**.

[0449] FIG. **27** is a top view of an example data processing system **700** in which a data processing chip **702** is mounted near optical/electrical communication interfaces **704a, 704b, 704c** (collectively referenced as **704**) to enable high bandwidth data paths (e.g., one, ten, or more Gigabits per second per data path) between the data processing chip **702** and each of the optical/electrical communication interfaces **704**. The data processing chip **702** is mounted on a first side of a circuit board **706** that is positioned near a front panel of an enclosure **710** of the system **700**, similar to the configuration of the system **650** (FIG. **25**). In some examples, the data processing chip **702** is mounted on a substrate (e.g., a ceramic substrate), and the substrate is attached to the circuit board **706**. The optical/electrical communication interfaces **704** are mounted on a second side of the circuit board **708**. In this example, the optical/electrical communication interfaces **704** pass through openings in the front panel **708**, allowing optical fibers to be easily coupled to the optical/electrical communication interfaces **704**.

[0450] The enclosure **710** has side panels **712** and **714**, a rear panel **716**, a top panel, and a bottom panel. In some examples, the circuit board **706** and the front panel **708** are oriented at an angle in a range -60° to 60° relative to a vertical direction of the bottom panel. In some examples, the circuit board **706** is substantially parallel to the front panel **708**, e.g., the angle between the surface of the circuit board **706** and the surface of the front panel **708** can be in a range of -5° to 5° . In some examples, the circuit board **706** is at an angle relative to the front panel **708**, in which the angle is in a range of -45° to 45° .

[0451] For example, the angle can refer to a rotation around an axis that is parallel to the larger dimension of the front panel (e.g., the width dimension in a typical 1U, 2U, or 4U rackmount device), or a rotation around an axis that is parallel to the shorter dimension of the front panel (e.g., the height dimension in the 1U, 2U, or 4U rackmount device). The angle can also refer to a rotation around an axis along any other direction. For example, the circuit board **706** is positioned relative to the front panel such that components such as the interconnection modules, including optical modules or photonic integrated circuits, mounted on or attached to the circuit board **706** can be accessed through the front side, either through one or more openings in the front panel, or by opening the front panel to expose the components, without the need to separate the top or side

panels from the bottom panel. Such orientation of the circuit board (or a substrate on which a data processing module is mounted) relative to the front panel also applies to the examples shown in FIGS. 21 to 26, 28B to 29B, 69A, 70, 71A, 72, 72A, 74A, 75A, 75C, 76, 77A, 77B, 78, 96 to 98, 100, 110, 112, 113, 115, 117 to 122, 125A to 127, 129, 136 to 149, 159, and 160.

[0452] Each of the optical/electrical communication interfaces 704 is electrically coupled to the data processing chip 702 by electrical connectors or traces 718 that pass through the circuit board 706 in the thickness direction, similar to those of the system 680 (FIG. 26). The signal paths between the data processing chip 702 and each of the optical/electrical communication interfaces 704 can be unidirectional or bidirectional, similar to those of the system 630 (FIG. 24), 650 (FIG. 25), and 680 (FIG. 26).

[0453] Optical connectors 716a, 716b, 716c (collectively referenced as 716) are provided to couple optical signals from the optical fibers to the optical/electrical communication interfaces 704a, 704b, 704c, respectively. The optical connectors 716 can be securely fixed, or releasably connected, to the optical/electrical communication interfaces 704, similar to those of the systems 630, 650, and 680.

[0454] The optical/electrical communication interface 704 can be similar to, e.g., the integrated communication device 210 (FIG. 2), 252 (FIG. 4), 374 (FIG. 11), 382 (FIG. 12), 402 (FIG. 13), 428 (FIG. 14), and 512 (FIG. 32), except that the optical/electrical communication interface 704 is mounted on the side of the circuit board 706 opposite to the side of the data processing chip 702. In some examples, the optical/electrical communication interface 704 can be similar to the integrated optical communication device 448, 462, 466, 472 (FIG. 17). The optical connector 716 can be similar to, e.g., the first optical connector part 213 (FIGS. 2, 4), the first optical connector 356 (FIGS. 11, 12), the first optical connector 404 (FIGS. 13, 14), the first optical connector part 456 (FIG. 17), and the first optical connector part 520 (FIG. 32). In some examples, the optical connector is not attached vertically to a photonic integrated circuit that is part of the optical/electrical communication interface 704 but rather can be attached in-plane to the photonic integrated circuit using, e.g., V-groove fiber attachments, tapered or un-tapered fiber edge coupling, etc., followed by a mechanism to direct the light interfacing to the photonic integrated circuit to a direction that is substantially perpendicular to the photonic integrated circuit, such as one or more substantially 90-degree turning mirrors, one or more substantially 90-degree bent optical fibers, etc. In some examples, a portion of the optical connector 716 can be part of the optical/electrical communication interface 704. In some examples, the optical connector 716 can also include the second optical connector part 223 (FIGS. 2, 4), 458 (FIG. 17) that is optically coupled to the optical fibers.

[0455] In some examples, the optical/electrical communication interfaces 704 are securely fixed (e.g., by soldering) to the circuit board 706. In some examples, the optical/electrical communication interfaces 704 are removably connected to the circuit board 706, e.g., by use of mechanical mechanisms such as one or more snap-on or screw-on mechanisms. An advantage of the system 700 is that in case of a malfunction at one of the optical/electrical communication interfaces 704, the faulty optical/electrical communication interface 704 can unplugged or decoupled from the circuit board 706 and replaced without opening the enclosure 710.

[0456] In some implementations, the optical/electrical communication interfaces 704 do not protrude through openings in the front panel 708. For example, each optical/electrical communication interface 704 can be at a distance behind the front panel 708, and a fiber patchcord or pigtail can connect the optical/electrical communication interface 704 to an optical connector on the front panel 708, similar to the examples shown in FIGS. 77A, 77B, 78, 125A, 125B, 129, and 159. In some examples, the front panel 708 is configured to be removable or to be able to open to allow servicing of communication interface 704, similar to the examples shown in FIGS. 77A, 125A, and 159.

[0457] FIG. 28A is a top view of an example data processing system 720 in which a data

processing chip **722** is mounted near an optical/electrical communication interface **724** to enable high bandwidth data paths (e.g., one, ten, or more Gigabits per second per data path) between the data processing chip **720** and the optical/electrical communication interface **724**. The data processing chip **722** is mounted on a first side of a circuit board **730** that functions as a front panel of an enclosure **732** of the system **720**. In some examples, the data processing chip **722** is mounted on a substrate (e.g., a ceramic substrate), and the substrate is attached to the circuit board **730**. The optical/electrical communication interface **724** is mounted on a second side of the circuit board **730**, in which the second side faces the exterior of the enclosure **732**. In this example, the optical/electrical communication interface **724** is mounted on an exterior side of the enclosure **732**, allowing optical fibers **734** to be easily coupled to the optical/electrical communication interface **724**.

[0458] The enclosure **732** has side panels **736** and **738**, a rear panel **740**, a top panel, and a bottom panel. In some examples, the circuit board **730** is perpendicular to the bottom panel. In some examples, the circuit board **730** is oriented at an angle in a range -60° to 60° relative to a vertical direction of the bottom panel.

[0459] The optical/electrical communication interface **724** includes a photonic integrated circuit **726** mounted on a substrate **728** that is electrically coupled to the circuit board **730**. The optical/electrical communication interface **724** is electrically coupled to the data processing chip **722** by electrical connectors or traces **742** that pass through the circuit board **730** in the thickness direction. For example, the electrical connectors or traces **742** can be configured as vias of the circuit board **730**. The signal paths between the data processing chip **722** and the optical/electrical communication interface **724** can be unidirectional or bidirectional, similar to those of the systems **630**, **650**, **680**, and **700**.

[0460] An optical connector **744** is provided to couple optical signals from the optical fibers **734** to the optical/electrical communication interface **724**. The optical connector **744** can be securely fixed, or removably connected, to the optical/electrical communication interface **744**, similar to those of the systems **630**, **650**, **680**, and **700**.

[0461] In some implementations, the optical/electrical communication interface **724** can be similar to, e.g., the integrated communication device **448**, **462**, **466**, and **472** of FIG. 17. The optical signals from the optical fibers are processed by the photonic integrated circuit **726**, which generates serial electrical signals based on the optical signals. For example, the serial electrical signals are amplified by a set of transimpedance amplifiers and drivers (which can be part of the photonic integrated circuit **726** or a serializers/deserializers module in the data processing chip **722**), which drives the output signals that are transmitted to the serializers/deserializers module embedded in the data processing chip **722**.

[0462] The optical connector **744** includes a first optical connector **746** and a second optical connector **748**, in which the second optical connector **748** is optically coupled to the optical fibers **734**. The first optical connector **746** can be similar to, e.g., the first optical connector part **213** (FIGS. 2, 4), the first optical connector **356** (FIGS. 11, 12), the first optical connector **404** (FIGS. 13, 14), the first optical connector part **456** (FIG. 17), and the first optical connector part **520** (FIG. 32). The second optical connector **748** can be similar to the second optical connector part **223** (FIGS. 2, 4) and **458** (FIG. 17). In some examples, the optical connectors **746** and **748** can form a single piece such that the optical/electrical communication interface **724** is securely or fixedly attached to a fiber bundle. In some examples, the optical connector is not attached vertically to the photonic integrated circuit **726** but rather can be attached in-plane to the photonic integrated circuit using, e.g., V-groove fiber attachments, tapered or un-tapered fiber edge coupling, etc., followed by a mechanism to direct the light interfacing to the photonic integrated circuit to a direction that is substantially perpendicular to the photonic integrated circuit, such as one or more substantially 90-degree turning mirrors, one or more substantially 90-degree bent optical fibers, etc.

[0463] In some examples, the optical/electrical communication interface **724** is securely fixed (e.g.,

by soldering) to the circuit board **730**. In some examples, the optical/electrical communication interface **724** is removably connected to the circuit board **730**, e.g., by use of mechanical mechanisms such as one or more snap-on or screw-on mechanisms. An advantage of the system **720** is that in case of a malfunction of the optical/electrical communication interface **724**, the faulty optical/electrical communication interface **724** can be replaced without opening the enclosure **732**. [0464] FIG. **28B** is a top view of an example data processing system **2800** that is similar to the system **720** of FIG. **28A**, except that the circuit board **730** that is recessed from a front panel **2802** of an enclosure **732** of the system **2800**. The photonic integrated circuit **726** is optically coupled through a fiber patchcord or pigtail **2804** to a first optical connector **2806** attached to the inner side of the front panel **2802**. The first optical connector **2806** is optically coupled to a second optical connector **2808** attached to the outer side of the front panel **2802**. The second optical connector **2808** is optically coupled to the exterior optical fibers **734**.

[0465] The technique of using a fiber patchcord or pigtail to optically couple the photonic integrated circuit to the optical connector attached to the inner side of the front panel can also be applied to the data processing system **700** of FIG. **27**. For example, the modified system can have a recessed substrate or circuit board, multiple co-packaged optical modules (e.g., **704**) mounted on the opposite side of the data processing chip **702** relative to the substrate or circuit board, and fiber jumpers (e.g., **2804**) optically coupling the co-packaged optical modules to the front panel.

[0466] In the examples of FIGS. **28A** and **28B**, the data processing chip **722** can be mounted on a substrate that is electrically coupled to the circuit board **730**, similar to the example shown in FIG. **150**.

[0467] In each of the examples in FIGS. **24**, **25**, **26**, **27**, and **28**, the optical/electrical communication interface **644**, **652**, **684**, **704**, and **724** can be electrically coupled to the circuit board **642**, **654**, **686**, **706**, and **730**, respectively, using electrical contacts that include one or more of spring-loaded elements, compression interposers, and/or land-grid arrays.

[0468] FIG. **29A** is a diagram of an example data processing system **750** that includes a vertically mounted circuit board **752** that enables high bandwidth data paths (e.g., one, ten, or more Gigabits per second per data path) between data processing chips **758** and optical/electrical communication interfaces **760**. The data processing chips **758** and the optical/electrical communication interfaces **760** are mounted on the circuit board **752**, in which each data processing chip **758** is electrically coupled to a corresponding optical/electrical communication interface **760**. The data processing chips **758** are electrically coupled to one another by electrical connectors (e.g., electrical signal lines on one or more layers of the circuit board **752**).

[0469] The data processing chips **758** can be similar to, e.g., the electronic processor integrated circuit, data processing chip, or host application specific integrated circuit **240** (FIGS. **2**, **4**, **6**, **7**, **11**, **12**), digital application specific integrated circuit **444** (FIG. **17**), data processor **502** (FIG. **20**), data processing chip **572** (FIGS. **22**, **23**), **640** (FIG. **24**), **670** (FIG. **25**), **681** (FIG. **26**), **702** (FIG. **27**), and **722** (FIG. **28**). Each of the data processing chips **758** can be, e.g., a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, or an application specific integrated circuit (ASIC).

[0470] Although the figure shows that the optical/electrical communication interfaces **760** are mounted on the side of the circuit board **752** facing the front panel **754**, the optical/electrical communication interfaces **760** can also be mounted on the side of the circuit board **752** facing the interior of the enclosure **756**. The optical/electrical communication interfaces **760** can be similar to, e.g., the integrated communication devices **210** (FIGS. **2**, **3**, **10**), **252** (FIGS. **4**, **5**), **262** (FIG. **6**), the integrated optical communication devices **282** (FIGS. **7-9**), **374** (FIG. **11**), **382** (FIG. **12**), **390** (FIG. **13**), **428** (FIG. **14**), **402** (FIGS. **15**, **16**), **448**, **462**, **466**, **472** (FIG. **17**), the integrated communication devices **574** (FIG. **22**), **612** (FIG. **23**), and the optical/electrical communication interfaces **644** (FIG. **24**), **652** (FIG. **25**), **684** (FIG. **26**), **704** (FIG. **27**).

[0471] The circuit board **752** is positioned near a front panel **754** of an enclosure **756**, and optical signals are coupled to the optical/electrical communication interfaces **760** through optical paths that pass through openings in the front panel **754**. This allows users to conveniently removably connect optical fiber cables **762** to the input/output interfaces **760**. The position and orientation of the circuit board **752** relative to the enclosure **756** can be similar to, e.g., those of the circuit board **654** (FIG. **25**) and **706** (FIG. **27**).

[0472] In some implementations, the data processing system **750** can include multiple types of optical/electrical communication interfaces **760**. For example, some of the optical/electrical communication interfaces **760** can be mounted on the same side of the circuit board **752** as the corresponding data processing chip **758**, and some of the optical/electrical communication interfaces **760** can be mounted on the opposite side of the circuit board **752** as the corresponding data processing chip **758**. Some of the optical/electrical communication interfaces **760** can include first and second serializers/deserializers modules, and the corresponding data processing chips **758** can include third serializers/deserializers modules, similar to the examples in FIGS. **2-8**, **11-14**, **20**, **22**, and **23**. Some of the optical/electrical communication interfaces **760** can include no serializers/deserializers module, and the corresponding data processing chips **758** can include serializers/deserializers modules, similar to the example of FIG. **17**. Some of the optical/electrical communication interfaces **760** can include sets of transimpedance amplifiers and drivers, either embedded in the photonic integrated circuits or in separate chips external to the photonic integrated circuits. Some of the optical/electrical communication interfaces **760** do not include transimpedance amplifiers and drivers, in which sets of transimpedance amplifiers and drivers are included in the corresponding data processing chips **758**. The data processing system **750** can also include electrical communication interfaces that interface to electrical cables, such as high speed PCIe cables, Ethernet cables, or Thunderbolt™ cables. The electrical communication interfaces can include modules that perform various functions, such as translation of communication protocols and/or conditioning of signals.

[0473] Other types of connections may be present and associated with circuit board **752** and other boards included in the enclosure **756**. For example, two or more circuit boards (e.g., vertically mounted circuit boards) can be connected which may or may not include the circuit board **752**. For instances in which circuit board **752** is connected to at least one other circuit board (e.g., vertically mounted in the enclosure **756**), one or more connection techniques can be employed. For example, an optical/electrical communication interface (e.g., similar to optical/electrical communication interfaces **760**) can be used to connect data processing chips **758** to other circuit boards. Interfaces for such connections can be located on the same side of the circuit board **752** that the processing chips **758** are mounted. In some implementations, interfaces can be located on another portion of the circuit board (e.g., a side that is opposite from the side that the processing chips **758** are mounted). Connections can utilize other portions of the circuit board **752** and/or one or more other circuit boards present in the enclosure **756**. For example, an interface can be located on an edge of one or more of the boards (e.g., an upper edge of a vertically mounted circuit board) and the interface can connect with one or more other interfaces (e.g., the optical/electrical communication interfaces **760**, another edge mounted interface, etc.). Through such connections, two or more circuit boards can connect, receive and send signals, etc.

[0474] In the example shown in FIG. **29A**, the circuit board **752** is placed near the front panel **754**. In some examples, the circuit board **752** can also function as the front panel, similar to the examples in FIGS. **22-24**, **26**, and **28**.

[0475] FIG. **29B** is a diagram of an example data processing system **2000** that illustrates some of the configurations described with respect to FIGS. **26A** to **26C** and FIG. **29A** along with other capabilities. The system **2000** includes a vertically mounted printed circuit board **2002** (or, e.g., a substrate) upon which is mounted a data processing chip **2004** (e.g., an ASIC), and a heat sink **2006** is thermally coupled to the data processing chip **2004**. Optical/electrical communication interfaces

are mounted on both sides of the printed circuit board **2002**. In particular, optical/electrical communication interface **2008** is mounted on the same side of the printed circuit board **2002** as the data processing chip **2004**. In this example, optical/electrical communication interfaces **2010**, **2012**, and **2014** are mounted on an opposite side of the printed circuit board **2002**. To send and receive signals (e.g., with other optical/electrical communication interfaces), each of the optical/electrical communication interfaces **2010**, **2012**, and **2014** connects to optical fibers **2016**, **2018**, **2020**, respectively. Electrical connection sockets/connectors can also be mounted to one or more sides of the printed circuit board **2002** for sending and receiving electrical signals, for example. In this example, two electrical connection sockets/connectors **2022** and **2024** are mounted to the side of the printed circuit board **2002** that the data processing chip **2004** is mounted and two electrical connection sockets/connectors **2026** and **2028** are mounted to the opposite side of the printed circuit board **2002**. In this example, electrical connection sockets/connector **2028** is connected (or includes) a timing module **2030** that provides various functionality (e.g., regenerate data, retime data, maintain signal integrity, etc.). To send and receive electrical signals, each of the electrical connection sockets/connectors **2022-2028** are connected to electrical connection cables **2032**, **2034**, **2036**, **2038**, respectively. One or more types of connection cables can be implemented, for example, fly-over cables can be employed for connecting to one or more of the electrical connection sockets/connectors **2022-2028**.

[0476] In this example, the system **2000** includes vertically mounted line cards **2040**, **2042**, **2044**. In this particular example, line card **2040** includes an electrical connection sockets/connector **2046** that is connected to electrical cable **2036**, and line card **2042** includes an electrical connection sockets/connector **2048** that is connected to electrical cable **2032**. Line card **2044** includes an electrical connection sockets/connector **2050**. Each of the line cards **2040**, **2042**, **2044** include pluggable optical modules **2052**, **2054**, **2056** that can implement various interface techniques (e.g., QSFP, QSFP-DD, XFP, SFP, CFP).

[0477] In this particular example, the printed circuit board **2002** is approximate to a forward panel **2058** of the system **2000**; however, the printed circuit board **2002** can be positioned in other locations within the system **2000**. Multiple printed circuit boards can also be included in the system **2000**. For example, a second printed circuit board **2060** (e.g., a backplane) is included in the system **2000** and is located approximate to a back panel **2062**. By locating the printed circuit board **2060** towards the rear, signals (e.g., data signals) can be sent to and received from other systems (e.g., another switch box) located, for example, in the same switch rack or other location as the system **2000**. In this example, a data processing chip **2064** is mounted to the printed circuit board **2060** that can perform various operations (e.g., data processing, prepare data for transmission, etc.). Similar to the printed circuit board **2002** located forward in the system **2000**, the printed circuit board **2060** includes an optical/electrical communication interface **2066** that communicates with the optical/electrical communication interface **2008** (located on the same side on printed circuit board **2002** as data processing chip **2004**) using optical fibers **2068**. The printed circuit board **2060** includes electrical connection sockets/connectors **2070** that uses the electrical connection cable **2034** to send electrical signals to and receive electrical signals from the electrical connection sockets/connectors **2024**. The printed circuit board **2060** can also communicate with other components of the system **2000**, for example, one or more of the line cards. As illustrated in the figure, electrical connection sockets/connectors **2072** located on the printed circuit board **2060** uses the electrical connection cable **2074** to send electrical signals to and/or receive electrical signals from the electrical connection sockets/connector **2050** of the line card **2044**. Similar to the printed circuit board **2002**, other portions of the system **2000** can include timing modules. For example, the line cards **2040**, **2042**, and **2044** can include timing modules (respectively identified with symbol “*”, “**”, and “***”). Similarly, the second circuit board **2060** can include timing modules such as timing modules **2076** and **2078** for regenerating data, re-timing data, maintaining signal integrity, etc.

[0478] A feature of some of the systems described in this document is that the main data processing module(s) of a system, such as switch chip(s) in a switch server, and the communication interface modules that support the main data processing module(s), are configured to allow convenient access by users. In the examples shown in FIGS. **21** to **29B**, **69A**, **70**, **71A**, **72**, **72A**, **74A**, **75A**, **75C**, **76**, **77A**, **77B**, **78**, **96** to **98**, **100**, **110**, **112**, **113**, **115**, **117** to **122**, **125A** to **127**, **129**, **136** to **149**, **159**, and **160**, the main data processing module and the communication interface modules are positioned near the front panel, the rear panel, or both, and allow easy access by the user through the front/rear panel. However, it is also possible to position the main data processing module and the communication interface modules near one or more side panels, the top panel, the bottom panel, or two or more of the above, depending on how the system is placed in the environment. In a system that includes multiple racks of rackmount devices (see e.g., FIGS. **76** and **86**), the communication interfaces (e.g., co-packaged optical modules) in each rackmount device can be conveniently accessed without the need to remove the rackmount device from the rack and opening up the housing in order to expose the inner components.

[0479] In some implementations, for a single rack of rackmount servers where there is open space at the front, rear, left, and right side of the rack, in each rackmount server, it is possible to place a first main data processing module and the communication interface modules supporting the first main data processing module near the front panel, place a second main data processing module and the communication interface modules supporting the second main data processing module near the left panel, place a third main data processing module and the communication interface modules supporting the third main data processing module near the right panel, and place a fourth main data processing module and the communication interface modules supporting the fourth main data processing module near the rear panel. The thermal solutions, including the placement of fans and heat dissipating devices, and the configuration of airflows around the main data processing modules and the communication interface modules, are adjusted accordingly.

[0480] For example, if a data processing server is mounted to the ceiling of a room or a vehicle, the main data processing module and the communication interface modules can be positioned near the bottom panel for easy access. For example, if a data processing server is mounted beneath the floor panel of a room or a vehicle, the main data processing module and the communication interface modules can be positioned near the top panel for easy access. The housing of the data processing system does not have to be in a box shape. For example, the housing can have curved walls, be shaped like a globe, or have an arbitrary three-dimensional shape.

[0481] FIG. **30** is a diagram of an example high bandwidth data processing system **800** that can be similar to, e.g., systems **200** (FIGS. **2**, **20**), **250** (FIG. **4**), **260** (FIG. **6**), **280** (FIG. **7**), **350** (FIG. **11**), **380** (FIG. **12**), **390** (FIG. **13**), **420** (FIG. **14**), **560** (FIG. **22**), **600** (FIG. **23**), **630** (FIG. **24**), and **650** (FIG. **25**) described above. A first optical signal **770** is transmitted from an optical fiber to a photonic integrated circuit **772**, which generates a first serial electrical signal **774** based on the first optical signal. The first serial electrical signal **774** is provided to a first serializers/deserializers module **776**, which converts the first serial electrical signal **774** to a third set of parallel signals **778**. The first serializers/deserializers module **776** conditions the serial electrical signal upon conversion into the parallel electrical signals, in which the signal conditioning can include, e.g., one or more of clock and data recovery, and signal equalization. The third set of parallel signals **778** is provided to a second serializers/deserializers module **780**, which generates a fifth serial electrical signal **782** based on the third set of parallel signals **778**. The fifth serial electrical signal **782** is provided to a third serializers/deserializers module **784**, which generates a seventh set of parallel signals **786** that is provided to a data processor **788**.

[0482] In some implementations, the photonic integrated circuit **772**, the first serializers/deserializers module **776**, and the second serializers/deserializers module **780** can be mounted on a substrate of an integrated communication device, an optical/electrical communication interface, or an input/output interface module. The first serializers/deserializers module **776** and the

second serializers/deserializers module **780** can be implemented in a single chip. In some implementations, the third serializers/deserializers module **784** can be embedded in the data processor **788**, or the third serializers/deserializers module **784** can be separate from the data processor **788**.

[0483] The data processor **788** generates an eighth set of parallel signals **790** that is sent to the third serializers/deserializers module **784**, which generates a sixth serial electrical signal **792** based on the eighth set of parallel signals **790**. The sixth serial electrical signal **792** is provided to the second serializers/deserializers module **780**, which generates a fourth set of parallel signals **794** based on the sixth serial electrical signal **792**. The second serializers/deserializers module **780** can condition the serial electrical signal **792** upon conversion into the fourth set of parallel electrical signals **794**. The fourth set of parallel signals **794** is provided to the first serializers/deserializers module **780**, which generates a second serial electrical signal **796** based on the fourth set of parallel signals **794** that is sent to the photonic integrated circuit **772**. The photonic integrated circuit **772** generates a second optical signal **798** based on the second serial electrical signal **796**, and sends the second optical signal **798** to an optical fiber. The first and second optical signals **770**, **798** can travel on the same optical fiber or on different optical fibers.

[0484] A feature of the system **800** is that the electrical signal paths traveled by the first, fifth, sixth, and second serial electrical signals **774**, **782**, **792**, **796** are short (e.g., less than 5 inches), to allow the first, fifth, sixth, and second serial electrical signals **782**, **792** to have a high data rate (e.g., up to 50 Gbps).

[0485] FIG. **31** is a diagram of an example high bandwidth data processing system **810** that can be similar to, e.g., systems **680** (FIG. **26**), **700** (FIG. **27**), and **750** (FIG. **29**) described above. The system **810** includes a data processor **812** that receives and sends signals from and to multiple photonic integrated circuits. The system **810** includes a second photonic integrated circuit **814**, a fourth serializers/deserializers module **816**, a fifth serializers/deserializers module **818**, and a sixth serializers/deserializers module **820**. The operations of the second photonic integrated circuit **814**, a fourth serializers/deserializers module **816**, a fifth serializers/deserializers module **818**, and a sixth serializers/deserializers module **820** can be similar to those of the first photonic integrated circuit **772**, the first serializers/deserializers module **776**, the second serializers/deserializers module **780**, and the third serializers/deserializers module **784**. The third serializers/deserializers module **784** and the sixth serializers/deserializers module **820** can be embedded in the data processor **812**, or be implemented in separate chips.

[0486] In some examples, the data processor **812** processes first data carried in the first optical signal received at the first photonic integrated circuit **772**, and generates second data that is carried in the fourth optical signal output from the second photonic integrated circuit **814**.

[0487] The examples in FIGS. **30** and **31** include three serializers/deserializers modules between the photonic integrated circuit and the data processor, it is understood that the same principles can be applied to systems that has only one serializers/deserializers module between the photonic integrated circuit and the data processor.

[0488] In some implementations, signals are transmitted unidirectionally from the photonic integrated circuit **772** to the data processor **788** (FIG. **30**). In that case, the first serializers/deserializers module **776** can be replaced with a serial-to-parallel converter, the second serializers/deserializers module **780** can be replaced with a parallel-to-serial converter, and the third serializers/deserializers module **784** can be replaced with a serial-to-parallel converter. In some implementations, signals are transmitted unidirectionally from the data processor **812** (FIG. **31**) to the second photonic integrated circuit **814**. In that case, the sixth serializers/deserializers module **820** can be replaced with a parallel-to-serial converter, the fifth serializers/deserializers module **818** can be replaced with a serial-to-parallel converter, and the fourth serializers/deserializers module **816** can be replaced with a parallel-to-serial converter.

[0489] It should be appreciated by those of ordinary skill in the art that the various embodiments

described herein in the context of coupling light from one or more optical fibers, e.g., **226** (FIGS. **2** and **4**) or **272** (FIGS. **6** and **7**) to the photonic integrated circuit, e.g., **214** (FIGS. **2** and **4**), **264** (FIG. **6**), or **296** (FIG. **7**) will be equally operable to couple light from the photonic integrated circuit to one or more optical fibers. This reversibility of the coupling direction is a general feature of at least some embodiments described herein, including some of those using polarization diversity.

[0490] The example optical systems disclosed herein should only be viewed as some of many possible embodiments that can be used to perform polarization demultiplexing and independent array pattern scaling, array geometry re-arrangement, spot size scaling, and angle-of-incidence adaptation using diffractive, refractive, reflective, and polarization-dependent optical elements, 3D waveguides and 3D printed optical components. Other implementations achieving the same set of functionalities are also covered by the spirit of this disclosure.

[0491] For example, the optical fibers can be coupled to the edges of the photonic integrated circuits, e.g., using fiber edge couplers. The signal conditioning (e.g., clock and data recovery, signal equalization, or coding) can be performed on the serial signals, the parallel signals, or both. The signal conditioning can also be performed during the transition from serial to parallel signals.

[0492] In some implementations, the data processing systems described above can be used in, e.g., data center switching systems, supercomputers, internet protocol (IP) routers, Ethernet switching systems, graphics processing work stations, and systems that apply artificial intelligence algorithms.

[0493] In the examples described above in which the figures show a first serializers/deserializers module (e.g., **216**) placed adjacent to a second serializers/deserializers module (e.g., **217**), it is understood that a bus processing unit **218** can be positioned between the first and second serializers/deserializers modules and perform, e.g., switching, re-routing, and/or coding functions described above.

[0494] In some implementations, the data processing systems described above includes multiple data generators that generate large amounts of data that are sent through optical fibers to the data processors for processing. For example, an autonomous driving vehicle (e.g., car, truck, train, boat, ship, submarine, helicopter, drone, airplane, space rover, or space ship) or a robot (e.g., an industrial robot, a helper robot, a medical surgery robot, a merchandise delivery robot, a teaching robot, a cleaning robot, a cooking robot, a construction robot, an entertainment robot) can include multiple high resolution cameras and other sensors (e.g., LIDARs (Light Detection and Ranging), radars) that generate video and other data that have a high data rate. The cameras and/or sensors can send the video data and/or sensor data to one or more data processing modules through optical fibers. The one or more data processing modules can apply artificial intelligence technology (e.g., using one or more neural networks) to recognize individual objects, collections of objects, scenes, individual sounds, collections of sounds, and/or situations in the environment of the vehicle and quickly determine appropriate actions for controlling the vehicle or robot.

[0495] FIG. **34** is a flow diagram of an example process for processing high bandwidth data. A process **830** includes receiving **832** a plurality of channels of first optical signals from a plurality of optical fibers. The process **830** includes generating **834** a plurality of first serial electrical signals based on the received optical signals, in which each first serial electrical signal is generated based on one of the channels of first optical signals. The process **830** includes generating **836** a plurality of sets of first parallel electrical signals based on the plurality of first serial electrical signals, and conditioning the electrical signals, in which each set of first parallel electrical signals is generated based on a corresponding first serial electrical signal. The process **830** includes generating **838** a plurality of second serial electrical signals based on the plurality of sets of first parallel electrical signals, in which each second serial electrical signal is generated based on a corresponding set of first parallel electrical signals.

[0496] In some implementations, a data center includes multiple systems, in which each system

incorporates the techniques disclosed in FIGS. 22 to 29 and the corresponding description. Each system includes a vertically mounted printed circuit board, e.g., **570** (FIG. 22), **610** (FIG. 23), **642** (FIG. 24), **654** (FIG. 25), **686** (FIG. 26), **706** (FIG. 27), **730** (FIG. 28), **752** (FIG. 29) that functions as the front panel of the housing or is substantially parallel to the front panel. At least one data processing chip and at least one integrated communication device or optical/electrical communication interface are mounted on the printed circuit board. The integrated communication device or optical/electrical communication interface can incorporate techniques disclosed in FIGS. 2-22 and 30-34 and the corresponding description. Each integrated communication device or optical/electrical communication interface includes a photonic integrated circuit that receives optical signals and generates electrical signals based on the optical signals. The optical signals are provided to the photonic integrated circuit through one or more optical paths (or spatial paths) that are provided by, e.g., cores of the fiber-optic cables, which can incorporate techniques described in U.S. patent application Ser. No. 16/822,103, now U.S. Pat. No. 11,194,109. A large number of parallel optical paths (or spatial paths) can be arranged in two-dimensional arrays using connector structures, which can incorporate techniques described in U.S. patent application Ser. No. 16/816,171, now U.S. Pat. No. 11,287,585.

[0497] FIG. 35A shows an optical communications system **1250** providing high-speed communications between a first chip **1252** and a second chip **1254** using co-packaged optical (CPO) interconnect modules **1258** similar to those shown in, e.g., FIGS. 2-5 and 17. Each of the first and second chips **1252**, **1254** can be a high-capacity chip, e.g., a high bandwidth Ethernet switch chip. The first and second chips **1252**, **1254** communicate with each other through an optical fiber interconnection cable **1734** that includes a plurality of optical fibers. In some implementations, the optical fiber interconnection cable **1734** can include optical fiber cores that transmit data and control signals between the first and second chips **802**, **804**. The optical fiber interconnection cable **1734** also includes one or more optical fiber cores that transmit optical power supply light from an optical power supply or photon supply to photonic integrated circuits that provide optoelectronic interfaces for the first and second chips **1252**, **1254**. The optical fiber interconnection cable **1734** can include single-core fibers or multi-core fibers. Each single-core fiber includes a cladding and a core, typically made from glasses of different refractive indices such that the refractive index of the cladding is lower than the refractive index of the core to establish a dielectric optical waveguide. Each multi-core optical fiber includes a cladding and multiple cores, typically made from glasses of different refractive indices such that the refractive index of the cladding is lower than the refractive index of the core. More complex refractive index profiles, such as index trenches, multi-index profiles, or gradually changing refractive index profiles can also be used. More complex geometric structures such as non-circular cores or claddings, photonic crystal structures, photonic bandgap structures, or nested antiresonant nodeless hollow core structures can also be used.

[0498] The example of FIG. 35A illustrates a switch-to-switch use case. An external optical power supply or photon supply **1256** provides optical power supply signals, which can be, e.g., continuous-wave light, one or more trains of periodic optical pulses, or one or more trains of non-periodic optical pulses. The power supply light is provided from the photon supply **1256** to the co-packaged optical interconnect modules **1258** through optical fibers **1730** and **1732**, respectively. For example, the optical power supply **1256** can provide continuous wave light, or both pulsed light for data modulation and synchronization, as described in U.S. patent application Ser. No. 16/847,705, now U.S. Pat. No. 11,153,670. This allows the first chip **1252** to be synchronized with the second chip **1254**.

[0499] For example, the photon supply **1256** can correspond to the optical power supply **103** of FIG. 1. The pulsed light from the photon supply **1256** can be provided to the link **102_6** of the data processing system **200** of FIG. 20. In some implementations, the photon supply **1256** can provide a sequence of optical frame templates, in which each of the optical frame templates includes a

respective frame header and a respective frame body, and the frame body includes a respective optical pulse train. The modulators **417** can load data into the respective frame bodies to convert the sequence of optical frame templates into a corresponding sequence of loaded optical frames that are output through optical fiber link **102_1**.

[0500] The implementation shown in FIG. **35A** uses a packaging solution corresponding to FIG. **35B**, whereby in contrast to FIG. **17** substrates **454** and **460** are not used and the photonic integrated circuit **464** is directly attached to the serializers/deserializers module **446**. FIG. **35C** shows an implementation similar to FIG. **5**, in which the photonic integrated circuit **464** is directly attached to the serializers/deserializers **216**.

[0501] FIG. **36** shows an example of an optical communications system **1260** providing high-speed communications between a high-capacity chip **1262** (e.g., an Ethernet switch chip) and multiple lower-capacity chips **1264a**, **1264b**, **1264c**, e.g., multiple network interface cards (NICs) attached to computer servers) using co-packaged optical interconnect modules **1258** similar to those shown in FIG. **35A**. The high-capacity chip **1262** communicates with the lower-capacity chips **1264a**, **1264b**, **1264c** through a high-capacity optical fiber interconnection cable **1740** that later branches out into several lower-capacity optical fiber interconnection cables **1742a**, **1742b**, **1742c** that are connected to the lower-capacity chips **1264a**, **1264b**, **1264c**, respectively. This example illustrates a switch-to-servers use case.

[0502] An external optical power supply or photon supply **1266** provides optical power supply signals, which can be continuous-wave light, one or more trains of periodic optical pulses, or one or more trains of non-periodic optical pulses. The power supply light is provided from the photon supply **1266** to the optical interconnect modules **1258** through optical fibers **1744**, **1746a**, **1746b**, **1746c**, respectively. For example, the optical power supply **1266** can provide both pulsed light for data modulation and synchronization, as described in U.S. patent application Ser. No. 16/847,705, now U.S. Pat. No. 11,153,670. This allows the high-capacity chip **1262** to be synchronized with the lower-capacity chips **1264a**, **1264b**, and **1264c**.

[0503] FIG. **37** shows an optical communications system **1270** providing high-speed communications between a high-capacity chip **1262** (e.g., an Ethernet switch chip) and multiple lower-capacity chips (**1264a**, **1264b**, e.g., multiple network interface cards (NICs) attached to computer servers) using a mix of co-packaged optical interconnect modules **1258** similar to those shown in FIG. **35** as well as conventional pluggable optical interconnect modules **1272**.

[0504] An external optical power supply or photon supply **1274** provides optical power supply signals, which can be continuous-wave light, one or more trains of periodic optical pulses, or one or more trains of non-periodic optical pulses. For example, the optical power supply **1274** can provide both pulsed light for data modulation and synchronization, as described in U.S. patent application Ser. No. 16/847,705, now U.S. Pat. No. 11,153,670. This allows the high-capacity chip **1262** to be synchronized with the lower-capacity chips **1264a** and **1264b**.

[0505] Some aspects of the systems **1250**, **1260**, and **1270** are described in more detail in connection with FIGS. **79** to **84B**.

[0506] FIG. **43** shows an exploded view of an example of a front-mounted module **860** of a data processing system that includes a vertically mounted printed circuit board **862** (or substrate made of, e.g., organic or ceramic material), a host application specific integrated circuit **864** mounted on the back-side of the circuit board **862**, and a heat sink **866**. In some examples, the host application specific integrated circuit **864** is mounted on a substrate (e.g., a ceramic substrate), and the substrate is attached to the circuit board **862**. The front-mounted module **860** can be, e.g., the front panel of the housing of the data processing system, similar to the configuration shown in FIGS. **26A**, **28A** or positioned near the front panel of the housing, similar to the configuration shown in FIGS. **27**, **28B**. Three optical modules with connectors, e.g., **868a**, **868b**, **868c**, collectively referenced as **868**, are shown in the figure. Additional optical modules with connectors can be used. The data processing system can be similar to, e.g., the data processing system **680** (FIG. **26A**) or

700 (FIG. 27). The printed circuit board **862** can be similar to, e.g., the printed circuit board **683** (FIG. 26) or **708** (FIG. 27). The application specific integrated circuit **864** can be similar to, e.g., the application specific integrated circuit **681** (FIG. 26) or **702** (FIG. 27). The heat sink **866** can be similar to, e.g., the heat sink **576** (FIG. 23). The optical modules with connectors **868** each include an optical module **880** (see FIGS. 44, 45) and a mechanical connector structure **900** (see FIGS. 46, 47). The optical module **880** can be similar to, e.g., the optical/electrical communication interfaces **682** (FIG. 26) or **704** (FIG. 27), or the integrated optical communication device **512** of FIG. 32.

[0507] The optical module with connector **868** can be inserted into a first grid structure **870**, which can function as both (i) a heat spreader/heat sink and (ii) a mechanical holding fixture for the optical modules with connectors **868**. The first grid structure **870** includes an array of receptors, and each receptor can receive an optical module with connector **868**. When assembled, the first grid structure **870** is connected to the printed circuit board **862**. The first grid structure **870** can be firmly held in place relative to the printed circuit board **862** by sandwiching the printed circuit board **862** in between the first grid structure **870** and a second structure **872** (e.g., a second grid structure) located on the opposite side of the printed circuit board **862** and connected to the first grid structure **870** through the printed circuit board **862**, e.g., by use of screws. Thermal vias between the first grid structure **870** and the second structure **872** can conduct heat from the front-side of the printed circuit board **862** to the heat sink **866** on the back-side of the printed circuit board **862**. Additional heat sinks can also be mounted directly onto the first grid structure **870** to provide cooling in the front.

[0508] The printed circuit board **862** includes electrical contacts **876** configured to electrically connect to the removable optical module with connectors **868** after the removable optical module with connectors **868** are inserted into the first grid structure **870**. The first grid structure **870** can include an opening **874** at the location in which the host application specific integrated circuit **864** is mounted on the other side of the printed circuit board **862** to allow for components such as voltage regulators, filters, and/or decoupling capacitors to be mounted on the printed circuit board **862** in immediate lateral vicinity to the host application specific integrated circuit **864**.

[0509] In some examples, the host application specific integrated circuit **864** is mounted on a substrate (e.g., a ceramic substrate), and the substrate is attached to the circuit board **862**, similar to the examples shown in FIGS. 136 to 159. The substrate can be similar to the substrate **13602** of FIGS. 136 to 159, the second grid structure **872** can be similar to the rear lattice structure **13626**, the circuit board **862** can be similar to the printed circuit board **13604**, the host application specific integrated circuit **864** can be similar to the data processing chip **12312**, and the heat sink **866** can be similar to the heat dissipating device **13610**. The first grid structure **870** can have an overall shape similar to the front lattice structure **13606** of FIGS. 136 to 159, except that the first grid structure **870** includes mechanisms for coupling to the removable optical module with connectors **868**.

[0510] FIGS. 44 and 45 show an exploded view and an assembled view, respectively, of an example optical module **880**, which can be similar to the integrated optical communication device **512** of FIG. 32. The optical module **880** includes an optical connector part **882** (which can be similar to the first optical connector **520** of FIG. 32) that can either directly or through an (e.g., geometrically wider) upper connector part **884** receive light from fibers embedded in a second optical connector part (not shown in FIGS. 44, 45), which can be similar to, e.g., the optical connector part **268** of FIGS. 6 and 7). In the example shown in FIGS. 44, 45, a matrix of fibers, e.g., 2×18 fibers, can be optically coupled to the optical connector part **882**. The matrix of fibers can have other configurations, such as a 3×12, 1×12, 3×12, 6×12, 12×12, 16×16, or 32×32 array of fibers. For example, the optical connector part **882** can have a configuration similar to the fiber coupling region **430** of FIG. 15 that is configured to couple 2×18 fibers, or any other number of fibers. The upper connector part **884** can also include alignment structures **886** (e.g., holes, grooves, posts) to receive corresponding mating structures of the second optical connector part.

[0511] The optical module **880** can have any of various configurations, including an optical module

containing silicon photonics integrated optics, indium phosphide integrated optics, one or more vertical-cavity surface-emitting lasers (VCSEL)s, one or more direct-detection optical receivers, or one or more coherent optical receivers. The optical module **880** can include any of the optical modules, co-packaged optical modules, integrated optical communication devices (e.g., **448**, **462**, **466**, or **472** of FIG. **17**, or **210** of FIG. **20**), integrated communication devices (e.g., **612** of FIG. **23**), or optical/electrical communication interfaces (e.g., **684** of FIG. **26**, **724** of FIG. **28**, or **760** of FIG. **29**) described in this specification and the documents incorporated by reference.

[0512] The optical connector part **882** is inserted through an opening **888** of a substrate **890** and optically coupled to a photonic integrated circuit **896** mounted on the underside of the substrate **890**. The substrate **890** can be similar to the substrate **514** of FIG. **32**, and the photonic integrated circuit **896** can be similar to the photonic integrated circuit **524**. A first serializers/deserializers chip **892** and a second serializers/deserializers chip **894** are mounted on the substrate **890**, in which the chip **892** is positioned on one side of the optical connector part **882**, and the chip **894** is positioned on the other side of the optical connector part **882**. The first serializers/deserializers chip **892** can include circuitry similar to, e.g., the third serializers/deserializers module **398** and the fourth serializers/deserializers module **400** of FIG. **32**. The second serializers/deserializers chip **894** can include circuitry similar to, e.g., the first serializers/deserializers module **394** and the second serializers/deserializers module **396**. A second slab **898** (which can be similar to the second slab **518** of FIG. **32**) can be provided on the underside of the substrate **890** to provide a removable connection to a package substrate (e.g., **230**).

[0513] FIGS. **46** and **47** show an exploded view and an assembled view, respectively, of a mechanical connector structure **900** built around the functional optical module **880** of FIGS. **44**, **45**. In this example embodiment, the mechanical connector structure **900** includes a lower mechanical part **902** and an upper mechanical part **904** that together receive the optical module **880**. Both lower and upper mechanical connector parts **902**, **904** can be made of a heat-conducting and rigid material, e.g., a metal.

[0514] In some implementations, the upper mechanical part **904**, at its underside, is brought in thermal contact with the first serializers/deserializers chip **892** and the second serializers/deserializers chip **894**. The upper mechanical part **904** is also brought in thermal contact with the lower mechanical part **902**. The lower mechanical part **902** includes a removable latch mechanism, e.g., two wings **906** that can be elastically bent inwards (the movement of the wings **906** are represented by a double-arrow **908** in FIG. **47**), and each wing **906** includes a tongue **910** on an outer side.

[0515] FIG. **48** is a diagram of a portion of the first grid structure **870** and the circuit board **862**. In some examples, a substrate (e.g., a ceramic substrate) can be used in place of the circuit board **862**. Grooves **920** are provided on the walls of the first grid structure **870**. As shown in the figure, the printed circuit board **862** (or substrate) has electrical contacts **876** that can be electrically coupled to electrical contacts on the second slab **898** of the optical module **880**. For example, the electrical contacts **876** can include an array of electrical contacts that has at least four rows and four columns of electrical contacts. For example, the array of electrical contacts can have ten or more rows or columns of electrical contacts. The electrical contacts **876** can be arranged in any two-dimensional pattern and do not necessarily have to be arranged in rows and columns. The circuit board **862** (or substrate) can also have three-dimensional features, such as on protruding elements or recessed elements, and the electrical contacts can be provided on the three-dimensional features. The optical module with connectors **868** can have three-dimensional features with electrical contacts that mate with the corresponding three-dimensional features with electrical contacts on the circuit board **862** (or substrate).

[0516] Referring to FIG. **49**, when the lower mechanical part **902** is inserted into the first grid structure **870**, the tongues **910** (on the wings **906** of the lower mechanical part **902**) can snap into corresponding grooves **920** within the first grid structure **870** to mechanically hold the optical

module **880** in place. The position of the tongues **910** on the wings **906** is selected such that when the mechanical connector structure **900** and the optical module **880** are inserted into the first grid structure **870**, the electrical connectors at the bottom of the second slab **898** are electrically coupled to the electrical contacts **876** on the printed circuit board **862** (or substrate). For example, the second slab **898** can include spring-loaded contacts that are mated with the contacts **876**.

[0517] FIG. **50** shows the front-view of an assembled front module **860**. Three optical module with connectors (e.g., **868a**, **868b**, **868c**) are inserted into the first grid structure **870**. In some embodiments, the optical modules **880** are arranged in a checkerboard pattern, whereby adjacent optical modules **880** and the corresponding mechanical connector structures **900** are rotated by 90 degrees such as to not allow any two wings to touch. This facilitates the removal of individual modules. In this example, the optical module with connector **868a** is rotated 90 degrees relative to the optical module with connectors **868b**, **868c**.

[0518] FIG. **51A** shows a first side view of the mechanical connector structure **900**. FIG. **51B** shows a cross-sectional view of the mechanical connector structure **900** along a plane **930** shown in FIG. **51A**. In some examples, the compression interposer (e.g., spring-loaded contacts) can be part of the receiving structure (e.g., mounted on the circuit board or substrate) as opposed to the removable module.

[0519] FIG. **52A** shows a first side view of the mechanical connector structure **900** mounted within the first grid structure **870**. FIG. **52B** shows a cross-sectional view of the mechanical connector structure **900** mounted within the first grid structure **870** along a plane **940** shown in FIG. **52A**.

[0520] FIG. **53** is a diagram of an assembly **958** that includes a fiber cable **956** that includes a plurality of optical fibers, an optical fiber connector **950**, the mechanical connector module **900**, and the first grid structure **870**. The optical fiber connector **95** can be inserted into the mechanical connector module **900**, which can be further inserted into the first grid structure **870**. The printed circuit board **862** (or substrate) is attached to the first grid structure **870**, in which the electrical contacts **876** face electrical contacts **954** on the bottom side of the second slab **898** of the optical module **880**.

[0521] FIG. **53** shows the individual components before they are connected. FIG. **54** is a diagram that shows the components after they are connected. The optical fiber connector **950** includes a lock mechanism **952** that disables the snap-in mechanism of the mechanical connector structure **900** so as to lock in place the mechanical connector structure **900** and the optical module **880**. In this example embodiment, the lock mechanism **952** includes studs on the optical fiber connector **95** that insert between the wings **906** and the upper mechanical part **904** of the mechanical connector module **900**, hence disabling the wings **906** from elastically bending inwards and consequentially locking the mechanical connector structure **900** and the optical module **880** in place. Further, the mechanical connector structure **900** includes a mechanism to hold the optical fiber connector **95** in place, such as a ball-detent mechanism as shown in the figure. When the optical fiber connector **95** is inserted into the mechanical connector structure **900**, spring-loaded balls **962** on the optical fiber connector **95** engage detents **964** in the wings **906** of the mechanical connector structure **900**. The springs push the balls **962** against the detents **964** and secure the optical fiber connector **95** in place. [0522] To remove the optical module **880** from the first grid structure **870**, the user can pull the optical fiber connector **95** and cause the balls **962** to disengage from the detents **964**. The user can then bend the wings **906** inwards so that the tongues **910** disengage from the grooves **920** on the walls of the first grid structure **870**.

[0523] FIGS. **55A** and **55B** show perspective views of the mechanisms shown in FIGS. **53** and **54** before the optical fiber connector **95** is inserted into the mechanical connector structure **900**. As shown in FIG. **55B**, the lower side of the optical connector **95** includes alignment structures **960** that mate with the alignment structures **886** (FIG. **44**) on the upper connector part **884** of the optical module **880**. FIG. **55B** also shows the photonic integrated circuit **896** and the second slab **898** that includes electrical contacts (e.g., spring-loaded electrical contacts).

[0524] FIG. 56 is a perspective view showing that the optical module **880** and the mechanical connector structure **900** are inserted into the first grid structure **870**, and the optical fiber connector **95** is separated from the mechanical connector structure **900**.

[0525] FIG. 57 is a perspective view showing that the optical fiber connector **95** is mated with the mechanical connector structure **900**, locking the optical module **880** within the mechanical connector structure **900**.

[0526] FIGS. 58A to 58D show an alternate embodiment in which an optical module with connector **970** includes a latch mechanism **972** that acts as a mechanical fastener that joins the optical module **880** to the printed circuit board **862** (or substrate) using the first grid structure **870** as a support. FIGS. 58A and 58B show various views of the optical module with connector **970** that includes the latch mechanism **972**. FIGS. 58C and 58D show various views of the optical module with connector **970** coupled to the printed circuit board **862** (or substrate) and the first grid structure **870**. For example, the user can easily attach or remove the optical module with connector **970** by pressing a lever **974** activating the latch mechanism **972**. The lever **974** is built in a way that it does not block the optical fibers (not shown in the figure) coming out of the optical module with connector **970**. Alternatively, an external tool can be used as a removable lever.

[0527] FIG. 59 is a view of an optical module **1030** that includes an optical engine with a latch mechanism used to realize the compression and attachment of the optical engine to the printed circuit board. The module **1030** is similar to the example shown in FIG. 58B but without the compression interposer. FIGS. 60A and 60B show an example latch mechanism that can be used for securing (with enough compression force) and removing the optical engine.

[0528] FIGS. 60A and 60B show an example implementation of the lever **974** and the latch mechanism **972** in the optical module **1030**. FIG. 60A shows an example in which the lever **974** is pushed down, causing the latch mechanism **972** to latch on to a support structure **976**, which can be part of the first grid structure **870**. FIG. 60B shows an example in which the lever **974** is pulled up, causing the latch mechanism **972** to be released from the support structure **976**.

[0529] FIG. 61 is a diagram of an example of a fiber cable connection design **980** that includes nested fiber optic cable and co-packaged optical module connections. In this design, a co-packaged optical module **982** is removably coupled to a co-packaged optical port **1000** formed in a support structure, such as the first grid structure **870**, and a fiber connector **983** is removably coupled to the co-packaged optical module **982**. The fiber connector **983** is coupled to a fiber cable **996** that includes a plurality of optical fibers. The fiber cable connection can be designed to be, e.g., MTP/MPO (Multi-fiber Termination Push-on/Multi-fiber Push On) compatible, or compatible to new standards as they emerge. Multi-fiber push on (MPO) connectors are commonly used to terminate multi-fiber ribbon connections in indoor environments and conforms to IEC-61754-7; EIA/TIA-604-5 (FOCIS 5) standards.

[0530] In some implementations, the co-packaged optical module **982** includes a mechanical connector structure **984** and a smart optical assembly **986**. The smart optical assembly **986** includes, e.g., a photonic integrated circuit (e.g., **896** of FIG. 44), and components for guiding light, power splitting, polarization management, optical filtering, and other light beam management before the photonic integrated circuit. The components can include, e.g., optical couplers, waveguides, polarization optics, filters, and/or lenses. Additional examples of the components that can be included in the co-packaged optical module **982** are described in U.S. patent application Ser. No. 16/816,171, now U.S. Pat. No. 11,287,585. The mechanical connector structure **984** includes one or more fiber connector latches **988** and one or more co-packaged optical module latches **990**. The mechanical connector structure **984** can be inserted into the co-packaged optical port **1000** (e.g., formed in the first grid structure **870**), in which the co-packaged optical module latches **990** engage grooves **992** in the walls of the first grid structure **870**, thus securing the co-packaged optical module **982** to the co-packaged optical port **1000**, and causing the electrical contacts of the smart optical assembly **986** to be electrically coupled to the electrical contacts **876** on the printed

circuit board **862** (or substrate). When the fiber connector **983** is inserted into the mechanical connector structure **984**, the fiber connector latches **988** engage grooves **994** in the fiber connector **983**, thus securing the fiber connector **983** to the co-packaged optical module **982**, and causing the fiber cable **996** to be optically coupled to the smart optical assembly **986**, e.g., through optical paths in the fiber connector **983**.

[0531] In some examples, the fiber connector **983** includes guide pins **998** that are inserted into holes in the smart optical assembly **986** to improve alignment of optical components (e.g., waveguides and/or lenses) in the fiber connector **983** to optical components (e.g., optical couplers and/or waveguides) in the smart optical assembly **986**. In some examples, the guide pins **998** can be chamfered shaped, or elliptical shaped that reduces wear.

[0532] In some implementations, after the fiber connector **983** is installed in the co-packaged optical module **982**, the fiber connector **983** prevents the co-packaged optical module latches **990** from bending inwards, thus preventing the co-packaged optical module **982** from being inserted into, or released from, the co-packaged optical port **1000**. To couple the fiber cable **996** to the data processing system, the co-packaged optical module **982** is first inserted into the co-packaged optical port **1000** without the fiber connector **983**, then the fiber connector **983** is inserted into the mechanical connector structure **984**. To remove the fiber cable **996** from the data processing system, the fiber connector **983** can be removed from the mechanical connector structure **984** while the co-packaged optical module **982** is still coupled to the co-packaged optical port **1000**.

[0533] In some implementations, the nested connection latches can be designed to allow the co-packaged optical module **982** to be inserted in, or removed from, the co-packaged optical port **1000** when a fiber cable is connected to the co-packaged optical module **982**.

[0534] FIGS. **62** and **63** are diagrams showing cross-sectional views of an example of a fiber cable connection design **1010** that includes nested fiber optic cable and co-packaged optical module connections. FIG. **62** shows an example in which a fiber connector **1012** is removably coupled to a co-packaged optical module **1014**. FIG. **63** shows an example in which the fiber connector **1012** is separated from the co-packaged optical module **1014**.

[0535] FIGS. **64** and **65** are diagrams showing additional cross-sectional views of the fiber cable connection design **1010**. The cross-sections are made along planes that vertically cut through the middle of the components shown in FIGS. **62** and **63**. FIG. **64** shows an example in which the fiber connector **1012** is removably coupled to the co-packaged optical module **1014**. FIG. **65** shows an example in which the fiber connector **1012** is separated from the co-packaged optical module **1014**.

[0536] The following describes rack unit thermal architectures for rackmount systems (e.g., **560** of FIG. **22**, **600** of FIG. **23**, **630** of FIG. **24**, **680** of FIG. **26**, **720** of FIG. **28**, **750** of FIG. **29**, **860** of FIG. **43**) that include data processing chips (e.g., **572** of FIGS. **22**, **23**, **640** of FIG. **24**, **681** of FIG. **26**, **722** of FIG. **28**, **758** of FIG. **29**, **864** of FIG. **43**) that are mounted on vertically oriented circuit boards that are substantially vertical to the bottom surfaces of the system housings or enclosures. In some implementations, the rack unit thermal architectures use air cooling to remove heat generated by the data processing chips. In these systems, the heat-generating data processing chips are positioned near the input/output interfaces, which can include, e.g., one or more of the integrated optical communication device **448**, **462**, **466**, or **472** of FIG. **17**, the integrated communication device **574** of FIG. **22** or **612** of FIG. **23**, the optical/electrical communication interface **644** of FIG. **24**, **684** of FIG. **26**, **724** of FIG. **28**, or **760** of FIG. **29**, or the optical module with connector **868** of FIG. **43**, that are positioned at or near the front panel to enable users to conveniently connect/disconnect optical transceivers to/from the rackmount systems. The rack unit thermal architectures described in this specification include mechanisms for increasing airflow across the surfaces of the data processing chips, or heat sinks thermally coupled to the data processing chips, taking into consideration that a substantial portion of the surface area on the front panel of the housing needs to be allocated to the input/output interfaces.

[0537] Referring to FIG. **67**, a data server **1140** suitable for installation in a standard server rack

can include a housing **1142** that has a front panel **1034**, a rear panel **1036**, a bottom panel **1038**, a top panel, and side panels **1040**. For example, the housing **1142** can have a 2 rack unit (RU) form factor, having a width of about 482.6 mm (19 inches) and a height of 2 rack units. One rack unit is about 44.45 mm (approximately 1.75 inches). A printed circuit board **1042** is mounted on the bottom panel **1038**, and at least one data processing chip **1044** is electrically coupled to the printed circuit board **1042**. A microcontroller unit **1046** is provided to control various modules, such as power supplies **1048** and exhaust fans **1050**. In this example, the exhaust fans **1050** are mounted at the rear panel **1036**. For example, single mode optical connectors **1052** are provided at the front panel **1034** for connection to external optical cables. Optical interconnect cables **1036** transmit signals between the single mode optical connectors **1052** and the at least one data processing chip **1044**. The exhaust fans **1050** mounted at the rear panel **1036** cause the air to flow from the front side to the rear side of the housing **1142**. The directions of air flow are represented by arrows **1058**. Warm air inside the housing **1142** is vented out of the housing **1142** through the exhaust fans **1050** at the rear panel **1036**. In this example, the front panel **1034** does not include any fan in order to maximize the area used for the single mode optical connectors **1052**.

[0538] For example, the data server **1300** can be a network switch server, and the at least one data processing chip **1044** can include at least one switch chip configured to process data having a total bandwidth of, e.g., about 51.2 Tbps. The at least one switch chip **1044** can be mounted on a substrate **1054** having dimensions of, e.g., about 100 mm×100 mm, and co-packaged optical modules **1056** can be mounted near the edges of the substrate **1054**. The co-packaged optical modules **1056** convert input optical signals received from the optical interconnect cables **1036** to input electrical signals that are provided to the at least one switch chip **1044**, and converts output electrical signals from the at least one switch chip **1044** to output optical signals that are provided to the optical interconnect cables **1036**. When any of the co-packaged optical modules **1056** fails, the user needs to remove the network switch server **1030** from the server rack and open the housing **1142** in order to repair or replace the faulty co-packaged optical module **1056**.

[0539] Referring to FIGS. **68A** and **68B**, in some implementations, a rackmount server **1060** includes a housing or case **1062** having a front panel **1064** (or face plate), a rear panel **1036**, a bottom panel **1038**, a top panel, and side panels **1040**. For example, the housing **1062** can have a form factor of 1RU, 2RU, 3RU, or 4RU, having a width of about 482.6 mm (19 inches) and a height of 1, 2, 3, or 4 rack units. A first printed circuit board **1066** is mounted on the bottom panel **1038**, and a microcontroller unit **1046** is electrically coupled to the first printed circuit board **1066** and configured to control various modules, such as power supplies **1048** and exhaust fans **1050**.

[0540] In some implementations, the front panel **1064** includes a second printed circuit board **1068** that is oriented in a vertical direction, e.g., substantially perpendicular to the first circuit board **1066** and the bottom panel **1038**. In the following, the second printed circuit board **1068** is referred to as the vertical printed circuit board **1068**. The figures shows that the second printed circuit board **1066** forms part of the front panel **1064**, but in some examples the second printed circuit board **1066** can also be attached to the front panel **1064**, in which the front panel **1064** includes openings to allow input/output connectors to pass through. The second printed circuit board **1066** includes a first side facing the front direction relative to the housing **1062** and a second side facing the rear direction relative to the housing **1062**. At least one data processing chip **1070** is electrically coupled to the second side of the vertical printed circuit board **1068**, and a heat dissipating device or heat sink **1072** is thermally coupled to the at least one data processing chip **1070**. In some examples, the at least one data processing chip **1070** is mounted on a substrate (e.g., a ceramic substrate), and the substrate is attached to the printed circuit board **1068**. FIG. **68C** is a perspective view of an example of the heat dissipating device or heat sink **1072**. For example, the heat dissipating device **1072** can include a vapor chamber thermally coupled to heat sink fins. The exhaust fans **1050** mounted at the rear panel **1036** cause the air to flow from the front side to the rear side of the housing **1142**. The directions of air flow are represented by arrows **1078**. Warm air inside the

housing **1142** is vented out of the housing **1142** through the exhaust fans **1050** at the rear panel **1036**.

[0541] Co-packaged optical modules **1074** (also referred to as the optical/electrical communication interfaces) are attached to the first side (i.e., the side facing the front exterior of the housing **1062**) of the vertical printed circuit board **1068** for connection to external fiber cables **1076**. Each fiber cable **1076** can include an array of optical fibers. By placing the co-packaged optical modules **1074** on the exterior side of the front panel **1064**, the user can conveniently service (e.g., repair or replace) the co-packaged optical modules **1074** when needed. Each co-packaged optical module **1074** is configured to convert input optical signals received from the external fiber cable **1076** into input electrical signals that are transmitted to the at least one data processing chip **1070** through signal lines in or on the vertical printed circuit board **1068**. The co-packaged optical module **1074** also converts output electrical signals from the at least one data processing chip **1070** into output optical signals that are provided to the external fiber cables **1076**. Warm air inside the housing **1062** is vented out of the housing **1062** through the exhaust fans **1050** mounted at the rear panel **1036**.

[0542] For example, the at least one data processing chip **1070** can include a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, or an application specific integrated circuit (ASIC). For example, each co-packaged optical module **1074** can include a module similar to the integrated optical communication device **448**, **462**, **466**, or **472** of FIG. **17**, the integrated optical communication device **210** of FIG. **20**, the integrated communication device **612** of FIG. **23**, the optical/electrical communication interface **684** of FIG. **26**, **724** of FIG. **28**, or **760** of FIG. **29**, the integrated optical communication device **512** of FIG. **32**, or the optical module with connector **868** of FIG. **43**. For example, each fiber cable **1076** can include the optical fibers **226** (FIGS. **2**, **4**), **272** (FIGS. **6**, **7**), **582** (FIGS. **22**, **23**), or **734** (FIG. **28**), or the optical fiber cable **762** (FIG. **762**), **956** (FIG. **53**), or **996** (FIG. **61**).

[0543] For example, the co-packaged optical module **1074** can include a first optical connector part (e.g., **456** of FIG. **17**, **578** of FIG. **22** or **23**, **746** of FIG. **28**) that is configured to be removably coupled to a second optical connector part (e.g., **458** of FIG. **17**, **580** of FIG. **22** or **23**, **748** of FIG. **28**) that is attached to the external fiber cable **1076**. For example, the co-packaged optical module **1074** includes a photonic integrated circuit (e.g., **450**, **464**, **468**, or **474** of FIG. **17**, **586** of FIG. **22**, **618** of FIG. **23**, or **726** of FIG. **28**) that is optically coupled to the first optical connector part. The photonic integrated circuit receives input optical signals from the first optical connector part and generates input electrical signals based on the input optical signals. At least a portion of the input electrical signals generated by the photonic integrated circuit are transmitted to the at least one data processing chip **1070** through electrical signal lines in or on the vertical printed circuit board **1068**. For example, the photonic integrated circuit can be configured to receive output electrical signals from the at least one data processing chip **1070** and generate output optical signals based on the output electrical signals. The output optical signals are transmitted through the first and second optical connector parts to the external fiber cable **1076**.

[0544] In some examples, the fiber cable **1076** can include, e.g., 10 or more cores of optical fibers, and the first optical connector part is configured to couple 10 or more channels of optical signals to the photonic integrated circuit. In some examples, the fiber cable **1076** can include 100 or more cores of optical fibers, and the first optical connector part is configured to couple 100 or more channels of optical signals to the photonic integrated circuit. In some examples, the fiber cable **1076** can include 500 or more cores of optical fibers, and the first optical connector part is configured to couple 500 or more channels of optical signals to the photonic integrated circuit. In some examples, the fiber cable **1076** can include 1000 or more cores of optical fibers, and the first optical connector part is configured to couple 1000 or more channels of optical signals to the photonic integrated circuit.

[0545] In some implementations, the photonic integrated circuit can be configured to generate first

serial electrical signals based on the received optical signals, in which each first serial electrical signal is generated based on one of the channels of first optical signals. Each co-packaged optical module **1074** can include a first serializers/deserializers module that includes serializer units and deserializer units, in which the first serializers/deserializers module is configured to generate sets of first parallel electrical signals based on the first serial electrical signals and condition the electrical signals, and each set of first parallel electrical signals is generated based on a corresponding first serial electrical signal. Each co-packaged optical module **1074** can include a second serializers/deserializers module that includes serializer units and deserializer units, in which the second serializers/deserializers module is configured to generate second serial electrical signals based on the sets of first parallel electrical signals, and each second serial electrical signal is generated based on a corresponding set of first parallel electrical signals.

[0546] In some examples, the rackmount server **1060** can include 4 or more co-packaged optical modules **1074** that are configured to be removably coupled to corresponding second optical connector parts that are attached to corresponding fiber cables **1076**. For example, the rackmount server **1060** can include 16 or more co-packaged optical modules **1074** that are configured to be removably coupled to corresponding second optical connector parts that are attached to corresponding fiber cables **1076**. In some examples, each fiber cable **1076** can include 10 or more cores of optical fibers. In some examples, each fiber cable **1076** can include 100 or more cores of optical fibers. In some examples, each fiber cable **1076** can include 500 or more cores of optical fibers. In some examples, each fiber cable **1076** can include 1000 or more cores of optical fibers. Each optical fiber can transmit one or more channels of optical signals. For example, the at least one data processing chip **1070** can include a network switch that is configured to receive data from an input port associated with a first one of the channels of optical signals, and forward the data to an output port associated with a second one of the channels of optical signals.

[0547] In some implementations, the co-packaged optical modules **1074** is removably coupled to the vertical printed circuit board **1068**. For example, the co-packaged optical modules **1074** can be electrically coupled to the vertical printed circuit board **1068** using electrical contacts that include, e.g., spring-loaded elements, compression interposers, or land-grid arrays.

[0548] Referring to FIGS. **69A** and **69B**, in some implementations, a rackmount server **1080** includes a housing **1082** having a front panel **1084**. The rackmount server **1080** is similar to the rackmount server **1060** of FIG. **68A**, except that one or more fans are mounted on the front panel **1084**, and one or more air louvers installed in the housing **1082** to direct air flow towards the heat dissipating device. For example, the rackmount server **1080** can include a first inlet fan **1086a** mounted on the front panel **1084** to the left of the vertical printed circuit board **1068**, and a second inlet fan **1086b** mounted on the front panel **1084** to the right of the vertical printed circuit board **1068**. The terms “right” and “left” refer to relative positions of components shown in the figure. It is understood that, depending on the orientation of a device having a first and second modules, a first module that is positioned to the “left” or “right” of a second module can in fact be to the “right” or “left” (or any other relative position) of the second module. The inlet and exhaust fans operate in a push-pull manner, in which the inlet fans **1086a** and **1086b** (collectively referenced as **1086**) pull cool air into the housing **1082**, and the exhaust fans **1050** push warm air out of the housing **1082**. The inlet fans **1086** in the front panel or face plate **1064** and the exhaust fans **1050** on the backside of the rack generate a pressure gradient through the housing or case to improve air cooling compared to standard 1RU implementations that include on backside exhaust fans.

[0549] In some implementations, a left air louver **1088a** and a right air louver **1088b** are installed in the housing **1082** to direct airflow toward the heat dissipating device **1072**. The air louvers **1088a**, **1088b** (collectively referenced as **1088**) partitions the space in the housing **1082** and forces air to flow from the inlet fans **1086a** and **1086b**, pass over surfaces of fins of the heat dissipating device **1072**, and towards an opening **1090** between distal ends of the air louvers **1088**. The directions of air flow near the inlet fans **1086a** and **1086b** are represented by arrows **1092a** and

1092b. The air louvers **1088** increase the amount of air flows across the surfaces of the heat sink fins and enhance the efficiency of heat removal. The heat sink fins are oriented to extend along planes that are substantially parallel to the bottom surface **1038** of the housing **1082**. For example, the air louvers **1088** can have a curved shape, e.g., an S-shape as shown in the figure. The curved shape of the air louvers **1088** can be configured to maximize the efficiency of the heat sink. In some examples, the air louvers **1088** can also have a linear shape.

[0550] For example, the heat sink can be a plate-fin heat sink, a pin-fin heat sink, or a plate-pin-fin heat sink. The pins can have a square or circular cross section. The heat sink configuration (e.g., pin pitch, length of pins or fins) and the louver configuration can be designed to optimize heat sink efficiency.

[0551] For example, the co-packaged optical modules **1074** can be electrically coupled to the vertical printed circuit board **1068** using electrical contacts that include, e.g., spring-loaded elements, compression interposers, or land-grid arrays. For example, when compression interposers are used, the vertical circuit board **1068** can be positioned such that the face of compression interposers of the co-packaged optical module **1074** is coplanar with the face plate **1064** and the inlet fans **1086**.

[0552] Referring to FIG. 70, in some implementations, a rackmount server **1090** is similar to the rackmount server **1080** of FIG. 69, which includes inlet fans mounted on the front panel. The inlet fans of the rackmount server **1090** are slightly rotated, as compared to the inlet fans of the rackmount server **1080** to improve efficiency of the heat sink. The rotational axes of the inlet fans, instead of being parallel to the front-to-rear direction relative to the housing **1082**, can be rotated slightly inwards. For example, the rotational axis of a left inlet fan **1092a** can be rotated slightly clockwise and the rotational axis of a right inlet fan **1092b** can be rotated slightly counter-clockwise, to enhance the air flow across the surfaces of the heat sink fins, further improving the efficiency of heat removal.

[0553] In some implementations heat removal efficiency can be improved by positioning the vertical circuit board **1068** and the heat dissipating device **1072** further toward the rear of the housing so that a larger amount of air flows across the surface of the fins of the heat dissipating device **1072**.

[0554] Referring to FIGS. 71A to 71B, a rackmount server **1100** includes a housing **1102** having a front panel or face plate **1104**, in which the portion of the face plate **1104** where the compression interposers for the co-packaged optical module **1074** are located are inset by a distance d with respect to the original face plate **1104**. The face plate **1104** has a recessed portion or an inset portion **1106** that is offset at a distance d (referred to as the “front panel inset distance”) toward the rear of the housing **1102** relative to the other portions (e.g., the portions on which the inlet fans **1086a** and **1086b** are mounted) of the front panel **1104**. The inset portion **1106** is referred to as the “recessed front panel,” “recessed face plate,” “front panel inset,” or “face plate inset.” The vertical printed circuit board **1068** is attached to the inset portion **1106**, which includes openings to allow the co-packaged optical modules **1074** to pass through. The inset portion **1106** is configured to have sufficient area to accommodate the co-packaged optical modules **1074**.

[0555] By providing the inset portion **1106** in the front panel **1104**, the fins of the heat dissipating device **1072** can be more optimally positioned to be closer to the main air flow generated by the inlet fans **1086**, while maintaining serviceability of the co-packaged optical modules **1074**, e.g., allowing the user to repair or replace damaged co-packaged optical modules **1074** without opening the housing **1102**. The heat sink configuration (e.g., pin pitch, length of pins or fins) and the louver configuration can be designed to optimize heat sink efficiency. In addition, the front panel inset distance d can be optimized to improve heat sink efficiency.

[0556] Referring to FIG. 72, in some implementations, a rackmount server **1110** is similar to the rackmount server **1100** of FIG. 71, except that the server **1110** includes a heat dissipating device **1112** that has fins **1114a** and **1114b** that extend beyond the edge of the vertical printed circuit board

1068 and closer to the inlet fans **1086a**, **1086b**, as compared to the fins in the example of FIG. **71**. The configuration of the fins (e.g., the shapes, sizes, and number of fins) can be selected to maximize the efficiency of heat removal.

[0557] Referring to FIGS. **73A** and **73B**, in some implementations, a rackmount server **1120** includes a housing **1122** having a front panel **1124**, a rear panel **1036**, a bottom panel **1038**, a top panel, and side panels **1040**. The width and height of the housing **1122** can be similar to those of the housing **1062** of FIG. **68A**. The server **1120** includes a first printed circuit board **1066** that extends parallel to the bottom panel **1038**, and one or more vertical printed circuit boards, e.g., **1126a** and **1126b** (collectively referenced as **1126**), that are mounted perpendicular to the first printed circuit board **1066**. The server **1120** includes one or more inlet fans **1086** mounted on the front panel **1124** and one or more exhaust fans **1050** mounted on the rear panel **1036**. The air flow in the housing **1122** is generally in the front-to-rear direction. The directions of the air flows are represented by the arrows **1134**.

[0558] Each vertical printed circuit board **1126** has a first surface and a second surface. The first surface defines the length and width of the vertical printed circuit board **1126**. The distance between the first and second surfaces defines the thickness of the vertical printed circuit board **1126**. The vertical printed circuit board **1126a** or **1126b** is oriented such that the first surface extends along a plane that is substantially parallel to the front-to-rear direction relative to the housing **1122**. At least one data processing chip **1128a** or **1128b** is electrically coupled to the first surface of the vertical printed circuit board **1126a** or **1126b**, respectively. In some examples, the at least one data processing chip **1128a** or **1128b** is mounted on a substrate (e.g., a ceramic substrate), and the substrate is attached to the printed circuit board **1126a** or **1126b**. A heat dissipating device **1130a** or **1130b** is thermally coupled to the at least one data processing chip **1128a** or **1128b**, respectively. The heat dissipating device **1130** includes fins that extend along planes that are substantially parallel to the bottom panel **1038** of the housing **1122**. The heat sinks **1130a** and **1130b** are positioned directly behind to the inlet fans **1086a** and **1086b**, respectively, to maximize air flow across the fins and/or pins of the heat sinks **1130**.

[0559] At least one co-packaged optical module **1132a** or **1132b** is mounted on the second side of the vertical printed circuit board **1126a** or **1126b**, respectively. The co-packaged optical modules **1132** are optically coupled, through optical interconnection links, to optical interfaces (not shown in the figure) mounted on the front panel **1124**. The optical interfaces are optically coupled to external fiber cables. The orientations of the vertical printed circuit boards **1126** and the fins of the heat dissipating devices **1130** are selected to maximize heat removal.

[0560] Referring to FIGS. **74A** to **74B**, in some implementations, a rackmount server **1150** includes vertical printed circuit boards **1152a** and **1152b** (collectively referenced as **1152**) that have surfaces that extend along planes substantially parallel to the front-to-rear direction relative to the housing or case, similar to the vertical printed circuit boards **1126a** and **1126b** of FIG. **73**. The rackmount server **1150** includes a housing **1154** that has a modified front panel or face plate **1156** that has an inset portion **1158** configured to improve access and field serviceability of co-packaged optical modules **1160a** and **1160b** (collectively referenced as **1160**) that are mounted on the vertical printed circuit boards **1152a** and **1152b**, respectively. The inset portion **1158** is referred to as the “front panel inset” or “face plate inset.” The inset portion **1158** has a width w that is selected to enable hot-swap, in-field serviceability of the co-packaged optical modules **1160** to avoid the need to take the rackmount server **1150** out of service for maintenance.

[0561] For example, the inset portion **1158** includes a first wall **1162**, a second wall **1164**, and a third wall **1166**. The first wall **1162** is substantially parallel to the second wall **1164**, and the third wall **1166** is positioned between the first wall **1162** and the second wall **1164**. For example, the first wall **1162** extends along a direction that is substantially parallel to the front-to-rear direction relative to the housing **1122**. The vertical printed circuit board **1152a** is attached to the first wall **1162** of the inset portion **1158**, and the vertical printed circuit board **1152b** is attached to the first

wall **1162** of the inset portion **1158**. The first wall **1162** includes openings to allow the co-packaged optical modules **1160a** to pass through, and the second wall **1164** includes openings to allow the co-packaged optical modules **1160b** to pass through. For example, an inlet fan **1086c** can be mounted on the third wall **1166**.

[0562] Each vertical printed circuit board **1152** has a first surface and a second surface. The first surface defines the length and width of the vertical printed circuit board **1152**. The distance between the first and second surfaces defines the thickness of the vertical printed circuit board **1152**. The vertical printed circuit board **1152a** or **1152b** is oriented such that the first surface extends along a plane that is substantially parallel to the front-to-rear direction relative to the housing **1154**. At least one data processing chip **1170a** or **1170b** is electrically coupled to the first surface of the vertical printed circuit board **1152a** or **1152b**, respectively. In some examples, the at least one data processing chip **1170a** or **1170b** is mounted on a substrate (e.g., a ceramic substrate), and the substrate is attached to the printed circuit board **1152a** or **1152b**. A heat dissipating device **1168a** or **1168b** is thermally coupled to the at least one data processing chip **1170a** or **1170b**, respectively. The heat dissipating device **1168** includes fins that extend along planes that are substantially parallel to the bottom panel **1038** of the housing **1154**. The heat sinks **1168a** and **1168b** are positioned directly behind to the inlet fans **1086a** and **1086b**, respectively, to maximize air flow across the fins and/or pins of the heat sinks **1168a** and **1168b**.

[0563] Referring to FIGS. 75A to 75B, in some implementations, a rackmount server **1180** includes a housing **1182** having a front panel **1184** that has an inset portion **1186** (referred to as the “front panel inset” or “face plate inset”). For example, the inset portion **1186** includes a first wall **1188** and a second wall **1190** that are oriented to make it easier for the user to connect or disconnect the fiber cables (e.g., **1076**) to the server **1180**, or to service the co-packaged optical modules **1074**. For example, the first wall **1188** can be at an angle $\theta_{\text{sub.1}}$ relative to a nominal plane **1192** of the front panel **1184**, in which $0 < \theta_{\text{sub.1}} < 90^\circ$. The second wall **1190** can be at an angle $\theta_{\text{sub.2}}$ relative to the nominal plane **1192** of the front panel, in which $0 < \theta_{\text{sub.2}} < 90^\circ$. The angles $\theta_{\text{sub.1}}$ and $\theta_{\text{sub.2}}$ can be the same or different. The nominal plane **1192** of the front panel **1184** is perpendicular to the side panels **1040** and the bottom panel.

[0564] For example, a first vertical printed circuit board **1152a** is attached to the first wall **1188**, and a second vertical printed circuit board **1152b** is attached to the second wall **1190**. Comparing the rackmount server **1180** with the rackmount servers **1060** of FIG. 68A, **1080** of FIG. 69A, and **1100** of FIG. 71, the server **1180** has a larger front panel area due to the angled front panel inset and can be connected to more fiber cables.

[0565] Positioning the first and second walls **1188**, **1190** at an angle between 0 and 90° relative to the nominal plane of the front panel improves access and field serviceability of the co-packaged optical modules. Comparing the rackmount server **1180** with the rackmount server **1150** of FIG. 74A, the server **1180** allows the user to more easily access the co-packaged optical modules that are positioned farther away from the nominal plane of the front panel. The angles $\theta_{\text{sub.1}}$ and $\theta_{\text{sub.2}}$ are selected to strike a balance between increasing the number of fiber cables that can be connected to the server and providing easy access to all of the co-packaged optical modules of the server. The front panel inset width and angle are configured to enable hot-swap, in-field serviceability to avoid taking the switch and rack out of service for maintenance.

[0566] For examples, intake fans **1086a** and **1086b** can be mounted on the front panel **1184**. Outside air is drawn in by the intake fans **1086a**, **1086b**, passes through the surfaces of the fins and/or pins of the heatsinks **1168a**, **1168b**, and flows towards the rear of the housing **1182**. Examples of the flow directions for the air entering through the intake fans **1086a** and **1086b** are represented by arrows **1198a**, **1198b**, **1198c**, and **1198d**.

[0567] Referring to FIGS. 75B and 75C, in some implementations, the front panel **1184** includes an upper air vent **1194a** and baffles to direct outside air to enter through the upper air vent **1194a**, flows downward and rearward such that the air passes over the surfaces of some of the fins and/or

pins of the heat sinks **1186** (e.g., including the fins and/or pins closer to the top of the heat sinks **1186**) and then flows toward an intake fan **1086c** mounted at or near the distal or rear end of the front panel inset portion **1186**. The front panel **1184** includes a lower air vent **1194b** and baffles to direct outside air to enter through the lower air vent **1194b**, flows upward and rearward such that the air passes over the surfaces of some of the fins and/or pins of the heat sinks **1186** (e.g., including the fins and/or pins closer to the bottom of the heat sinks **1186**) and then flows toward the intake fan **1086c**. Examples of the air flows through the upper and lower air vents **1194a**, **1194b** to the intake fan **1086c** are represented by arrows **1196a**, **1196b**, **1196c**, and **1196d** in FIG. 75C.

[0568] For example, fiber cables connected to the co-packaged optical modules **1074** can block air flow for the intake fan **1086c** if the intake fan **1086c** is configured to receive air through openings directly in front of the intake fan **1086c**. By using the upper air vent **1194a**, the lower air vent **1194b**, and the baffles to direct air flow as described above, the heat dissipating efficiency of the system can be improved (as compared to not having the air vents **1194** and the baffles).

[0569] Referring to FIG. 76, in some implementations, a network switch system **1210** includes a plurality of rackmount switch servers **1212** installed in a server rack **1214**. The network switch rack includes a top of the rack switch **1216** that routes data among the switch servers **1212** within the network switch system **1210**, and serves as a gateway between the network switch system **1210** and other network switch systems. The rackmount switch servers **1212** in the network switch system **1210** can be configured in a manner similar to any of the rackmount servers described above or below.

[0570] In some implementations, the examples of rackmount servers shown in in FIGS. 68A, 69A, and 70 can be modified by positioning the vertical printed circuit board behind the front panel. The co-packaged optical modules can be optically connected to fiber connector parts mounted on the front panel through short optical connection paths, e.g., fiber jumpers.

[0571] Referring to FIGS. 77A and 77B, in some implementations, a rackmount server **1220** includes a housing **1222** having a front panel **1224**, a rear panel **1036**, a top panel **1226**, a bottom panel **1038**, and side panels **1040**. The front panel **1224** can be opened to allow the user to access components without removing the rackmount server **1220** from the rack. A vertically mounted printed circuit board **1230** is positioned substantially parallel to the front panel **1224** and recessed from the front panel **1224**, i.e., spaced apart at a small distance (e.g., less than 6 inches, or less than 3 inches, or less than 2 inches) to the rear of the front panel **1224**. The printed circuit board **1230** includes a first side facing the front direction relative to the housing **1222** and a second side facing the rear direction relative to the housing **1222**. At least one data processing chip **1070** is electrically coupled to the second side of the vertical printed circuit board **1226**, and a heat dissipating device or heat sink **1072** is thermally coupled to the at least one data processing chip **1070**. In some examples, the at least one data processing chip **1070** is mounted on a substrate (e.g., a ceramic substrate), and the substrate is attached to the printed circuit board **1226**.

[0572] Co-packaged optical modules **1074** (also referred to as the optical/electrical communication interfaces) are attached to the first side (i.e., the side facing the front exterior of the housing **1222**) of the vertical printed circuit board **1230**. In some examples, the co-packaged optical modules **1074** are mounted on a substrate that is attached to the vertical printed circuit board **1230**, in which electrical contacts on the substrate are electrically coupled to corresponding electrical contacts on the vertical printed circuit board **1230**. In some examples, the at least one data processing chip **1070** is mounted on the rear side of the substrate, and the co-packaged optical modules **1074** are removably attached to the front side of the substrate, in which the substrate provides high speed connections between the at least one data processing chip **1070** and the co-packaged optical modules **1074**. For example, the substrate can be attached to a front side of the printed circuit board **1068**, in which the printed circuit board **1068** includes one or more openings that allow the at least one data processing chip **1070** to be mounted on the rear side of the substrate. The printed circuit board **1068** can provide from a motherboard electrical power to the substrate (and hence to the at

least one data processing chip **1070** and the co-packaged optical modules **1074**, and allow the at least one data processing chip **1070** and the co-packaged optical modules **1074** to connect to the motherboard using low-speed electrical links. An array of co-packaged optical modules **1074** can be mounted on the vertical printed circuit board **1230** (or the substrate), similar to the examples shown in FIGS. **69B** and **71B**. The electrical connections between the co-packaged optical modules **1074** and the vertical printed circuit board **1070** (or the substrate) can be removable, e.g., by using land-grid arrays and/or compression interposers. The co-packaged optical modules **1074** are optically connected to first fiber connector parts **1232** mounted on the front panel **1224** through short fiber jumpers **1234a**, **1234b** (collectively referenced as **1234**). When the front panel **1224** is closed, the user can plug a second fiber connector part **1236** into the first fiber connector part **1232** on the front panel **1224**, in which the second fiber connector part **1236** is connected to an optical fiber cable **1238** that includes an array of optical fibers.

[0573] In some implementations, the rackmount server **1220** is pre-populated with co-packaged optical modules **1074**, and the user does not need to access the co-packaged optical modules **1074** unless the modules need maintenance. During normal operation of the rackmount server **1220**, the user mostly accesses the first fiber connector parts **1232** on the front panel **1224** to connect to fiber cables **1238**.

[0574] One or more intake fans, e.g., **1086a**, **1086b**, can be mounted on the front panel **1224**, similar to the examples shown in FIGS. **69A** and **70**. The positions and configurations of the intake fans **1086**, the heat sink **1072**, and the air louvers **1088a**, **1088b** are selected to maximize the heat transfer efficiency of the heat sink **1072**.

[0575] The rackmount server **1220** can have a number of advantages. By placing the vertical printed circuit board **1230** at a recessed position inside the housing **1222**, the vertical printed circuit board **1230** is better protected by the housing **1222**, e.g., preventing users from accidentally bumping into the circuit board **1230**. By orienting the vertical printed circuit board **1230** substantially parallel to the front panel **1224** and mounting the co-packaged optical modules **1074** on the side of the circuit board **1230** facing the front direction, the co-packaged optical modules **1074** can be accessible to users for maintenance without the need to remove the rackmount server **1220** from the rack.

[0576] In some implementations, the front panel **1224** is coupled to the bottom panel **1038** using a hinge **1228** and configured such that the front panel **1224** can be securely closed during normal operation of the rackmount server **1220** and easily opened for maintenance. For example, if a co-packaged optical module **1074** fails, a technician can open and rotate the front panel **1224** down to a horizontal position to gain access to the co-packaged optical module **1074** to repair or replace it. For example, the movements of the front panel **1224** is represented by the bi-directional arrow **1250**. In some implementations, different fiber jumpers **1234** can have different lengths, depending on the distance between the parts that are connected by the fiber jumpers **1234**. For example, the distance between the co-packaged optical module **1074** and the first fiber connector part **1232** connected by the fiber jumper **1234a** is less than the distance between the co-packaged optical module **1074** and the first fiber connector part **1232** connected by the fiber jumper **1234b**, so the fiber jumper **1234a** can be shorter than the fiber jumper **1234b**. This way, by using fiber jumpers with appropriate lengths, it is possible to reduce the clutter caused by the fiber jumpers **1234** inside the housing **1222** when the front panel **1224** is closed and in its vertical position.

[0577] In some implementations, the front panel **1224** can be configured to be opened and lifted upwards using lift-up hinges. This can be useful when the rackmount server is positioned near the top of the rack. In some examples, the front panel **1224** can be coupled to the side panel **1040** by using a hinge so that the front panel **1224** can be opened and rotated sideways. In some examples, the front panel can include a left front subpanel and a right front subpanel, in which the left front subpanel is coupled to the left side panel **1040** by using a first hinge, and the right front subpanel is coupled to the right panel **1040** by using a second hinge. The left front subpanel can be opened and

rotated towards the left side, and the right front subpanel can be opened and rotated towards the right side. These various configurations for the front panel enable protection of the vertical printed circuit board **1230** and convenient access to the co-packaged optical modules **1074**.

[0578] In some examples, the front panel can have an inset portion, similar to the example shown in FIG. **71A**, in which the vertical printed circuit board is in a recessed position relative to the inset portion of the front panel, i.e., at a small distance to the rear of the inset portion of the front panel. The front panel inset distance, the distance between the vertical printed circuit board and the front panel inset portion, and the air louver configuration can be selected to maximize the heat sink efficiency.

[0579] Referring to FIG. **78**, in some implementations, a rackmount server **1240** can be similar to the rackmount server **1150** of FIG. **74A**, except that the vertical printed circuit boards are at recessed positions relative to the walls of the inset portion of the front panel. For example, a vertical printed circuit board **1152a** is in a recessed position relative to a first wall **1242a** of an inset portion **1244**, i.e., the vertical printed circuit board **1152a** is spaced apart a small distance to the left from the first wall **1242a**. A vertical printed circuit board **1152b** is in a recessed position relative to a second wall **1242b** of the inset portion **1244**, i.e., the vertical printed circuit board **1152b** is spaced apart a small distance to the right from the second wall **1242b**.

[0580] For example, the first wall **1242a** can be coupled to the bottom or top panel through hinges so that the first wall **1242a** can be closed during normal operation of the rackmount server **1240** and opened for maintenance of the server **1240**. The distance w_2 between the first wall **1242a** and the second wall **1242b** is selected to be sufficiently large to enable the first wall **1242a** and the second wall **1242b** to be opened properly. This design has advantages similar to those of the rackmount server **1220** in FIGS. **77A**, **77B**.

[0581] In some implementations, a rackmount server can be similar to the rackmount server **1180** shown in FIGS. **75A** to **75C**, except that the vertical printed circuit boards are at recessed positions relative to the walls of the inset portion of the front panel. For example, a first vertical printed circuit board is in a recessed position relative to the first wall **1188** of the inset portion **1186**, and a second vertical printed circuit board is in a recessed position relative to the second wall **1190** of the inset portion **1186**. For example, the first wall **1188** can be coupled to the bottom or top panel through hinges so that the first wall **1188** can be closed during normal operation of the rackmount server and opened for maintenance of the server. The angles $\theta_{sub.1}$ and $\theta_{sub.2}$ are selected to enable the first wall **1188** and the second wall **1190** to be opened properly. This design has advantages similar to those of the rackmount server **1220** in FIGS. **77A**, **77B**.

[0582] A feature of the thermal architecture for the rackmount units (e.g., the rackmount servers **1060** of FIG. **68A**, **1090** of FIGS. **69A**, **70**, **1100** of FIGS. **71A**, **72**, **1120** of FIG. **73A**, **1150** of FIG. **74A**, **1180** of FIG. **75A**, **1220** of FIG. **77B**, and **1240** of FIG. **78**) described above is the use of co-packaged optical modules or optical/electrical communication interfaces that have higher bandwidth per module or interface, as compared to conventional designs. For example, each co-packaged optical module or optical/electrical communication interface can be coupled to a fiber cable that carries a large number of densely packed optical fiber cores. FIG. **9** shows an example of the integrated optical communication device **282** in which the optical signals provided to the photonic integrated circuit can have a total bandwidth of about 12.8 Tbps. By using co-packaged optical modules or optical/electrical communication interfaces that have higher bandwidth per module or interface, the number of co-packaged optical modules or optical/electrical communication interfaces required for a given total bandwidth for the rackmount unit is reduced, so the amount of area on the front panel of the housing reserved for connecting to optical fibers can be reduced. Therefore, it is possible to add one or more inlet fans on the front panel to improve thermal management while still maintaining or even increasing the total bandwidth of the rackmount unit, as compared to conventional designs.

[0583] In some implementations, for the examples shown in FIGS. **72**, **74A**, **75A**, and **78**, and the

variations in which the vertical printed circuit boards are at recessed positions relative to the front panel, the shape of each of the top and bottom panels of the housing can have an inset portion at the front that corresponds to the inset portion of the front panel. This makes it more convenient to access the co-packaged optical modules or the optical connector parts mounted on the front panel without being hindered by the top and bottom panels. In some implementations, the server rack (e.g., **1214** of FIG. **76**) is designed such that front support structures of the server rack also have inset portions that correspond to the inset portions of the front panels of the rackmount servers installed in the server rack. For example, a custom server rack can be designed to install rackmount servers that all have the inset portions similar to the inset portion **1158** of FIG. **74A**. For example, a custom server rack can be designed to install rackmount servers that all have the inset portions similar to the inset portion **1186** of FIG. **75A**. In such examples, the inset portions extend vertically from the bottom-most server to the top-most server without any obstruction, making it easier for the user to access the co-packaged optical modules or optical connector parts.

[0584] In some implementations, for the examples shown in FIGS. **72**, **74A**, **75A**, and **78**, and the variations in which the vertical printed circuit boards are at recessed positions relative to the front panel, the shape of the top and bottom panels of the housing can be similar to standard rackmount units, e.g., the top and bottom panels can have a generally rectangular shape.

[0585] In the examples shown in FIGS. **68A**, **68B**, **69A** to **75C**, and **77A** to **78**, a grid structure similar to the grid structure **870** shown in FIG. **43** can be attached to the vertical printed circuit board. The grid structure can function as both (i) a heat spreader/heat sink and (ii) a mechanical holding fixture for the co-packaged optical modules (e.g., **1074**) or optical/electrical communication interfaces.

[0586] FIGS. **96** to **97B** are diagrams of an example of a rackmount server **1820** that includes a vertically oriented circuit board **1822** positioned at a front portion of the rackmount server **1820**. FIG. **96** shows a top view of the rackmount server **1820**, FIG. **97A** shows a perspective view of the rackmount server **1820**, and FIG. **97B** shows a perspective view of the rackmount server **1820** with the top panel removed. The rackmount server **1820** has an active airflow management system that is configured to remove heat from a data processor during operation of the rackmount server **1820**. [0587] Referring to FIGS. **96**, **97A**, and **97B**, in some implementations, the rackmount server **1820** includes a housing **1824** that has a front panel **1826**, a left side panel **1828**, a right side panel **1840**, a bottom panel **1841**, a top panel **1843**, and a rear panel **1842**. The front panel **1826** can be similar to the front panels in the examples shown in FIGS. **68A**, **68B**, **69A** to **72**, **77A**, and **77B**. For example, the vertically oriented circuit board **1822** can be part of the front panel **1826**, or attached to the front panel **1826**, or positioned in a vicinity of the front panel **1826**, in which a distance between the circuit board **1822** and the front panel **1826** is not more than, e.g., 6 inches. A data processor **1844** (which can be, e.g., a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, or an application specific integrated circuit)(see FIG. **99**) is mounted on the circuit board **1822**.

[0588] A heat dissipating module **1846**, e.g., a heat sink, is thermally coupled to the data processor **1844** and configured to dissipate heat generated by the data processor **1828** during operation. The heat dissipating module **1846** can be similar to the heat dissipating device **1072** of FIGS. **68A**, **68C**, **69A**, **70**, and **71A**. In some examples, the heat dissipating module **1846** includes heat sink fins or pins having heat dissipating surfaces configured to optimize heat dissipation. In some examples, the heating dissipating module **1846** includes a vapor chamber thermally coupled to heat sink fins or pins. The rackmount server **1820** can include other components, such as power supply units, rear outlet fans, one or more additional horizontally oriented circuit boards, one or more additional data processors mounted on the horizontally oriented circuit boards, and one or more additional air louvers, that have been previously described in other embodiments of rackmount servers and are not repeated here.

[0589] In some implementations, the active airflow management system includes an inlet fan **1848** that is positioned at a left side of the heat dissipating module **1846** and oriented to blow incoming air to the right toward the heat dissipating module **1846**. A front opening **1850** provides incoming air for the inlet fan **1848**. The front opening **1850** can be positioned to the left of the inlet fan **1848**. In the example of FIG. **96**, the circuit board **1822** is substantially parallel to the front panel **1826**, and the rotational axis of the inlet fan **1848** is substantially parallel to the plane of the circuit board **1822**. The inlet fan **1848** can also be oriented slightly differently. For example, the rotational axis of the inlet fan **1848** can be at an angle θ relative to the plane of the front panel **1826**, the angle θ being measured along a plane parallel to the bottom panel **1841**, in which $\theta \leq 45^\circ$, or in some examples $\theta \leq 25^\circ$, or in some examples $\theta \leq 5^\circ$, or in some examples $\theta = 0^\circ$.

[0590] In some implementations, a baffle or an air louver **1852** (or internal panel or internal wall) is provided to guide the air entering the opening **1850** towards the inlet fan **1848**. An arrow **1854** shows the general direction of airflow from the opening **1850** to the inlet fan **1848**. In some examples, the air louver **1852** extends from the left side panel **1828** of the housing **1840** to a rear edge of the inlet fan **1848**. The air louver **1852** can be straight or curved. In some examples, the air louver **1852** can be configured to guide the inlet air blown from the inlet fan **1848** towards the heat dissipating module **1846**. For example, the air louver **1852** can extend from the left side panel **1828** to the left edge of the heat dissipating module **1846**. For example, the air louver **1852** can extend from the left side panel **1828** to a position at or near the rear of the heat dissipating module **1846**, in which the position can be anywhere from the left rear portion of the heat dissipating module **1846** to the right rear portion of the heat dissipating module **1846**. The air louver **1852** can extend from the bottom panel **1841** to the top panel **1843** in the vertical direction. An arrow **1856** shows the general direction of air flow through and out of the heating dissipating module **1846**.

[0591] For example, the air louver **1852**, a front portion of the left side panel **1828**, the front panel **1826**, the circuit board **1822**, a front portion of the bottom panel **1841**, and a front portion of the top panel **1843** can form an air duct that guides the incoming cool air to flow across the heat dissipating surface of the heat dissipating module **1846**. Depending on the design, the air duct can extend to the left edge of the heat dissipating module **1846**, to a middle portion of the heat dissipating module **1846**, or extend approximately the entire length (from left to right) of the heat dissipating module **1846**.

[0592] The inlet fan **1848** and the air louver **1852** are designed to improve airflow across the heat dissipating surface of the heat dissipating module **1846** to optimize or maximize heat dissipation from the data processor **1844** through the heat dissipating module **1846** to the ambient air. Different rackmount servers can have vertically mounted circuit boards with different lengths, can have data processors with different heat dissipation requirements, and can have heat dissipating modules with different designs. For example, the heat sink fins and/or pins can have different configurations. The inlet fan **1848** and the air louver **1852** can also have any of various configurations in order to optimize or maximize the heat dissipation from the data processor **1844**. In the example of FIG. **96**, the inlet fan **1848** directs air to flow generally in a direction (in this example, from left to right) that is parallel to the front panel across the heat dissipating surface of the heat dissipating module **1846**. In some implementations, the front opening can be positioned to the right side of the front panel, and the inlet fan can be positioned to the right side of the heat dissipating module and direct air to flow from right to left across the heat dissipating surface of the heat dissipating module. The air louver can be modified accordingly to optimize airflow and heat dissipation from the data processor.

[0593] FIG. **98** is a diagram showing the front portion of the rackmount server **1820**. The baffle or air louver **1852**, a portion of the bottom panel **1841**, a portion of the top panel **1843**, and a portion of the left side panel **1828** form a duct that directs external air toward the inlet fan **1848**. A safety mechanism (not shown in the figure), such as a protective mesh, that allows air to substantially freely pass through while blocking larger objects, such as users' fingers, can be placed across the

opening **1850**.

[0594] In some examples, orienting the inlet fan to face towards the side direction instead of the front direction (as in the examples shown in FIGS. **69A** and **71A**) can improve the safety and comfort of users operating the rackmount server **1820**. In some examples, orienting the inlet fan towards the side direction instead of the front direction can avoid the presence of a region in the heat dissipating module having little to no air flow. In the example of FIG. **71A**, the left and right inlet fans blow air toward the left and right side regions, respectively, of the heat dissipating device **1072**. The incoming air is drawn toward the rear of the heat dissipating module due to the air pressure gradient generated by the front and rear inlet fans. In some cases, the incoming air entering the left side of the heat dissipating device **1072** is drawn toward the rear of the heat dissipating device **1072** before reaching the middle part of the heat dissipating device **1072**. Similarly, the incoming air entering the right side of the heat dissipating device **1072** is drawn toward the rear of the heat dissipating device **1072** before reaching the middle part of the heat dissipating device **1072**. As a result, near the middle or middle-front region of the heat dissipating device **1072** there may be a region having little to no air flow, reducing the efficiency of heat dissipation. The design shown in FIGS. **96** to **98** avoids or reduces this problem.

[0595] The front panel **1826** includes openings or interface ports **1860** that allow the rackmount server **1820** to be coupled to optical fiber cables and/or electrical cables. In some implementations, co-packaged optical modules **1870** can be inserted into the interface ports **1860**, in which the co-packaged optical modules **1870** function as optical/electrical communication interfaces for the data processor **1844**. The co-packaged optical modules have been described earlier in this document.

[0596] FIG. **99** includes an upper diagram **1880** that shows a perspective front view of an example of the front panel **1826**, and a lower diagram **1882** that shows a perspective rear view of the front panel **1826**. The lower diagram **1882** shows the data processor **1844** mounted to the back side of the vertically oriented circuit board **1822**. The front panel **1826** includes openings or interface ports **1860** that allow insertion of communication interface modules, such as co-packaged optical modules, that provide interfaces between the data processor **1844** and external optical or electrical cables. The optical and electrical signal paths between the data processor **1844** and the co-packaged optical modules have been previously described in this document.

[0597] FIG. **100** is a diagram of a top view of an example of a rackmount server **1890** that includes a vertically oriented circuit board **1822** positioned at a front portion of the rackmount server **1890**. A data processor **1844** is mounted on the circuit board **1822**, and a heat dissipating module **1846** is thermally coupled to the data processor **1844**. The rackmount server **1890** has an active airflow management system that is configured to remove heat from the data processor **1844** during operation. The rackmount server **1890** includes components that are similar to those of the rackmount server **1820** (FIG. **96**) and are not otherwise described here.

[0598] In some implementations, the active airflow management system includes an inlet fan **1894** that is positioned at a left side of the heat dissipating module **1846** and oriented to blow inlet air to the right toward the heat dissipating module **1846**. A front opening **1850** allows incoming air to pass to the inlet fan **1894**. The front opening **1850** can be positioned to the left of the inlet fan **1894**. For example, the inlet fan **1894** can have a rotational axis that is at an angle θ relative to the front panel **1826**, in which $\theta \leq 45^\circ$. In some examples, $\theta \leq 25^\circ$. In some examples, $\theta \leq 5^\circ$. In some examples, the circuit board **1822** is substantially parallel to the front panel **1826**, and the rotational axis of the inlet fan **1894** is substantially parallel to the circuit board **1822**. An inlet fan **1894**,

[0599] In some implementations, a first baffle or air louver **1892** is provided to guide air from the opening **1850** towards the inlet fan **1894**, and from the inlet fan **1894** towards the heat dissipating module **1846**. A second baffle or air louver **1908** is provided to guide air from the right portion of the heat dissipating module **1846** toward the rear of the rackmount server **1890**. The first and second air louvers **1892**, **1894** can extend from the bottom panel to the top panel in the vertical direction.

[0600] An arrow **1902** shows a general direction of airflow from the opening **1850** to the inlet fan **1894**. An arrow **1904** shows a general direction of airflow from the inlet fan **1894** to, and through, a center portion the heat dissipating module **1846**. An arrow **1906** shows a general direction of airflow through, and exiting, the right portion of the heat dissipating module **1846**. The first air louver **1892**, a front portion of the left panel, a front portion of the top panel, a front portion of the bottom panel, the front panel **1826**, the circuit board **1822**, and the second air louver **1908** in combination form a duct that channels the air to flow through the entire heat dissipating module **1846**, or a substantial portion of the heat dissipating module **1846**, thereby increasing the efficiency of heat dissipation from the data processor **1844**.

[0601] In this example, the first air louver **1892** includes a left curved section **1896**, a middle straight section **1898**, and a right curved section **1900**. The left curved section **1896** extends from the left side panel to the inlet fan **1894**. The left curved section **1896** directs incoming air to turn from flowing in the rear direction to flowing in the left-to-right direction. The middle straight section **1898** is positioned to the rear of the heat dissipating module **1846** and extends from the inlet fan **1894** to beyond the center portion of the heat dissipating module **1846**. The middle straight section **1898** directs the air to flow generally in a left-to-right direction through a substantial portion (e.g., more than half) of the heat dissipating module **1846**. The right curved section **1900** and the second air louver **1908** in combination guide the air to turn from flowing in the left-to-right direction to flowing in a rear direction. The designs of the first and second air louvers **1892**, **1908** are selected to optimize the heat dissipation efficiency. The heat dissipating module **1846** can have a design that is different from what is shown in the figure, and the first and second air louvers **1892**, **1908** can also be modified accordingly.

[0602] In the example of FIG. **100**, the inlet fan **1894** directs air to flow generally in a direction (in this example, from left to right) that is parallel to the front panel **1826** across the heat dissipating surface of the heat dissipating module **1846**. In some implementations, the front opening can be positioned to the left side of the front panel, and the inlet fan can be positioned to the right side of the heat dissipating module and direct air to flow from right to left across the heat dissipating surface of the heat dissipating module. The first and second air louvers can be modified accordingly to optimize airflow and heat dissipation from the data processor.

[0603] FIGS. **35A** to **37** show examples of optical communications systems **1250**, **1260**, **1270** in which in each system an optical power supply or photon supply provides optical power supply light to photonic integrated circuits hosted in multiple communication devices (e.g., optical transponders), and the optical power supply is external to the communication devices. The optical power supply can have its own housing, electrical power supply, and control circuitry, independent of the housings, electrical power supplies, and control circuitry of the communication devices. This allows the optical power supply to be serviced, repaired, or replaced independent of the communication devices. Redundant optical power supplies can be provided so that a defective external optical power supply can be repaired or replaced without taking the communication devices off-line. The external optical power supply can be placed at a convenient centralized location with a dedicated temperature environment (as opposed to being crammed inside the communication devices, which may have a high temperature). The external optical power supply can be built more efficiently than individual power supply units, as certain common parts such as monitoring circuitry and thermal control units can be amortized over many more communication devices. The following describes implementations of the fiber cabling for remote optical power supplies.

[0604] FIG. **79** is a system functional block diagram of an example of an optical communication system **1280** that includes a first communication transponder **1282** and a second communication transponder **1284**. Each of the first and second communication transponders **1282**, **1284** can include one or more co-packaged optical modules described above. Each communication transponder can include, e.g., one or more data processors, such as network switches, central

processing units, graphics processor units, tensor processing units, digital signal processors, and/or other application specific integrated circuits (ASICs). In this example, the first communication transponder **1282** sends optical signals to, and receives optical signals from, the second communication transponder **1284** through a first optical communication link **1290**. The one or more data processors in each communication transponder **1282**, **1284** process the data received from the first optical communication link **1290** and outputs processed data to the first optical communication link **1290**. The optical communication system **1280** can be expanded to include additional communication transponders. The optical communication system **1280** can also be expanded to include additional communication between two or more external photon supplies, which can coordinate aspects of the supplied light, such as the respectively emitted wavelengths or the relative timing of the respectively emitted optical pulses.

[0605] A first external photon supply **1286** provides optical power supply light to the first communication transponder **1282** through a first optical power supply link **1292**, and a second external photon supply **1288** provides optical power supply light to the second communication transponder **1284** through a second optical power supply link **1294**. In one example embodiment, the first external photon supply **1286** and the second external photon supply **1288** provide continuous wave laser light at the same optical wavelength. In another example embodiment, the first external photon supply **1286** and the second external photon supply **1288** provide continuous wave laser light at different optical wavelengths. In yet another example embodiment, the first external photon supply **1286** provides a first sequence of optical frame templates to the first communication transponder **1282**, and the second external photon supply **1288** provides a second sequence of optical frame templates to the second communication transponder **1284**. For example, as described in U.S. patent Ser. No. 16/847,705, each of the optical frame templates can include a respective frame header and a respective frame body, and the frame body includes a respective optical pulse train. The first communication transponder **1282** receives the first sequence of optical frame templates from the first external photon supply **1286**, loads data into the respective frame bodies to convert the first sequence of optical frame templates into a first sequence of loaded optical frames that are transmitted through the first optical communication link **1290** to the second communication transponder **1284**. Similarly, the second communication transponder **1284** receives the second sequence of optical frame templates from the second external photon supply **1288**, loads data into the respective frame bodies to convert the second sequence of optical frame templates into a second sequence of loaded optical frames that are transmitted through the first optical communication link **1290** to the first communication transponder **1282**.

[0606] FIG. **80A** is a diagram of an example of an optical communication system **1300** that includes a first switch box **1302** and a second switch box **1304**. Each of the switch boxes **1302**, **1304** can include one or more data processors, such as network switches. The first and second switch boxes **1302**, **1304** can be separated by a distance greater than, e.g., 1 foot, 3 feet, 10 feet, 100 feet, or 1000 feet. The figure shows a diagram of a front panel **1306** of the first switch box **1302** and a front panel **1308** of the second switch box **1304**. In this example, the first switch box **1302** includes a vertical ASIC mount grid structure **1310**, similar to the grid structure **870** of FIG. **43**. A co-packaged optical module **1312** is attached to a receptor of the grid structure **1310**. The second switch box **1304** includes a vertical ASIC mount grid structure **1314**, similar to the grid structure **870** of FIG. **43**. A co-packaged optical module **1316** is attached to a receptor of the grid structure **1314**. The first co-packaged optical module **1312** communicates with the second co-packaged optical module **1316** through an optical fiber bundle **1318** that includes multiple optical fibers. Optional fiber connectors **1320** can be used along the optical fiber bundle **1318**, in which shorter sections of optical fiber bundles are connected by the fiber connectors **1320**.

[0607] In some implementations, each co-packaged optical module (e.g., **1312**, **1316**) includes a photonic integrated circuit configured to convert input optical signals to input electrical signals that are provided to a data processor, and convert output electrical signals from the data processor to

output optical signals. The co-packaged optical module can include an electronic integrated circuit configured to process the input electrical signals from the photonic integrated circuit before the input electrical signals are transmitted to the data processor, and to process the output electrical signals from the data processor before the output electrical signals are transmitted to the photonic integrated circuit. In some implementations, the electronic integrated circuit can include a plurality of serializers/deserializers configured to process the input electrical signals from the photonic integrated circuit, and to process the output electrical signals transmitted to the photonic integrated circuit. The electronic integrated circuit can include a first serializers/deserializers module having multiple serializer units and deserializer units, in which the first serializers/deserializers module is configured to generate a plurality of sets of first parallel electrical signals based on a plurality of first serial electrical signals provided by the photonic integrated circuit, and condition the electrical signals, in which each set of first parallel electrical signals is generated based on a corresponding first serial electrical signal. The electronic integrated circuit can include a second serializers/deserializers module having multiple serializer units and deserializer units, in which the second serializers/deserializers module is configured to generate a plurality of second serial electrical signals based on the plurality of sets of first parallel electrical signals, and each second serial electrical signal is generated based on a corresponding set of first parallel electrical signals. The plurality of second serial electrical signals can be transmitted toward the data processor.

[0608] The first switch box **1302** includes an external optical power supply **1322** (i.e., external to the co-packaged optical module) that provides optical power supply light through an optical connector array **1324**. In this example, the optical power supply **1322** is located internal of the housing of the switch box **1302**. Optical fibers **1326** are optically coupled to an optical connector **1328** (of the optical connector array **1324**) and the co-packaged optical module **1312**. The optical power supply **1322** sends optical power supply light through the optical connector **1328** and the optical fibers **1326** to the co-packaged optical module **1312**. For example, the co-packaged optical module **1312** includes a photonic integrated circuit that modulates the power supply light based on data provided by a data processor to generate a modulated optical signal, and transmits the modulated optical signal to the co-packaged optical module **1316** through one of the optical fibers in the fiber bundle **1318**.

[0609] In some examples, the optical power supply **1322** is configured to provide optical power supply light to the co-packaged optical module **1312** through multiple links that have built-in redundancy in case of malfunction in some of the optical power supply modules. For example, the co-packaged optical module **1312** can be designed to receive N channels of optical power supply light (e.g., N1 continuous wave light signals at the same or at different optical wavelengths, or N1 sequences of optical frame templates), N1 being a positive integer, from the optical power supply **1322**. The optical power supply **1322** provides N1+M1 channels of optical power supply light to the co-packaged optical module **1312**, in which M1 channels of optical power supply light are used for backup in case of failure of one or more of the N1 channels of optical power supply light, M1 being a positive integer.

[0610] The second switch box **1304** receives optical power supply light from a co-located optical power supply **1330**, which is, e.g., external to the second switch box **1304** and located near the second switch box **1304**, e.g., in the same rack as the second switch box **1304** in a data center. The optical power supply **1330** includes an array of optical connectors **1332**. Optical fibers **1334** are optically coupled to an optical connector **1336** (of the optical connectors **1332**) and the co-packaged optical module **1316**. The optical power supply **1330** sends optical power supply light through the optical connector **1336** and the optical fibers **1334** to the co-packaged optical module **1316**. For example, the co-packaged optical module **1316** includes a photonic integrated circuit that modulates the power supply light based on data provided by a data processor to generate a modulated optical signal, and transmits the modulated optical signal to the co-packaged optical module **1312** through one of the optical fibers in the fiber bundle **1318**.

[0611] In some examples, the optical power supply **1330** is configured to provide optical power supply light to the co-packaged optical module **1316** through multiple links that have built-in redundancy in case of malfunction in some of the optical power supply modules. For example, the co-packaged optical module **1316** can be designed to receive $N2$ channels of optical power supply light (e.g., $N2$ continuous wave light signals at the same or at different optical wavelengths, or $N2$ sequences of optical frame templates), $N2$ being a positive integer, from the optical power supply **1322**. The optical power supply **1322** provides $N2+M2$ channels of optical power supply light to the co-packaged optical module **1312**, in which $M2$ channels of optical power supply light are used for backup in case of failure of one or more of the $N2$ channels of optical power supply light, $M2$ being a positive integer.

[0612] FIG. **80B** is a diagram of an example of an optical cable assembly **1340** that can be used to enable the first co-packaged optical module **1312** to receive optical power supply light from the first optical power supply **1322**, enable the second co-packaged optical module **1316** to receive optical power supply light from the second optical power supply **1330**, and enable the first co-packaged optical module **1312** to communicate with the second co-packaged optical module **1316**. FIG. **80C** is an enlarged diagram of the optical cable assembly **1340** without some of the reference numbers to enhance clarity of illustration.

[0613] The optical cable assembly **1340** includes a first optical fiber connector **1342**, a second optical fiber connector **1344**, a third optical fiber connector **1346**, and a fourth optical fiber connector **1348**. The first optical fiber connector **1342** is designed and configured to be optically coupled to the first co-packaged optical module **1312**. For example, the first optical fiber connector **1342** can be configured to mate with a connector part of the first co-packaged optical module **1312**, or a connector part that is optically coupled to the first co-packaged optical module **1312**. The first, second, third, and fourth optical fiber connectors **1342**, **1344**, **1346**, **1348** can comply with an industry standard that defines the specifications for optical fiber interconnection cables that transmit data and control signals, and optical power supply light.

[0614] The first optical fiber connector **1342** includes optical power supply (PS) fiber ports, transmitter (TX) fiber ports, and receiver (RX) fiber ports. The optical power supply fiber ports provide optical power supply light to the co-packaged optical module **1312**. The transmitter fiber ports allow the co-packaged optical module **1312** to transmit output optical signals (e.g., data and/or control signals), and the receiver fiber ports allow the co-packaged optical module **1312** to receive input optical signals (e.g., data and/or control signals). Examples of the arrangement of the optical power supply fiber ports, the transmitter ports, and the receiver ports in the first optical fiber connector **1342** are shown in FIGS. **80D**, **89**, and **90**.

[0615] FIG. **80D** shows an enlarged upper portion of the diagram of FIG. **80B**, with the addition of an example of a mapping of fiber ports **1750** of the first optical fiber connector **1342** and a mapping of fiber ports **1752** of the third optical fiber connector **1346**. The mapping of fiber ports **1750** shows the positions of the transmitter fiber ports (e.g., **1753**), receiver fiber ports (e.g., **1755**), and power supply fiber ports (e.g., **1751**) of the first optical fiber connector **1342** when viewed in the direction **1754** into the first optical fiber connector **1342**. The mapping of fiber ports **1752** shows the positions of the power supply fiber ports (e.g., **1757**) of the third optical fiber connector **1346** when viewed in the direction **1756** into the third optical fiber connector **1346**.

[0616] The second optical fiber connector **1344** is designed and configured to be optically coupled to the second co-packaged optical module **1316**. The second optical fiber connector **1344** includes optical power supply fiber ports, transmitter fiber ports, and receiver fiber ports. The optical power supply fiber ports provide optical power supply light to the co-packaged optical module **1316**. The transmitter fiber ports allow the co-packaged optical module **1316** to transmit output optical signals, and the receiver fiber ports allow the co-packaged optical module **1316** to receive input optical signals. Examples of the arrangement of the optical power supply fiber ports, the transmitter ports, and the receiver ports in the second optical fiber connector **1344** are shown in FIGS. **80E**, **89**,

and **90**.

[0617] FIG. **80E** shows an enlarged lower portion of the diagram of FIG. **80B**, with the addition of an example of a mapping of fiber ports **1760** of the second optical fiber connector **1344** and a mapping of fiber ports **1762** of the fourth optical fiber connector **1348**. The mapping of fiber ports **1760** shows the positions of the transmitter fiber ports (e.g., **1763**), receiver fiber ports (e.g., **1765**), and power supply fiber ports (e.g., **1761**) of the second optical fiber connector **1344** when viewed in the direction **1764** into the second optical fiber connector **1344**. The mapping of fiber ports **1762** shows the positions of the power supply fiber ports (e.g., **1767**) of the fourth optical fiber connector **1348** when viewed in the direction **1766** into the fourth fiber connector **1348**.

[0618] The third optical connector **1346** is designed and configured to be optically coupled to the power supply **1322**. The third optical connector **1346** includes optical power supply fiber ports (e.g., **1757**) through which the power supply **1322** can output the optical power supply light. The fourth optical connector **1348** is designed and configured to be optically coupled to the power supply **1330**. The fourth optical connector **1348** includes optical power supply fiber ports (e.g., **1762**) through which the power supply **1322** can output the optical power supply light.

[0619] In some implementations, the optical power supply fiber ports, the transmitter fiber ports, and the receiver fiber ports in the first and second optical fiber connectors **1342**, **1344** are designed to be independent of the communication devices, i.e., the first optical fiber connector **1342** can be optically coupled to the second switch box **1304**, and the second optical fiber connector **1344** can be optically coupled to the first switch box **1302** without any re-mapping of the fiber ports. Similarly, the optical power supply fiber ports in the third and fourth optical fiber connectors **1346**, **1348** are designed to be independent of the optical power supplies, i.e., if the first optical fiber connector **1342** is optically coupled to the second switch box **1304**, the third optical fiber connector **1346** can be optically coupled to the second optical power supply **1330**. If the second optical fiber connector **1344** is optically coupled to the first switch box **1302**, the fourth optical fiber connector **1348** can be optically coupled to the first optical power supply **1322**.

[0620] The optical cable assembly **1340** includes a first optical fiber guide module **1350** and a second optical fiber guide module **1352**. The optical fiber guide module depending on context is also referred to as an optical fiber coupler or splitter because the optical fiber guide module combines multiple bundles of fibers into one bundle of fibers, or separates one bundle of fibers into multiple bundles of fibers. The first optical fiber guide module **1350** includes a first port **1354**, a second port **1356**, and a third port **1358**. The second optical fiber guide module **1352** includes a first port **1360**, a second port **1362**, and a third port **1364**. The fiber bundle **1318** extends from the first optical fiber connector **1342** to the second optical fiber connector **1344** through the first port **1354** and the second port **1356** of the first optical fiber guide module **1350** and the second port **1362** and the first port **1360** of the second optical fiber guide module **1352**. The optical fibers **1326** extend from the third optical fiber connector **1346** to the first optical fiber connector **1342** through the third port **1358** and the first port **1354** of the first optical fiber guide module **1350**. The optical fibers **1334** extend from the fourth optical fiber connector **1348** to the second optical fiber connector **1344** through the third port **1364** and the first port **1360** of the second optical fiber guide module **1352**.

[0621] A portion (or section) of the optical fibers **1318** and a portion of the optical fibers **1326** extend from the first port **1354** of the first optical fiber guide module **1350** to the first optical fiber connector **1342**. A portion of the optical fibers **1318** extend from the second port **1356** of the first optical fiber guide module **1350** to the second port **1362** of the second optical fiber guide module **1352**, with optional optical connectors (e.g., **1320**) along the paths of the optical fibers **1318**. A portion of the optical fibers **1326** extend from the third port **1358** of the first optical fiber connector **1350** to the third optical fiber connector **1346**. A portion of the optical fibers **1334** extend from the third port **1364** of the second optical fiber connector **1352** to the fourth optical fiber connector **1348**.

[0622] The first optical fiber guide module **1350** is designed to restrict bending of the optical fibers such that the bending radius of any optical fiber in the first optical fiber guide module **1350** is greater than the minimum bending radius specified by the optical fiber manufacturer to avoid excess optical light loss or damage to the optical fiber. For example, the minimum bend radii can be 2 cm, 1 cm, 5 mm, or 2.5 mm. Other bend radii are also possible. For example, the fibers **1318** and the fibers **1326** extend outward from the first port **1354** along a first direction, the fibers **1318** extend outward from the second port **1356** along a second direction, and the fibers **1326** extend outward from the third port **1358** along a third direction. A first angle is between the first and second directions, a second angle is between the first and third directions, and a third angle is between the second and third directions. The first optical fiber guide module **1350** can be designed to limit the bending of optical fibers so that each of the first, second, and third angles is in a range from, e.g., 30° to 180°.

[0623] For example, the portion of the optical fibers **1318** and the portion of the optical fibers **1326** between the first optical fiber connector **1342** and the first port **1354** of the first optical fiber guide module **1350** can be surrounded and protected by a first common sheath **1366**. The optical fibers **1318** between the second port **1356** of the first optical fiber guide module **1350** and the second port **1362** of the second optical fiber guide module **1352** can be surrounded and protected by a second common sheath **1368**. The portion of the optical fibers **1318** and the portion of the optical fibers **1334** between the second optical fiber connector **1344** and the first port **1360** of the second optical fiber guide module **1352** can be surrounded and protected by a third common sheath **1369**. The optical fibers **1326** between the third optical fiber connector **1346** and the third port **1358** of the first optical fiber guide module **1350** can be surrounded and protected by a fourth common sheath **1367**. The optical fibers **1334** between the fourth optical fiber connector **1348** and the third port **1364** of the second optical fiber guide module **1352** can be surrounded and protected by a fifth common sheath **1370**. Each of the common sheaths can be laterally flexible and/or laterally stretchable, as described in, e.g., U.S. patent application Ser. No. 16/822,103, now U.S. Pat. No. 11,194,109.

[0624] One or more optical cable assemblies **1340** (FIGS. **80B**, **80C**) and other optical cable assemblies (e.g., **1400** of FIG. **82B**, **82C**, **1490** of FIG. **84B**, **84C**) described in this document can be used to optically connect switch boxes that are configured differently compared to the switch boxes **1302**, **1304** shown in FIG. **80A**, in which the switch boxes receive optical power supply light from one or more external optical power supplies. For example, in some implementations, the optical cable assembly **1340** can be attached to a fiber-optic array connector mounted on the outside of the front panel of an optical switch, and another fiber-optic cable then connects the inside of the fiber connector to a co-packaged optical module that is mounted on a circuit board positioned inside the housing of the switch box. The co-packaged optical module (which includes, e.g., a photonic integrated circuit, optical-to-electrical converters, such as photodetectors, and electrical-to-optical converters, such as laser diodes) can be co-packaged with a switch ASIC and mounted on a circuit board that can be vertically or horizontally oriented. For example, in some implementations, the front panel is mounted on hinges and a vertical ASIC mount is recessed behind it. See the examples in FIGS. **77A**, **77B**, and **78**. The optical cable assembly **1340** provides optical paths for communication between the switch boxes, and optical paths for transmitting power supply light from one or more external optical power supplies to the switch boxes. The switch boxes can have any of a variety of configurations regarding how the power supply light and the data and/or control signals from the optical fiber connectors are transmitted to or received from the photonic integrated circuits, and how the signals are transmitted between the photonic integrated circuits and the data processors.

[0625] One or more optical cable assemblies **1340** and other optical cable assemblies (e.g., **1400** of FIG. **82B**, **82C**, **1490** of FIG. **84B**, **84C**) described in this document can be used to optically connect computing devices other than switch boxes. For example, the computing devices can be

server computers that provide a variety of services, such as cloud computing, database processing, audio/video hosting and streaming, electronic mail, data storage, web hosting, social network, supercomputing, scientific research computing, healthcare data processing, financial transaction processing, logistics management, weather forecast, or simulation, to list a few examples. The optical power light required by the optoelectronic modules of the computing devices can be provided using one or more external optical power supplies. For example, in some implementations, one or more external optical power supplies that are centrally managed can be configured to provide the optical power supply light for hundreds or thousands of server computers in a data center, and the one or more optical power supplies and the server computers can be optically connected using the optical cable assemblies (e.g., **1340**, **1400**, **1490**) described in this document and variations of the optical cable assemblies using the principles described in this document.

[0626] FIG. **81** is a system functional block diagram of an example of an optical communication system **1380** that includes a first communication transponder **1282** and a second communication transponder **1284**, similar to those in FIG. **79**. The first communication transponder **1282** sends optical signals to, and receives optical signals from, the second communication transponder **1284** through a first optical communication link **1290**. The optical communication system **1380** can be expanded to include additional communication transponders.

[0627] An external photon supply **1382** provides optical power supply light to the first communication transponder **1282** through a first optical power supply link **1384**, and provides optical power supply light to the second communication transponder **1284** through a second optical power supply link **1386**. In one example, the external photon supply **1382** provides continuous wave light to the first communication transponder **1282** and to the second communication transponder **1284**. In one example, the continuous wave light can be at the same optical wavelength. In another example, the continuous wave light can be at different optical wavelengths. In yet another example, the external photon supply **1382** provides a first sequence of optical frame templates to the first communication transponder **1282**, and provides a second sequence of optical frame templates to the second communication transponder **1284**. Each of the optical frame templates can include a respective frame header and a respective frame body, and the frame body includes a respective optical pulse train. The first communication transponder **1282** receives the first sequence of optical frame templates from the external photon supply **1382**, loads data into the respective frame bodies to convert the first sequence of optical frame templates into a first sequence of loaded optical frames that are transmitted through the first optical communication link **1290** to the second communication transponder **1284**. Similarly, the second communication transponder **1284** receives the second sequence of optical frame templates from the external photon supply **1382**, loads data into the respective frame bodies to convert the second sequence of optical frame templates into a second sequence of loaded optical frames that are transmitted through the first optical communication link **1290** to the first communication transponder **1282**.

[0628] FIG. **82A** is a diagram of an example of an optical communication system **1390** that includes a first switch box **1302** and a second switch box **1304**, similar to those in FIG. **80A**. The first switch box **1302** includes a vertical ASIC mount grid structure **1310**, and a co-packaged optical module **1312** is attached to a receptor of the grid structure **1310**. The second switch box **1304** includes a vertical ASIC mount grid structure **1314**, and a co-packaged optical module **1316** is attached to a receptor of the grid structure **1314**. The first co-packaged optical module **1312** communicates with the second co-packaged optical module **1316** through an optical fiber bundle **1318** that includes multiple optical fibers.

[0629] As discussed above in connection with FIGS. **80A** to **80E**, the first and second switch boxes **1302**, **1304** can have other configurations. For example, horizontally mounted ASICs can be used. A fiber-optic array connector attached to a front panel can be used to optically connect the optical cable assembly **1340** to another fiber-optic cable that connects to a co-packaged optical module

mounted on a circuit board inside the switch box. The front panel can be mounted on hinges and a vertical ASIC mount can be recessed behind it. The switch boxes can be replaced by other types of server computers.

[0630] In an example embodiment, the first switch box **1302** includes an external optical power supply **1322** that provides optical power supply light to both the co-packaged optical module **1312** in the first switch box **1302** and the co-packaged optical module **1316** in the second switch box **1304**. In another example embodiment, the optical power supply can be located outside the switch box **1302** (cf. **1330**, FIG. **80A**). The optical power supply **1322** provides the optical power supply light through an optical connector array **1324**. Optical fibers **1392** are optically coupled to an optical connector **1396** and the co-packaged optical module **1312**. The optical power supply **1322** sends optical power supply light through the optical connector **1396** and the optical fibers **1392** to the co-packaged optical module **1312** in the first switch box **1302**. Optical fibers **1394** are optically coupled to the optical connector **1396** and the co-packaged optical module **1316**. The optical power supply **1322** sends optical power supply light through the optical connector **1396** and the optical fibers **1394** to the co-packaged optical module **1316** in the second switch box **1304**.

[0631] FIG. **82B** shows an example of an optical cable assembly **1400** that can be used to enable the first co-packaged optical module **1312** to receive optical power supply light from the optical power supply **1322**, enable the second co-packaged optical module **1316** to receive optical power supply light from the optical power supply **1322**, and enable the first co-packaged optical module **1312** to communicate with the second co-packaged optical module **1316**. FIG. **82C** is an enlarged diagram of the optical cable assembly **1400** without some of the reference numbers to enhance clarity of illustration.

[0632] The optical cable assembly **1400** includes a first optical fiber connector **1402**, a second optical fiber connector **1404**, and a third optical fiber connector **1406**. The first optical fiber connector **1402** is similar to the first optical fiber connector **1342** of FIGS. **80B**, **80C**, **80D**, and is designed and configured to be optically coupled to the first co-packaged optical module **1312**. The second optical fiber connector **1404** is similar to the second optical fiber connector **1344** of FIGS. **80B**, **80C**, **80E**, and is designed and configured to be optically coupled to the second co-packaged optical module **1316**. The third optical connector **1406** is designed and configured to be optically coupled to the power supply **1322**. The third optical connector **1406** includes first optical power supply fiber ports (e.g., **1770**, FIG. **82D**) and second optical power supply fiber ports (e.g., **1772**). The power supply **1322** outputs optical power supply light through the first optical power supply fiber ports to the optical fibers **1392**, and outputs optical power supply light through the second optical power supply fiber ports to the optical fibers **1394**. The first, second, and third optical fiber connectors **1402**, **1404**, **1406** can comply with an industry standard that defines the specifications for optical fiber interconnection cables that transmit data and control signals, and optical power supply light.

[0633] FIG. **82D** shows an enlarged upper portion of the diagram of FIG. **82B**, with the addition of an example of a mapping of fiber ports **1774** of the first optical fiber connector **1402** and a mapping of fiber ports **1776** of the third optical fiber connector **1406**. The mapping of fiber ports **1774** shows the positions of the transmitter fiber ports (e.g., **1778**), receiver fiber ports (e.g., **1780**), and power supply fiber ports (e.g., **1782**) of the first optical fiber connector **1402** when viewed in the direction **1784** into the first optical fiber connector **1402**. The mapping of fiber ports **1776** shows the positions of the power supply fiber ports (e.g., **1770**, **1772**) of the third optical fiber connector **1406** when viewed in the direction **1786** into the third optical fiber connector **1406**. In this example, the third optical fiber connector **1406** includes 8 optical power supply fiber ports.

[0634] In some examples, optical connector array **1324** of the optical power supply **1322** can include a first type of optical connectors that accept optical fiber connectors having 4 optical power supply fiber ports, as in the example of FIG. **80D**, and a second type of optical connectors that accept optical fiber connectors having 8 optical power supply fiber ports, as in the example of FIG.

82D. In some examples, if the optical connector array **1324** of the optical power supply **1322** only accepts optical fiber connectors having 4 optical power supply fiber ports, then a converter cable can be used to convert the third optical fiber connector **1406** of FIG. **82D** to two optical fiber connectors, each having 4 optical power supply fiber ports, that is compatible with the optical connector array **1324**.

[0635] FIG. **82E** shows an enlarged lower portion of the diagram of FIG. **82B**, with the addition of an example of a mapping of fiber ports **1790** of the second optical fiber connector **1404**. The mapping of fiber ports **1790** shows the positions of the transmitter fiber ports (e.g., **1792**), receiver fiber ports (e.g., **1794**), and power supply fiber ports (e.g., **1796**) of the second optical fiber connector **1404** when viewed in the direction **1798** into the second optical fiber connector **1404**.

[0636] The port mappings of the optical fiber connectors shown in FIGS. **80D**, **80E**, **82D**, and **82E** are merely examples. Each optical fiber connector can include a greater number or a smaller number of transmitter fiber ports, a greater number or a smaller number of receiver fiber ports, and a greater number or a smaller number of optical power supply fiber ports, as compared to those shown in FIGS. **80D**, **80E**, **82D**, and **82E**. The arrangement of the relative positions of the transmitter, receiver, and optical power supply fiber ports can also be different from those shown in FIGS. **80D**, **80E**, **82D**, and **82E**.

[0637] The optical cable assembly **1400** includes an optical fiber guide module **1408**, which includes a first port **1410**, a second port **1412**, and a third port **1414**. The optical fiber guide module **1408** depending on context is also referred as an optical fiber coupler (for combining multiple bundles of optical fibers into one bundle of optical fiber) or an optical fiber splitter (for separating a bundle of optical fibers into multiple bundles of optical fibers). The fiber bundle **1318** extends from the first optical fiber connector **1402** to the second optical fiber connector **1404** through the first port **1410** and the second port **1412** of the optical fiber guide module **1408**. The optical fibers **1392** extend from the third optical fiber connector **1406** to the first optical fiber connector **1402** through the third port **1414** and the first port **1410** of the optical fiber guide module **1408**. The optical fibers **1394** extend from the third optical fiber connector **1406** to the second optical fiber connector **1404** through the third port **1414** and the second port **1412** of the optical fiber guide module **1408**.

[0638] A portion of the optical fibers **1318** and a portion of the optical fibers **1392** extend from the first port **1410** of the optical fiber guide module **1408** to the first optical fiber connector **1402**. A portion of the optical fibers **1318** and a portion of the optical fibers **1394** extend from the second port **1412** of the optical fiber guide module **1408** to the second optical fiber connector **1404**. A portion of the optical fibers **1394** extend from the third port **1414** of the optical fiber connector **1408** to the third optical fiber connector **1406**.

[0639] The optical fiber guide module **1408** is designed to restrict bending of the optical fibers such that the radius of curvature of any optical fiber in the optical fiber guide module **1408** is greater than the minimum radius of curvature specified by the optical fiber manufacturer to avoid excess optical light loss or damage to the optical fiber. For example, the optical fibers **1318** and the optical fibers **1392** extend outward from the first port **1410** along a first direction, the optical fibers **1318** and the optical fibers **1394** extend outward from the second port **1412** along a second direction, and the optical fibers **1392** and the optical fibers **1394** extend outward from the third port **1414** along a third direction. A first angle is between the first and second directions, a second angle is between the first and third directions, and a third angle is between the second and third directions. The optical fiber guide module **1408** is designed to limit the bending of optical fibers so that each of the first, second, and third angles is in a range from, e.g., 30° to 180°.

[0640] For example, the portion of the optical fibers **1318** and the portion of the optical fibers **1392** between the first optical fiber connector **1402** and the first port **1410** of the optical fiber guide module **1408** can be surrounded and protected by a first common sheath **1416**. The optical fibers **1318** and the optical fibers **1394** between the second optical fiber connector **1404** and the second port **1412** of the optical fiber guide module **1408** can be surrounded and protected by a second

common sheath **1418**. The optical fibers **1392** and the optical fibers **1394** between the third optical fiber connector **1406** and the third port **1414** of the optical fiber guide module **1408** can be surrounded and protected by a third common sheath **1420**. Each of the common sheaths can be laterally flexible and/or laterally stretchable.

[0641] FIG. **83** is a system functional block diagram of an example of an optical communication system **1430** that includes a first communication transponder **1432**, a second communication transponder **1434**, a third communication transponder **1436**, and a fourth communication transponder **1438**. Each of the communication transponders **1432**, **1434**, **1436**, **1438** can be similar to the communication transponders **1282**, **1284** of FIG. **79**. The first communication transponder **1432** communicates with the second communication transponder **1434** through a first optical link **1440**. The first communication transponder **1432** communicates with the third communication transponder **1436** through a second optical link **1442**. The first communication transponder **1432** communicates with the fourth communication transponder **1438** through a third optical link **1444**.

[0642] An external photon supply **1446** provides optical power supply light to the first communication transponder **1432** through a first optical power supply link **1448**, provides optical power supply light to the second communication transponder **1434** through a second optical power supply link **1450**, provides optical power supply light to the third communication transponder **1436** through a third optical power supply link **1452**, and provides optical power supply light to the fourth communication transponder **1438** through a fourth optical power supply link **1454**.

[0643] FIG. **84A** is a diagram of an example of an optical communication system **1460** that includes a first switch box **1462** and a remote server array **1470** that includes a second switch box **1464**, a third switch box **1466**, and a fourth switch box **1468**. The first switch box **1462** includes a vertical ASIC mount grid structure **1310**, and a co-packaged optical module **1312** is attached to a receptor of the grid structure **1310**. The second switch box **1464** includes a co-packaged optical module **1472**, the third switch box **1466** includes a co-packaged optical module **1474**, and the fourth switch box **1468** includes a co-packaged optical module **1476**. The first co-packaged optical module **1312** communicates with the co-packaged optical modules **1472**, **1474**, **1476** through an optical fiber bundle **1478** that later branches out to the co-packaged optical modules **1472**, **1474**, **1476**.

[0644] In one example embodiment, the first switch box **1462** includes an external optical power supply **1322** that provides optical power supply light through an optical connector array **1324**. In another example embodiment, the optical power supply can be located external to switch box **1462** (cf. **1330**, FIG. **80A**). Optical fibers **1480** are optically coupled to an optical connector **1482**, and the optical power supply **1322** sends optical power supply light through the optical connector **1482** and the optical fibers **1480** to the co-packaged optical modules **1312**, **1472**, **1474**, **1476**.

[0645] FIG. **84B** shows an example of an optical cable assembly **1490** that can be used to enable the optical power supply **1322** to provide optical power supply light to the co-packaged optical modules **1312**, **1472**, **1474**, **1476**, and enable the co-packaged optical module **1312** to communicate with the co-packaged optical modules **1472**, **1474**, **1476**. The optical cable assembly **1490** includes a first optical fiber connector **1492**, a second optical fiber connector **1494**, a third optical fiber connector **1496**, a fourth optical fiber connector **1498**, and a fifth optical fiber connector **1500**. The first optical fiber connector **1492** is configured to be optically coupled to the co-packaged optical module **1312**. The second optical fiber connector **1494** is configured to be optically coupled to the co-packaged optical module **1472**. The third optical fiber connector **1496** is configured to be optically coupled to the co-packaged optical module **1474**. The fourth optical fiber connector **1498** is configured to be optically coupled to the co-packaged optical module **1476**. The fifth optical fiber connector **1500** is configured to be optically coupled to the optical power supply **1322**. FIG. **84C** is an enlarged diagram of the optical cable assembly **1490**.

[0646] Optical fibers that are optically coupled to the optical fiber connectors **1500** and **1492** enable the optical power supply **1322** to provide the optical power supply light to the co-packaged

optical module **1312**. Optical fibers that are optically coupled to the optical fiber connectors **1500** and **1494** enable the optical power supply **1322** to provide the optical power supply light to the co-packaged optical module **1472**. Optical fibers that are optically coupled to the optical fiber connectors **1500** and **1496** enable the optical power supply **1322** to provide the optical power supply light to the co-packaged optical module **1474**. Optical fibers that are optically coupled to the optical fiber connectors **1500** and **1498** enable the optical power supply **1322** to provide the optical power supply light to the co-packaged optical module **1476**.

[0647] Optical fiber guide modules **1502**, **1504**, **1506**, and common sheaths are provided to organize the optical fibers so that they can be easily deployed and managed. The optical fiber guide module **1502** is similar to the optical fiber guide module **1408** of FIG. **82B**. The optical fiber guide modules **1504**, **1506** are similar to the optical fiber guide module **1350** of FIG. **80B**. The common sheaths gather the optical fibers in a bundle so that they can be more easily handled, and the optical fiber guide modules guide the optical fibers so that they extend in various directions toward the devices that need to be optically coupled by the optical cable assembly **1490**. The optical fiber guide modules restrict bending of the optical fibers such that the bending radiuses are greater than minimum values specified by the optical fiber manufacturers to prevent excess optical light loss or damage to the optical fibers.

[0648] The optical fibers **1480** that extend from the include optical fibers that extend from the optical **1482** are surrounded and protected by a common sheath **1508**. At the optical fiber guide module **1502**, the optical fibers **1480** separate into a first group of optical fibers **1510** and a second group of optical fibers **1512**. The first group of optical fibers **1510** extend to the first optical fiber connector **1492**. The second group of optical fibers **1512** extend toward the optical fiber guide modules **1504**, **1506**, which together function as a 1:3 splitter that separates the optical fibers **1512** into a third group of optical fibers **1514**, a fourth group of optical fibers **1516**, and a fifth group of optical fibers **1518**. The group of optical fibers **1514** extend to the optical fiber connector **1494**, the group of optical fibers **1516** extend to the optical fiber connector **1496**, and the group of optical fibers **1518** extend to the optical fiber connector **1498**. In some examples, instead of using two 1:2 split optical fiber guide modules **1504**, **1506**, it is also possible to use a 1:3 split optical fiber guide module that has four ports, e.g., one input port and three output ports. In general, separating the optical fibers in a 1:N split (N being an integer greater than 2) can occur in one step or multiple steps.

[0649] FIG. **85** is a diagram of an example of a data processing system (e.g., data center) **1520** that includes N servers **1522** spread across K racks **1524**. In this example, there are 6 racks **1524**, and each rack **1524** includes 15 servers **1522**. Each server **1522** directly communicates with a tier 1 switch **1526**. The left portion of the figure shows an enlarged view of a portion **1528** of the system **1520**. A server **1522a** directly communicates with a tier 1 switch **1526a** through a communication link **1530a**. Similarly, servers **1522b**, **1522c** directly communicate with the tier 1 switch **1526a** through communication links **1530b**, **1530c**, respectively. The server **1522a** directly communicates with a tier 1 switch **1526b** through a communication link **1532a**. Similarly, servers **1522b**, **1522c** directly communicate with the tier 1 switch **1526b** through communication links **1532b**, **1532c**, respectively. Each communication link can include a pair of optical fibers to allow bi-directional communication. The system **1520** bypasses the conventional top-of-rack switch and can have the advantage of higher data throughput. The system **1520** includes a point-to-point connection between every server **1522** and every tier 1 switch **1526**. In this example, there are 4 tier 1 switches **1526**, and 4 fiber pairs are used per server **1522** for communicating with the tier 1 switches **1526**. Each tier-1 switch **1526** is connected to N servers, so there are N fiber pairs connected to each tier-1 switch **1526**.

[0650] Referring to FIG. **86**, in some implementations, a data processing system (e.g., data center) **1540** includes tier-1 switches **1526** that are co-located in a rack **1540** separate from the N servers **1522** that are spread across K racks **1524**. Each server **1522** has a direct link to each of the tier-1

switches **1526**. In some implementations, there is one fiber cable **1542** (or a small number $\ll N/K$ of fiber cables) from the tier-1 switch rack **1540** to each of the K server racks **1524**.
[0651] FIG. **87A** is a diagram of an example of a data processing system **1550** that includes $N=1024$ servers **1552** spread across $K=32$ racks **1554**, in which each rack **1554** includes $N/K=1024/32=32$ servers **1552**. There are 4 tier-1 switches **1556** and an optical power supply **1558** that is co-located in a rack **1560**.

[0652] Optical fibers connect the servers **1552** to the tier-1 switches **1556** and the optical power supply **1558**. In this example, a bundle **1562** of 9 optical fibers is optically coupled to a co-packaged optical module **1564** of a server **1552**, in which 1 optical fiber provides the optical power supply light, and 4 pairs of (a total of 8) optical fibers provide 4 bi-directional communication channels, each channel having a 100 Gbps bandwidth, for a total of 4×100 Gbps bandwidth in each direction. Because there are 32 servers **1552** in each rack **1554**, there are a total of $256+32=288$ optical fibers that extend from each rack **1554** of servers **1552**, in which 32 optical fibers provide the optical power supply light, and 256 optical fibers provide 128 bi-directional communication channels, each channel having a 100 Gbps bandwidth.

[0653] For example, at the server rack side, optical fibers **1566** (that are connected to the servers **1552** of a rack **1554**) terminate at a server rack connector **1568**. At the switch rack side, optical fibers **1578** (that are connected to the switch boxes **1556** and the optical power supply **1558**) terminate at a switch rack connector **1576**. An optical fiber extension cable **1572** is optically coupled to the server rack side and the switch rack side. The optical fiber extension cable **1572** includes $256+32=288$ optical fibers. The optical fiber extension cable **1572** includes a first optical fiber connector **1570** and a second optical fiber connector **1574**. The first optical fiber connector **1570** is connected to the server rack connector **1568**, and the second optical fiber connector **1574** is connected to the switch rack connector **1576**. At the switch rack side, the optical fibers **1578** include 288 optical fibers, of which 32 optical fibers **1580** are optically coupled to the optical power supply **1558**. The 256 optical fibers that carry 128 bi-directional communication channels (each channel having a 100 Gbps bandwidth in each direction) are separated into four groups of 64 optical fibers, in which each group of 64 optical fibers is optically coupled to a co-packaged optical module **1582** in one of the switch boxes **1556**. The co-packaged optical module **1582** is configured to have a bandwidth of 32×100 Gbps = 3.2 Tbps in each direction (input and output). Each switch box **1556** is connected to each server **1552** of the rack **1554** through a pair of optical fibers that carry a bandwidth of 100 Gbps in each direction.

[0654] The optical power supply **1558** provides optical power supply light to co-packaged optical modules **1582** at the switch boxes **1556**. In this example, the optical power supply **1558** provides optical power supply light through 4 optical fibers to each co-packaged optical module **1582**, so that a total of 16 optical fibers are used to provide the optical power supply light to the 4 switch boxes **1556**. A bundle of optical fibers **1584** is optically coupled to the co-packaged optical module **1582** of the switch box **1556**. The bundle of optical fibers **1584** includes $64+16=80$ fibers. In some examples, the optical power supply **1558** can provide additional optical power supply light to the co-packaged optical module **1582** using additional optical fibers. For example, the optical power supply **1558** can provide optical power supply light to the co-packaged optical module **1582** using 32 optical fibers with built-in redundancy.

[0655] Referring to FIG. **87B**, the data processing system **1550** includes an optical fiber guide module **1590** that helps organize the optical fibers so that they are directed to the appropriate directions. The optical fiber guide module **1590** also restricts bending of the optical fibers to be within the specified limits to prevent excess optical light loss or damage to the optical fibers. The optical fiber guide module **1590** includes a first port **1592**, a second port **1594**, and a third port **1596**. The optical fibers that extend outward from the first port **1592** are optically coupled to the switch rack connector **1576**. The optical fibers that extend outward from the second port **1594** are optically coupled to the switch boxes. The optical fibers that extend outward from the third port

1596 are optically coupled to the optical power supply **1558**.

[0656] FIG. **88** is a diagram of an example of the connector port mapping for an optical fiber interconnection cable **1600**, which includes a first optical fiber connector **1602**, a second optical fiber connector **1604**, optical fibers **1606** that transmit data and/or control signals between the first and second optical fiber connectors **1602**, **1604**, and optical fibers **1608** that transmit optical power supply light. Each optical fiber terminates at an optical fiber port **1610**, which can include, e.g., lenses for focusing light entering or exiting the optical fiber port **1610**. The first and second optical fiber connectors **1602**, **1604** can be, e.g., the optical fiber connectors **1342** and **1344** of FIGS. **80B**, **80C**, the optical fiber connectors **1402** and **1404** of FIGS. **82B**, **82C**, or the optical fiber connectors **1570** and **1574** of FIG. **87A**. The principles for designing the optical fiber interconnection cable **1600** can be used to design the optical cable assembly **1340** of FIGS. **80B**, **80C**, the optical cable assembly **1400** of FIGS. **82B**, **82C**, and the optical cable assembly **1490** of FIGS. **84B**, **84C**.

[0657] In the example of FIG. **88**, each optical fiber connector **1602** or **1604** includes 3 rows of optical fiber ports, each row including 12 optical fiber ports. Each optical fiber connector **1602** or **1604** includes 4 power supply fiber ports that are connected to optical fibers **1608** that are optically coupled to one or more optical power supplies. Each optical fiber connector **1602** or **1604** includes 32 fiber ports (some of which are transmitter fiber ports, and some of which are receiver fiber ports) that are connected to the optical fibers **1606** for data transmission and reception.

[0658] In some implementations, the mapping of the fiber ports of the optical fiber connectors **1602**, **1604** are designed such that the interconnection cable **1600** can have the most universal use, in which each fiber port of the optical fiber connector **1602** is mapped to a corresponding fiber port of the optical fiber connector **1604** with a 1-to-1 mapping and without transponder-specific port mapping that would require fibers **1606** to cross over. This means that for an optical transponder that has an optical fiber connector compatible with the interconnection cable **1600**, the optical transponder can be connected to either the optical fiber connector **1602** or the optical fiber connector **1604**. The mapping of the fiber ports is designed such that each transmitter port of the optical fiber connector **1602** is mapped to a corresponding receiver port of the optical fiber connector **1604**, and each receiver port of the optical fiber connector **1602** is mapped to a corresponding transmitter port of the optical fiber connector **1604**.

[0659] FIG. **89** is a diagram showing an example of the fiber port mapping for an optical fiber interconnection cable **1660** that includes a pair of optical fiber connectors, i.e., a first optical fiber connector **1662** and a second optical fiber connector **1664**. The optical fiber connectors **1662** and **1664** are designed such that either the first optical fiber connector **1662** or the second optical fiber connector **1664** can be connected to a given communication transponder that is compatible with the optical fiber interconnection cable **1660**. The diagram shows the fiber port mapping when viewed from the outer edge of the optical fiber connector into the optical fiber connector (i.e., toward the optical fibers in the interconnection cable **1660**).

[0660] The first optical fiber connector **1662** includes transmitter fiber ports (e.g., **1614a**, **1616a**), receiver fiber ports (e.g., **1618a**, **1620a**), and optical power supply fiber ports (e.g., **1622a**, **1624a**). The second optical fiber connector **1664** includes transmitter fiber ports (e.g., **1614b**, **1616b**), receiver fiber ports (e.g., **1618b**, **1620b**), and optical power supply fiber ports (e.g., **1622b**, **1624b**). For example, assume that the first optical fiber connector **1662** is connected to a first optical transponder, and the second optical fiber connector **1664** is connected to a second optical transponder. The first optical transponder transmits first data and/or control signals through the transmitter ports (e.g., **1614a**, **1616a**) of the first optical fiber connector **1662**, and the second optical transponder receives the first data and/or control signals from the corresponding receiver fiber ports (e.g., **1618b**, **1620b**) of the second optical fiber connector **1664**. The transmitter ports **1614a**, **1616a** are optically coupled to the corresponding receiver fiber ports **1618b**, **1620b** through optical fibers **1628**, **1630**, respectively. The second optical transponder transmits second data and/or control signals through the transmitter ports (e.g., **1614b**, **1616b**) of the second optical fiber

connector **1664**, and the first optical transponder receives the second data and/or control signals from the corresponding receiver fiber ports (**1618a**, **1620a**) of the first optical fiber connector **1662**. The transmitter port **1616b** is optically coupled to the corresponding receiver fiber port **1620a** through an optical fiber **1632**.

[0661] A first optical power supply transmits optical power supply light to the first optical transponder through the power supply fiber ports of the first optical fiber connector **1662**. A second optical power supply transmits optical power supply light to the second optical transponder through the power supply fiber ports of the second optical fiber connector **1664**. The first and second power supplies can be different (such as the example of FIG. **80B**) or the same (such as the example of FIG. **82B**).

[0662] In the following description, when referring to the rows and columns of fiber ports of the optical fiber connector, the uppermost row is referred to as the 1.sup.st row, the second uppermost row is referred to as the 2.sup.nd row, and so forth. The leftmost column is referred to as the 1.sup.st column, the second leftmost column is referred to as the 2.sup.nd column, and so forth.

[0663] For an optical fiber interconnection cable having a pair of optical fiber connectors (i.e., a first optical fiber connector and a second optical fiber connector) to be universal, i.e., either one of the pair of optical fiber connectors can be connected to a given optical transponder, the arrangement of the transmitter fiber ports, the receiver fiber ports, and the power supply fiber ports in the optical fiber connectors have a number of properties. These properties are referred to as the “universal optical fiber interconnection cable port mapping properties.” The term “mapping” here refers to the arrangement of the transmitter fiber ports, the receiver fiber ports, and the power supply fiber ports at particular locations within the optical fiber connector. The first property is that the mapping of the transmitter, receiver, and power supply fiber ports in the first optical fiber connector is the same as the mapping of the transmitter, receiver, and power supply fiber ports in the second optical fiber connector (as in the example of FIG. **89**).

[0664] In the example of FIG. **89**, the individual optical fibers connecting the transmitter, receiver, and power supply fiber ports in the first optical fiber connector to the transmitter, receiver, and power supply fiber ports in the second optical fiber connector are parallel to one another.

[0665] In some implementations, each of the optical fiber connectors includes a unique marker or mechanical structure, e.g., a pin, that is configured to be at the same spot on the co-packaged optical module, similar to the use of a “dot” to denote “pin 1” on electronic modules. In some examples, such as those shown in FIGS. **89** and **90**, the larger distance from the bottom row (the third row in the examples of FIGS. **89** and **90**) to the connector edge can be used as a “marker” to guide the user to attach the optical fiber connector to the co-packaged optical module connector in a consistent manner.

[0666] The mapping of the fiber ports of the optical fiber connectors of a “universal optical fiber interconnection cable” has a second property: When mirroring the port map of an optical fiber connector and replacing each transmitter port with a receiver port as well as replacing each receiver port with a transmitter port in the mirror image, the original port mapping is recovered. The mirror image can be generated with respect to a reflection axis at either connector edge, and the reflection axis can be parallel to the row direction or the column direction. The power supply fiber ports of the first optical fiber connector are mirror images of the power supply fiber ports of the second optical fiber connector.

[0667] The transmitter fiber ports of the first optical fiber connector and the receiver fiber ports of the second optical fiber connector are pairwise mirror images of each other, i.e., each transmitter fiber port of the first optical fiber connector is mirrored to a receiver fiber port of the second optical fiber connector. The receiver fiber ports of the first optical fiber connector and the transmitter fiber ports of the second optical fiber connector are pairwise mirror images of each other, i.e., each receiver fiber port of the first optical fiber connector is mirrored to a transmitter fiber port of the second optical fiber connector.

[0668] Another way of looking at the second property is as follows: Each optical fiber connector is transmitter port-receiver port (TX-RX) pairwise symmetric and power supply port (PS) symmetric with respect to one of the main or center axes, which can be parallel to the row direction or the column direction. For example, if an optical fiber connector has an even number of columns, the optical fiber connector can be divided along a center axis parallel to the column direction into a left half portion and a right half portion. The power supply fiber ports are symmetric with respect to the main axis, i.e., if there is a power supply fiber port in the left half portion of the optical fiber connector, there will also be a power supply fiber port at the mirror location in the right half portion of the optical fiber connector. The transmitter fiber ports and the receiver fiber ports are pairwise symmetric with respect to the main axis, i.e., if there is a transmitter fiber port in the left half portion of the optical fiber connector, there will be a receiver fiber port at a mirror location in the right half portion of the optical fiber connector. Likewise, if there is a receiver fiber port in the left half portion of the optical fiber connector, there will be a transmitter fiber port at a mirror location in the right half portion of the optical fiber connector.

[0669] For example, if an optical fiber connector has an even number of rows, the optical fiber connector can be divided along a center axis parallel to the row direction into an upper half portion and a lower half portion. The power supply fiber ports are symmetric with respect to the main axis, i.e., if there is a power supply fiber port in the upper half portion of the optical fiber connector, there will also be a power supply fiber port at the mirror location in the lower half portion of the optical fiber connector. The transmitter fiber ports and the receiver fiber ports are pairwise symmetric with respect to the main axis, i.e., if there is a transmitter fiber port in the upper half portion of the optical fiber connector, there will be a receiver fiber port at a mirror location in the lower half portion of the optical fiber connector. Likewise, if there is a receiver fiber port in the upper half portion of the optical fiber connector, there will be a transmitter fiber port at a mirror location in the lower half portion of the optical fiber connector.

[0670] The mapping of the transmitter fiber ports, receiver fiber ports, and power supply fiber ports follow a symmetry requirement that can be summarized as follows: [0671] (i) Mirror all ports on either one of the two connector edges. [0672] (ii) Swap TX (transmitter) and RX (receiver) functionality on the mirror image. [0673] (iii) Leave mirrored PS (power supply) ports as PS ports. [0674] (iv) The resulting port map is the same as the original one.

[0675] Essentially, a viable port map is TX-RX pairwise symmetric and PS symmetric with respect to one of the main axes.

[0676] The properties of the mapping of the fiber ports of the optical fiber connectors can be mathematically expressed as follows:

[0677] Port matrix M with entries $PS=0$, $TX=+1$, $RX=-1$;

[0678] Column-mirror operation ;

[0679] Row-mirror operation  M ;

[0680] A viable port map either satisfies $-\text{custom-character}=M$ or $-\text{custom-character}M=M$.

[0681] In some implementations, if a universal optical fiber interconnection cable has a first optical fiber connector and a second optical fiber connector that are mirror images of each other after swapping the transmitter fiber ports to receiver fiber ports and swapping the receiver fiber ports to transmitter fiber ports in the mirror image, and the mirror image is generated with respect to a reflection axis parallel to the column direction, as in the example of FIG. 89, then each optical fiber connector should be TX-RX pairwise symmetric and PS symmetric with respect to a center axis parallel to the column direction. If a universal optical fiber interconnection cable has a first optical fiber connector and a second optical fiber connector that are mirror images of each other after swapping the transmitter and receiver fiber ports in the mirror image, and the mirror image is generated with respect to a reflection axis parallel to the row direction, as in the example of FIG. 90, then each optical fiber connector should be TX-RX pairwise symmetric and PS symmetric with respect to a center axis parallel to the row direction.

[0682] In some implementations, a universal optical fiber interconnection cable: [0683] a. Comprises n_{trx} strands of TX/RX fibers and n_{p} strands of power supply fibers, in which $0 \leq n_{\text{p}} \leq n_{\text{trx}}$. [0684] b. The n_{trx} strands of TX/RX fibers are mapped 1:1 from a first optical fiber connector to the same port positions on a second optical fiber connector through the optical fiber cable, i.e., the optical fiber cable can be laid out in a straight manner without leading to any cross-over fiber strands. [0685] c. Those connector ports that are not 1:1 connected by TX/RX fibers may be connected to power supply fibers via a break-out cable.

[0686] In some implementations, a universal optical module connector has the following properties: [0687] a. Starting from a connector port map PM0. [0688] b. First mirror port map PM0 either across the row dimension or across the column dimension. [0689] c. Mirroring can be done either across a column axis or across a row axis. [0690] d. Replace TX ports by RX ports and vice versa. [0691] e. If at least one mirrored and replaced version of the port map again results in the starting port map PM0, the connector is called a universal optical module connector.

[0692] In FIG. 89, the arrangement of the transmitter, receiver, and power supply fiber ports in the first optical fiber connector **1662**, and the arrangement of the transmitter, receiver, and power supply fiber ports in the second optical fiber connector **1664** have the two properties described above. First property: When looking into the optical fiber connector (from the outer edge of the connector inward toward the optical fibers), the mapping of the transmitter, receiver, and power supply fiber ports in the first optical fiber connector **1662** is the same as the mapping of the transmitter, receiver, and power supply fiber ports in the optical fiber connector **1664**. Row 1, column 1 of the optical fiber connector **1662** is a power supply fiber port (**1622a**), and row 1, column 1 of the optical fiber connector **1664** is also a power supply fiber port (**1622b**). Row 1, column 3 of the optical fiber connector **1662** is a transmitter fiber port (**1614a**), and row 1, column 3 of the optical fiber connector **1664** is also a transmitter fiber port (**1614b**). Row 1, column 10 of the optical fiber connector **1662** is a receiver fiber port (**1618a**), and row 1, column 10 of the optical fiber connector **1664** is also a receiver fiber port (**1618b**), and so forth.

[0693] The optical fiber connectors **1662** and **1664** have the second universal optical fiber interconnection cable port mapping property described above. The port mapping of the optical fiber connector **1662** is a mirror image of the port mapping of the optical fiber connector **1664** after swapping each transmitter port to a receiver port and swapping each receiver port to a transmitter port in the mirror image. The mirror image is generated with respect to a reflection axis **1626** at the connector edge that is parallel to the column direction. The power supply fiber ports (e.g., **1662a**, **1624a**) of the optical fiber connector **1662** are mirror images of the power supply fiber ports (e.g., **1622b**, **1624b**) of the optical fiber connector **1664**. The transmitter fiber ports (e.g., **1614a**, **1616a**) of the optical fiber connector **1662** and the receiver fiber ports (e.g., **1618b**, **1620b**) of the optical fiber connector **1664** are pairwise mirror images of each other, i.e., each transmitter fiber port (e.g., **1614a**, **1616a**) of the optical fiber connector **1662** is mirrored to a receiver fiber port (e.g., **1618b**, **1620b**) of the optical fiber connector **1664**. The receiver fiber ports (e.g., **1618a**, **1620a**) of the optical fiber connector **1662** and the transmitter fiber ports (e.g., **1618b**, **1620b**) of the optical fiber connector **1664** are pairwise mirror images of each other, i.e., each receiver fiber port (e.g., **1618a**, **1620a**) of the optical fiber connector **1662** is mirrored to a transmitter fiber port (e.g., **1618b**, **1620b**) of the optical fiber connector **1664**.

[0694] For example, the power supply fiber port **1622a** at row 1, column 1 of the optical fiber connector **1662** is a mirror image of the power supply fiber port **1624b** at row 1, column 12 of the optical fiber connector **1664** with respect to the reflection axis **1626**. The power supply fiber port **1624a** at row 1, column 12 of the optical fiber connector **1662** is a mirror image of the power supply fiber port **1622b** at row 1, column 1 of the optical fiber connector **1664**. The transmitter fiber port **1614a** at row 1, column 3 of the optical fiber connector **1662** and the receiver fiber port **1618b** at row 1, column 10 of the optical fiber connector **1604** are pairwise mirror images of each other. The receiver fiber port **1618a** at row 1, column 10 of the optical fiber connector **1662** and the

transmitter fiber port **1614b** at row 1, column 3 of the optical fiber connector **1664** are pairwise mirror images of each other. The transmitter fiber port **1616a** at row 3, column 3 of the optical fiber connector **1662** and the receiver fiber port **1620b** at row 3, column 10 of the optical fiber connector **1664** are pairwise mirror images of each other. The receiver fiber port **1620a** at row 3, column 10 of the optical fiber connector **1662** and the transmitter fiber port **1616b** at row 3, column 3 of the optical fiber connector **1664** are pairwise mirror images of each other.

[0695] In addition, and as an alternate view of the second property, each optical fiber connector **1662**, **1664** is TX-RX pairwise symmetric and PS symmetric with respect to the center axis that is parallel to the column direction. Using the first optical fiber connector **1662** as an example, the power supply fiber ports (e.g., **1622a**, **1624a**) are symmetric with respect to the center axis, i.e., if there is a power supply fiber port in the left half portion of the first optical fiber connector **1662**, there will also be a power supply fiber port at the mirror location in the right half portion of the first optical fiber connector **1662**. The transmitter fiber ports and the receiver fiber ports are pairwise symmetric with respect to the main axis, i.e., if there is a transmitter fiber port in the left half portion of the first optical fiber connector **1662**, there will be a receiver fiber port at a mirror location in the right half portion of the first optical fiber connector **1662**. Likewise, if there is a receiver fiber port in the left half portion of the optical fiber connector **1662**, there will be a transmitter fiber port at a mirror location in the right half portion of the optical fiber connector **1662**.

[0696] If the port mapping of the first optical fiber connector **1662** is represented by port matrix M with entries $PS=0$, $TX=+1$, $RX=-1$, then $- \text{custom-character}=M$, in which custom-character represents the column-mirror operation, e.g., generating a mirror image with respect to the reflection axis **1626**.

[0697] FIG. **90** is a diagram showing another example of the fiber port mapping for an optical fiber interconnection cable **1670** that includes a pair of optical fiber connectors, i.e., a first optical fiber connector **1672** and a second optical fiber connector **1674**. In the diagram, the port mapping for the second optical fiber connector **1674** is the same as that of optical fiber connector **1672**. The optical fiber interconnection cable **1670** has the two universal optical fiber interconnection cable port mapping properties described above.

[0698] First property: The mapping of the transmitter, receiver, and power supply fiber ports in the first optical fiber connector **1672** is the same as the mapping of the transmitter, receiver, and power supply fiber ports in the second optical fiber connector **1674**.

[0699] Second property: The port mapping of the first optical fiber connector **1672** is a mirror image of the port mapping of the second optical fiber connector **1674** after swapping each transmitter port to a receiver port and swapping each receiver port to a transmitter port in the mirror image. The mirror image is generated with respect to a reflection axis **1640** at the connector edge parallel to the row direction.

[0700] Alternative view of the second property: Each of the first and second optical fiber connectors **1672**, **1674** is TX-RX pairwise symmetric and PS symmetric with respect to the central axis that is parallel to the row direction. For example, the optical fiber connector **1672** can be divided in two halves along a central axis parallel to the row direction. The power supply fiber ports (e.g., **1678**, **1680**) are symmetric with respect to the center axis. The transmitter fiber ports (e.g., **1682**, **1684**) and the receiver fiber ports (e.g., **1686**, **1688**) are pairwise symmetric with respect to the center axis, i.e., if there is a transmitter fiber port (e.g., **1682** or **1684**) in the upper half portion of the first optical fiber connector **1672**, then there will be a receiver fiber port (e.g., **1686**, **1688**) at a mirror location in the lower half of the optical fiber connector **1672**. Likewise, if there is a receiver fiber port in the upper half portion of the optical fiber connector **1672**, then there is a transmitter fiber port at a mirror location in the lower half portion of the optical fiber connector **1672**. In the example of FIG. **90**, the middle row **1690** should all be power supply fiber ports.

[0701] In general, if the port mapping of the first optical fiber connector is a mirror image of the

port mapping of the second optical fiber connector after swapping the transmitter and receiver ports in the mirror image, the mirror image is generated with respect to a reflection axis at the connector edge parallel to the row direction (as in the example of FIG. 90), and there is an odd number of rows in the port matrix, then the center row should all be power supply fiber ports. If the port mapping of the first optical fiber connector is a mirror image of the port mapping of the second optical fiber connector after swapping the transmitter and receiver ports in the mirror image, the mirror image is generated with respect to a reflection axis at the connector edge parallel to the column direction, and there is an odd number of columns in the port matrix, then the center column should all be power supply fiber ports.

[0702] FIG. 91 is a diagram of an example of a viable port mapping for an optical fiber connector 1700 of a universal optical fiber interconnection cable. The optical fiber connector 1700 includes power supply fiber ports (e.g., 1702), transmitter fiber ports (e.g., 1704), and receiver fiber ports (e.g., 1706). The optical fiber connector 1700 is TX-RX pairwise symmetric and PS symmetric with respect to the center axis that is parallel to the column direction.

[0703] FIG. 92 is a diagram of an example of a viable port mapping for an optical fiber connector 1710 of a universal optical fiber interconnection cable. The optical fiber connector 1710 includes power supply fiber ports (e.g., 1712), transmitter fiber ports (e.g., 1714), and receiver fiber ports (e.g., 1716). The optical fiber connector 1710 is TX-RX pairwise symmetric and PS symmetric with respect to the center axis that is parallel to the column direction.

[0704] FIG. 93 is a diagram of an example of a port mapping for an optical fiber connector 1720 that is not appropriate for a universal optical fiber interconnection cable. The optical fiber connector 1720 includes power supply fiber ports (e.g., 1722), transmitter fiber ports (e.g., 1724), and receiver fiber ports (e.g., 1726). The optical fiber connector 1720 is not TX-RX pairwise symmetric with respect to the center axis that is parallel to the column direction, or the center axis that is parallel to the row direction.

[0705] FIG. 94 is a diagram of an example of a viable port mapping for a universal optical fiber interconnection cable that includes a pair of optical fiber connectors, i.e., a first optical fiber connector 1800 and a second optical fiber connector 1802. The mapping of the transmitter, receiver, and power supply fiber ports in the first optical fiber connector 1800 is the same as the mapping of the transmitter, receiver, and power supply fiber ports in the second optical fiber connector 1802. The port mapping of the first optical fiber connector 1800 is a mirror image of the port mapping of the second optical fiber connector 1802 after swapping the transmitter and receiver ports in the mirror image. The mirror image is generated with respect to a reflection axis 1804 at the connector edge parallel to the column direction. The optical fiber connector 1800 is TX-RX pairwise symmetric and PS symmetric with respect to the center axis 1806 that is parallel to the column direction.

[0706] FIG. 95 is a diagram of an example of a viable port mapping for a universal optical fiber interconnection cable that includes a pair of optical fiber connectors, i.e., a first optical fiber connector 1810 and a second optical fiber connector 1812. The mapping of the transmitter, receiver, and power supply fiber ports in the first optical fiber connector 1810 is the same as the mapping of the transmitter, receiver, and power supply fiber ports in the second optical fiber connector 1812. The port mapping of the first optical fiber connector 1810 is a mirror image of the port mapping of the second optical fiber connector 1812 after swapping the transmitter and receiver ports in the mirror image. The mirror image is generated with respect to a reflection axis 1814 at the connector edge parallel to the column direction. The optical fiber connector 1810 is TX-RX pairwise symmetric and PS symmetric with respect to the center axis 1816 that is parallel to the column direction.

[0707] In the example of FIG. 95, the first, third, and fifth rows each has an even number of fiber ports, and the second and fourth rows each has an odd number of fiber ports. In general, a viable port mapping for a universal optical fiber interconnection cable can be designed such that an

optical fiber connector includes (i) rows that all have even numbers of fiber ports, (ii) rows that all have odd numbers of fiber ports, or (iii) rows that have mixed even and odd numbers of fiber ports. A viable port mapping for a universal optical fiber interconnection cable can be designed such that an optical fiber connector includes (i) columns that all have even numbers of fiber ports, (ii) columns that all have odd numbers of fiber ports, or (iii) columns that have mixed even and odd numbers of fiber ports.

[0708] The optical fiber connector of a universal optical fiber interconnection cable does not have to be a rectangular shape as shown in the examples of FIGS. **89, 90, 92 to 95**. The optical fiber connectors can also have an overall triangular, square, pentagonal, hexagonal, trapezoidal, circular, oval, or n-sided polygon shape, in which n is an integer larger than 6, as long as the arrangement of the transmitter, receiver, and power supply fiber ports in the optical fiber connectors have the three universal optical fiber interconnection cable port mapping properties described above.

[0709] In the examples of FIGS. **80A, 82A, 84A, and 87A**, the switch boxes (e.g., **1302, 1304**) includes co-packaged optical modules (e.g., **1312, 1316**) that is optically coupled to the optical fiber interconnection cables or optical cable assemblies (e.g., **1340, 1400, 1490**) through fiber array connectors. For example, the fiber array connector can correspond to the first optical connector part **213** in FIG. **20**. The optical fiber connector (e.g., **1342, 1344, 1402, 1404, 1492, 1498**) of the optical cable assembly can correspond to the second optical connector part **223** in FIG. **20**. The port map (i.e., mapping of power supply fiber ports, transmitter fiber ports, and receiver fiber ports) of the fiber array connector (which is optically coupled to the photonic integrated circuit) is a mirror image of the port map of the optical fiber connector (which is optically coupled to the optical fiber interconnection cable). The port map of the fiber array connector refers to the arrangement of the power supply, transmitter, and receiver fiber ports when viewed from an external edge of the fiber array connector into the fiber array connector.

[0710] As described above, universal optical fiber connectors have symmetrical properties, e.g., each optical fiber connector is TX-RX pairwise symmetric and PS symmetric with respect to one of the main or center axes, which can be parallel to the row direction or the column direction. The fiber array connector also has the same symmetrical properties, e.g., each fiber array connector is TX-RX pairwise symmetric and PS symmetric with respect to one of the main or center axes, which can be parallel to the row direction or the column direction.

[0711] In some implementations, a restriction can be imposed on the port mapping of the optical fiber connectors of the optical cable assembly such that the optical fiber connector can be pluggable when rotated by 180 degrees, or by 90 degrees in the case of a square connector. This results in further port mapping constraints.

[0712] FIG. **101** is a diagram of an example of an optical fiber connector **1910** having a port map **1912** that is invariant against a 180-degree rotation. Rotating the optical fiber connector **1910** 180 degrees results in a port map **1914** that is the same as the port map **1912**. The port map **1912** also satisfies the second universal optical fiber interconnection cable port mapping property, e.g., the optical fiber connector is TX-RX pairwise symmetric and PS symmetric with respect to the center axis parallel to the column direction.

[0713] FIG. **102** is a diagram of an example of an optical fiber connector **1920** having a port map **1922** that is invariant against a 90-degree rotation. Rotating the optical fiber connector **1920** 180 degrees results in a port map **1924** that is the same as the port map **1922**. The port map **1922** also satisfies the second universal optical fiber interconnection cable port mapping property, e.g., the optical fiber connector is TX-RX pairwise symmetric and PS symmetric with respect to the center axis parallel to the column direction.

[0714] FIG. **103A** is a diagram of an example of an optical fiber connector **1930** having a port map **1932** that is TX-RX pairwise symmetric and PS symmetric with respect to the center axis parallel to the column direction. When mirroring the port map **1932** to generate a mirror image **1934** and replacing each transmitter port with a receiver port as well as replacing each receiver port with a

transmitter port in the mirror image **1934**, the original port map **1932** is recovered. The mirror image **1934** is generated with respect to a reflection axis at the connector edge parallel to the column direction.

[0715] Referring to FIG. **103B**, the port map **1932** of the optical fiber connector **1930** is also TX-RX pairwise symmetric and PS symmetric with respect to the center axis parallel to the row direction. When mirroring the port map **1932** to generate a mirror image **1936** and replacing each transmitter port with a receiver port as well as replacing each receiver port with a transmitter port in the mirror image **1936**, the original port map **1932** is recovered. The mirror image **1936** is generated with respect to a reflection axis at the connector edge parallel to the row direction.

[0716] In the examples of FIGS. **69A** to **78**, **96** to **98**, and **100**, one or more fans (e.g., **1086**, **1092**, **1848**, **1894**) blow air across the heatsink (e.g., **1072**, **1114**, **1130**, **1168**, **1846**) thermally coupled to the data processor (e.g., **1844**). The co-packaged optical modules can generate heat, in which some of the heat can be directed toward the heatsink and dissipated through the heatsink. To further improve heat dissipation from the co-packaged optical modules, in some implementations, the rackmount system includes two fans placed side-by-side, in which a first fan blows air toward the co-packaged optical modules that are mounted on a front side of the printed circuit board (e.g., **1068**), and a second fan blows air toward the heatsink that is thermally coupled to the data processor mounted on a rear side of the printed circuit board.

[0717] In some implementations, the one or more fans can have a height that is smaller than the height of the housing (e.g., **1824**) of the rackmount server (e.g., **1820**). The co-packaged optical modules (e.g., **1074**) can occupy a region on the printed circuit board (e.g., **1068**) that extends in the height direction greater than the height of the one or more fans. One or more baffles can be provided to guide the cool air from the one or more fans or intake air duct to the heatsink and the co-packaged optical modules. One or more baffles can be provided to guide the warm air from the heatsink and the co-packaged optical modules to an air duct that directs the air toward the rear of the housing.

[0718] When the one or more fans have a height that is smaller than the height of the housing (e.g., **1824**), the space above and/or below the one or more fans can be used to place one or more remote laser sources. The remote laser sources can be positioned near the front panel and also near the co-packaged optical modules. This allows the remote laser sources to be serviced conveniently.

[0719] FIG. **104** shows a top view of an example of a rackmount device **1940**. The rackmount device **1940** includes a vertically oriented printed circuit board **1230** positioned at a distance behind a front panel **1224** that can be closed during normal operation of the device, and opened for maintenance of the device, similar to the configuration of the rackmount server **1220** of FIG. **77A**. A data processing chip **1070** is electrically coupled to the rear side of the vertical printed circuit board **1230**, and a heat dissipating device or heat sink **1072** is thermally coupled to the data processing chip **1070**. Co-packaged optical modules **1074** are attached to the front side (i.e., the side facing the front exterior of the housing **1222**) of the vertical printed circuit board **1230**. A first fan **1942** is provided to blow air across the co-packaged optical modules **1074** at the front side of the printed circuit board **1230**. A second fan **1944** is provided to blow air across the heatsink **1072** to the rear of the printed circuit board **1230**. The first and second fans **1942**, **1944** are positioned at the left of the printed circuit board **1230**. Cooler air (represented by arrows **1946**) is directed from the first and second fans **1942**, **1944** toward the heatsink **1072** and the co-packaged optical modules **1074**. Warmer air (represented by arrows **1948**) is directed from the heatsink **1072** and the co-packaged optical modules **1074** through an air duct **1950** positioned at the right of the printed circuit board **1230** toward the rear of the housing.

[0720] FIG. **105** shows a front view of the rackmount device **1940** when the front panel **1224** is opened to allow access to the co-packaged optical modules **1074**. The first and second fans **1942**, **1944** have a height that is smaller than the height of the region occupied by the co-packaged optical modules **1074**. A first baffle **1952** directs the air from the fan **1942** to the region where the co-

packaged optical modules **1074** are mounted, and a second baffle **1954** directs the air from the region where the co-packaged optical modules **1074** are mounted to the air duct **1950**.

[0721] In this example, the first and second fans **1942**, **1944** have a height that is smaller than the height of the housing of the rackmount device **1940**. Remote laser sources **1956** can be positioned above and below the fans. Remote laser sources **1956** can also be positioned above and below the air duct **1950**.

[0722] For example, a switch device having a 51.2 Tbps bandwidth can use thirty-two 1.6 Tbps co-packaged optical modules. Two to four power supply fibers (e.g., **1326** in FIG. **80A**) can be provided for each co-packaged optical module, and a total of 64 to 128 power supply fibers can be used to provide optical power to the 32 co-packaged optical modules. One or two laser modules at 500 mW each can be used to provide the optical power to each co-packaged optical module, and 32 to 64 laser modules can be used to provide the optical power to the 32 co-packaged optical modules. The 32 to 64 laser modules can be fitted in the space above and below the fans **1942**, **1944** and the air duct **1950**.

[0723] For example, the area **1958a** above the fans **1942**, **1944** can have an area (measured along a plane parallel to the front panel) of about 16 cm×5 cm and can fit about 28 QSFP cages, and the area **1958b** below the fans can have an area of about 16 cm×5 cm and can fit about 28 QSFP cages. The area **1958c** above the air duct **1950** can have an area of about 8 cm×5 cm and can fit about 12 QSFP cages, and the area **1958d** below the air duct **1950** can have an area of about 8 cm×5 cm and can fit about 12 QSFP cages. Each QSFP cage can include a laser module. In this example, a total of 80 QSFP cages can be fit above and below the fans and the air duct, allowing 80 laser modules to be positioned near the front panel and near the co-packaged optical modules, making it convenient to service the laser modules in the event of malfunction or failure.

[0724] Referring to FIGS. **106** and **107**, an optical cable assembly **1960** includes a first fiber connector **1962**, a second fiber connector **1964**, and a third fiber connector **1966**. The first fiber connector **1962** can be optically connected to the co-packaged optical module **1074**, the second fiber connector **1964** can be optically connected to the laser module, and the third fiber connector **1966** can be optically connected to the fiber connector part (e.g., **1232** of FIG. **77A**) at the front panel **1224**. The first fiber connector **1962** can have a configuration similar to that of the fiber connector **1342** of FIGS. **80C**, **80D**. The second fiber connector **1964** can have a configuration similar to that of the fiber connector **1346**. The third fiber connector **1964** can have a configuration similar to that of the first fiber connector **1962** but without the power supply fiber ports. The optical fibers **1968** between the first fiber connector **1962** and the third fiber connector **1966** perform the function of the fiber jumper **1234** of FIG. **77A**.

[0725] FIG. **108** is a diagram of an example of a rackmount device **1970** that is similar to the rackmount device **1940** of FIGS. **104**, **105**, **107**, except that the optical axes of the laser modules **1956** are oriented at an angle θ relative to the front-to-rear direction, $0<\theta<90^\circ$. This can reduce the bending of the optical fibers that are optically connected to the laser modules **1956**.

[0726] FIG. **109** is a diagram showing the front view of the rackmount device **1970**, with the optical cable assembly **1960** optically connected to modules of the rackmount device **1970**. When the laser modules **1956** are oriented at an angle θ relative to the front-to-rear direction, $0<\theta<90^\circ$, fewer laser modules **1956** can be placed in the spaces above and below the fans **1942**, **1944** and the air duct **1950**, as compared to the example of FIGS. **104**, **105**, **107**, in which the optical axes of the laser modules **1956** are oriented parallel to the front-to-rear direction. In the example of FIG. **109**, a total of 64 laser modules are placed in the spaces above and below the fans **1942**, **1944** and the air duct **1950**.

[0727] FIG. **110** is a top view diagram of an example of a rackmount device **1980** that is similar to the rackmount device **1940** of FIGS. **104**, **105**, **107**, except that the optical axes of the laser modules **1956** are oriented parallel to the front panel **1224**. This can reduce the bending of the optical fibers that are optically connected to the laser modules **1956**.

[0728] FIG. **111** is a front view diagram of the rackmount device **1980**, with the optical cable assembly **1960** optically connected to modules of the rackmount device **1980**. The laser modules **1956a** are positioned to the left side of the space above and below the fans **1942**, **1944**. Sufficient space (e.g., **1982**) is provided at the right of the laser modules **1956a** to allow the user to conveniently connect or disconnect the fiber connectors **1964** to the laser modules **1956a**. The laser modules **1956b** are positioned above and below the air duct **1950**. Sufficient space (e.g., **1984**) is provided at the left of the laser modules **1956b** to allow the user to conveniently connect or disconnect the fiber connectors **1964** to the laser modules **1956b**.

[0729] Referring to FIG. **112**, a table **1990** shows example parameter values of the rackmount device **1940**.

[0730] FIGS. **113** and **114** show another example of a rackmount device **2000** and example parameter values.

[0731] FIGS. **115** and **116** are a top view and a front view, respectively, of the rackmount device **2000**. An upper baffle **2002** and a lower baffle **2004** are provided to guide the air flowing from the fans **1942**, **1944** to the heatsink **1072** and the co-packaged optical modules **1074**, and from the heatsink **1072** and the co-packaged optical modules **1074** to the air duct **1950**. In this example, portions of the upper and lower baffles **2002**, **2004** form portions of the upper and lower walls of the air duct **1950**.

[0732] The upper baffle **2002** includes a cutout or opening **2007** that allows optical fibers **2008** to pass through. As shown in FIG. **116**, the optical fibers **2008** extend from the co-packaged optical modules **1074a** upward, through the cutout or opening **2007** in the upper baffle **2002**, and extend toward the laser modules **1956** along the space above the upper baffle **2002**. The upper baffle **2002** allows the optical fibers **2008** to be better organized to reduce the obstruction to the air flow caused by the optical fibers **2008**. The lower baffle **2004** has a similar cutout or opening to help organize the optical fibers that are optically connected to the laser modules located in the space below the fans **1942**, **1944**.

[0733] FIG. **117** is a top view diagram of a system **2010** that includes a front panel **2012**, which can be rotatably coupled to the lower panel by a hinge. The front panel **2012** includes an air inlet grid **2014** and an array of fiber connector parts **2016**. Each fiber connector part **2016** can be optically coupled to the third fiber connector **1966** of the cable assembly **1960** of FIG. **106**. In some implementations, the hinged front panel includes a mechanism that shuts off the remote laser source modules **1956**, or reduces the power to the remote laser source modules **1956**, once the flap is opened. This prevents the technicians from being exposed to harmful radiation.

[0734] FIG. **118** is a diagram of an example of a system **2120** that includes a recirculating reservoir that circulates a coolant to carry heat away from the data processor, which for example can be a switch integrated circuit. In this example, the data is immersed in the coolant, and the inlet fan is used to blow air across the surface of the co-packaged optical modules to a heat dissipating device thermally coupled to the co-packaged optical modules.

[0735] FIGS. **119** to **122** are examples that provide heat dissipating solutions for co-packaged optical modules, taking into consideration the locations of “hot aisles” in data centers. In case it is desirable that fiber cabling be done on the back side of a rack (where hot air is blown out, hence “hot aisle”), one can either use a duct inside the box to transfer cold air to the co-packaged optical modules that are now mounted on the back side (FIG. **121**) or one can use fiber jumper cables to connect the co-packaged optical modules that are still facing the front aisle (towards the cold aisle) to connect to a “back-panel” facing the hot aisle (FIG. **122**).

[0736] Referring to FIG. **123**, in some implementations, a vertically mounted processor blade **12300** can include a substrate **12302** having a first side **12304** and a second side **12306**. The substrate **12302** can be, e.g., a printed circuit board. An electronic processor **12308** is mounted on the first side **12304** of the substrate **12302**, in which the electronic processor **12308** is configured to process or store data. For example, the electronic processor **12308** can be a network switch, a

central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, or an application specific integrated circuit (ASIC). For example, the electronic processor **12308** can be a memory device or a storage device. In this context, processing of data includes writing data to, or reading data from, the memory or storage device, and optionally performing error correction. The memory device can be, e.g., random access memory (RAM), which can include, e.g., dynamic RAM (DRAM) or static RAM (SRAM). The storage device can include, e.g., solid state memory or drive, which can include, e.g., one or more non-volatile memory (NVM) Express® (NVMe) SSD (solid state drive) modules, or Intel® Optane™ persistent memory. The example of FIG. **123** shows one electronic processor **12308**, through there can also be multiple electronic processors **12308** mounted on the substrate **12302**.

[0737] The vertically mounted processor blade **12300** includes one or more optical interconnect modules or co-packaged optical modules **12310** mounted on the second side **12306** of the substrate **12302**. For example, the optical interconnect module **12310** includes an optical port configured to receive optical signals from an external optical fiber cable, and a photonic integrated circuit configured to generate electrical signals based on the received optical signals, and transmit the electrical signals to the electronic processor **12308**. The photonic integrated circuit can also be configured to generate optical signals based on electrical signals received from the electronic processor **12308**, and transmit the optical signals to the external optical fiber cable. The optical interconnect module or co-packaged optical module **12310** can be similar to, e.g., the integrated optical communication device **262** of FIG. **6**; **282** of FIGS. **7-9**; **462, 466, 448, 472** of FIG. **17**; **612** of FIG. **23**; **684** of FIG. **26**; **704** of FIG. **27**; **724** of FIG. **28**; the co-packaged optical module **1074** of FIGS. **68A, 69A, 70, 71A; 1132** of FIG. **73A**; **1160** of FIG. **74A**; **1074** of FIGS. **75A, 75B, 77A, 77B, 104, 107, 109, 116; 1312** of FIGS. **80A, 82A, 84A**; or **1564, 1582** of FIG. **87A**. In the example of FIG. **123**, the optical interconnect module or co-packaged optical module **12310** does not necessarily have to include serializers/deserializers (SerDes), e.g., **216, 217** of FIGS. **2 to 8 and 10 to 12**. The optical interconnect module or co-packaged optical module **12310** can include the photonic integrated circuit **12314** without any serializers/deserializers. For example, the serializers/deserializers can be mounted on the substrate separate from the optical interconnect module or co-packaged optical module **12310**.

[0738] For example, the substrate **12302** can include electrical connectors that extend from the first side **12304** to the second side **12306** of the substrate **12302**, in which the electrical connectors pass through the substrate **12302** in a thickness direction. For example, the electrical connectors can include vias of the substrate **12302**. The optical interconnect module **12310** is electrically coupled to the electronic processor **12308** by the electrical connectors.

[0739] For example, the vertically mounted processor blade **12300** can include an optional optical fiber connector **12312** for connection to an optical fiber cable bundle. The optical fiber connector **12312** can be optically coupled to the optical interconnector modules **12310** through optical fiber cables **12314**. The optical fiber cables **12314** can be connected to the optical interconnect modules **12310** through a fixed connector (in which the optical fiber cable **12314** is securely fixed to the optical interconnect module **12310**) or a removable connector in which the optical fiber cable **12314** can be easily detached from the optical interconnect module **12310**, such as with the use of an optical connector part **266** as shown in FIG. **6**. The removable connector can include a structure similar to the mechanical connector structure **900** of FIGS. **46, 47 and 51A to 57**.

[0740] For example, the substrate **12302** can be positioned near the front panel of the housing of the server that includes the vertically mounted processor blade **12300**, or away from the front panel and located anywhere inside the housing. For example, the substrate **12302** can be parallel to the front panel of the housing, perpendicular to the front panel, or oriented in any angle relative to the front panel. For example, the substrate **12302** can be oriented vertically to facilitate the flow of hot air and improve dissipation of heat generated by the electronic processor **12308** and/or the optical

interconnect modules **12310**.

[0741] For example, the optical interconnect module or co-packaged optical module **12310** can receive optical signals through vertical or edge coupling. FIG. **123** shows an example in which the optical fiber cables are vertically coupled to the optical interconnect modules or co-packaged optical modules **12310**. It is also possible to connect the optical fiber cables to the edges of the optical interconnect modules or co-packaged optical modules **12310**. For example, optical fibers in the optical fiber cable can be attached in-plane to the photonic integrated circuit using, e.g., V-groove fiber attachments, tapered or un-tapered fiber edge coupling, etc., followed by a mechanism to direct the light interfacing to the photonic integrated circuit to a direction that is substantially perpendicular to the photonic integrated circuit, such as one or more substantially 90-degree turning mirrors, one or more substantially 90-degree bent optical fibers, etc.

[0742] For example, the optical interconnect modules **12310** can receive optical power from an optical power supply, such as **1322** of FIG. **80A**, **1558** of FIG. **87A**. For example, the optical interconnect modules **12310** can include one or more of optical coupling interfaces **414**, demultiplexers **419**, splitters **415**, multiplexers **418**, receivers **421**, or modulators **417** of FIG. **20**.

[0743] FIG. **124** is a top view of an example of a rack system **12400** that includes several vertically mounted processor blades **12300**. The vertically mounted processor blades **12300** can be positioned such that the optical fiber connectors **12312** are near the front of the rack system **12400** (which allows external optical fiber cables to be optically coupled to the front of the rack system **12400**), or near the back of the rack system **12400** (which allows external optical fiber cables to be optically coupled to the back of the rack system **12400**). Several rack systems **12400** can be stacked vertically similar to the example shown in FIG. **76**, in which the server rack **1214** includes several servers **1212** stacked vertically, or the example shown in FIG. **87A**, in which several servers **1552** are stacked vertically in a rack **1554**. For example, the optical interconnect modules **12310** can receive optical power from an optical power supply, such as **1558** of FIG. **87A**.

[0744] In some implementations, the vertically mounted processor blades **12300** can include blade pairs, in which each blade pair includes a switch blade and a processor blade. The electronic processor of the switch blade includes a switch, and the electronic processor of the processor blade is configured to process data provided by the switch. For example, the electronic processor of the processor blade is configured to send processed data to the switch, which switches the processed data with other data, e.g., data from other processor blades.

[0745] In the examples shown in FIGS. **123** and **124**, the optical interconnect module or co-packaged optical module **12310** is mounted on the second side of the substrate **12302**. In some implementations, the optical interconnect module **12310** or the optical fiber cable **12314** extends through or partially through an opening in the substrate **12302**, similar to the example shown in FIGS. **35A** to **35C**. The photonic integrated circuit in the optical interconnect module **12310** is electrically coupled to the electronic processor **12308** or to another electronic circuit, such as a serializers/deserializers module positioned at or near the first side of the substrate **12302**. The optical interconnect module **12310** and the optical fiber cable **12314** define a signal path that allows a signal from the optical fiber cable **12314** to be transmitted from the second side of the substrate **12302** through the opening to the electronic processor **12308**. The signal is converted from an optical signal to an electric signal by the photonic integrated circuit, which defines part of the signal path. This allows the optical fiber cables to be positioned on the second side of the substrate **12302**.

[0746] In the example of FIG. **104**, the printed circuit board **1230** is positioned a short distance from the front panel **1224** to improve air flow between the printed circuit board **1230** and the front panel **1224** to help dissipate heat generated by the co-packaged optical modules **1074**. The following describes a mechanism that allows the user to conveniently connect the co-packaged optical module to an optical fiber cable using a pluggable module that has a rigid structure that spans the distance between the co-packaged optical modules and the front panel.

[0747] Referring to FIG. 125A, in some implementations, a rackmount server **12300** can have a hinge-mounted front panel, similar to the example shown in FIG. 77A. The rackmount server **12300** includes a housing **12302** having a top panel **12304**, a bottom panel **12306**, and a front panel **12308** that is coupled to the bottom panel **12306** using a hinge **12324**. A vertically mounted substrate **12310** is positioned substantially perpendicular to the bottom panel **12306** and recessed from the front panel **12308**. The substrate **12310** includes a first side facing the front direction relative to the housing **12302** and a second side facing the rear direction relative to the housing **12302**. At least one electronic processor or data processing chip **12312** is electrically coupled to the second side of the vertical substrate **12310**, and a heat dissipating device or heat sink **12314** is thermally coupled to the at least one data processing chip **12312**. Co-packaged optical modules **12316** (or optical interconnect modules) are attached to the first side of the vertical substrate **12310**. The substrate **12310** provides high-speed connections between the co-packaged optical modules **12316** and the data processing chip **12312**. The co-packaged optical module **12316** is optically connected to a first fiber connector part **12318**, which is optically connected through a fiber pigtail **12320** to one or more second fiber connector parts **12322** mounted on the front panel **12308**.

[0748] In the example of FIG. 125A, the front panel **12308** is rotatably connected to the bottom panel by the hinge **12324**. In other examples, the front panel can be rotatably connected to the top panel or the side panel so as to flap upwards or to flap sideways when opened.

[0749] For example, the electronic processor **12312** can be a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, or an application specific integrated circuit (ASIC). For example, the electronic processor **12312** can be a memory device or a storage device. In this context, processing of data includes writing data to, or reading data from, the memory or storage device, and optionally performing error correction. The memory device can be, e.g., random access memory (RAM), which can include, e.g., dynamic RAM (DRAM) or static RAM (SRAM). The storage device can include, e.g., solid state memory or drive, which can include, e.g., one or more non-volatile memory (NVM) Express® (NVMe) SSD (solid state drive) modules, or Intel® Optane™ persistent memory. The example of FIG. 125A shows one electronic processor **12312**, through there can also be multiple electronic processors **12312** mounted on the substrate **12310**. In some examples, the substrate **12310** can also be replaced by a circuit board.

[0750] The co-packaged optical module (or optical interconnect module) **12316** can be similar to, e.g., the integrated optical communication device **262** of FIG. 6; **282** of FIGS. 7-9; **462**, **466**, **448**, **472** of FIG. 17; **612** of FIG. 23; **684** of FIG. 26; **704** of FIG. 27; **724** of FIG. 28; the co-packaged optical module **1074** of FIGS. 68A, 69A, 70, 71A; **1132** of FIG. 73A; **1160** of FIG. 74A; **1074** of FIGS. 75A, 75B, 77A, 77B, **104**, **107**, **109**, **116**; **1312** of FIGS. 80A, 82A, 84A; or **1564**, **1582** of FIG. 87A. In the example of FIG. 125A, the optical interconnect module or co-packaged optical module **12316** does not necessarily have to include serializers/deserializers (SerDes), e.g., **216**, **217** of FIGS. 2 to 8 and 10 to 12. The optical interconnect module or co-packaged optical module **12316** can include the photonic integrated circuit without any serializers/deserializers. For example, the serializers/deserializers can be mounted on the circuit board separate from the optical interconnect module or co-packaged optical module **12316**.

[0751] FIG. 159 is a side view of an example of a rackmount server **15900** that has a hinge-mounted front panel. The rackmount server **15900** includes a housing **15902** having a top panel **15904**, a bottom panel **15906**, and an upper swivel front panel **15908** that is coupled to a lower fixed front panel **15930** using a hinge **15910**. In some examples, the hinge can be attached to the side panel so that the front panel is opened horizontally. A horizontally mounted host printed circuit board **15912** is attached to the bottom panel **15906**. A vertically mounted printed circuit board **15914**, which can be, e.g., a daughter-card, is positioned substantially vertically and perpendicular to the bottom panel **15906** and recessed from the front panel **15908**. A package substrate **15916** is attached to the front side of the vertical printed circuit board **15914**. At least one electronic

processor or data processing chip **15918** is electrically coupled to the rear side of the package substrate **15916**, and a heat dissipating device or heat sink **15920** is thermally coupled to the at least one data processing chip **15918**. Co-packaged optical modules **15922** (or optical interconnect modules) are removably attached to the front side of the package substrate **15916**. The package substrate **15916** provides high-speed connections between the co-packaged optical modules **15922** and the data processing chip **15918**. The co-packaged optical module **15922** is optically connected to a first fiber connector part **15924**, which is optically connected through a fiber pigtail **15926** to one or more second fiber connector parts **15928** attached to the back side of the front panel **15908**. The second fiber connector parts **15928** can be optically connected to optical fiber cables that pass through openings in the hinged front panel **15908**.

[0752] For example, the fiber connector **15928** can be connected to the backside of the front panel **15908** during replacement of the CPO module **15922**. The CPO module **15922** can be unplugged from the connector (e.g., an LGA socket) on the package substrate **15916**, and be disconnected from the first fiber connector part **15924**.

[0753] For example, one or more rows of pluggable external laser sources (ELS) **15932** can be in standard pluggable form factor accessible from the lower fixed part **15930** of the front panel with rear blind-mate connectors. Optical fibers **15934** transmit the power supply light from the laser sources **15932** to the CPO modules **15922**. The external laser sources **15932** are electrically connected to a conventionally (horizontal) oriented system printed circuit board or the vertically oriented daughterboard. In this example, the row(s) of pluggable external laser sources **15932** is/are positioned below the datapath optical connection. The pluggable external laser sources **15932** do not need to connect to the CPO substrate because there are no high-speed signals that require proximity.

[0754] In some implementations, as shown in FIG. **160**, external laser sources can be located behind the hinged front panel (not user accessible without opening the door) and can then be front-mating similar to typical optical pluggables. FIG. **160** is a top view of an example of a rackmount server **16000** that is similar to the rackmount server **15900** of FIG. **159** except that one or more rows of external laser sources **16002** are placed inside the housing **15902**. Optical fibers **15934** transmit the power supply light from the laser sources **16002** to the CPO modules **15922**.

[0755] FIG. **161** is a diagram of an example of the optical cable **15926** that optically couples the CPO modules **15922** to the optical fiber cables at the front panel **15908**. The optical cable **15926** includes a first multi-fiber push on (MPO) connector **16100**, a laser supply MPO connector **16102**, four datapath MPO connectors **16104**, and a jumper cable **16106** that includes optical fibers that optically connect the MPO connectors. In this example, the optical cable **15926** supports a total bandwidth of 1.6 Tb/s, including 16 full-duplex 400G DR4+ signals (100G per fiber) plus 4 ELS connections.

[0756] The first MPO connector **16100** is optically coupled to the CPO module **15922** and includes, e.g., 36 fiber ports (e.g., 3 rows of fiber ports, each row having 12 fiber ports, similar to the fiber ports shown in FIGS. **80D**, **80E**, **82D**, **82E**, **89** to **93**), which includes 4 power supply fiber ports and 32 data fiber ports. The laser supply MPO connector **16102** is optically coupled to the external laser source, such as **15932** (FIG. **159**) or **16002** (FIG. **160**). The datapath MPO connectors **16104** are optically coupled to external optical fiber cables. For example, each external optical fiber cable can support a 400GBASE-DR4 link, so the four datapath MPO connectors **16104** can support 16 full-duplex 400G DR4+ signals (100G per fiber). The jumper cable **16106** fans the MPO connector **16100** out to datapath MPOs **16104** on the front panel **15908** (e.g., 4×400G DR4+ using 4×1×12 MPOs or 2×800G DR8+ using 2×2×12 MPOs) and the laser supply MPO **16102**. For example, the optical cable **15926** can be DR-16+ (e.g., 1.6 Tb/s at 100G per fiber, gray optics, -2 km reach). This architecture also supports FR-n (WDM).

[0757] In this example, the CPO module **15922** is configured to support 4×400 Gb/s=1.6 Tb/s data rate. The jumper cable **16106** includes four (4) power supply optical fibers **15934** that optically

connect four (4) power supply fiber ports of the laser supply MPO connector **16102** to the corresponding power supply fiber ports of the first MPO connector **16100**. The jumper cable **16106** includes four (4) sets of eight (8) data optical fibers. The eight (8) data optical fibers **16106** optically connect eight (8) transmit or receive fiber ports of each datapath MPO connector **16104** to the corresponding transmit or receive fiber ports of the first MPO connector **16100**. For example, the power supply optical fibers **15934** can be polarization maintaining optical fibers. The fan-out cable **16106** can handle multiple functions including merging the external laser source and data paths, splitting of external light source between multiple CPO modules **15922**, and handling polarization. Regarding the force requirement on the CPO module's connector, the optical connector leverages an MPO type connection and can have a similar or smaller force as compared to a standard MPO connector.

[0758] Referring to FIG. **125B**, in some implementations, a rackmount server **12400** has a front panel **12402** (which can be, e.g., fixed) and a vertically mounted substrate **12310** recessed from the front panel **12402**. The front panel **12402** has openings that allow pluggable modules **12404** to be inserted. Each pluggable module **12404** includes a co-packaged optical module **12316**, one or more multi-fiber push on (MPO) connectors **12406**, a fiber guide **12408** that mechanically connects the co-packaged optical module **12316** to the one or more multi-fiber push on connectors **12406**, and a fiber pigtail **12410** that optically connects the co-packaged optical module **12316** to the one or more multi-fiber push on connectors **12406**. For example, the length of the fiber guide **12408** is designed such that when the pluggable module **12404** is inserted into the opening of the front panel **12402** and the co-packaged optical module **12316** is electrically coupled to the vertically mounted substrate **12310**, the one or more multi-fiber push on connectors **12406** are near the front panel, e.g., flush with, or slightly protrude from, the front panel **12402** so that the user can conveniently attach external fiber optic cables. For example, the front face of the connectors **12406** can be within an inch, or half an inch, or one-fourth of an inch, of the front surface of the front panel **12402**.

[0759] For example, the housing **12302** can include guide rails or guide cage **12412** that help guide the pluggable modules **12404** so that the electrical connectors of the co-packaged optical modules **12316** are aligned with the electrical connectors on the printed circuit board.

[0760] In some implementations, the rackmount server **12400** has inlet fans mounted near the front panel **12402** and blow air in a direction substantially parallel to the front panel **12402**, similar to the examples shown in FIGS. **96** to **98**, **100**, **104**, **105**, **107** to **116**. The height h_1 of the fiber guide **12408** (measured along a direction perpendicular to the bottom panel) can be designed to be smaller than the height h_2 of the multi-fiber push on connectors **12406** so that there is space **12414** between adjacent fiber guides **12408** (in the vertical direction) to allow air to flow between the fiber guides **12408**. The fiber guide **12408** can be a hollow tube with inner dimensions sufficiently large to accommodate the fiber pigtail **12410**. The fiber guide **12408** can be made of metal or other thermally conductive material to help dissipate heat generated by the co-packaged optical module **12316**. The fiber guide **12408** can have arbitrary shapes, e.g., to optimize thermal properties. For example, the fiber guide **12408** can have side openings, or a web structure, to allow air to flow pass the fiber guide **12408**. The fiber guide **12408** is designed to be sufficiently rigid to enable the pluggable module **12404** to be inserted and removed from the rackmount server **12400** multiple times (e.g., several hundred times, several thousand times) under typical usage without deformation.

[0761] FIG. **126A** includes various views of an example of a rackmount server **12500** that includes CPO front-panel pluggable modules **12502**. Each pluggable module **12502** includes a co-packaged optical module **12504** that is optically coupled to one or more array connectors, such as multi-fiber push on connectors **12506**, through a fiber pigtail **12508**. In this example, each co-packaged optical module **12504** is optically coupled to 2 array connectors **12506**. The pluggable module **12502** includes a rigid fiber guide **12510** that approximately spans the distance between the front panel and the vertically mounted printed circuit board.

[0762] A front view **12512** (at the upper right of FIG. **126A**) shows an example of a front panel **12514** with an upper group of array connectors **12516**, a lower group of array connectors **12518**, a left group of array connectors **12520**, and a right group of array connectors **12522**. Each rectangle in the front view **12512** represents an array connector **12506**. In this example, each group of array connectors **12516**, **12518**, **12520**, **12522** includes 16 array connectors **12506**.

[0763] A front view **12524** (at the middle right of FIG. **126A**) shows an example of a recessed vertically mounted printed circuit board **12526** on which an application specific integrated circuit (ASIC) or data processing chip **12312** is mounted on the rear side and not shown in the front view **12524**. The printed circuit board **12526** has an upper group of electrical contacts **12528**, a lower group of electrical contacts **12530**, a left group of electrical contacts **12532**, and a right group of electrical contacts **12534**. Each rectangle in the front view **12524** represents an array of electrical contacts associated with one co-packaged optical module **12504**. In this example, each group of electrical contacts **12528**, **12530**, **12532**, **12534** includes 8 arrays of electrical contacts that are configured to be electrically coupled to the electrical contacts of 8 co-packaged optical modules **12504**. In this example, each co-packaged optical module **12504** is optically coupled to two array connectors **12506**, so the number of rectangles shown in the front view **12512** is twice the number of squares shown in the front view **12524**. The front panel **12514** includes openings that allow insertion of the pluggable modules **12502**. In this example, each opening has a size that can accommodate two array connectors **12506**.

[0764] A top view **12536** (at the lower right of FIG. **126A**) of the front portion of the rackmount server **12500** shows a top view of the pluggable modules **12506**. In the top view **12536**, the two left-most pluggable modules **12538** include co-packaged optical modules **12504** that are electrically coupled to the electrical contacts in the left group of electrical contacts **12532** shown in the front view **12524**, and include array connectors **12506** in the left group of array connectors **12520** shown in the front view **12512**. In the top view **12536**, the two right-most pluggable modules **12540** include co-packaged optical modules **12504** that are electrically coupled to the electrical contacts in the right group of electrical contacts **12534** shown in the front view **12524**, and include array connectors **12506** in the right group of array connectors **12522** shown in the front view **12512**. In the top view **12536**, the four middle pluggable modules **12542** include co-packaged optical modules **12504** that are electrically coupled to the electrical contacts in the upper group of electrical contacts **12528** shown in the front view **12524**, and include array connectors **12506** in the upper group of array connectors **12516** shown in the front view **12512**.

[0765] The front view **12524** (at the middle right of FIG. **126A**) shows a first inlet fan **12544** that blows air from left to right across the space between the front panel **12514** and the printed circuit board **12526**. The top view **12536** (at the lower right of FIG. **126A**) shows the first inlet fan **12544** and a second inlet fan **12546**. The first inlet fan **12544** is mounted at the front side of the printed circuit board **12526** and blows air across the pluggable modules **12502** to help dissipate the heat generated by the co-packaged optical modules **12504**. The second inlet fan **12546** is mounted at the rear side of the printed circuit board **12526** and blows air across the data processing chip **12312** and the heat dissipating device **12314**.

[0766] As shown in the front view **12512** (at the upper right of the FIG. **126A**), the front panel **12514** includes an opening **12548** that provides incoming air for the front inlet fans **12544**, **12546**. A protective mesh or grid can be provided at the opening **12548**.

[0767] A left side view **12550** (at the middle left of FIG. **126A**) of the front portion of the rackmount server **12500** shows pluggable modules **12552** that correspond to the upper group of array connectors **12516** in the front view **12512** and the upper group of electrical contacts **12528** in the front view **12524**. The left side view **12550** also shows pluggable modules **12554** that correspond to the lower group of array connectors **12518** in the front view **12512** and the lower group of electrical contacts **12530** in the front view **12524**. As shown in the left side view **12550**, guide rails or guide cage **12556** can be provided to help guide the pluggable modules **12502** so that

the electrical connectors of the co-packaged optical modules **12504** are aligned with the electrical contacts on the printed circuit board **12526**. The pluggable modules **12502** can be fastened at the front panel **12514**, e.g., using clip mechanisms.

[0768] A left side view **12558** of the front portion of the rackmount server **12500** shows pluggable modules **12560** that correspond to the left group of array connectors **12520** in the front view **12512** and the left group of electrical contacts **12532** in the front view **12524**.

[0769] In this example, the fiber guides **12510** for the pluggable modules **12502** that correspond to the left and right groups of array connectors **12520**, **12522**, and the left and right groups of electrical contacts **12532**, **12534** are designed to have smaller heights so that there are gaps between adjacent fiber guides **12510** in the vertical direction to allow air to flow through.

[0770] In some implementations, each co-packaged optical module can receive optical signals from a large number of fiber cores, and each co-packaged optical module can be optically coupled to external fiber optic cables through three or more array connectors that occupy an overall area at the front panel that is larger than the overall area occupied by the co-packaged optical module on the printed circuit board.

[0771] Referring to FIG. **126B**, in some implementations, a rackmount server **12600** is designed to use pluggable modules **12602** having a spatial fan-out design. Each pluggable module **12602** includes a co-packaged optical module **12604** that is optically coupled, through a fiber pigtail **12606**, to one or more array connectors **12608** that have an overall area larger than the area of the co-packaged optical module **12604**. The area is measured along the plane parallel to the front panel. In this example, each co-packaged optical module **12604** is optically coupled to 4 array connectors **12608**. The pluggable module **12602** includes a tapered fiber guide **12610** that is narrower near the co-packaged optical module **12604** and wider near the array connectors **12608**.

[0772] A front view **12612** (at the upper right of FIG. **126B**) shows an example of a front panel **12614** that can accommodate an array of 128 array connectors **12608** arranged in 16 rows and 8 columns. The front view **12524** (at the middle right of FIG. **126B**) of the recessed printed circuit board **12526** and the top view (at the lower right of FIG. **126B**) of the front portion of the rackmount server **12600** are similar to corresponding views in FIG. **126A**.

[0773] A left side view **12616** (at the middle left of FIG. **126B**) shows an example of pluggable modules **12602** that have co-packaged optical modules that are connected to the upper and lower groups of electrical contacts on the printed circuit board **12526**. A left side view **12618** (at the lower left of FIG. **126B**) shows an example of pluggable modules **12602** that have co-packaged optical modules that are connected to the left group of electrical contacts on the printed circuit board **12526**. As shown in the left side view **12618**, guide rails or guide cage **12620** can be provided to help guide the pluggable modules **12602** so that the electrical contacts of the co-packaged optical modules **12604** are aligned with corresponding electrical contacts on the printed circuit board **12526**.

[0774] For example, the rackmount server **12400**, **12500**, **12600** can be provided to customers with or without the pluggable modules. The customer can insert as many pluggable modules as needed.

[0775] Referring to FIG. **127**, in some implementations, a CPO front panel pluggable module **12700** can include a blind mate connector **12702** that is designed receive optical power supply light. A portion of the fiber pigtail **12714** is optically coupled to the blind mate connector **12702**. FIG. **127** includes a side view **12704** of a rackmount server **12706** that includes laser sources **12708** that provide optical power supply light to the co-packaged optical modules **12710** in the pluggable modules **12700**. The laser sources **12708** are optically coupled, through optical fibers **12712**, to optical connectors **12714** that are configured to mate with the blind-mate connectors **12702** on the pluggable modules **12700**. When the pluggable module **12700** is inserted into the rackmount server **12706**, the electrical contacts of the co-packaged optical module **12710** contacts the corresponding electrical contacts on the printed circuit board **12526**, and the blind-mate connector **12702** mates with the optical connector **12714**. This allows the co-packaged optical module **12710** to receive

optical signals from external fiber optic cables and the optical power supply light through the fiber pigtail **12714**.

[0776] In some implementations, to prevent the light from the laser source **12708** from harming operators of the rackmount server **12706**, a safety shut-off mechanism is provided. For example, a mechanical shutter can be provided on disconnection of the blind-mate connector **12702** from the optical connector **12712**. As another example, electrical contact sensing can be used, and the laser can be shut off upon detecting disconnection of the blind-mate connector **12702** from the optical connector **12712**.

[0777] Referring to FIG. **128**, in some implementations, one or more photon supplies **12800** can be provided in the fiber guide **12408** to provide power supply light to the co-packaged optical module **12316** through one or more power supply optical fibers **12802**. The one or more photon supplies **12800** can be selected to have a wavelength (or wavelengths) and power level (or power levels) suitable for the co-packaged optical module **12316**. Each photon supply **12800** can include, e.g., one or more diode lasers having the same or different wavelengths.

[0778] Electrical connections (not shown in the figure) can be used to provide electrical power to the one or more photon supplies **12800**. In some implementations, the electrical connections are configured such that when the co-packaged optical module **12316** is removed from the substrate **12310**, the electrical power to the one or more photon supplies **12800** is turned off. This prevents light from the one or more photon supplies **12800** from harming operators. Additional signals lines (not shown in the figure) can provide control signals to the photon supply **12800**. In some embodiments, electrical connections to the photon supplies **12800** are made to the system through the CPO module **12316**. In some embodiments, electrical connections to the photon supplies **12800** use parts of the fiber guide **12408**, which in some embodiments is made from electrically conductive materials. In some embodiments, the fiber guide **12408** is made of multiple parts, some of which are made from electrically conductive materials and some of which are made from electrically insulating materials. In some embodiments, two electrically conductive parts are mechanically connected but electrically separated by an electrical insulating part.

[0779] For example, the photon supply **12800** is thermally coupled to the fiber guide **12408**, and the fiber guide **12408** can help dissipate heat from the photon supply **12800**.

[0780] In some examples, the CPO module **12316** is coupled to spring-loaded elements or compression interposers mounted on the substrate **12310**. The force required to press the CPO module **12316** into the spring-loaded elements or the compression interposers can be large. The following describes mechanisms to facilitate pressing the CPO module **12361** into the spring-loaded elements or the compression interposers.

[0781] Referring to FIG. **129**, in some implementations, a rackmount server includes a substrate **12310** that is attached to a printed circuit board **12906**, which has an opening to allow the data processing chip **12312** to protrude or partially protrude through the opening and be attached to the substrate **12310**. The printed circuit board **12906** can have many functions, such as providing support for a large number of electrical power connections for the data processing chip **12312**. The CPO module **12316** can be mounted on the substrate **12310** through a CPO mount or a front lattice **12902**. A bolster plate **12914** is attached to the rear side of the printed circuit board **12906**. Both the substrate **12310** and the printed circuit board **12906** are sandwiched between the CPO mount or front lattice **12902** and the bolster plate **12914** to provide mechanical strength so that CPO modules **12316** can exert the required pressure onto the substrate **12310**. Guide rails/cage **12900** extend from the front panel **12904** or the front portion of the fiber guide **12408** to the bolster plate **12914** and provide rigid connections between the CPO mount **12902** and the front panel **12904** or the front portion of the fiber guide **12408**.

[0782] Clamp mechanisms **12908**, such as screws, are used to fasten the guide rails/cage **12900** to the front portion of the fiber guide **12408**. After the CPO module **12316** is initially pressed into the spring-loaded elements or the compression interposers, the screws **12908** are tightened, which pulls

the guide rails/cage **12900** forward, thereby pulling the bolster plate **12914** forward and provide a counteracting force that pushes the spring-loaded elements or the compression interposers in the direction of the CPO module **12316**. Springs **12910** can be provided between the guide rails **12900** and the front portion of the fiber guide **12408** to provide some tolerance in the positioning of the front portion of the fiber guide **12408** relative to the guide rails **12900**.

[0783] The right side of FIG. **129** shows front views of the guide rails/cage **12900**. For example, the guide rails **12900** can include multiple rods (e.g., four rods) that are arranged in a configuration based on the shape of the front portion of the fiber guide **12408**. If the front portion of the fiber guide **12408** has a square shape, the four rods of the guide rails **12900** can be positioned near the four corners of the front portion of the squared-shaped fiber guide **12408**. In some examples, a guide cage **12912** can be provided to enclose the guide rails **12900**. The guide rails **12900** can also be used without the guide cage **12912**.

[0784] As described above, in some examples, the CPO module **12316** (FIG. **0.123**) is coupled to spring-loaded elements or compression interposers mounted on the substrate **12310**, and the force required to press the CPO module **12316** into the spring-loaded elements or the compression interposers can be large. The following describes a press plate insert to lock (PPIL) technique that makes it easier to attach and detach the CPO modules.

[0785] Referring to FIG. **130**, in some implementations, a compression plate **13000** is used to apply a force to press the CPO module **12316** against a compression socket **13002**, and a U-shaped bolt **13010** is used to fasten the compression plate **13000** to a front lattice structure **13008**. An example of the compression plate **13000** is shown in FIG. **131**, an example of the U-shaped bolt is shown in FIG. **132**, and an example of the front lattice structure **13008** is shown in FIGS. **134** and **135**. For example, the compression socket **13002** is mounted on a substrate **12310**, and the compression socket **13002** includes compression interposers. The CPO module **12316** includes a photonic integrated circuit **13004** that is mounted on a substrate **13006**. For example, the photonic integrated circuit **13004** can be similar to the photonic integrated circuit **214** (FIGS. **2** to **5**), **450**, or **464** (FIG. **17**), and the substrate **13006** can be similar to the substrate **211** (FIGS. **2** to **5**) or **454** (FIG. **17**). The bottom side of the substrate **13006** includes electrical contacts that are electrically coupled to electrical contacts in the compression socket **13002**.

[0786] The front lattice structure **13008** is attached to the substrate **12310**, and the U-shaped bolt **13010** is inserted into holes in the sidewalls of the front lattice structure **13008** and holes in the compression plate **13000** to secure the compression plate **13000** in place relative to the front lattice structure **13008**. In this example, the front lattice structure **13008** includes a first sidewall **13008a** and a second sidewall **13008b**. The first sidewall **13008a** includes two through-holes. As shown in the example of FIG. **135B**, the second sidewall **13008b** includes two partial-through-holes that do not entirely pass through the second sidewall **13008b**. This allows another CPO module to be inserted in the space to the right of the second sidewall **13008b**, and another U-shaped bolt **13010b** to secure the other CPO module to the sidewalls of the front lattice structure **13008**. In this example, the U-shaped bolt **13010a** is inserted from the left of the first sidewall **13008a**, through the two through-holes in the first sidewall **13008a**, through the two through-holes in the compression plate **13000**, and into the two partial-through-holes in the second sidewall **13008b** of the front lattice structure **13008**.

[0787] Alternatively, as shown in the example of FIG. **135C**, the second sidewall **13008b** can include full through-holes and the U-shaped bolt **13010a** can completely pass through the second sidewall **13008b**. A second CPO module can be inserted in the space to the right of the second sidewall **13008b** using another U-shaped bolt **13010b** to secure the second CPO module to the sidewalls of the front lattice structure **13008**. In this example, the through-holes in the second sidewall **13008b** for securing the second CPO module can be laterally offset from the through-holes in the second sidewall **13008b** securing the first CPO module.

[0788] In some implementations, a wave spring **13012** is positioned between the compression plate

13000 and the CPO module **12316** to distribute the compression load to the CPO module **12316**. A groove can be cut on the bottom side of the compression plate **13000** to prevent the wave spring **13012** from sliding around on the top surface of the outer shell of the photonic integrated circuit **13004** during assembly. An example of the wave spring **13012** is shown in FIG. **133**. The wave spring **13012** can also provide tolerance in the positioning and dimensions of the CPO module **12316**.

[0789] FIG. **131** is a diagram of an example of the compression plate **13000**. The compression plate **13000** can be made of a stiff material, e.g., steel, titanium, copper, or brass. The compression plate **13000** defines an opening **13100** to allow an optical fiber cable to pass through and be connected to the CPO module **12316**. The compression plate **13000** defines two through-holes **13102a** and **13102b** (collectively referenced as **13102**) that allow two arms of the U-shaped bolt **13010** to pass through. In this figure, the through-holes **13102** are not drawn to scale. The hole diameter is configured to be smaller than the plate thickness. The compression plate **13000** can be made relatively thick (e.g., 1 mm to 5 mm) to enhance rigidity.

[0790] FIG. **132** is a diagram of an example of the U-shaped bolt **13010**. The U-shaped bolt **13010** can be made of, e.g., stainless steel, titanium, copper, or brass, and includes two arms **13200a** and **13200b** (collectively referenced as **13200**) that can be inserted into the through-holes and partial-through-holes in the sidewalls **13008a**, **13008b** of the front lattice structure **13008**, and the through-holes **13102a** and **13102b** in the compression plate **13000** to lock the compression plate **13000** in place. The U-shaped bolt **13010** can have a one-piece design, e.g., made by bending an elongated thin rod to the required shape.

[0791] FIG. **133** is a diagram of an example of the wave spring **13012**. The wave spring **13012** can also have other configurations.

[0792] FIG. **134** is a perspective view of an example of the front lattice structure **13008**. FIG. **135** is a top view of a portion of the front lattice structure **13008**. In this example, the front lattice structure **13008** defines a larger opening **13400** near the center region, and several smaller openings **13402** around the larger opening **13400**. When the front lattice structure **13008** is attached to the substrate **12310** as shown in FIG. **129**, the position of the center opening **13400** corresponds to the position of the data processing chip **12312** on the other side (e.g., rear side) of the substrate **12310**. One or more components can be mounted on the front side of the substrate **12310** to support the data processor chip **12312** on the rear side of the substrate **12310**. For example, the one or more components can include one or more capacitors, one or more filters, and/or one or more power converters. The one or more components have certain thicknesses and protrude through or partially through the opening **13400**.

[0793] Each of the openings **13402** allows a CPO module **12316** to pass through and be coupled to a corresponding compression socket **13002**. In the example shown in FIG. **134**, the front lattice structure **13008** defines 32 openings **13402** that allow the insertion of 32 CPO modules **12316**. The dimensions of this configuration support a half width 2U rack with 12 mm square optical module footprint. The openings **13402** are spaced apart at distances to support XSR channel compliance.

[0794] FIGS. **134**, **135A**, and **135B** show an example in which an outer CPO module is locked in place using a compression plate **13000a** and a U-shaped bolt **13010a**, and an inner CPO module is locked in place using a compression plate **13000b** and a U-shaped bolt **13010b** without a lateral offset between the bolts (e.g., **13010a**, **13010b**) and hence requiring partial-through-holes in the portion of the lattice between the CPO modules. FIG. **135C** shows an example in which a lateral offset is provided between the bolts and allowing the bolts to pass through complete through-holes in the portion of the lattice between the CPO modules. The term “outer CPO module” refers to a CPO module positioned closer to the outer edges of the front lattice structure **13008**, and the term “inner CPO module” refers to a CPO module positioned closer to the inner edges of the front lattice structure **13008**.

[0795] In some implementations, instead using a bolt (or clip) having arms that pass through holes

in the sidewalls of the front lattice structure **13008** and holes in the compression plate **13000**, a clamp or screws (e.g., spring-loaded screws) can be used to fasten or lock the compression plate **13000** in place relative to the front lattice structure **13008**.

[0796] FIG. **136** is an exploded front perspective view of an example of an assembly **13600** in a rackmount system **13630**. In some implementations, the assembly **13600** includes the data processing chip **12312** mounted on a substrate **13602**, a printed circuit board **13604**, a front lattice structure **13606**, a rear lattice structure **13608**, and a heat dissipating device **13610**. The printed circuit board **13604** is positioned between the substrate **13602** and the front lattice structure **13606**. The rear lattice structure **13608** is positioned between the substrate **13602** and the heat dissipating device **13610**. The assembly **13600** can be placed in a housing **13634** of the rackmount system **13630**. The housing **13634** has a front panel, and the substrate **13602** has a main surface (e.g., the front surface) that is at an angle in a range from 0 to 45° relative to the plane of the front panel. In some examples, the main surface of the substrate **13602** is substantially parallel to (e.g., in a range from 0 to 5°) relative to the plane of the front panel.

[0797] As discussed in more detail below in connection with FIG. **151**, in an alternative embodiment, the printed circuit board **13604** can be positioned between the substrate **13602** and the rear lattice structure **13626**.

[0798] For example, the printed circuit board **13604** is used to facilitate the provision of electrical power, control signals, and/or data signals to the data processing chip **12312**. The substrate **13602** can be, e.g., a ceramic substrate that is more expensive than a printed circuit board of comparable size, and it may be difficult to cost effectively manufacture the ceramic substrate sufficiently large to accommodate all the necessary connectors. The outer dimensions of the substrate **13602** can be smaller than the outer dimensions of the printed circuit board **13604**. Connectors **13612** can be mounted on the printed circuit board **13604** for receiving electrical power, control signals, and/or data signals. The connectors **13612** can have a size sufficiently large that can be conveniently handled by an operator. For example, the connectors **13612** can be Molex connectors or other types of connectors. The front surface of the substrate **13602** has electrical contacts **13632** that are electrically coupled to electrical contacts on the rear surface of the printed circuit board **13604**. The electrical contacts allow the electrical power, control signals, and/or data signals to be transmitted from the printed circuit board **13604** to the data processing chip **12312** through the substrate **13602**. In some examples, the connectors **13612** are configured to mate with external connectors in a direction parallel to the plane of the printed circuit board **13604**. In some examples, the connectors **13612** are configured to mate with external connectors in a direction perpendicular to the plane of the printed circuit board **13604**, and the signal lines extend in a rearward direction. This can reduce the spaces to the left and to the right of the printed circuit board **13604** that are needed to accommodate the signal wires. The connectors **13612** and the signal lines connected to the connectors **13612** can also be used to transmit signals from the data processing chip **12312** to other parts of the system.

[0799] This construction enables the delivery of power and other signals external to the system, maintaining the ASIC and module attachment directly to the package substrate. The delivery of power and other signals can be achieved through, e.g., land grid arrays, ball grid arrays, pin grid arrays, or sockets on the front side of the package substrate **13602** that connect to the printed circuit board **13604**. The printed circuit board **13604** can include any of the usual printed circuit board components, including the connectors **13612**. The printed circuit board connectors **13612** enable power and signal delivery through the connectors **13612**, which are then transferred to the package substrate **13602**. The package substrate **13602** is preferably attached to the printed circuit board **13604** during assembly and then placed in the rear lattice structure assembly.

[0800] The front lattice structure **13606** defines several openings **13614** that allow CPO modules **12316** to pass through and be coupled to electrical contacts or sockets **13616** mounted on the front side of the substrate **13602**. The printed circuit board **13604** defines an opening **13618** to allow the

CPO modules **12316** to pass through. The front lattice structure **13606** has an overhang **13700** (FIG. **137**) that extends through the opening **13618** and is attached to the front side of the substrate **13602**. The front lattice structure **13606** can be made of, e.g., steel or copper. The figure shows that the printed circuit board **13604** defines a single large central opening **13618**, similar to a “picture frame.” In other examples, it is also possible to divide the opening **13618** into two or more smaller openings.

[0801] Electrical components can be mounted on the front side of the substrate **13602** in a first region occupying approximately the same footprint as the data processing chip **12312**, which is on the rear side of the substrate **13600**. The electrical components support the data processing chip **12312** and can include, e.g., one or more capacitors, one or more filters, and/or one or more power converters. The front lattice structure **13606** defines a larger opening **13620** in the central region that occupies a slightly larger footprint than the first region. The electrical components mounted on the front surface of the substrate **13602** protrude through or partially through the opening **13618** in the printed circuit board **13604**, and protrude through or partially through the opening **13620** in the front lattice structure **13606**.

[0802] In some implementations, the front lattice structure **13606** can have a configuration similar to that of the front lattice structure **13008** of FIG. **134**, and the CPO modules **12316** can be pressed by compression plates **13000** against corresponding sockets **13002**. U-shaped bolts **13010** can be used to secure the compression plates **13000** to the sidewalls of the front lattice structure **13606**.

[0803] The rear lattice structure **13608** defines a central opening **13622** that is slightly larger than the data processing chip **12312**. The data processing chip **12312** protrudes through or partially through the opening **13622** and is thermally coupled to the heat dissipating device **13610**. The rear lattice structure **13608** defines several openings **13624** that generally correspond to the openings **13614** in the front lattice structure **13606**. Electronic components **13702** (FIG. **137**) can be mounted on the rear side of the substrate **13602** to support the CPO modules **12316** that are coupled to the front side of the substrate **13612**. The electronic components **13702** can protrude through or partially through the openings **13624** in the rear lattice structure **13608**. The electronic components **13702** can include, e.g., capacitors for power integrity, microcontrollers, and/or separately regulated power supplies that can isolate the optical module power domains.

[0804] In some implementations, screws **13628** are used to fasten the front lattice structure **13606**, the printed circuit board **13604**, the substrate **13602**, the rear lattice structure **13608**, and the heat dissipating device **13610** together. The rear lattice structure **13608** has lips **13626** that function as a backstop to prevent crushing of the interface (e.g., land grid arrays, pin grid arrays, ball grid arrays, sockets, or other electrical connectors) between the substrate **13602** and the printed circuit board **13604** when force is applied to fasten the front lattice structure **13606**, the printed circuit board **13604**, the substrate **13602**, the rear lattice structure **13608**, and the heat dissipating device **13610** together. In this example, the lips **13626** are formed near the upper and lower edges on the front side of the rear lattice structure **13608**. It is also possible to form the lips **13626** near the right and left edges on the front side of the rear lattice structure **13608**, or at other locations on the front side of the rear lattice structure **13608**.

[0805] FIG. **137** is an exploded rear perspective view of an example of the assembly **13600**. The front lattice structure **13606** has an overhang **13700** that extends through the opening **13618** in the printed circuit board **13604** and is attached to the front side of the substrate **13602**. The data processing chip **12312** mounted on the rear side of the substrate **13602** extends through or partially through the opening **13622** in the rear lattice structure **13608** and is thermally coupled to the heat dissipating device **13610**. For example, a thermally conductive gel or pad can be positioned between the data processing chip **12312** and the heat dissipating device **13610**. The electronic components **13702** mounted on the rear side of the substrate **13602** extends through or partially through the openings **13624** in the rear lattice structure **13608**. The upper lip **13626** extends over the upper edge of the substrate **13602** and contacts the rear side of the printed circuit board **13604**,

and the lower lip **13626** extends under the lower edge of the substrate **13602** and contacts the rear side of the printed circuit board **13604**.

[0806] In this example, the connectors **13612** include male Molex connectors configured to receive female Molex connectors along a direction parallel to the plane of the printed circuit board **13604**. It is also possible to configure the connectors **13612** to receive connectors along a direction perpendicular to the plane of the printed circuit board **13604** so that the signal lines extend in a rearward direction.

[0807] FIG. **138** is an exploded top view of an example of the assembly **13600**. In this example, the width of the overhang **13700** of the front lattice structure **13606** is selected to be slightly smaller than that of the opening **13618** of the printed circuit board **13604**. The width of the printed circuit board **13604** can be almost as wide as the inner width of the housing **13634**. The connectors **13612** are positioned near the left and right edges of the printed circuit board **13604** at locations to provide sufficient space to accommodate the signal lines that are connected to the connectors **13612**. The width of the substrate **13602** and the width of the rear lattice structure **13608** are selected so that they fit in the space between the connectors **13612** near the left edge of the printed circuit board **13604** and the connectors **13612** near the right edge of the printed circuit board **13604**.

[0808] FIG. **139** is an exploded side view of an example of the assembly **13600**. In this example, the height of the overhang **13700** of the front lattice structure **13606** is selected to be slightly smaller than that of the opening **13618** of the printed circuit board **13604**. The height of the printed circuit board **13604** can be almost as tall as the inner height of the housing **13634**. The height of the substrate **13602** is selected so that the substrate **13602** fits in the space between the upper lip **13626** and the lower lip **13626**.

[0809] FIG. **140** is a front perspective view of an example of the assembly **13600** that has been fastened together. The overhang **13700** of the front lattice structure **13606** contacts the front surface of the substrate **13604**, and the electronic components that support the data processing chip **12312** extend through or partially through the opening **13618** in the printed circuit board **13604** and the opening **13620** in the front lattice structure **13606**. The sidewalls of the front lattice structure **13606** function as guides for aligning the CPO modules **12316** to the sockets **13616** on the front surface of the substrate **13602**. The large printed circuit board **13604** has more surface area to mount connectors **13612** for providing electrical power, control signals, and/or data signals to the data processing chip **12312**. The assembly **13600** is vertically mounted, e.g., the substrate **13602** is substantially vertical with respect to the top or bottom panel of the housing **13634** and substantially parallel to the front panel. The assembly **13600** is positioned near the front panel, e.g., not more than 12 inches from the front panel. The front panel can be opened to allow an operator to easily access the CPO modules **12316**, e.g., to insert or remove the CPO modules **12316** into or from the sockets **13616**.

[0810] FIG. **141** is a front perspective view of an example of the assembled assembly **13600** without the front lattice structure **13606**. The printed circuit board **13604** is shaped similar to a “picture frame” and the opening **13618** is configured to allow the CPO modules **12316** to be coupled to the sockets **13616**, and to provide space to accommodate the various electronic components mounted on the front side of the substrate **13602** that support the data processing chip **12312** on the rear side of the substrate **13602**.

[0811] FIG. **142** is a front perspective view of an example of the assembled assembly **13600** without the printed circuit board **13604** and the front lattice structure **13606**. Electrical contacts or sockets **13616** (each socket can include a plurality of electrical contacts) are provided on the front side of the substrate **13602**, in which the electrical contacts or sockets **13616** are configured to be coupled to the CPO modules **12316**. In this example, arrays of electrical contacts **13632** are provided at the left and right regions of the substrate **13602**. For example, power converters can be mounted on the printed circuit board **13604** to receive electric power that has a higher voltage (e.g., 12V or 24V) and a lower current, and output electric power that has a lower voltage (e.g., 1.5V)

and a higher current. In some implementations, the data processing chip **12312** can require more than 100 A of peak current during certain periods of time. By providing a large number of electrical contacts **13632**, the overall resistance to the higher current can be made smaller.

[0812] FIG. **143** is a front perspective view of an example of the assembled rear lattice structure **13608** and the heat dissipating device **13610**. The rear lattice structure **13608** defines an opening **13622** to provide space for the data processing chip **12312** mounted on the rear side of the substrate **13602**. The rear lattice structure **13608** defines openings **13624** to provide space for the components **13702** mounted on the rear side of the substrate **13602**, in which the components support the CPO modules **12316** coupled to the electrical contacts **13616** on the front side of the substrate **13602**. The upper and lower lips **13626** prevent crushing of the interface (e.g., land grid arrays, pin grid arrays, ball grid arrays, sockets, or other electrical connectors) between the substrate **13602** and the printed circuit board **13604** when force is applied to fasten the front lattice structure **13606**, the printed circuit board **13604**, the substrate **13602**, the rear lattice structure **13608**, and the heat dissipating device **13610** together.

[0813] FIG. **144** is a front perspective view of an example of the heat dissipating device **13610** and the screws **13628**. The heat dissipating device **13610** can include fins that extend in the horizontal direction. For example, an inlet fan (e.g., **12546** of FIG. **125**) blows air in the horizontal direction across the fins to help carry away the heat generated by the data processing chip **12312**.

[0814] FIG. **145** is a rear perspective view of an example of the assembly **13600** in which the front lattice structure **13606**, the printed circuit board **13604**, the substrate **13602**, the rear lattice structure **13608**, and the heat dissipating device **13610** have been fastened together. The heat dissipating device **13610** as shown in the figure includes horizontal fins, but can also have other configurations, such as having pins or posts, such as those shown in FIG. **68C**. The heating dissipating device **13610** can include a vapor chamber thermally coupled to the heat sink fins or pins.

[0815] FIG. **146** is a rear perspective view of an example of the assembly **13600** without the rear lattice structure **13608**. The data processing chip **12312** protrudes through or partially through the opening **13622** in the rear lattice structure **13608**. The components **13702** protrude through or partially through the openings **13624** in the rear lattice structure **13608**.

[0816] FIG. **147** is a rear perspective view of an example of the front lattice structure **13606**, the printed circuit board **13604**, and the substrate **13602** that have been fastened together. FIG. **148** is a rear perspective view of an example of the front lattice structure **13606** and the printed circuit board **13604** that have been fastened together. The overhang **13700** of the front lattice structure **13606** extends into the opening **13618** in the printed circuit board **13604**. FIG. **149** is a rear perspective view of an example of the front lattice structure **13606**.

[0817] Referring to FIG. **150**, in some implementations, a data processing chip **15000** is mounted on a substrate (e.g., a ceramic substrate) **15002**, which is electrically coupled to a first side of a printed circuit board **15004**. A CPO module **15006** is mounted on a substrate (e.g., a ceramic substrate) **15008**, which is electrically coupled to a second side of the printed circuit board **15004**. The configuration shown in FIG. **150** can be used in any of the systems or assemblies described above that includes a data processing chip communicating with one or more CPO modules.

[0818] FIG. **151** shows, in the right portion of the figure, a top view of an example of an assembly **15100**, suitable for use in a rackmount system, that includes a vertical printed circuit board **13604** (e.g., a daughter card) that is positioned between a package substrate **13602** (also referred to as a CPO substrate) and a rear lattice structure **13626**. The package substrate **13602** is positioned between the printed circuit board **13604** and a front lattice structure **13606**. In this example, each CPO module **12316** is removably attached to a high-speed LGA socket **15104** that is mounted on the front side of the package substrate **13602**. The data processing chip **13612** (which in this example is a switch ASIC) is mounted on the rear side of the package substrate **13602**. The high-speed LGA socket **15104** is electrically coupled to high-speed LGA pads **15106** on the front surface

of the package substrate **13602**. High speed traces **15102** within the package substrate **13602** provides high speed signal connections between the CPO modules **12316** and the data processing chip **13612**.

[0819] In this example, the printed circuit board **13604** defines an opening that allows the data processing chip **13612** to pass through to be thermally coupled to a heat dissipating device **13610**. The printed circuit board **13604** is a “picture frame” with a cut-out for the switch ASIC **13612**. The package substrate **13602** has power and low-speed contact pads **15108** on the rear side for attaching to the vertical printed circuit board **13604** (the “picture frame” daughter card) for receiving electrical power and low-speed control signals from the printed circuit board **13604**. The power and low-speed contact pads **15108** are relatively large (e.g., about 1 mm), as compared to the high-speed LGA pads **15106**. The power and low-speed contact pads **15108** is positioned between the CPO substrate **13602** and the printed circuit board **13604**, and do not impact the mounting of the heat sink **13610** to the data processing chip **13612**.

[0820] In some implementations, the printed circuit board **13604** defines an opening that is that is large enough to accommodate the data processor (e.g., switch ASIC) **13612** and additional components that are mounted on the rear side of the substrate **13602**, in which the additional components support the CPO modules **12316**. The additional components can include, e.g., one or more capacitors, filters, power converters, or voltage regulators. In some examples, instead of having one large opening, the printed circuit board **13604** can define multiple openings that are positioned to allow the data processor **13612** and the additional components to protrude through or partially through.

[0821] FIG. **151** shows, in the left portion of the figure, a perspective rear view of the package substrate **13602**, the CPO module **12316**, and compression plates **15110**. As shown in this diagram, in some implementations, there can be a large number (e.g., several hundred or thousand) of power and low-speed contact pads **15108** to allow routing of a large amount of power to the data processing chip **13612** and the CPO modules **12316**. In this example, each compression plate **15110** has an integrated heat sink **15112** for dissipating the heat generated by the CPO module **12316**.

[0822] Referring to FIG. **152**, in some implementations, the CPO modules **12316** can easily be removed from the package substrate **13602** for replacement or repair. For example, a fiber connector is attached to the CPO module **12316**, which is attached to the LGA socket **15104**, which is removably attached to the package substrate **13602**. The compression plate **15110** presses down on the CPO module **12316** and is secured relative to the front and rear lattice structures **13606**, **13626** using the U-shaped bolts **13010** and spring-loaded screws **15200**. The compression plate **15110** can have a latch for latching the fiber connector **12318**. If a CPO module **12316** malfunctions, the technician can remove the screws **15200**, remove the U-shaped bolts **13010**, and detach the CPO module **12316** from the LGA socket **15104**, or detach the LGA socket from the package substrate **13602**.

[0823] FIG. **153** is a diagram showing an example of a process **15300** for assembling the assembly **15100**. The front lattice structure **13606** is attached **15302** to the CPO substrate **13602**, and the CPO substrate **13602** is attached **15304** to the printed circuit board **13604**. The heat sink **13610** is thermally coupled to the data processing chip **13612**. This diagram shows the front side of the CPO substrate **13602**, the data processing chip **13612** is mounted on the other side of the CPO substrate **13602** and not shown in the figure. The diagram **15306** shows the assembly **15100** ready for insertion of the CPO modules **12316**. The diagram **15308** shows CPO modules **12316** with compression plates **15110** inserted into the front lattice structure **13606**, and before attachment of the optical fibers.

[0824] FIG. **154** is a diagram showing an example of a CPO module **12316** having a lid **15400** to protect the CPO module **12316**. Also shown is a compression plate **15110** with an integrated heat sink **15112**. In this example, screws **15402** are used to secure the compression plate **15110** to the

front lattice structure **13606** and/or the package substrate **13602** and/or the vertical printed circuit board **13604** and/or the rear lattice structure **13626**.

[0825] FIG. **155A** is a rear perspective view of an example of the LGA socket **15104**, the optical module **12316**, and the compression plate **15110**. FIG. **155B** is a front perspective view of an example of the LGA socket **15104**, the optical module **12316**, and the compression plate **15110**. In FIGS. **155A** and **155B**, the LGA socket **15104** has been inserted into the front lattice structure **13606**, ready for insertion or attachment of the optical module **12316** and the compression plate **15110**.

[0826] FIG. **156** is a front view (assuming the printed circuit board **13604** is vertically mounted in a rackmount server) of an example of an array of compression plates **15110** mounted on the front lattice structure **13606**. The front lattice structure **13606** includes an opening **13400** for placing components that support the data processor chip **12312** on the rear side of the substrate **12310**. For example, the one or more components can include one or more decoupling capacitors, one or more filters, and/or one or more voltage regulators, if needed. The one or more components have certain thicknesses and protrude through or partially through the opening **13400**.

[0827] FIG. **157** is a front perspective view of an example of the assembly **15100**. Several CPO modules **12316** with lids **15400** are mounted on the front side of the package substrate **13602**. The CPO modules **12316** are pressed against the package substrate **13602** by compression plates **15110** having integrated heat sinks **15112**.

[0828] FIG. **158** is a top view of an example of the assembly **15100**. The switch ASIC **13612** is mounted on the rear side of the package substrate **13602**. Several CPO modules **12316** with lids **15400** are mounted on the front side of the package substrate **13602**. The CPO modules **12316** are pressed against the package substrate **13602** by compression plates **15110** having integrated heat sinks **15112**.

[0829] FIGS. **156** to **158** show compression plates **15110** on top of (or in front of) the optical modules **12316** showing the fiber connector receptacles. Under the compression plates is a baseplate (which is referred to as the lattice or honeycomb structure) that is mounted with screws through the system printed circuit board **13602** to the rear lattice **13626** or ASIC heatsink **13610** on the backside. In addition, or alternatively, a clip-based or bolt-based design similar to the design shown in FIGS. **130** to **135C** can be used to secure the compression plates **15110** to the front lattice structure **13606**.

[0830] In the examples shown in FIGS. **2**, **4**, **6**, **7**, **12**, **17**, **20**, **22** to **31**, **35A** to **37**, **43**, **68A**, **69A**, **70**, **71A**, **72**, **73A**, **74A**, **75A**, **75C**, **77A** to **78**, **99**, **100**, **104**, **108**, **110**, **112**, **113**, **115**, **117**, **118** to **125B**, **129**, **136** to **153**, and **158** to **160**, one or more data processing modules are mounted on a substrate or circuit board that is positioned near the front panel (or any panel that is accessible to the user), and the communication interfaces such as co-packaged optical modules support the one or more data processing modules. Each data processing module can be, e.g., a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, a storage device, or an application specific integrated circuit (ASIC). Each data processing module can include an electronic processor and/or a photonic processor. The data processing modules can be mounted on the substrate or circuit board using various types of contacts, such as ball grid arrays or sockets. The data processing modules can also be mounted on smaller substrates or circuit boards that are in turn mounted on larger substrates or circuit boards. The following describes an example in which the communication interface(s) support memory modules mounted in smaller circuit boards that are electrically coupled to a larger circuit board positioned near the front panel.

[0831] FIG. **162** shows a top view of an example of a system **16200** that includes a vertically oriented circuit board **16202** (also referred to as a carrier card) that is substantially parallel to the front panel **16204**. Several memory modules **16206** are electrically coupled to the circuit board **16202**, e.g., using sockets, such as DIMM (dual in line memory module) sockets. Each memory

module **16206** includes a circuit board **16208** and one or more memory integrated circuits **16210**, which can be mounted on one side or both sides of the circuit board **16208**. One or more optical interface modules **16212** (e.g., co-packaged optical modules) are electrically coupled to the circuit board **16202** and function as the interface between the memory modules **16206** and one or more communication optical fiber cables **16214**. For example, each optical interface module **16214** can support up to 1.6 Tbps bandwidth. When N optical interface modules **16214** are used (N being a positive integer), the total bandwidth can be up to $N \times 1.6$ Tbps. One or more fans **16216** can be mounted near the front panel **16204** to assist in removing heat generated by the various components (e.g., the optical interface modules **16212** and the memory modules **16206**) coupled to the circuit board **16202**. The technologies for implementing the optical interface modules **16212** and configuring the fans **16216** and airflows for optimizing heat removal have been described above and not repeated here.

[0832] FIG. **163** is an enlarged diagram of the carrier card **16202**, the optical interface module(s) **16212**, and the memory modules **16206**. In this example, the memory modules **16206** are mounted on both the front side and the rear side of the carrier card **16202**. It is also possible mount the memory modules **16206** to just the front side, or just the rear side, of the carrier card **16202**. In some examples, heat sinks are thermally attached to the memory chips **16210**.

[0833] In some implementations, the memory modules **16206** on the carrier card **16202** can be used as, e.g., computer memory, disaggregated memory, or a memory pool. For example, the system **16200** can provide a large memory bank or memory pool that is accessible by more than one central processing unit. A data processing system can be implemented as a spatially co-located solution, e.g., 4 sets of the memory modules **16206** supporting 4 processors sitting in a common box or housing. A data processing system can also be implemented as a spatially separated solution, e.g., a rack full of processors, connected by optical fiber cables to another rack full of DIMMs (or other memory). In this example, the rack full of memory modules can includes multiple systems **16200**. For example, the system **16200** is useful for implementing memory disaggregation to decouple physical memory allocated to virtual servers (e.g., virtual machines or containers or executors) at their initialization time from the runtime management of the memory. The decoupling allows a server under high memory usage to use the idle memory either from other servers hosted on the same physical node (node level memory disaggregation) or from remote nodes in the same cluster (cluster level memory disaggregation).

[0834] FIG. **164** is a front view of an example of the carrier card **16202**, the optical interface module(s) **16212**, and the memory modules **16206**. In this example, three rows of memory modules **16206** are attached to the circuit board **16202**. The number of memory modules **16206** can vary depending on application. The orientation of the memory modules **16206** can also be modified depending on how the system is configured. For example, instead of orienting the memory modules **16206** to extend in the vertical direction as shown in FIG. **164**, the memory modules **16206** can also be oriented to extend in the horizontal direction, or at an angle between 0° to 90° relative to the horizontal direction, in order to optimize air flow and heat dissipation.

[0835] FIG. **165** is a front view of an example of the carrier card **16202** with two optical interface modules **16212**, and memory modules **16206**. FIGS. **164** and **165**, as well as many other figures, are not drawn to scale. The optical interface modules **16212** can be much smaller than what is shown in the figure, and many more optical interface modules **16212** can be attached to the circuit board **16202**. For example, the optical interface module **16212** can be positioned in the space **16218** (shown in dashed lines) between the four memory modules **16206**. In some examples, the memory modules **16206** can interface directly with the optical interface module **16212**.

[0836] Referring to FIG. **166**, in some implementations, one or more memory controllers or switches **16600** (e.g., Compute Express Link (CXL) controller(s)) is/are electrically coupled to the carrier card **16202** and configured to aggregate the traffic from the memory modules **16206**. For example, the memory controller(s) or switch(es) **16600** can be implemented as an integrated circuit

mounted on the rear side of the carrier card **16202** opposite to the optical interface module(s) **16212**. Electrical traces are provided on or in the circuit board **16202** to connect the memory modules **16206** to the CXL controller/switch(es) **16600**, and the CXL controller/switch(es) **16600** then aggregate the traffic from the memory modules **16206** and interface them to the CPO module **16212**.

[0837] The carrier card **16202** and the memory modules **16206** can be any of a variety of sizes depending on the available space in the housing. The capacity of the memory modules **16206** can vary depending on application. As memory technology improves in the future, it is expected that the capacity of the memory modules **16206** will increase in the future. For example, the carrier card **16202** can have dimensions of 20 cm×20 cm, each memory module **16206** can have dimensions of 10 cm×2 cm, and each memory module can have a capacity of 64 GB. A spacing of 6 mm can be provided between memory modules **16206**. The memory modules **16206** can occupy both sides of the carrier card **16202**. In this example, the carrier card **16202** has a height of 20 cm and can support 2 rows of memory modules **16206**, with each memory module **16206** extending 10 cm in the vertical direction. With a carrier card width of 20 cm and a 6 mm spacing between memory modules **16206**, there can be about 32 memory modules per row, and about 64 memory modules per side of the carrier card **16202**. When the memory modules are mounted on both sides of the carrier card **16202**, there can be up to a total of about 128 memory modules **16206** per carrier card. With up to 64 GB capacity for each memory module **16206**, the carrier card **16202** can support up to about 8 TB memory in a space approximately the size of 1,600 cm.^{sup.3}

[0838] In some implementations, in the examples shown in FIGS. **136** to **149** and **151** to **153**, the printed circuit board **13604** and the front grid structure **13606** and/or the rear grid structure **13608** can be modified to accommodate memory modules **16206** that are attached to the printed circuit board **13604**. For example, a portion of the surface area of the printed circuit board **13604** can support the memory modules **16206**, and another portion of the surface area of the printed circuit board **13604** can overlap the front/rear grid structure. In some examples, the front/rear grid structure has openings that allow the memory modules **16206** on the printed circuit board **13604** to protrude through the openings.

[0839] In the examples shown in FIGS. **6** and **23**, an optical fiber cable is optically coupled to the top side of the photonic integrated circuit, and the bottom side of the photonic integrated circuit is mounted on a substrate. One or more electronic integrate circuits, such as a serializer/deserialize module, is/are mounted on or partially on the photonic integrated circuit adjacent to or near the optical fiber cable or the optical connector that connects to the optical fiber cable. In the examples shown in FIGS. **7** and **32**, the photonic integrated circuit and the electronic integrated circuit(s) are mounted on opposite sides of the substrate, in which the electronic integrated circuit(s) is/are mounted adjacent to or near the optical fiber cable or the optical connector that connects to the optical fiber cable. In the examples shown in FIGS. **35A** to **37**, an optical fiber cable is optically coupled to the bottom side of the photonic integrated circuit, and the electronic integrated circuit is coupled to the top side of the photonic integrated circuit. These examples illustrate how one or more electronic integrated circuits can be vertically stacked on a photonic integrated circuit (either directly or indirectly through a substrate) in a way that accommodates the optical path from the optical fiber cable to the photonic integrated circuit. The following describes such packaging for the co-packaged optical module in which ASICs are placed adjacent to, near, or around the vertical fiber connector.

[0840] Referring to FIG. **167**, a co-packaged optical module **16700** includes a substrate **16702** and a photonic integrated circuit **16704** mounted on the substrate **16702**. A lens array **16706** and a micro optics connector **16708** optically couples the photonic integrated circuit **16704** to an optical fiber cable. The lens array **16706** and the micro optics connector **16708** will be referred to as the optical connector. A first set of one or more integrated circuits **16710** are mounted on the top side of the photonic integrated circuit **16704** using, e.g., copper pillars, or solder bumps. The first set of

one or more integrated circuits **16710** is positioned adjacent to or near the optical connector. For example, two or more integrated circuits **16710** can be positioned on two or more sides of the optical connector, surrounding or partially surrounding the optical connector. A second set of integrated circuits **16712** is mounted on the substrate **16702** and electrically coupled to the photonic integrated circuit **16704**.

[0841] For example, each integrated circuit **16710** (mounted on the photonic integrated circuit **16704**) can include an electrical drive amplifier or a transimpedance amplifier. Each integrated circuits **16712** (mounted on the substrate) can include a SerDes or a DSP chip or a combination of SerDes/DSP chips.

[0842] FIG. **168** shows perspective views of an example of the co-packaged optical module **16700**. The diagram on the left of the figure shows the substrate **16702**, the photonic integrated circuit **16704**, the first set of electrical integrated circuits **16710** mounted on the photonic integrate circuit **16704**, and a second set of electrical integrated circuits **16712** mounted on the substrate **16702**. The diagram on the right of the figure shows the same components as those shown in the left diagram, with the addition of a smart connector **16800** that connects to an optical fiber cable, and a socket **16802** that electrically couples to the electrical contacts on the bottom side of the substrate **16702**.

[0843] FIGS. **169A** and **169B** shows additional examples of perspective views of the co-packaged optical module **16700**. FIG. **170** shows a top view of an example of the placement of the electrical integrated circuits **16710** on the photonic integrated circuit **16704**. In this example, the lens array **16706** is positioned near the center of the photonic integrated circuit **16704**, and the electrical integrated circuits **16710** are placed at the north, south, east, and west positions relative to the lens array **16706**. By placing the electrical integrated circuits **16710** on top of the photonic integrated circuit **16704** and surrounding the lens array **16706** (or any other type of optical connector), the co-packaged optical module **16700** can be made more compact. Furthermore, the conductive traces between the electrical integrated circuits **16710** and active components in the photonic integrated circuit **16704** can be made shorter, resulting in better performance, e.g., higher data rate, higher signal-to-noise ratio, and lower power required to transmit the signals, as compared to a configuration in which the electrical signals have to travel longer distances.

[0844] There are several ways to package the electrical integrated circuits and the photonic integrated circuit in order to achieve a compact, small-size, and energy efficient co-packaged optical module. FIG. **171A** shows an example in which a photonic integrated circuit **16704** has an active layer **17100** that is positioned near the top surface of the photonic integrated circuit **16704**. The fiber connection **17102** (which can include, e.g., a 2D array of focusing lenses) is coupled to the fiber connection **17102** from the top side. For example, grating couplers in the active PIC layer **17100** can be positioned under the fiber connection **17102** to couple the optical signals from the fiber connection **17102** into optical waveguides on the active PIC layer **17100**, and from the optical waveguides out to the fiber connection **17102**. The electrical integrated circuits **16710** are mounted on the top side of the photonic integrated circuit **16704** and are coupled to the active PIC layer **17100** through contact pads and optionally short conductive traces. For example, the active PIC layer **17100** can include photodetectors that convert the optical signals received from the fiber connection **17102** to electrical current signals that are transmitted to the drivers and transimpedance amplifiers in the electrical integrated circuits **16710**. Similarly, the electrical integrated circuits **16710** can send electrical signals to the electro-optic modulators in the active PIC layer **17100** that convert the electrical signals to optical signals that are output through the fiber connection **17102**.

[0845] FIG. **171B** shows an example in which the electrical integrated circuits **16710** are coupled to the bottom surface of the photonic integrated circuit **16704** and electrically coupled to the active PIC layer **17100** using through silicon vias **17104**. The through silicon vias **17104** provide signal conduction paths in the thickness direction through the silicon die or substrate of the photonic integrated circuit **16704**. The drivers and transimpedance amplifiers in the electrical integrated circuits **16710** can be positioned directly under the photonic integrated circuit active components,

such as the photodiodes and the electro-optic modulators, so that the shortest electrical signal paths can be used between the photonic integrated circuit **16704** and the electrical integrated circuits **16710**.

[0846] FIG. **171C** shows an example in which the fiber connection **17102** is coupled to the photonic integrated circuit **16704** through the bottom side (in a configuration referred to as “backside illumination”), such that the optical signals from the fiber connection **17102** pass through the silicon die or substrate before being received by the photodetectors in the active PIC layer **17100**. Likewise, the modulators in the active PIC layer **17100** transmit modulated optical signals through the silicon die or substrate to the fiber connection **17102**. The portion of the active PIC layer **17100** directly above the fiber connection **17102** can include grating couplers. The photodetectors and modulators are positioned at a distance from the grating couplers. The electrical integrated circuits **16710** are positioned directly above or near the photodetectors and the modulators, so the locations of the electrical integrated circuits **16710** relative to the active PIC layer **17100** in the example of FIG. **171C** will be similar to those in the example of FIG. **171A**.

[0847] FIG. **171D** shows an example in which backside illumination is used, and the electrical integrated circuits **16710** are coupled to the bottom side of the photonic integrated circuit **16704**. The electrical integrated circuits **16710** are electrically coupled to the active components (e.g., photodetectors and electro-optic modulators) in the active PIC layer **17100** using through silicon vias **17104**, similar to the example in FIG. **171B**.

[0848] In some implementations, an integrated circuit is configured to surround or partially surround the vertical fiber connector. For example, the integrated circuit can have an L-shape that surrounds two sides of the vertical fiber connector (e.g., two of north, east, south, and west sides). For example, the integrated circuit can have a U-shape that surrounds three sides of the vertical fiber connector (e.g., three of north, east, south, and west sides). For example, the integrated circuit can have an opening in the center region to allow the vertical fiber connector to pass through, in which the integrated circuit completely surrounds the vertical fiber connector. The dimensions of the opening in the integrated circuit are selected to allow the optical fiber connector to pass through to enable an optical fiber to be optically coupled to the photonic integrated circuit. For example, the integrated circuit with an opening in the center region can have a circular or polygonal shape at the outer perimeter. A feature of the integrated circuit mounted on the same surface as the vertical fiber connector is that it takes advantage of the space available on the surface of the photonic integrated circuit that is not occupied by the vertical fiber connector so that the electrical integrated circuit can be placed near or adjacent to the active components (e.g., photodetectors and/or modulators) of the photonic integrated circuit.

[0849] In some implementations, an integrated circuit defining an opening can be manufactured by the following process: [0850] Step 1: Use semiconductor lithography to form an integrated circuit on a semiconductor die (or wafer or substrate), in which a first interior region of the semiconductor die does not have integrated circuit component intended to be used for the final integrated circuit (but can have components intended to be used for other products). [0851] Step 2: Use a laser (or any other suitable cutting tool) to cut an opening in the first interior region of the semiconductor die. [0852] Step 3: Place the semiconductor die on a lower mold resin that defines an opening in an interior region. A lead frame or electrical connectors are attached to the lower mold resin. [0853] Step 4: Wire bond electrical contacts on the semiconductor die to the lead frame or electrical connectors attached to the lower mold resin. [0854] Step 5: Attach an upper mold resin to the lower mold resin, and enclose the semiconductor die between the lower and upper mold resins. The upper mold resin defines an opening in an interior region that corresponds to the opening in the lower mold resin. In some examples, the footprint of the semiconductor die is within the footprint of the lower/upper mold resins so that the semiconductor die is completely enclosed inside the lower and upper mold resins. In some examples, the lower and/or upper mold resin can have additional openings, and the opening(s) in the lower and/or upper mold resins can be configured to expose

one or more portions of the semiconductor die.

[0855] An integrated circuit having an L-shape or a U-shape can be manufactured using a similar process. For example, in step 1, circuitry is formed in an L-shaped or U-shaped footprint. In step 2, the laser or cutting tool cuts the die according to the L-shape or U-shape footprint. In steps 3 and 5, a lower mold resin and an upper mold resin having the desired L-shape or U-shape are used.

[0856] While this disclosure includes references to illustrative embodiments, this specification is not intended to be construed in a limiting sense. Various modifications of the described embodiments, as well as other embodiments within the scope of the disclosure, which are apparent to persons skilled in the art to which the disclosure pertains are deemed to lie within the principle and scope of the disclosure, e.g., as expressed in the following claims.

[0857] For example, the techniques described above for improving the operations of systems that include rackmount servers (see FIGS. 76, 85 to 87B) can also be applied to systems that include blade servers.

[0858] Some embodiments can be implemented as circuit-based processes, including possible implementation on a single integrated circuit.

[0859] Unless explicitly stated otherwise, each numerical value and range should be interpreted as being approximate as if the word “about” or “approximately” preceded the value or range.

[0860] It will be further understood that various changes in the details, materials, and arrangements of the parts which have been described and illustrated in order to explain the nature of this disclosure can be made by those skilled in the art without departing from the scope of the disclosure, e.g., as expressed in the following claims.

[0861] The use of figure numbers and/or figure reference labels in the claims is intended to identify one or more possible embodiments of the claimed subject matter in order to facilitate the interpretation of the claims. Such use is not to be construed as necessarily limiting the scope of those claims to the embodiments shown in the corresponding figures.

[0862] Although the elements in the following method claims, if any, are recited in a particular sequence with corresponding labeling, unless the claim recitations otherwise imply a particular sequence for implementing some or all of those elements, those elements are not necessarily intended to be limited to being implemented in that particular sequence.

[0863] Reference herein to “one embodiment” or “an embodiment” means that a particular feature, structure, or characteristic described in connection with the embodiment can be included in at least one embodiment of the disclosure. The appearances of the phrase “in one embodiment” in various places in the specification are not necessarily all referring to the same embodiment, nor are separate or alternative embodiments necessarily mutually exclusive of other embodiments. The same applies to the term “implementation.”

[0864] Unless otherwise specified herein, the use of the ordinal adjectives “first,” “second,” “third,” etc., to refer to an object of a plurality of like objects merely indicates that different instances of such like objects are being referred to, and is not intended to imply that the like objects so referred-to have to be in a corresponding order or sequence, either temporally, spatially, in ranking, or in any other manner.

[0865] Also for purposes of this description, the terms “couple,” “coupling,” “coupled,” “connect,” “connecting,” or “connected” refer to any manner known in the art or later developed in which energy is allowed to be transferred between two or more elements, and the interposition of one or more additional elements is contemplated, although not required. Conversely, the terms “directly coupled,” “directly connected,” etc., imply the absence of such additional elements.

[0866] As used herein in reference to an element and a standard, the term compatible means that the element communicates with other elements in a manner wholly or partially specified by the standard, and would be recognized by other elements as sufficiently capable of communicating with the other elements in the manner specified by the standard. The compatible element does not need to operate internally in a manner specified by the standard.

[0867] The described embodiments are to be considered in all respects as only illustrative and not restrictive. In particular, the scope of the disclosure is indicated by the appended claims rather than by the description and figures herein. All changes that come within the meaning and range of equivalency of the claims are to be embraced within their scope.

[0868] The description and drawings merely illustrate the principles of the disclosure. It will thus be appreciated that those of ordinary skill in the art will be able to devise various arrangements that, although not explicitly described or shown herein, embody the principles of the disclosure and are included within its spirit and scope. Furthermore, all examples recited herein are principally intended expressly to be only for pedagogical purposes to aid the reader in understanding the principles of the disclosure and the concepts contributed by the inventor(s) to furthering the art, and are to be construed as being without limitation to such specifically recited examples and conditions. Moreover, all statements herein reciting principles, aspects, and embodiments of the disclosure, as well as specific examples thereof, are intended to encompass equivalents thereof.

[0869] The functions of the various elements shown in the figures, including any functional blocks labeled or referred to as “processors” and/or “controllers,” can be provided through the use of dedicated hardware as well as hardware capable of executing software in association with appropriate software. When provided by a processor, the functions can be provided by a single dedicated processor, by a single shared processor, or by a plurality of individual processors, some of which can be shared. Moreover, explicit use of the term “processor” or “controller” should not be construed to refer exclusively to hardware capable of executing software, and can implicitly include, without limitation, digital signal processor (DSP) hardware, network processor, application specific integrated circuit (ASIC), field programmable gate array (FPGA), read only memory (ROM) for storing software, random access memory (RAM), and non-volatile storage. Other hardware, conventional and/or custom, can also be included. Similarly, any switches shown in the figures are conceptual only. Their function can be carried out through the operation of program logic, through dedicated logic, through the interaction of program control and dedicated logic, or even manually, the particular technique being selectable by the implementer as more specifically understood from the context.

[0870] As used in this application, the term “circuitry” can refer to one or more or all of the following: (a) hardware-only circuit implementations (such as implementations in only analog and/or digital circuitry); (b) combinations of hardware circuits and software, such as (as applicable): (i) a combination of analog and/or digital hardware circuit(s) with software/firmware and (ii) any portions of hardware processor(s) with software (including digital signal processor(s)), software, and memory(ies) that work together to cause an apparatus, such as a mobile phone or server, to perform various functions); and (c) hardware circuit(s) and or processor(s), such as a microprocessor(s) or a portion of a microprocessor(s), that requires software (e.g., firmware) for operation, but the software does not need to be present when it is not needed for operation.” This definition of circuitry applies to all uses of this term in this application, including in any claims. As a further example, as used in this application, the term circuitry also covers an implementation of merely a hardware circuit or processor (or multiple processors) or portion of a hardware circuit or processor and its (or their) accompanying software and/or firmware. The term circuitry also covers, for example and if applicable to the particular claim element, a baseband integrated circuit or processor integrated circuit for a mobile device or a similar integrated circuit in server, a cellular network device, or other computing or network device.

[0871] It should be appreciated by those of ordinary skill in the art that any block diagrams herein represent conceptual views of illustrative circuitry embodying the principles of the disclosure.

[0872] Although the present invention is defined in the attached claims, it should be understood that the present invention can also be defined in accordance with the following sets of embodiments:

[0873] First set of embodiments:

[0874] Apparatus 1: An apparatus comprising: [0875] an optical interconnect module comprising: [0876] an optical input port configured to receive a plurality of channels of first optical signals; [0877] a photonic integrated circuit configured to generate a plurality of first serial electrical signals based on the received optical signals, in which each first serial electrical signal is generated based on one of the channels of first optical signals; [0878] a first serializers/deserializers module comprising multiple serializer units and deserializer units, in which the first serializers/deserializers module is configured to generate a plurality of sets of first parallel electrical signals based on the plurality of first serial electrical signals, and condition the electrical signals, in which each set of first parallel electrical signals is generated based on a corresponding first serial electrical signal; and [0879] a second serializers/deserializers module comprising multiple serializer units and deserializer units, in which the second serializers/deserializers module is configured to generate a plurality of second serial electrical signals based on the plurality of sets of first parallel electrical signals, in which each second serial electrical signal is generated based on a corresponding set of first parallel electrical signals.

[0880] Embodiment 2: The apparatus of embodiment 1, comprising: [0881] a third serializers/deserializers module comprising multiple serializer units and deserializer units, in which the third serializers/deserializers module is configured to generate a plurality of sets of second parallel electrical signals based on the plurality of second serial electrical signals, in which each set of second parallel electrical signals is generated based on a corresponding second serial electrical signal; and [0882] at least one of a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, or an application specific integrated circuit (ASIC) that is configured to process the plurality of sets of second parallel electrical signals.

[0883] Embodiment 3: The apparatus of embodiment 1 in which the first serializers/deserializers module is configured to perform signal conditioning on the electrical signals, the signal conditioning comprising at least one of (i) clock and data recovery, or (ii) signal equalization.

[0884] Embodiment 4: The apparatus of embodiment 1 in which the photonic integrated circuit comprises at least one of waveguides, photodetectors, vertical grating couplers, or fiber edge couplers.

[0885] Embodiment 5: The apparatus of embodiment 1 in which each of the first serializers/deserializers module and the second serializers/deserializers module comprises at least one of (i) a multiplexer, (ii) a demultiplexer, (iii) a serial data port, (iv) a parallel data bus, (v) an equalizer, (vi) a clock recovery unit, or (vii) a data recovery unit.

[0886] Embodiment 6: The apparatus of embodiment 1, comprising a bus processing module configured to process signals transmitted between the first serializers/deserializers module and the second serializers/deserializers module, in which the bus processing module performs at least one of switching of data, re-shuffling data, or coding of data.

[0887] Embodiment 7: The apparatus of embodiment 6, in which the first serializers/deserializers module comprises a first serializer/deserializer unit and a second serializer/deserializer unit that are configured to serially interface with N electrical signals at a first serial interface; [0888] the second serializers/deserializers module comprises a third serializer/deserializer unit that is configured to serially interface with M electrical signals at a second serial interface, M and N are positive integers, and N is different from M; and [0889] the bus processing module is configured to route signals among the first, second, and third serializer/deserializer units to enable the N serial electrical signals at the first serial interface to be mapped to the M serial electrical signals at the second serial interface.

[0890] Embodiment 8: The apparatus of embodiment 7 in which the first serializer/deserializer unit and the second serializer/deserializer unit are configured to serially interface with N lanes of P Gbps electrical signals, and [0891] the third serializer/deserializer unit is configured to serially interface with N/Q lanes of P*Q Gbps electrical signals, and P and Q are positive numbers.

[0892] Embodiment 9: The apparatus of embodiment 6, in which the first serializers/deserializers module comprises a first serializer/deserializer unit and a second serializer/deserializer unit that are configured to serially interface with N lanes of $T \times N / (N - k)$ Gbps electrical signals at a first serial interface, N and k are positive integers, and T is a real value; [0893] the second serializers/deserializers module comprises a third serializer/deserializer unit that is configured to serially interface with N lanes of T Gbps electrical signals at a second serial interface; and [0894] the bus processing module is configured to route signals among the first, second, and third serializer/deserializer units to enable $N - k$ out of the N lanes serially interfacing the first and second serializer/deserializer units to be mapped to the N lanes of T Gbps electrical signals at the second serial interface.

[0895] Embodiment 10: The apparatus of embodiment 6, in which the first serializers/deserializers module comprises a first serializer/deserializer unit and a second serializer/deserializer unit that are configured to serially interface with N lanes of $T \times N / (N - k)$ Gbps electrical signals at a first serial interface, N and k are positive integers, and T is a real value; [0896] the second serializers/deserializers module comprises a third serializer/deserializer unit that is configured to serially interface with N/M lanes of $M \times T$ Gbps electrical signals at a second serial interface, M is different from N ; and [0897] the bus processing module is configured to route signals among the first, second, and third serializer/deserializer units to enable $N - k$ out of the N lanes serially interfacing the first and second serializer/deserializer units to be mapped to the N/M lanes of $M \times T$ Gbps electrical signals at the second serial interface.

[0898] Embodiment 11: The apparatus of embodiment 1 in which the photonic integrated circuit configured to generate N serial electrical signals, and the second serializers/deserializers module is configured to generate M serial electrical signals based on the plurality of sets of first parallel electrical signals, M and N are positive integers, and M is different from N .

[0899] Embodiment 12: The apparatus of embodiment 11 in which the photonic integrated circuit is configured to generate N lanes of P Gbps serial electrical signals, the second serializers/deserializers module is configured to generate N/Q lanes of $P \times Q$ Gbps serial electrical signals, and P and Q are positive numbers.

[0900] Embodiment 13: The apparatus of embodiment 1 in which the first serial electrical signals are modulated according to a first modulation format, and the second serial electrical signals are modulated according to a second modulation format that is different from the first modulation format.

[0901] Embodiment 14: The apparatus of embodiment 1, further comprising an optical output port; [0902] the second serializers/deserializers module is configured to receive a plurality of third serial electrical signals, and generate a plurality of sets of third parallel electrical signals based on the plurality of third serial electrical signals, in which each set of third parallel electrical signals is generated based on a corresponding third serial electrical signal; [0903] the first serializers/deserializers module is configured to generate a plurality of fourth serial electrical signals based on the plurality of sets of third parallel signals, in which each fourth serial electrical signal is generated based on a corresponding set of fourth parallel electrical signals; [0904] the photonic integrated circuit is configured to generate a plurality of channels of second optical signals based on the plurality of fourth serial electrical signals; and [0905] the optical output port is configured to output the plurality of channels of second optical signals.

[0906] Embodiment 15: The apparatus of embodiment 14, comprising: [0907] at least one of a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, or an application specific integrated circuit (ASIC) that is configured to generate a plurality of sets of fourth parallel electrical signals; and [0908] a third serializers/deserializers module comprising multiple serializer units and deserializer units, in which the third serializers/deserializers module is configured to generate the third serial electrical signals based on

the sets of fourth parallel electrical signals.

[0909] Embodiment 16: The apparatus of embodiment 15 in which the third serializers/deserializers module is configured to generate M serial electrical signals based on the sets of fourth parallel electrical signals, the first serializers/deserializers module is configured to generate N serial electrical signals based on the sets of third parallel signals, M and N are positive integers, and N is different from M.

[0910] Embodiment 17: The apparatus of embodiment 16 in which the third serializers/deserializers module is configured to generate N/Q lanes of P*Q Gbps serial electrical signals, the first serializers/deserializers module is configured to generate N lanes of P Gbps serial electrical signals, and P and Q are positive numbers.

[0911] Embodiment 18: The apparatus of embodiment 14 in which the third serial electrical signals are modulated according to a first modulation format, and the fourth serial electrical signals are modulated according to a second modulation format that is different from the first modulation format.

[0912] Embodiment 19: The apparatus of embodiment 14 in which the second serializers/deserializers module is configured to perform signal conditioning on the electrical signals, the signal conditioning comprising at least one of (i) clock and data recovery, or (ii) signal equalization.

[0913] Embodiment 20: The apparatus of embodiment 14 in which the photonic integrated circuit comprises at least one of waveguides, vertical grating couplers, fiber edge couplers, modulators, optical power splitters, or optical polarization splitters.

[0914] Embodiment 21: The apparatus of embodiment 14 in which the first serializers/deserializers module comprises an interpolator or an electrical phase adjustment element that aligns serial electrical output signals with respective optical pulse trains that power respective optical modulators for modulating output optical signals based on the serial electric output signals.

[0915] Embodiment 22: The apparatus of embodiment 14, comprising a bus processing module configured to process signals transmitted between the first serializers/deserializers module and the second serializers/deserializers module, in which the bus processing module performs at least one of switching of data, re-shuffling data, or coding of data.

[0916] Embodiment 23: The apparatus of embodiment 1 in which the optical interconnect module comprises a first circuit board or a first substrate, [0917] wherein the photonic integrated circuit, the first serializers/deserializers module, and the second serializers/deserializers module are mounted on the first circuit board or the first substrate.

[0918] Embodiment 24: The apparatus of embodiment 23 in which the optical interconnect module comprises first electrical terminals arranged on the first circuit board or the first substrate, and the first electrical terminals are configured to mate with second electrical terminals arranged on a second circuit board or a second substrate.

[0919] Embodiment 25: The apparatus of embodiment 24 in which the first electrical terminals are removably coupled to the second electrical terminals of the second circuit board or the second substrate.

[0920] Embodiment 26: The apparatus of embodiment 25 in which at least one of the first electrical terminals or the second electrical terminals comprise at least one of spring loaded connectors, compression interposers, or land-grid arrays.

[0921] Embodiment 27: The apparatus of embodiment 24 in which the first electrical terminals are arranged on a second side of the first circuit board or the first substrate, the photonic integrated circuit is also mounted on the second side of the first circuit board or the first substrate, and at least a portion of the photonic integrated circuit is positioned between the first circuit board (or the first substrate) and the second circuit board (or the second substrate) when the first electrical terminals of the first circuit board mate (or the first substrate) with the second electrical terminals of the second circuit board (or the second substrate).

[0922] Embodiment 28: The apparatus of embodiment 24 in which the second serializers/deserializers module is electrically coupled to the first circuit board or the first substrate through third electrical terminals that have a first minimum spacing between the terminals, the first electrical terminals arranged on the first circuit board or the first substrate have a second minimum spacing between terminals, and the second minimum spacing is larger than the first minimum spacing.

[0923] Embodiment 29: The apparatus of embodiment 28 in which the second minimum spacing is at least twice the first minimum spacing.

[0924] Embodiment 30: The apparatus of embodiment 28 in which the first minimum spacing is less than or equal to 200 μm .

[0925] Embodiment 31: The apparatus of embodiment 28 in which the first minimum spacing is less than or equal to 100 μm .

[0926] Embodiment 32: The apparatus of embodiment 28 in which the first minimum spacing is less than or equal to 50 μm .

[0927] Embodiment 33: The apparatus of embodiment 1 in which the optical input port comprises a first optical connector configured to mate with a second optical connector coupled to an optical fiber cable that provides a plurality of optical paths.

[0928] Embodiment 34: The apparatus of embodiment 33 in which each optical path is provided by a core of an optical fiber in the optical fiber cable.

[0929] Embodiment 35: The apparatus of embodiment 33 in which the first optical connector is configured to couple optical signals propagating along at least two optical paths to the photonic integrated circuit.

[0930] Embodiment 36: The apparatus of embodiment 35 in which the photonic integrated circuit is configured to process the at least two channels of optical signals and generate at least two first serial electrical signals.

[0931] Embodiment 37: The apparatus of embodiment 36 in which the first serializers/deserializers module is configured to convert the at least two first serial electrical signals into at least two sets of parallel electrical signals, and each set of parallel electrical signals comprises at least two parallel electrical signals.

[0932] Embodiment 38: The apparatus of embodiment 37 in which each set of parallel electrical signals comprises at least four parallel electrical signals.

[0933] Embodiment 39: The apparatus of embodiment 38 in which each set of parallel electrical signals comprises at least eight parallel electrical signals.

[0934] Embodiment 40: The apparatus of embodiment 33 in which the first optical connector is configured to couple optical signals propagating along at least four optical paths to the photonic integrated circuit.

[0935] Embodiment 41: The apparatus of embodiment 33 in which the first optical connector is configured to couple optical signals propagating along at least eight optical paths to the photonic integrated circuit.

[0936] Embodiment 42: The apparatus of embodiment 33 in which the optical fiber cable comprises at least 10 cores of optical fibers, and the first optical connector is configured to couple at least 10 channels of optical signals to the photonic integrated circuit.

[0937] Embodiment 43: The apparatus of embodiment 42 in which the photonic integrated circuit is configured to process the at least 10 channels of optical signals and generate at least 10 first serial electrical signals.

[0938] Embodiment 44: The apparatus of embodiment 43 in which the first serializers/deserializers module is configured to convert the at least 10 first serial electrical signals into at least 10 sets of parallel electrical signals, and each set of parallel electrical signals comprises at least two parallel electrical signals.

[0939] Embodiment 45: The apparatus of embodiment 44 in which each set of parallel electrical

signals comprises at least four parallel electrical signals.

[0940] Embodiment 46: The apparatus of embodiment 45 in which each set of parallel electrical signals comprises at least eight parallel electrical signals.

[0941] Embodiment 47: The apparatus of embodiment 46 in which each set of parallel electrical signals comprises at least 32 parallel electrical signals.

[0942] Embodiment 48: The apparatus of embodiment 47 in which each set of parallel electrical signals comprises at least 64 parallel electrical signals.

[0943] Embodiment 49: The apparatus of embodiment 33 in which the optical fiber cable comprises at least 100 cores of optical fibers, and the first optical connector is configured to couple at least 100 channels of optical signals to the photonic integrated circuit.

[0944] Embodiment 50: The apparatus of embodiment 49 in which the photonic integrated circuit is configured to process the at least 100 channels of optical signals and generate at least 100 first serial electrical signals.

[0945] Embodiment 51: The apparatus of embodiment 50 in which the first serializers/deserializers module is configured to convert the at least 100 first serial electrical signals into at least 100 sets of parallel electrical signals, and each set of parallel electrical signals comprises at least two parallel electrical signals.

[0946] Embodiment 52: The apparatus of embodiment 51 in which each set of parallel electrical signals comprises at least four parallel electrical signals.

[0947] Embodiment 53: The apparatus of embodiment 52 in which each set of parallel electrical signals comprises at least eight parallel electrical signals.

[0948] Embodiment 54: The apparatus of embodiment 53 in which each set of parallel electrical signals comprises at least 32 parallel electrical signals.

[0949] Embodiment 55: The apparatus of embodiment 54 in which each set of parallel electrical signals comprises at least 64 parallel electrical signals.

[0950] Embodiment 56: The apparatus of embodiment 49 in which the optical fiber cable comprises at least 500 cores of optical fibers, and the first optical connector is configured to couple at least 500 channels of optical signals to the photonic integrated circuit.

[0951] Embodiment 57: The apparatus of embodiment 56 in which the first serializers/deserializers module is configured to convert the at least 500 first serial electrical signals into at least 500 sets of parallel electrical signals, and each set of parallel electrical signals comprises at least two parallel electrical signals.

[0952] Embodiment 58: The apparatus of embodiment 56 in which the optical fiber cable comprises at least 1000 cores of optical fibers, and the first optical connector is configured to couple at least 1000 channels of optical signals to the photonic integrated circuit.

[0953] Embodiment 59: The apparatus of embodiment 58 in which the first serializers/deserializers module is configured to convert the at least 1000 first serial electrical signals into at least 1000 sets of parallel electrical signals, and each set of parallel electrical signals comprises at least two parallel electrical signals.

[0954] Embodiment 60: The apparatus of embodiment 1 in which the optical interconnect module comprises a first circuit board or a first substrate that has a first side and a second side, the second serializers/deserializers module has a first side and a second side, the optical interconnect module comprises first electrical terminals arranged on the first side of the first circuit board or the first substrate, the optical interconnect module comprises second electrical terminals arranged on the second side of the second serializers/deserializers module, the second electrical terminals are electrically coupled to the first electrical terminals, and [0955] the optical interconnect module comprises third electrical terminals arranged on the second side of the first circuit board or the first substrate, the third electrical terminals are configured to be electrically coupled to fourth electrical terminals that are arranged on a second circuit board or a second substrate.

[0956] Embodiment 61: The apparatus of embodiment 60 in which the photonic integrated circuit

has a first side and a second side, the photonic integrated circuit comprises fifth electrical terminals arranged on the first side of the photonic integrated circuit, and the fifth electrical terminals are electrically coupled to sixth electrical terminals of the first serializers/deserializers module.

[0957] Embodiment 62: The apparatus of embodiment 61 in which the fifth electrical terminals arranged on the first side of the photonic integrated circuit are directly soldered to the sixth electrical terminals of the first serializers/deserializers module.

[0958] Embodiment 63: The apparatus of embodiment 61 in which the first serializers/deserializers module and the photonic integrated circuit are mounted on opposite sides of the first circuit board or the first substrate, and the fifth electrical terminals arranged on the first side of the photonic integrated circuit are electrically coupled to the sixth electrical terminals of the first serializers/deserializers module through electrical connectors that pass through the first circuit board or the first substrate.

[0959] Embodiment 64: The apparatus of embodiment 61 in which the optical input port comprises a first optical connector configured to mate with a second optical connector that is coupled to an optical fiber cable that comprises a plurality of optical fibers, and the first optical connector is optically coupled to the second side of the photonic integrated circuit.

[0960] Embodiment 65: The apparatus of embodiment 64 in which the second side of the first circuit board or the first substrate includes an opening, and [0961] at least one of the first optical connector, the second optical connector, or the optical cable passes the opening of the second side of the first circuit board or the first substrate.

[0962] Embodiment 66: The apparatus of embodiment 65, comprising: [0963] the second circuit board or the second substrate, [0964] a second integrated circuit configured to convert the plurality of channels of second serial electrical signals to a plurality of sets of second parallel electrical signals, in which each channel of second serial electrical signal is converted to a set of second parallel electrical signals, the second integrated circuit comprises a deserializers module or a third serializers/deserializers module, and [0965] a third integrated circuit configured to process the sets of second parallel electrical signals, [0966] wherein the third integrated circuit is mounted on the second circuit board or the second substrate, and the second integrated circuit is mounted on the second circuit board or the second substrate or embedded in the third integrated circuit.

[0967] Embodiment 67: The apparatus of embodiment 66 in which the third integrated circuit comprises at least one of a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, or an application specific integrated circuit (ASIC).

[0968] Embodiment 68: The apparatus of embodiment 66 in which the second circuit board or the second substrate defines an opening, and [0969] at least one of the first optical connector, the second optical connector, or the optical cable passes the opening in the second circuit board or the second substrate.

[0970] Embodiment 69: The apparatus of embodiment 1, comprising: [0971] a second circuit board or a second substrate; [0972] a second integrated circuit configured to convert the plurality of channels of second serial electrical signals to a plurality of sets of second parallel signals, in which each channel of second serial electrical signal is converted to a set of second parallel electrical signals, the second integrated circuit comprises a deserializers module or a third serializers/deserializers module, and [0973] a third integrated circuit configured to process the plurality of sets of second parallel electrical signals, [0974] wherein the third integrated circuit is mounted on the second circuit board or the second substrate, and the second integrated circuit is mounted on the second circuit board or the second substrate or embedded in the third integrated circuit.

[0975] Embodiment 70: The apparatus of embodiment 69 in which the third integrated circuit comprises at least one of a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital

signal processor, a microcontroller, or an application specific integrated circuit (ASIC).

[0976] Embodiment 71: The apparatus of embodiment 1 in which the optical interconnect module comprises a first circuit board or a first substrate that has a first side and a second side, the first serializers/deserializers module is mounted on the first side of the first circuit board or the first substrate, and the photonic integrated circuit is mounted in a recess at the first side of the first circuit board or the first substrate.

[0977] Embodiment 72: The apparatus of embodiment 1 in which the optical input port comprises a first optical connector configured to mate with a second optical connector that is coupled to an optical fiber cable that comprises a plurality of optical fibers, [0978] the photonic integrated circuit has a first side and a second side, [0979] the first serializers/deserializers module is electrically coupled to the first side of the photonic integrated circuit, and [0980] the first optical connector is optically coupled to the first side of the photonic integrated circuit.

[0981] Embodiment 73: The apparatus of embodiment 1 in which the optical interconnect module comprises a first circuit board or a first substrate that has a first side and a second side, [0982] the first serializers/deserializers module has first electrical terminals that are electrically coupled to second electrical terminals arranged on the first side of the first circuit board or the first substrate, [0983] the photonic integrated circuit has a first side and a second side, the photonic integrated circuit has third electrical terminals arranged on the first side, and the third electrical terminals are electrically coupled to fourth electrical terminals arranged on the second side of the first circuit board or the first substrate, [0984] the second electrical terminals are electrically coupled to the fourth electrical terminals by electrical connectors that pass through the first circuit board or the first substrate, [0985] the optical input port comprises a first optical connector configured to mate with a second optical connector that is coupled to an optical fiber cable that comprises a plurality of optical fibers, and [0986] the first optical connector is optically coupled to the first side of the photonic integrated circuit.

[0987] Embodiment 74: The apparatus of embodiment 1 in which the optical input port comprises a first optical connector configured to mate with a second optical connector that is coupled to an optical fiber cable that comprises a plurality of optical fibers, [0988] the photonic integrated circuit has a first side and a second side, [0989] the first serializers/deserializers module is electrically coupled to the first side of the photonic integrated circuit, and the first optical connector is optically coupled to the second side of the photonic integrated circuit.

[0990] Embodiment 75: The apparatus of embodiment 1 in which the optical interconnect module comprises a first circuit board or a first substrate that has a first side and a second side, [0991] wherein the photonic integrated circuit has a first side and a second side, the photonic integrated circuit includes first electrical terminals arranged on the second side, the first electrical terminals are electrically coupled to second electrical terminals arranged on the first side of the first circuit board or the first substrate, [0992] the first serializers/deserializers module is mounted on the first side of the first circuit board or the first substrate, and [0993] the optical input port is optically coupled to the second side of the photonic integrated circuit.

[0994] Embodiment 76: The apparatus of embodiment 75 comprising third electrical terminals that are arranged on the second side of the first circuit board or the first substrate, in which the third electrical terminals are configured to be mated with fourth electrical terminals arranged on a second circuit board or a second substrate.

[0995] Embodiment 77: The apparatus of embodiment 76 in which the third electrical terminals are removably coupled to the fourth electrical terminals, and the third electrical terminals are connected to the fourth electrical terminals without using solder.

[0996] Embodiment 78: The apparatus of embodiment 1 in which the optical interconnect module comprises a first circuit board or a first substrate, the photonic integrated circuit has a first footprint on the first circuit board or the first substrate, the first serializers/deserializers module has a second footprint on the first circuit board or the first substrate, and the second footprint overlaps the first

footprint.

[0997] Embodiment 79: The apparatus of embodiment 78 in which the first footprint is completely within the second footprint.

[0998] Embodiment 80: The apparatus of embodiment 78 in which a portion of the first footprint does not overlap the second footprint.

[0999] Embodiment 81: The apparatus of embodiment 1 in which the first optical signals provided to the photonic integrated circuit have a total bit rate of at least 1 Tbps.

[1000] Embodiment 82: The apparatus of embodiment 1 in which the first optical signals provided to the photonic integrated circuit have a total bit rate of at least 10 Tbps.

[1001] Embodiment 83: The apparatus of embodiment 1 in which the photonic integrated circuit comprises a photo detector, the optical interconnect module comprises at least one of a driver or a transimpedance amplifier, the driver is configured to drive an optical modulator, and the transimpedance amplifier is configured to amplify a signal output from the photo detector.

[1002] Embodiment 84: The apparatus of embodiment 1 in which the optical interconnect module comprises a first circuit board or a first substrate; [1003] the photonic integrated circuit, the first serializers/deserializers module, and the second serializers/deserializers module are mounted on the first circuit board or the first substrate; [1004] the photonic integrated circuit has a top surface, a bottom surface, a first side surface extending from a first edge of the top surface to a first edge of the bottom surface, and a second side surface extending from a second edge of the top surface to a second edge of the bottom surface; [1005] a first portion of the first serializers/deserializers module is disposed on the first circuit board or the first substrate closer to the first side surface of the photonic integrated circuit than the second side surface of the photonic integrated circuit; and [1006] a second portion of the first serializer/deserializers module is disposed on the first circuit board or the first substrate closer to the second side surface of the photonic integrated circuit than the first side surface of the photonic integrated circuit.

[1007] Embodiment 85: The apparatus of embodiment 84 in which the photonic integrated circuit comprises first electrical terminals arranged on the bottom side, the first electrical terminals are electrically coupled to second electrical terminals arranged on the first circuit board or the first substrate, the first circuit board or the first substrate defines an opening, the optical input port passes the opening in the first circuit board or the first substrate and is optical coupled to the bottom side of the photonic integrated circuit.

[1008] Embodiment 86: The apparatus of embodiment 84 in which a first subset of the second serializer/deserializers is disposed on the first circuit board or the first substrate in a vicinity of the first portion of the first serializers/deserializers module, and a second portion of the second serializers/deserializers module is disposed on the first circuit board or the first substrate in a vicinity of the second portion of the first serializers/deserializers module.

[1009] Embodiment 87: The apparatus of embodiment 86 in which the first portion of the first serializers/deserializers module is disposed between the photonic integrated circuit and the first portion of the second serializers/deserializers module, and the second portion of the first serializers/deserializers module is disposed between the photonic integrated circuit and the second portion of the second serializers/deserializers module.

[1010] Embodiment 88: The apparatus of embodiment 86 in which the first circuit board or the first substrate comprises a top surface and a bottom surface, [1011] the second serializers/deserializers module is mounted on the top surface of the first circuit board or the first substrate, [1012] the optical interconnect module comprises first electrical terminals arranged on the top surface of the first circuit board or the first substrate and second electrical terminals arranged on the bottom surface of the first circuit board or the first substrate, the first electrical terminals have a first spacing between the terminals, the second electrical terminals have a second spacing between the terminals, the first spacing is smaller than the second spacing, [1013] the first electrical terminals are configured to electrically couple to third electrical terminals arranged on the second

serializers/deserializers module, and [1014] the second electrical terminals are configured to electrically couple to fourth electrical terminals arranged on a second circuit board or a second substrate.

[1015] Embodiment 89: The apparatus of embodiment 88 in which the second electrical terminals are removably coupled to the fourth electrical terminals, and at least one of the second electrical terminals or the fourth electrical terminals comprise at least one of spring loaded connectors, compression interposers, or land-grid arrays.

[1016] Embodiment 90: The apparatus of embodiment 1 in which the optical interconnect module comprises a first circuit board or a first substrate and a plurality of driver/transimpedance amplifiers; [1017] the photonic integrated circuit, the first serializers/deserializers module, the second serializers/deserializers module, the first driver/transimpedance amplifier, and the second driver/transimpedance amplifier are mounted on the first circuit board or the first substrate; [1018] the photonic integrated circuit has a top surface, a bottom surface, a first side surface extending from a first edge of the top surface to a first edge of the bottom surface, and a second side surface extending from a second edge of the top surface to a second edge of the bottom surface; [1019] a first subset of the driver/transimpedance amplifiers are disposed on the first circuit board or the first substrate closer to the first side surface of the photonic integrated circuit than the second side surface of the photonic integrated circuit; and [1020] a second subset of the driver/transimpedance amplifiers are disposed on the first circuit board or the first substrate closer to the second side surface of the photonic integrated circuit than the first side surface of the photonic integrated circuit.

[1021] Embodiment 91: The apparatus of embodiment 90 in which the photonic integrated circuit comprises first electrical terminals arranged on the bottom side, the first electrical terminals are electrically coupled to second electrical terminals arranged on the first circuit board or the first substrate, the first circuit board or the first substrate defines an opening, and the optical input port passes the opening in the first circuit board or the first substrate and is optical coupled to the bottom side of the photonic integrated circuit.

[1022] Embodiment 92: The apparatus of embodiment 90 in which a first subset of the first serializer/deserializers are disposed on the first circuit board or the first substrate in a vicinity of [1023] the first subset of the driver/transimpedance amplifiers, the first subset of the driver/transimpedance amplifiers are configured to amplify a first set of electrical signals from the photonic integrated circuit and drive a first set of optical modulators, [1024] a second subset of the first serializer/deserializers are disposed on the first circuit board or the first substrate in a vicinity of the second subset of the driver/transimpedance amplifiers, and [1025] the second subset of the driver/transimpedance amplifiers are configured to amplify a second set of electrical signals from the photonic integrated circuit and drive a second set of optical modulators.

[1026] Embodiment 93: The apparatus of embodiment 92 in which the first subset of the driver/transimpedance amplifiers are disposed between the photonic integrated circuit and the first subset of the first serializer/deserializers, and the second subset of the driver/transimpedance amplifiers are disposed between the photonic integrated circuit and the second subset of the first serializer/deserializers.

[1027] Embodiment 94: The apparatus of embodiment 92 in which a first subset of the second serializer/deserializers are disposed on the first circuit board or the first substrate in a vicinity of the first subset of the first serializer/deserializers, and a second subset of the second serializer/deserializers are disposed on the first circuit board or the first substrate in a vicinity of the second subset of the first serializer/deserializers.

[1028] Embodiment 95: The apparatus of embodiment 1 in which the second serializers/deserializers module is configured to receive a plurality of third serial electrical signals, and generate a plurality of sets of second parallel electrical signals, [1029] wherein the first serializers/deserializers module is configured to generate a plurality of fourth serial electrical

signals based on the plurality of sets of second parallel electrical signals, and [1030] wherein the photonic integrated circuit is configured to generate second optical signals based on the plurality of fourth serial electrical signals.

[1031] Embodiment 96: The apparatus of embodiment 95 in which the photonic integrated circuit comprises at least one of waveguides, vertical grating couplers, fiber edge couplers, modulators, optical power splitters, or optical polarization splitters.

[1032] Embodiment 97: The apparatus of embodiment 95 in which the first serializers/deserializers module comprises an interpolator or an electrical phase adjustment element that aligns multiple serial output signals that are transmitted to the photonic integrated circuit.

[1033] Embodiment 98: An apparatus comprising: [1034] an optical interconnect module comprising: [1035] an optical input port configured to receive an optical signal; [1036] a photonic integrated circuit configured to generate a first serial electrical signal based on the received optical signal; [1037] a first serializer/deserializer configured to generate a set of first parallel electrical signals based on the first serial electrical signals, and condition the electrical signals; and [1038] a second serializer/deserializer configured to generate a second serial electrical signal based on the set of first parallel electrical signals.

[1039] Embodiment 99: An apparatus comprising: [1040] an optical interconnect module comprising: [1041] an optical input port configured to receive a plurality of channels of optical signals; [1042] a photonic integrated circuit configured to process the optical signals and generate a plurality of first serial electrical signals, in which each first serial electrical signal is generated based on one of the channels of optical signals; [1043] a first deserializer configured to convert the plurality of first serial electrical signals to a plurality of sets of first parallel electrical signals, and condition the electrical signals, in which each first serial electrical signal is converted to a corresponding set of first parallel electrical signals; and [1044] a first serializer configured to convert the plurality of sets of first parallel electrical signals to a plurality of second serial electrical signals, in which each set of first parallel electrical signals is converted to a corresponding second serial electrical signal.

[1045] Embodiment 100: The apparatus of embodiment 99, comprising: [1046] a second deserializer configured to generate a plurality of sets of second parallel electrical signals based on the plurality of second serial electrical signals, in which each set of second parallel electrical signals is generated based on a corresponding second serial electrical signal; and [1047] at least one of a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, or an application specific integrated circuit (ASIC) that is configured to process the plurality of sets of second parallel electrical signals.

[1048] Embodiment 101: The apparatus of embodiment 99 in which the first deserializer is configured to perform signal conditioning on the electrical signals, the signal conditioning comprising at least one of (i) clock and data recovery, or (ii) signal equalization.

[1049] Embodiment 102: The apparatus of embodiment 99 in which the photonic integrated circuit comprises at least one of waveguides, photodetectors, vertical grating couplers, or fiber edge couplers.

[1050] Embodiment 103: The apparatus of embodiment 99 in which each of the first deserializer and the first serializer comprises at least one of (i) a multiplexer, (ii) a demultiplexer, (iii) a serial data port, (iv) a parallel data bus, (v) an equalizer, (vi) a clock recovery unit, or (vii) a data recovery unit.

[1051] Embodiment 104: The apparatus of embodiment 99, comprising a bus processing module configured to process signals transmitted from the first deserializer to the first serializer, in which the bus processing module performs at least one of switching of data, re-shuffling data, or coding of data.

[1052] Embodiment 105: The apparatus of embodiment 99 in which the photonic integrated circuit

configured to generate N serial electrical signals, and the first serializer is configured to generate M serial electrical signals based on the plurality of sets of first parallel electrical signals, M and N are positive integers, and M is different from N.

[1053] Embodiment 106: The apparatus of embodiment 105 in which the photonic integrated circuit is configured to generate N lanes of P Gbps serial electrical signals, the second serializers/deserializers module is configured to generate N/Q lanes of P*Q Gbps serial electrical signals, and P and Q are positive numbers.

[1054] Embodiment 107: The apparatus of embodiment 99 in which the first serial electrical signals are modulated according to a first modulation protocol, and the second serial electrical signals are modulated according to a second modulation protocol that is different from the first modulation protocol.

[1055] Embodiment 108: The apparatus of embodiment 99 in which the optical interconnect module comprises a first circuit board or a first substrate, [1056] wherein the photonic integrated circuit, the first deserializer, and the first serializer are mounted on the first circuit board or the first substrate.

[1057] Embodiment 109: The apparatus of embodiment 108 in which the optical interconnect module comprises first electrical terminals arranged on the first circuit board or the first substrate, and the first electrical terminals are configured to mate with second electrical terminals arranged on a second circuit board or a second substrate.

[1058] Embodiment 110: The apparatus of embodiment 109 in which the first electrical terminals are removably coupled to the second electrical terminals of the second circuit board or the second substrate.

[1059] Embodiment 111: The apparatus of embodiment 109 in which at least one of the first electrical terminals or the second electrical terminals comprise at least one of spring loaded connectors, compression interposers, or land-grid arrays.

[1060] Embodiment 112: The apparatus of embodiment 109 in which the first electrical terminals are arranged on a second side of the first circuit board or the first substrate, the photonic integrated circuit is also mounted on the second side of the first circuit board or the first substrate, and at least a portion of the photonic integrated circuit is positioned between the first circuit board or the first substrate and the second circuit board or the second substrate when the first electrical terminals of the first circuit board or the first substrate mate with the second electrical terminals of the second circuit board or the second substrate.

[1061] Embodiment 113: The apparatus of embodiment 109 in which the first serializer is electrically coupled to the first circuit board or the first substrate through third electrical terminals that have a first minimum spacing between the terminals, the first electrical terminals arranged on the first circuit board or the first substrate have a second minimum spacing between terminals, and the second minimum spacing is larger than the first minimum spacing.

[1062] Embodiment 114: The apparatus of embodiment 113 in which the second minimum spacing is at least twice the first minimum spacing.

[1063] Embodiment 115: The apparatus of embodiment 113 in which the first minimum spacing is less than or equal to 200 μm .

[1064] Embodiment 116: The apparatus of embodiment 113 in which the first minimum spacing is less than or equal to 100 μm .

[1065] Embodiment 117: The apparatus of embodiment 113 in which the first minimum spacing is less than or equal to 50 μm .

[1066] Embodiment 118: The apparatus of embodiment 99 in which the optical input port comprises a first optical connector configured to mate with a second optical connector coupled to an optical fiber cable that provides a plurality of optical paths.

[1067] Embodiment 119: The apparatus of embodiment 118 in which each optical path is provided by a core of an optical fiber in the optical fiber cable.

[1068] Embodiment 120: The apparatus of embodiment 118 in which the first optical connector is configured to couple optical signals propagating along at least two optical paths to the photonic integrated circuit.

[1069] Embodiment 121: The apparatus of embodiment 120 in which the photonic integrated circuit is configured to process the at least two channels of optical signals and generate at least two first serial electrical signals.

[1070] Embodiment 122: The apparatus of embodiment 121 in which the first serializers/deserializers module is configured to convert the at least two first serial electrical signals into at least two sets of parallel electrical signals, and each set of parallel electrical signals comprises at least two parallel electrical signals.

[1071] Embodiment 123: The apparatus of embodiment 122 in which each set of parallel electrical signals comprises at least four parallel electrical signals.

[1072] Embodiment 124: The apparatus of embodiment 123 in which each set of parallel electrical signals comprises at least eight parallel electrical signals.

[1073] Embodiment 125: The apparatus of embodiment 124 in which each set of parallel electrical signals comprises at least 32 parallel electrical signals.

[1074] Embodiment 126: The apparatus of embodiment 125 in which each set of parallel electrical signals comprises at least 64 parallel electrical signals.

[1075] Embodiment 127: The apparatus of embodiment 118 in which the first optical connector is configured to couple optical signals propagating along at least four optical paths to the photonic integrated circuit.

[1076] Embodiment 128: The apparatus of embodiment 118 in which the first optical connector is configured to couple optical signals propagating along at least eight optical paths to the photonic integrated circuit.

[1077] Embodiment 129: The apparatus of embodiment 118 in which the optical fiber cable comprises at least 10 cores of optical fibers, and the first optical connector is configured to couple at least 10 channels of optical signals to the photonic integrated circuit.

[1078] Embodiment 130: The apparatus of embodiment 129 in which the photonic integrated circuit is configured to process the at least 10 channels of optical signals and generate at least 10 first serial electrical signals.

[1079] Embodiment 131: The apparatus of embodiment 130 in which the first serializers/deserializers module is configured to convert the at least 10 first serial electrical signals into at least 10 sets of parallel electrical signals, and each set of parallel electrical signals comprises at least two parallel electrical signals.

[1080] Embodiment 132: The apparatus of embodiment 131 in which each set of parallel electrical signals comprises at least four parallel electrical signals.

[1081] Embodiment 133: The apparatus of embodiment 132 in which each set of parallel electrical signals comprises at least eight parallel electrical signals.

[1082] Embodiment 134: The apparatus of embodiment 118 in which the optical fiber cable comprises at least 100 cores of optical fibers, and the first optical connector is configured to couple at least 100 channels of optical signals to the photonic integrated circuit.

[1083] Embodiment 135: The apparatus of embodiment 134 in which the photonic integrated circuit is configured to process the at least 100 channels of optical signals and generate at least 100 first serial electrical signals.

[1084] Embodiment 136: The apparatus of embodiment 135 in which the first deserializer is configured to convert the at least 100 first serial electrical signals into at least 100 sets of parallel electrical signals, and each set of parallel electrical signals comprises at least two parallel electrical signals.

[1085] Embodiment 137: The apparatus of embodiment 136 in which each set of parallel electrical signals comprises at least four parallel electrical signals.

[1086] Embodiment 138: The apparatus of embodiment 137 in which each set of parallel electrical signals comprises at least eight parallel electrical signals.

[1087] Embodiment 139: The apparatus of embodiment 138 in which each set of parallel electrical signals comprises at least 32 parallel electrical signals.

[1088] Embodiment 140: The apparatus of embodiment 139 in which each set of parallel electrical signals comprises at least 64 parallel electrical signals.

[1089] Embodiment 141: The apparatus of embodiment 134 in which the optical fiber cable comprises at least 500 optical fibers, and the first optical connector is configured to couple at least 500 channels of optical signals to the photonic integrated circuit.

[1090] Embodiment 142: The apparatus of embodiment 141 in which the first deserializer is configured to convert the at least 500 first serial electrical signals into at least 500 sets of parallel electrical signals, and each set of parallel electrical signals comprises at least two parallel electrical signals.

[1091] Embodiment 143: The apparatus of embodiment 141 in which the optical fiber cable comprises at least 1000 optical fibers, and the first optical connector is configured to couple at least 1000 channels of optical signals to the photonic integrated circuit.

[1092] Embodiment 144: The apparatus of embodiment 143 in which the first deserializer is configured to convert the at least 1000 first serial electrical signals into at least 1000 sets of parallel electrical signals, and each set of parallel electrical signals comprises at least two parallel electrical signals.

[1093] Embodiment 145: The apparatus of embodiment 99 in which the optical interconnect module comprises a first circuit board or a first substrate that has a first side and a second side, the first serializer has a first side and a second side, the optical interconnect module comprises first electrical terminals arranged on the first side of the first circuit board or the first substrate, the optical interconnect module comprises second electrical terminals arranged on the second side of the first serializer, the second electrical terminals are electrically coupled to the first electrical terminals, and [1094] the optical interconnect module comprises third electrical terminals arranged on the second side of the first circuit board or the first substrate, the third electrical terminals are configured to be electrically coupled to fourth electrical terminals that are arranged on a second circuit board or a second substrate.

[1095] Embodiment 146: The apparatus of embodiment 145 in which the photonic integrated circuit has a first side and a second side, the photonic integrated circuit comprises fifth electrical terminals arranged on the first side of the photonic integrated circuit, and the fifth electrical terminals are electrically coupled to sixth electrical terminals of the first deserializer.

[1096] Embodiment 147: The apparatus of embodiment 146 in which the fifth electrical terminals arranged on the first side of the photonic integrated circuit are directly soldered to the sixth electrical terminals of the first deserializer.

[1097] Embodiment 148: The apparatus of embodiment 146 in which the first deserializer and the photonic integrated circuit are mounted on opposite sides of the first circuit board or the first substrate, and the fifth electrical terminals arranged on the first side of the photonic integrated circuit are electrically coupled to the sixth electrical terminals of the first deserializer through electrical connectors that pass through the first circuit board or the first substrate.

[1098] Embodiment 149: The apparatus of embodiment 146 in which the optical input port comprises a first optical connector configured to mate with a second optical connector that is coupled to an optical fiber cable that comprises a plurality of optical fibers, and the first optical connector is optically coupled to the second side of the photonic integrated circuit.

[1099] Embodiment 150: The apparatus of embodiment 149 in which the second side of the first circuit board or the first substrate includes an opening, and [1100] at least one of the first optical connector, the second optical connector, or the optical cable passes the opening of the second side of the first circuit board or the first substrate.

[1101] Embodiment 151: The apparatus of embodiment 150, comprising: [1102] the second circuit board or the second substrate, [1103] a second deserializer configured to convert the plurality of channels of second serial electrical signals to a plurality of sets of second parallel electrical signals, in which each channel of second serial electrical signal is converted to a set of second parallel electrical signals, and [1104] a third integrated circuit configured to process the sets of second parallel electrical signals, [1105] wherein the third integrated circuit is mounted on the second circuit board or the second substrate, and the second deserializer is mounted on the second circuit board or the second substrate or embedded in the third integrated circuit.

[1106] Embodiment 152: The apparatus of embodiment 151 in which the third integrated circuit comprises at least one of a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, or an application specific integrated circuit (ASIC).

[1107] Embodiment 153: The apparatus of embodiment 151 in which the second circuit board or the second substrate defines an opening, and [1108] at least one of the first optical connector, the second optical connector, or the optical cable passes the opening in the second circuit board or the second substrate.

[1109] Embodiment 154: The apparatus of embodiment 99, comprising: [1110] a second circuit board or a second substrate; [1111] a second deserializer configured to convert the plurality of channels of second serial electrical signals to a plurality of sets of second parallel signals, in which each channel of second serial electrical signal is converted to a set of second parallel electrical signals, and [1112] a third integrated circuit configured to process the plurality of sets of second parallel electrical signals, [1113] wherein the third integrated circuit is mounted on the second circuit board or the second substrate, and the second deserializer is mounted on the second circuit board or the second substrate or embedded in the third integrated circuit.

[1114] Embodiment 155: The apparatus of embodiment 154 in which the third integrated circuit comprises at least one of a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, or an application specific integrated circuit (ASIC).

[1115] Embodiment 156: The apparatus of embodiment 99 in which the optical interconnect module comprises a first circuit board or a first substrate that has a first side and a second side, the first deserializer is mounted on the first side of the first circuit board or the first substrate, and the photonic integrated circuit is mounted in a recess at the first side of the first circuit board or the first substrate.

[1116] Embodiment 157: The apparatus of embodiment 99 in which the optical input port comprises a first optical connector configured to mate with a second optical connector that is coupled to an optical fiber cable that comprises a plurality of optical fibers, [1117] the photonic integrated circuit has a first side and a second side, [1118] the first deserializer is electrically coupled to the first side of the photonic integrated circuit, and [1119] the first optical connector is optically coupled to the first side of the photonic integrated circuit.

[1120] Embodiment 158: The apparatus of embodiment 99 in which the optical interconnect module comprises a first circuit board or a first substrate that has a first side and a second side, [1121] the first deserializer has first electrical terminals that are electrically coupled to second electrical terminals arranged on the first side of the first circuit board or the first substrate, [1122] the photonic integrated circuit has a first side and a second side, the photonic integrated circuit has third electrical terminals arranged on the first side, and the third electrical terminals are electrically coupled to fourth electrical terminals arranged on the second side of the first circuit board or the first substrate, [1123] the second electrical terminals are electrically coupled to the fourth electrical terminals by electrical connectors that pass through the first circuit board or the first substrate, [1124] the optical input port comprises a first optical connector configured to mate with a second optical connector that is coupled to an optical fiber cable that comprises a plurality of optical fibers,

and [1125] the first optical connector is optically coupled to the first side of the photonic integrated circuit.

[1126] Embodiment 159: The apparatus of embodiment 99 in which the optical input port comprises a first optical connector configured to mate with a second optical connector that is coupled to an optical fiber cable that comprises a plurality of optical fibers, [1127] the photonic integrated circuit has a first side and a second side, [1128] the first deserializer is electrically coupled to the first side of the photonic integrated circuit, and the first optical connector is optically coupled to the second side of the photonic integrated circuit.

[1129] Embodiment 160: The apparatus of embodiment 99 in which the optical interconnect module comprises a first circuit board or a first substrate that has a first side and a second side, [1130] wherein the photonic integrated has a first side and a second side, the photonic integrated circuit includes first electrical terminals arranged on the second side, the first electrical terminals are electrically coupled to second electrical terminals arranged on the first side of the first circuit board or the first substrate, [1131] the first deserializer is mounted on the first side of the first circuit board or the first substrate, and [1132] the optical input port is optically coupled to the second side of the photonic integrated circuit.

[1133] Embodiment 161: The apparatus of embodiment 160, comprising third electrical terminals that are arranged on the second side of the first circuit board or the first substrate, in which the third electrical terminals are configured to be mated with fourth electrical terminals arranged on a second circuit board or a second substrate.

[1134] Embodiment 162: The apparatus of embodiment 161 in which the third electrical terminals are removably coupled to the fourth electrical terminals, and the third electrical terminals are connected to the fourth electrical terminals without using solder.

[1135] Embodiment 163: The apparatus of embodiment 99 in which the optical interconnect module comprises a first circuit board or a first substrate, the photonic integrated circuit has a first footprint on the first circuit board or the first substrate, the first deserializer has a second footprint on the first circuit board or the first substrate, and the second footprint overlaps the first footprint.

[1136] Embodiment 164: The apparatus of embodiment 163 in which the first footprint is completely within the second footprint.

[1137] Embodiment 165: The apparatus of embodiment 163 in which a portion of the first footprint does not overlap the second footprint.

[1138] Embodiment 166: The apparatus of embodiment 99 in which the first optical signals provided to the photonic integrated circuit have a total bit rate of at least 1 Tbps.

[1139] Embodiment 167: The apparatus of embodiment 99 in which the first optical signals provided to the photonic integrated circuit have a total bit rate of at least 10 Tbps.

[1140] Embodiment 168: The apparatus of embodiment 99 in which the photonic integrated circuit comprises a photo detector, the optical interconnect module comprises a transimpedance amplifier, and the transimpedance amplifier is configured to amplify a signal output from the photo detector.

[1141] Embodiment 169: The apparatus of embodiment 99 in which the optical interconnect module comprises a first circuit board or a first substrate; [1142] the photonic integrated circuit, the first deserializer, and the first serializer are mounted on the first circuit board or the first substrate;

[1143] the photonic integrated circuit has a top surface, a bottom surface, a first side surface extending from a first edge of the top surface to a first edge of the bottom surface, and a second side surface extending from a second edge of the top surface to a second edge of the bottom surface; [1144] a first portion of the first deserializer is disposed on the first circuit board or the first substrate closer to the first side surface of the photonic integrated circuit than the second side surface of the photonic integrated circuit; and [1145] a second portion of the first deserializer is disposed on the first circuit board or the first substrate closer to the second side surface of the photonic integrated circuit than the first side surface of the photonic integrated circuit.

[1146] Embodiment 170: The apparatus of embodiment 169 in which the photonic integrated

circuit comprises first electrical terminals arranged on the bottom side, the first electrical terminals are electrically coupled to second electrical terminals arranged on the first circuit board or the first substrate, the first circuit board or the first substrate defines an opening, the optical input port passes the opening in the first circuit board or the first substrate and is optical coupled to the bottom side of the photonic integrated circuit.

[1147] Embodiment 171: The apparatus of embodiment 169 in which a first portion of the first serializer is disposed on the first circuit board or the first substrate in a vicinity of the first portion of the first deserializer, and a second portion of the first serializer is disposed on the first circuit board or the first substrate in a vicinity of the second portion of the first deserializer.

[1148] Embodiment 172: The apparatus of embodiment 171 in which the first portion of the first deserializer is disposed between the photonic integrated circuit and the first portion of the first serializer, and the second portion of the first deserializer is disposed between the photonic integrated circuit and the second portion of the first serializer.

[1149] Embodiment 173: The apparatus of embodiment 171 in which the first circuit board or the first substrate comprises a top surface and a bottom surface, [1150] the first serializer is mounted on the top surface of the first circuit board or the first substrate, [1151] the optical interconnect module comprises first electrical terminals arranged on the top surface of the first circuit board or the first substrate and second electrical terminals arranged on the bottom surface of the first circuit board or the first substrate, the first electrical terminals have a first spacing between the terminals, the second electrical terminals have a second spacing between the terminals, the first spacing is smaller than the second spacing, [1152] the first electrical terminals are configured to electrically couple to third electrical terminals arranged on the first serializer, and [1153] the second electrical terminals are configured to electrically couple to fourth electrical terminals arranged on a second circuit board or a second substrate.

[1154] Embodiment 174: The apparatus of embodiment 173 in which the second electrical terminals are removably coupled to the fourth electrical terminals, and at least one of the second electrical terminals or the fourth electrical terminals comprise at least one of spring loaded connectors, compression interposers, or land-grid arrays.

[1155] Embodiment 175: The apparatus of embodiment 99 in which the optical interconnect module comprises a first circuit board or a first substrate and a plurality of transimpedance amplifiers; [1156] the photonic integrated circuit, the first deserializer, the first serializer, and the transimpedance amplifiers are mounted on the first circuit board or the first substrate; [1157] the photonic integrated circuit has a top surface, a bottom surface, a first side surface extending from a first edge of the top surface to a first edge of the bottom surface, and a second side surface extending from a second edge of the top surface to a second edge of the bottom surface; [1158] a first subset of the transimpedance amplifiers are disposed on the first circuit board or the first substrate closer to the first side surface of the photonic integrated circuit than the second side surface of the photonic integrated circuit; and [1159] a second subset of the transimpedance amplifiers are disposed on the first circuit board or the first substrate closer to the second side surface of the photonic integrated circuit than the first side surface of the photonic integrated circuit.

[1160] Embodiment 176: The apparatus of embodiment 175 in which the photonic integrated circuit comprises first electrical terminals arranged on the bottom side, the first electrical terminals are electrically coupled to second electrical terminals arranged on the first circuit board or the first substrate, the first circuit board or the first substrate defines an opening, and the optical input port passes the opening in the first circuit board or the first substrate and is optical coupled to the bottom side of the photonic integrated circuit.

[1161] Embodiment 177: The apparatus of embodiment 175 in which a first portion of the first deserializer is disposed on the first circuit board or the first substrate in a vicinity of the first subset of the transimpedance amplifiers, [1162] the first subset of the transimpedance amplifiers are

configured to amplify a first set of electrical signals from the photonic integrated circuit, [1163] a second portion of the first deserializer is disposed on the first circuit board or the first substrate in a vicinity of the second subset of the transimpedance amplifiers, and [1164] the second subset of the transimpedance amplifiers are configured to amplify a second set of electrical signals from the photonic integrated circuit.

[1165] Embodiment 178: The apparatus of embodiment 177 in which the first subset of the transimpedance amplifiers are disposed between the photonic integrated circuit and the first portion of the first deserializer, and the second subset of the transimpedance amplifiers are disposed between the photonic integrated circuit and the second portion of the first deserializer.

[1166] Embodiment 179: The apparatus of embodiment 177 in which a first portion of the first serializer disposed on the first circuit board or the first substrate in a vicinity of the first portion of the first deserializer, and a second portion of the first serializer is disposed on the first circuit board or the first substrate in a vicinity of the second portion of the first deserializer.

[1167] Embodiment 180: The apparatus of embodiment 99 in which the first serializer is configured to receive a plurality of third serial electrical signals, and generate a plurality of sets of second parallel electrical signals, [1168] wherein the first deserializer is configured to generate a plurality of fourth serial electrical signals based on the plurality of sets of second parallel electrical signals, and [1169] wherein the photonic integrated circuit is configured to generate second optical signals based on the plurality of fourth serial electrical signals.

[1170] Embodiment 181: An apparatus comprising: [1171] an optical interconnect module comprising: [1172] an optical input port configured to receive an optical signal; [1173] a photonic integrated circuit configured to generate a first serial electrical signal based on the received optical signal; [1174] a first deserializer configured to generate a set of first parallel electrical signals based on the first serial electrical signals, and condition the electrical signals; and [1175] a first serializer configured to generate a second serial electrical signal based on the set of first parallel electrical signals.

[1176] Embodiment 182: The apparatus of embodiment 181, comprising: [1177] a second deserializer configured to generate a set of second parallel electrical signals based on the second serial electrical signal; and [1178] at least one of a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, or an application specific integrated circuit (ASIC) that is configured to process the set of second parallel electrical signals.

[1179] Embodiment 183: The apparatus of embodiment 181 in which the first deserializer is configured to perform signal conditioning on the electrical signals, the signal conditioning comprising at least one of (i) clock and data recovery, or (ii) signal equalization.

[1180] Embodiment 184: The apparatus of embodiment 181 in which the photonic integrated circuit comprises at least one of waveguides, photodetectors, vertical grating couplers, or fiber edge couplers.

[1181] Embodiment 185: The apparatus of embodiment 181 in which each of the first deserializer and the first serializer comprises at least one of (i) a multiplexer, (ii) a demultiplexer, (iii) a serial data port, (iv) a parallel data bus, (v) an equalizer, (vi) a clock recovery unit, or (vii) a data recovery unit.

[1182] Embodiment 186: The apparatus of embodiment 181, comprising a bus processing module configured to process signals transmitted from the first deserializer to the first serializer, in which the bus processing module performs at least one of switching of data, re-shuffling data, or coding of data.

[1183] Embodiment 187: An apparatus comprising: [1184] an optical interconnect module comprising: [1185] a first deserializer configured to receive a plurality of first serial electrical signals, and generate a plurality of sets of first parallel electrical signals based on the plurality of first serial electrical signals, in which each set of first parallel electrical signal is generated based

on a corresponding first serial electrical signal; [1186] a first serializer configured to generate a plurality of second serial electrical signals based on the plurality of sets of first parallel signals, in which each second serial electrical signal is generated based on a corresponding set of first parallel electrical signals; [1187] a photonic integrated circuit configured to generate a plurality of channels of optical signals based on the plurality of second serial electrical signals; and [1188] an optical output port configured to output the plurality of channels of optical signals.

[1189] Embodiment 188: The apparatus of embodiment 187, further comprising: [1190] at least one of a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, or an application specific integrated circuit (ASIC) that is configured to generate a plurality of sets of second parallel electrical signals; and [1191] a second serializer configured to generate the first serial electrical signals based on the sets of second parallel electrical signals.

[1192] Embodiment 189: An apparatus comprising: [1193] an optical interconnect module comprising: [1194] a first circuit board or a first substrate having a length, a width, and a thickness, in which the length is at least twice the thickness, and the width is at least twice the thickness, the first circuit board or the first substrate has a first surface defined by the length and the width; [1195] an optical input port configured to receive a plurality of channels of optical signals; [1196] a photonic integrated circuit mounted on the first circuit board or the first substrate and configured to generate a plurality of first serial electrical signals based on the received optical signals; and [1197] an array of first electrical terminals arranged on the first surface of the first circuit board or the first substrate, in which the array of first electrical terminals comprises at least two electrical terminals distributed along the length direction and at least two electrical terminals distributed along the width direction, the first electrical terminals are configured to output the first serial electrical signals.

[1198] Embodiment 190: The apparatus of embodiment 189 in which the first electrical terminals extend along directions substantially perpendicular to the first surface of the first circuit board or the first substrate.

[1199] Embodiment 191: The apparatus of embodiment 189 in which the first electrical terminals comprise at least one of spring loaded connectors, compression interposers, or land-grid arrays.

[1200] Embodiment 192: The apparatus of embodiment 189, comprising: [1201] a second circuit board or a second substrate; [1202] a deserializer or a serializer/deserializer configured to generate a plurality of sets of parallel electrical signals based on the first serial electrical signals, in which each set of parallel electrical signals is generated based on a corresponding first serial electrical signal; and [1203] a second integrated circuit mounted on the second circuit board or the second substrate and configured to process the plurality of sets of parallel electrical signals.

[1204] Embodiment 193: The apparatus of embodiment 192 in which the second integrated circuit comprises at least one of a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, or an application specific integrated circuit (ASIC).

[1205] Embodiment 194: The apparatus of embodiment 192 in which the deserializer or the serializer/deserializer is embedded in the second integrated circuit.

[1206] Embodiment 195: The apparatus of embodiment 189 in which the photonic integrated circuit is mounted on the first surface of the first circuit board or the first substrate.

[1207] Embodiment 196: The apparatus of embodiment 189 in which the first circuit board or the first substrate has a second surface defined by the length and the width, the second surface is spaced apart from the first surface by the thickness; [1208] the photonic integrated circuit is mounted on the second surface of the first circuit board or the first substrate; [1209] the photonic integrated circuit comprise second electrical terminals that are electrically coupled to the first electrical terminals through electrical connectors that pass through the first circuit board or the first substrate in the thickness direction.

[1210] Embodiment 197: The apparatus of embodiment 189 in which the optical interconnect module comprises at least one of a driver or a transimpedance amplifier, the driver is configured to drive an optical modulator, and the transimpedance amplifier is configured to amplify a signal output from a photo detector; [1211] the first circuit board or the first substrate has a second surface defined by the length and the width, the second surface is spaced apart from the first surface by the thickness; [1212] the photonic integrated circuit and the at least one of the driver or the transimpedance amplifier are mounted on the second surface of the first circuit board or the first substrate; [1213] the at least one of the driver or transimpedance amplifier has second electrical terminals that are electrically coupled to the first electrical terminals through electrical connectors that pass through the first circuit board or the first substrate in the thickness direction.

[1214] Embodiment 198: The apparatus of embodiment 189 in which the photonic integrated circuit has a length, a width, and a thickness, the length is at least twice the thickness, and the width is at least twice the thickness; [1215] the photonic integrated circuit has a first surface defined by the length and the width; [1216] the photonic integrated circuit comprises second electrical terminals arranged on the first surface, the second electrical terminals are electrical coupled to the first electrical terminals on the first circuit board or the first substrate; and [1217] the optical input port is optical coupled to the first surface of the photonic integrated circuit.

[1218] Embodiment 199: The apparatus of embodiment 189 in which the photonic integrated circuit has a length, a width, and a thickness, the length is at least twice the thickness, and the width is at least twice the thickness; [1219] the photonic integrated circuit has a first surface defined by the length and the width, the photonic integrated circuit has a second surface defined by the length and the width, the second surface is spaced apart from the first surface by the thickness; [1220] the photonic integrated circuit comprises second electrical terminals arranged on the first surface, the second electrical terminals are electrical coupled to the first electrical terminals on the first circuit board or the first substrate; [1221] the optical input port is optical coupled to the second surface of the photonic integrated circuit.

[1222] Embodiment 200: The apparatus of embodiment 189 in which the optical interconnect module comprises at least one of a driver or a transimpedance amplifier, the driver is configured to drive an optical modulator, and the transimpedance amplifier is configured to amplify an electrical signal from a photo detector; [1223] the photonic integrated circuit and at least one of the driver or the transimpedance amplifier are mounted on the first surface of the first circuit board or the first substrate; and [1224] the at least one of the driver or transimpedance amplifier has second electrical terminals that are electrically coupled to the first electrical terminals.

[1225] Embodiment 201: The apparatus of embodiment 189 in which the optical input port comprises a first optical connector configured to mate with a second optical connector coupled to an optical fiber cable that comprises a plurality of optical fibers.

[1226] Embodiment 202: The apparatus of embodiment 201 in which the photonic integrated circuit comprises vertical-coupling elements configured to couple light from the optical input port to the photonic integrated circuit.

[1227] Embodiment 203: The apparatus of embodiment 202 in which the first optical connector comprises one or more lenses configured to project light onto the vertical coupling elements.

[1228] Embodiment 204: The apparatus of embodiment 202 in which the first optical connector and the second optical connector comprise one or more optical components configured to couple M spatial paths of the optical fibers and an array of N vertical-coupling elements of the photonic integrated circuit, N is a positive integer, M is a positive integer, and N is equal to or different from M .

[1229] Embodiment 205: The apparatus of embodiment 204 in which the one or more optical components of the first and second optical connectors are configured to implement at least one of [1230] (i) magnifying or de-magnifying by a first factor a minimum core-to-core spacing of the optical fibers at a fiber end face plane to match a minimum spacing between the vertical-coupling

elements at a coupling plane; [1231] (ii) magnifying or de-magnifying by a second factor a maximum core-to-core spacing of optical fibers at a fiber end face plane to match a maximum spacing between the vertical-coupling elements at a coupling plane; [1232] (iii) magnifying or de-magnifying by a third factor an effective core diameter of optical fibers at a fiber end face plane to match an effective size of the vertical coupling elements at a coupling plane; [1233] (iv) magnifying or de-magnifying by a fourth factor an effective core diameter of optical fibers at a fiber end face plane to achieve a different effective beam diameter at a connector mating plane than at the fiber end face plane; or [1234] (v) changing an effective cross-sectional geometrical layout of the plurality of spatial paths at at least one of a fiber end face plane, a connector mating plane, or a coupling plane.

[1235] Embodiment 206: A system comprising: [1236] a housing comprising a bottom surface; [1237] a first circuit board or a first substrate comprising a first surface at an angle relative to the bottom surface of the housing, in which the angle is in a range from 30° to 150°; [1238] at least one data processor mounted on the first circuit board or the first substrate; and [1239] at least one optical interconnect module mounted on the first surface of the first circuit board or the first substrate, in which each optical interconnect module comprises a first optical connector configured to connect to an external optical link, each optical interconnect module comprises a photonic integrated circuit configured to generate a first serial electrical signal based on an optical signal received from the first optical connector; [1240] wherein the at least one data processor is configured to process data carried in the first serial electrical signal.

[1241] Embodiment 207: The system of embodiment 206 in which at least one of the at least one optical interconnect module is removably coupled to the first circuit board or the first substrate.

[1242] Embodiment 208: The system of embodiment 206 in which at least one of the at least one optical interconnect module is removably coupled to the first circuit board or the first substrate.

[1243] Embodiment 209: The system of embodiment 206 in which the first optical connector is removably coupled to the external optical link.

[1244] Embodiment 210: The system of embodiment 206 in which the first optical connector is fixedly coupled to the external optical link.

[1245] Embodiment 211: The system of embodiment 206 in which one or more optical fibers are directly attached to the photonic integrated circuit.

[1246] Embodiment 212: The system of embodiment 206 in which one or more optical fibers are removably attached to the at least one optical interconnect module.

[1247] Embodiment 213: The system of embodiment 206 in which the at least one data processor comprises at least a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, or an application specific integrated circuit (ASIC).

[1248] Embodiment 214: The system of embodiment 206 in which the housing comprises a front panel having a front surface; [1249] the first circuit board or the first substrate has a length, a width, and a thickness, wherein the length is at least twice the thickness, and the width is at least twice the thickness, the first circuit board or the first substrate has a first surface defined by the length and the width; and [1250] the first circuit board or the first substrate is oriented such that the first surface is substantially parallel to the front surface of the front panel.

[1251] Embodiment 215: The system of embodiment 214 in which the at least one data processor is also mounted on the first surface of the first circuit board or the first substrate.

[1252] Embodiment 216: The system of embodiment 215, comprising one or more optical paths that pass a first opening of the first circuit board or the first substrate and a second opening of the front panel, in which the one or more optical paths enable one or more optical signals from the external optical link to be coupled through the first optical connector to the photonic integrated circuit.

[1253] Embodiment 217: The system of embodiment 214 in which the at least one data processor is

mounted on a second surface of the first circuit board or the first substrate, the second surface is opposite to the first surface.

[1254] Embodiment 218: The system of embodiment 217, comprising one or more optical paths that pass an opening of the front panel, in which the optical path enables one or more optical signals from the external optical link to be coupled through the first optical connector to the photonic integrated circuit.

[1255] Embodiment 219: The system of embodiment 217 in which at least one of the at least one optical interconnect module passes an opening of the front panel.

[1256] Embodiment 220: The system of embodiment 206 in which the first circuit board or the first substrate is configured as a front panel or a portion of a front panel of the housing.

[1257] Embodiment 221: The system of embodiment 206, comprising a rackmount module, in which the rackmount module comprises the housing, the first circuit board or the first substrate, the at least one data processor, and the at least one optical interconnect module, [1258] wherein the first circuit board or the first substrate is configured as a front panel or a portion of the front panel of the rackmount module or is oriented substantially parallel to a front panel of the rackmount module.

[1259] Embodiment 222: The system of embodiment 206 in which the optical interconnect module comprises: [1260] a first serializer/deserializer configured to generate a set of first parallel electrical signals based on the first serial electrical signal, and condition the electrical signals; and [1261] a second serializer/deserializer configured to generate a second serial electrical signal based on the set of first parallel electrical signals; [1262] wherein the at least one data processor is configured to process data carried in the second serial electrical signal.

[1263] Embodiment 223: The system of embodiment 222, comprising a third serializer/deserializer configured to generate a set of second parallel electrical signal based on the second serial electrical signal; [1264] wherein the at least one data processor is configured to process data carried in the set of second parallel electrical signal.

[1265] Embodiment 224: The system of embodiment 223 in which the third serializer/deserializer is embedded in the at least one data processor.

[1266] Embodiment 225: The system of embodiment 206, comprising a first serializer/deserializer configured to generate a set of first parallel electrical signals based on the first serial electrical signal; [1267] wherein the at least one data processor is configured to process data carried in the set of first parallel electrical signal.

[1268] Embodiment 226: The system of embodiment 225 in which the first serializer/deserializer is embedded in the at least one data processor.

[1269] Embodiment 227: The system of embodiment 206 in which the first circuit board or the first substrate has a first surface and a second surface, [1270] the second surface of the first circuit board or the first substrate faces a front side of the housing; [1271] the at least one data processor and the at least one optical interconnect module are mounted on the first surface of the first circuit board or the first substrate; [1272] the first circuit board or the first substrate defines an opening, the optical signal is transmitted through an optical path that passes through the opening of the first circuit board or the first substrate to the photonic integrated circuit.

[1273] Embodiment 228: The system of embodiment 227 in which the first circuit board or the first substrate functions as a front panel or a portion of a front panel of the housing.

[1274] Embodiment 229: The system of embodiment 227 in which the housing comprises a front panel, and the first circuit board or the first substrate is substantially parallel to the front panel.

[1275] Embodiment 230: The system of embodiment 229 in which the first circuit board or the first substrate is spaced apart from the front panel, and a distance between the first circuit board or the first substrate and the front panel is in a range from 0.1 to 2 inches.

[1276] Embodiment 231: The system of embodiment 206 in which the first circuit board or the first substrate has a first surface and a second surface, [1277] the second surface of the first circuit board

or the first substrate faces a front side of the housing; [1278] the at least one data processor is mounted on the first surface of the first circuit board or the first substrate; and [1279] the at least one optical interconnect module is mounted on the second surface of the first circuit board or the first substrate, the at least one optical interconnect module is electrically coupled to the at least one data process through electrical connections that pass through the first circuit board or the first substrate in a thickness direction of the first circuit board or the first substrate.

[1280] Embodiment 232: The system of embodiment 231 in which the first circuit board or the first substrate functions as a front panel or a portion of a front panel of the housing.

[1281] Embodiment 233: The system of embodiment 231 in which the housing comprises a front panel, and the first circuit board or the first substrate is substantially parallel to the front panel.

[1282] Embodiment 234: The system of embodiment 233 in which the first circuit board or the first substrate is spaced apart from the front panel, and a distance between the first circuit board or the first substrate and the front panel is in a range from 0.1 to 2 inches.

[1283] Embodiment 235: The system of embodiment 206, comprising a second circuit board or a second substrate comprising a second surface oriented substantially parallel to the bottom surface of the housing, and a plurality of electronic components mounted on the second surface of the second circuit board or the second substrate, [1284] wherein the first circuit board or the first substrate is electrically coupled to the second circuit board or the second substrate.

[1285] Embodiment 236: The system of embodiment 206 in which the first optical connector is configured to releasably connect with a second optical connector that is coupled to a bundle of optical fibers.

[1286] Embodiment 237: The system of embodiment 236 in which the first optical connector is configured to provide at least 2 optical paths to enable optical signals from the bundle of optical fibers to be coupled to the at least one data processor.

[1287] Embodiment 238: The system of embodiment 236 in which the first optical connector is configured to provide at least 4 optical paths to enable optical signals from the bundle of optical fibers to be coupled to the at least one data processor.

[1288] Embodiment 239: The system of embodiment 236 in which the first optical connector is configured to provide at least 8 optical paths to enable optical signals from the bundle of optical fibers to be coupled to the at least one data processor.

[1289] Embodiment 240: The system of embodiment 236 in which the first optical connector is configured to provide at least 16 optical paths to enable optical signals from the bundle of optical fibers to be coupled to the at least one data processor.

[1290] Embodiment 241: The system of embodiment 236 in which the first optical connector is configured to provide at least 32 optical paths to enable optical signals from the bundle of optical fibers to be coupled to the at least one data processor.

[1291] Embodiment 242: The system of embodiment 236 in which the first optical connector is configured to provide at least 64 optical paths to enable optical signals from the bundle of optical fibers to be coupled to the at least one data processor.

[1292] Embodiment 243: The system of embodiment 236 in which the plurality of optical fibers comprise at least 100 cores of optical fibers.

[1293] Embodiment 244: The system of embodiment 236 in which the plurality of optical fibers comprise at least 500 cores of optical fibers.

[1294] Embodiment 245: The system of embodiment 236 in which the plurality of optical fibers comprise at least 1000 cores of optical fibers.

[1295] Embodiment 246: A system comprising: [1296] a housing comprising a front panel, in which the front panel comprises a first circuit board or a first substrate; [1297] at least one data processor mounted on the first circuit board or the first substrate; and [1298] at least one optical/electrical communication interface mounted on the first circuit board or the first substrate.

[1299] Embodiment 247: The system of embodiment 246 in which each optical/electrical

communication interface comprises: [1300] a first optical connector configured to connect to an external optical link, and [1301] a photonic integrated circuit configured to generate an electrical signal based on an optical signal received from the first optical connector.

[1302] Embodiment 248: The system of embodiment 247 in which at least one of the at least one data processor and at least one of the at least one photonic integrated circuit are mounted on a same side of the first circuit board or the first substrate.

[1303] Embodiment 249: The system of embodiment 247 in which at least one of the at least one data processor is mounted on a first side of the first circuit board or the first substrate, at least one of the at least one photonic integrated circuit is mounted on a second side of the first circuit board or the first substrate, and the second side is opposite the first side.

[1304] Embodiment 250: A system comprising: [1305] a plurality of rack mount systems, each rack mount system comprising: [1306] a housing comprising a front panel, in which the front panel comprises a first circuit board or a first substrate; [1307] at least one data processor mounted on the first circuit board or the first substrate; and [1308] at least one optical/electrical communication interface mounted on the first circuit board or the first substrate.

[1309] Embodiment 251: The system of embodiment 250 in which the at least one optical/electrical communication interface is configured to receive one or more optical signals from an external optical link, and generate one or more electrical signals based on the one or more optical signals; and [1310] the at least one data processor is configured to process the one or more electrical signals provided by the at least one optical/electrical communication interface.

[1311] Embodiment 252: A system comprising: [1312] a housing comprising a front panel; [1313] a first circuit board or a first substrate oriented at a first angle relative to the front panel, in which the first angle is in a range from -30° to 30° ; [1314] at least one data processor mounted on the first circuit board or the first substrate; and [1315] at least one optical/electrical communication interface mounted on the first circuit board or the first substrate.

[1316] Embodiment 253: The system of embodiment 252 in which the first angle is in a range from -5° to 5° .

[1317] Embodiment 254: The system of embodiment 252 in which the first circuit board or the first substrate is spaced apart from the front panel at a distance in a range from 0.1 to 2 inches.

[1318] Embodiment 255: The system of embodiment 252 in which each optical/electrical communication interface comprises: [1319] a first optical connector configured to connect to an external optical link, and [1320] a photonic integrated circuit configured to generate an electrical signal based on an optical signal received from the first optical connector.

[1321] Embodiment 256: The system of embodiment 255 in which at least one of the at least one data processor and at least one of the at least one photonic integrated circuit are mounted on a same side of the first circuit board or the first substrate.

[1322] Embodiment 257: The system of embodiment 255 in which at least one of the at least one data processor is mounted on a first side of the first circuit board or the first substrate, at least one of the at least one photonic integrated circuit is mounted on a second side of the first circuit board or the first substrate, and the second side is opposite the first side.

[1323] Embodiment 258: A system comprising: [1324] a plurality of rack mount systems, each rack mount system comprising: [1325] a housing comprising a front panel; [1326] a first circuit board or a first substrate oriented at a first angle relative to the front panel, in which the first angle is in a range from -30° to 30° ; [1327] at least one data processor mounted on the first circuit board or the first substrate; and [1328] at least one optical/electrical communication interface mounted on the first circuit board or the first substrate.

[1329] Embodiment 259: The system of embodiment 258 in which the at least one optical/electrical communication interface is configured to receive one or more optical signals from an external optical link, and generate one or more electrical signals based on the one or more optical signals; and [1330] the at least one data processor is configured to process the one or more electrical signals

provided by the at least one optical/electrical communication interface.

[1331] Embodiment 260: A system comprising: [1332] a first optical interconnect module comprising: [1333] a first optical input/output port configured to at least one of (i) receive a plurality of channels of first optical signals from a first plurality of optical fibers, or (ii) transmit a plurality of channels of second optical signals to the first plurality of optical fibers; [1334] a first photonic integrated circuit configured to at least one of (i) generate a plurality of first serial electrical signals based on the first optical signals, or (ii) generate the second optical signals based on a plurality of second serial electrical signals; [1335] a plurality of first serializer/deserializers configured to at least one of (i) generate a plurality of sets of third parallel electrical signals based on the plurality of first serial electrical signals, and condition the electrical signals, in which each set of third parallel electrical signals is generated based on a corresponding first serial electrical signal, or (ii) generate the plurality of second serial electrical signals based on a plurality of sets of fourth parallel electrical signals, in which each second serial electrical signal is generated based on a corresponding set of fourth parallel electrical signals; and [1336] a plurality of second serializer/deserializers configured to at least one of (i) generate a plurality of fifth serial electrical signals based on the plurality of sets of third parallel electrical signals, in which each fifth serial electrical signal is generated based on a corresponding set of third parallel electrical signals, or (ii) generate the plurality of sets of fourth parallel electrical signals based on a plurality of sixth serial electrical signals, in which each set of fourth parallel electrical signal is generated based on a corresponding sixth serial signal; [1337] a plurality of third serializer/deserializers configured to at least one of (i) generate a plurality of sets of seventh parallel electrical signals based on the plurality of fifth serial electrical signals, and condition the electrical signals, in which each set of seventh parallel electrical signals is generated based on a corresponding fifth serial electrical signal, or (ii) generate the plurality of sixth serial electrical signals based on a plurality of sets of eighth parallel electrical signals, in which each sixth serial electrical signal is generated based on a corresponding set of eighth parallel electrical signals; and [1338] a data processor configured to at least one of (i) process the plurality of sets of seventh parallel electrical signals, or (ii) output the plurality of sets of eighth parallel electrical signals.

[1339] Embodiment 261: The system of embodiment 260 in which the data processor comprises a network switch configured to switch signals transmitted in a first subset of the optical fibers to a second subset of the optical fibers.

[1340] Embodiment 262: The system of embodiment 260 in which the plurality of optical fibers comprise at least 100 optical fibers.

[1341] Embodiment 263: The system of embodiment 260 in which the plurality of optical fibers comprise at least 500 optical fibers.

[1342] Embodiment 264: The system of embodiment 260 in which the plurality of optical fibers comprise at least 1000 optical fibers.

[1343] Embodiment 265: The system of embodiment 260 in which the plurality of optical fibers are bundled in an optical cable having a cross section, for at least a portion of the cross section the optical cable has at least 4 optical fibers per square millimeter.

[1344] Embodiment 266: The system of embodiment 260 in which the plurality of optical fibers are bundled in an optical cable having a cross section, for at least a portion of the cross section the optical cable has at least 8 optical fibers per square millimeter.

[1345] Embodiment 267: The system of embodiment 260 in which the plurality of optical fibers are bundled in an optical cable having a cross section, for at least a portion of the cross section the optical cable has at least 16 optical fibers per square millimeter.

[1346] Embodiment 268: The system of embodiment 260, comprising a first circuit board (or a first substrate) and a second circuit board (or a second substrate); [1347] wherein the first photonic integrated circuit, the first serializer/deserializers, and the second serializer/deserializers are mounted on the first circuit board or the first substrate, the third serializer/deserializers and the data

processor are mounted on the second circuit board or the second substrate, and the first circuit board or the first substrate is electrically coupled to the second circuit board or the second substrate.

[1348] Embodiment 269: The system of embodiment 260 in which the plurality of third serializer/deserializers are embedded in the data processor.

[1349] Embodiment 270: The system of embodiment 260 in which the data processor comprises at least one of a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, or an application specific integrated circuit (ASIC).

[1350] Embodiment 271: The system of embodiment 260, comprising a second optical interconnect module comprising: [1351] a second optical input/output port configured to at least one of (i) receive a plurality of channels of third optical signals from a second plurality of optical fibers, or (ii) transmit a plurality of channels of fourth optical signals to the second plurality of optical fibers; [1352] a second photonic integrated circuit configured to at least one of (i) generate a plurality of ninth serial electrical signals based on the third optical signals, or (ii) generate the fourth optical signals based on a plurality of tenth serial electrical signals; [1353] a plurality of fourth serializer/deserializers configured to at least one of (i) generate a plurality of sets of eleventh parallel electrical signals based on the plurality of ninth serial electrical signals, and condition the electrical signals, in which each set of eleventh parallel electrical signals is generated based on a corresponding ninth serial electrical signal, or (ii) generate the plurality of tenth serial electrical signals based on a plurality of sets of twelfth parallel electrical signals, in which each tenth serial electrical signal is generated based on a corresponding set of twelfth parallel electrical signals; and [1354] a plurality of fifth serializer/deserializers configured to at least one of (i) generate a plurality of thirteenth serial electrical signals based on the plurality of sets of eleventh parallel electrical signals, in which each thirteenth serial electrical signal is generated based on a corresponding set of eleventh parallel electrical signals, or (ii) generate the plurality of sets of twelfth parallel electrical signals based on a plurality of fourteenth serial electrical signals, in which each set of twelfth parallel electrical signal is generated based on a corresponding fourteenth serial signal; and [1355] a plurality of sixth serializer/deserializers configured to at least one of (i) generate a plurality of sets of fifteenth parallel electrical signals based on the plurality of thirteenth serial electrical signals, and condition the electrical signals, in which each set of fifteenth parallel electrical signals is generated based on a corresponding thirteenth serial electrical signal, or (ii) generate the plurality of fourteenth serial electrical signals based on a plurality of sets of sixteenth parallel electrical signals, in which each fourteenth serial electrical signal is generated based on a corresponding set of sixteenth parallel electrical signals; [1356] wherein the data processor is configured to at least one of (i) process the plurality of sets of fifteenth parallel electrical signals, or (ii) output the plurality of sets of sixteenth parallel electrical signals.

[1357] Embodiment 272: An apparatus comprising: [1358] a substrate comprising: [1359] a first main surface and a second main surface; [1360] a first array of electrical contacts arranged on the first main surface and having a first minimum spacing between the contacts; [1361] a second array of electrical contacts arranged on the second main surface and having a second minimum spacing between the contacts, in which the first minimum spacing is larger than the second minimum spacing; and [1362] electrical connections between the first array of electrical contacts and the second array of electrical contacts; [1363] a photonic integrated circuit having a first main surface and a second main surface; [1364] a first optical connector part configured to couple light to the first main surface of the photonic integrated circuit; [1365] an electronic integrated circuit having a first main surface that has a first portion and a second portion, in which the first portion of the first main surface is electrically coupled to the second main surface of the photonic integrated circuit, and the second portion of the first main surface is electrically coupled to the second array of electrical contacts arranged on the second main surface of the substrate.

[1366] Embodiment 273: The system of embodiment 272 in which the substrate is configured to be removably connectable to a printed circuit board.

[1367] Embodiment 274: The system of embodiment 272, further comprising a second optical connector part that is configured to couple light from an array of optical fibers to the first optical connector part.

[1368] Embodiment 275: The system of embodiment 272 in which the electronic integrated circuit comprises a first serializers/deserializers and a second serializers/deserializers; [1369] the first serializers/deserializers has a serial communication portion that is electrically coupled to the photonic integrated circuit, the second serializers/deserializers has a serial communication portion that is electrically coupled to electrical terminals arranged on the substrate; and [1370] the first serializers/deserializers has a parallel communication portion, the second serializers/deserializers has a parallel communication portion, and the parallel communication portion of the first serializers/deserializers is electrically coupled to the parallel communication portion of the second serializers/deserializers.

[1371] Embodiment 276: The system of embodiment 272, further comprising a third serializers/deserializers that has a serial communication portion that is electrically coupled to the serial communication portion of the second serializers/deserializers.

[1372] Embodiment 277: The system of embodiment 272 in which the photonic integrated circuit is configured to receive and process optical signals provided from an external signal source, in which the optical signals are carried by at least one of continuous wave light or pulsed light.

[1373] Embodiment 278: The system of embodiment 272, further comprising a connector module configured to removably attach the substrate to a printed circuit board.

[1374] Embodiment 279: An apparatus comprising: [1375] a printed circuit board having a first main surface and a second main surface; [1376] a substrate comprising: [1377] a first main surface and a second main surface; [1378] a first array of electrical contacts arranged on the first main surface and having a first minimum spacing between the contacts; [1379] a second array of electrical contacts arranged on the second main surface and having a second minimum spacing between the contacts, in which the first minimum spacing is larger than the second minimum spacing; and [1380] electrical connections between the first array of electrical contacts and the second array of electrical contacts; [1381] wherein the first main surface of the substrate is configured to be removably connectable to the second main surface of the printed circuit board; [1382] a photonic integrated circuit having a second main surface; [1383] a first optical connector part that is optically coupled to the second main surface of the photonic integrated circuit; and [1384] an electronic integrated circuit that is electrically coupled to the second main surface of the photonic integrated circuit and the second array of electrical contacts arranged on the second main surface of the substrate.

[1385] Embodiment 280: The apparatus of embodiment 279, further comprising a second optical connector part that is optically coupled to an array of optical fibers and configured to couple light from the optical fibers to the first optical connector part.

[1386] Embodiment 281: The apparatus of embodiment 279 in which the electronic integrated circuit comprises a first serializers/deserializers and a second serializers/deserializers; [1387] the first serializers/deserializers has a serial communication portion that is electrically coupled to the photonic integrated circuit, the second serializers/deserializers has a serial communication portion that is electrically coupled to electrical terminals arranged on the substrate; and [1388] the first serializers/deserializers has a parallel communication portion, the second serializers/deserializers has a parallel communication portion, and the parallel communication portion of the first serializers/deserializers is electrically coupled to the parallel communication portion of the second serializers/deserializers.

[1389] Embodiment 282: The apparatus of embodiment 281, further comprising a third serializers/deserializers that has a serial communication portion that is electrically coupled to the

serial communication portion of the second serializers/deserializers.

[1390] Embodiment 283: The apparatus of embodiment 282, further comprising a digital application specific integrated circuit mounted on the printed circuit board, in which the third serializers/deserializers are embedded in the application specific integrated circuit, and the serial communication portion of the third serializers/deserializers is electrically coupled to the serial communication portion of the second serializers/deserializers through electrical connections on or in the printed circuit board.

[1391] Embodiment 284: The apparatus of embodiment 279 in which the photonic integrated circuit is configured to receive and process optical signals provided from an external signal source, in which the optical signals are carried by at least one of continuous wave light or pulsed light.

[1392] Embodiment 285: The apparatus of embodiment 279, further comprising a connector module configured to removably attach the substrate to a printed circuit board.

[1393] Embodiment 286: An apparatus comprising: [1394] a plurality of serializer units; [1395] a plurality of deserializer units; and [1396] a bus processing unit electrically coupled to the serializer units and deserializer units; [1397] wherein the bus processing unit is configured to enable switching of the signals at the serializer units and deserializer units.

[1398] Embodiment 287: An apparatus comprising: [1399] a first array of serializers/deserializers configured to convert one or more first serial signals to one or more sets of parallel signals; [1400] a second array of serializers/deserializers configured to convert one or more sets of parallel signals to one or more second serial signals; and [1401] a bus processing unit electrically coupled to the first array of serializers/deserializers and the second array of serializers/deserializers, in which the bus processing unit is configured to processing the one or more sets of parallel signals, and send one or more sets of processed parallel signals to the second array of serializers/deserializers.

[1402] Embodiment 288: An apparatus comprising: [1403] a printed circuit board having a first main surface and a second main surface; [1404] a substrate comprising: [1405] a first main surface and a second main surface; [1406] a first array of electrical contacts arranged on the first main surface and having a first minimum spacing between the contacts; [1407] a second array of electrical contacts arranged on the second main surface and having a second minimum spacing between the contacts, in which the first minimum spacing is larger than the second minimum spacing; [1408] a third array of electrical contacts arranged on the first main surface; [1409] first electrical connections between the first array of electrical contacts and a first subset of the second array of electrical contacts; and [1410] second electrical connections between the third array of electrical contacts and a second subset of the second array of electrical contacts; [1411] wherein the first main surface of the substrate is configured to be removably connectable to the second main surface of the printed circuit board; [1412] an electronic integrated circuit that is electrically coupled to the second array of electrical contacts arranged on the second main surface of the substrate; [1413] a photonic integrated circuit having a second main surface and electrical contacts arranged on the second main surface that are electrically coupled to the third array of electrical contacts arranged on the first main surface of the substrate; [1414] a first optical connector part that is optically coupled to the photonic integrated circuit.

[1415] Embodiment 289: The apparatus of embodiment 288 in which the first optical connector part is optically coupled to the second main surface of the photonic integrated circuit.

[1416] Embodiment 290: The apparatus of embodiment 289, further comprising a second optical connector part that is optically coupled to an array of optical fibers and configured to couple light from the optical fibers to the first optical connector part.

[1417] Embodiment 291: The apparatus of embodiment 288 in which the first optical connector part is optically coupled to the first main surface of the photonic integrated circuit.

[1418] Embodiment 292: The apparatus of embodiment 288 in which the electronic integrated circuit comprises a first serializers/deserializers and a second serializers/deserializers; [1419] the first serializers/deserializers comprises a serial communication portion that is electrically coupled

to the photonic integrated circuit, the second serializers/deserializers comprises a serial communication portion that is electrically coupled to electrical terminals arranged on the substrate; and [1420] the first serializers/deserializers comprises a parallel communication portion, the second serializers/deserializers comprises a parallel communication portion, and the parallel communication portion of the first serializers/deserializers is electrically coupled to the parallel communication portion of the second serializers/deserializers.

[1421] Embodiment 293: The apparatus of embodiment 292, further comprising a third serializers/deserializers that comprises a serial communication portion that is electrically coupled to the serial communication portion of the second serializers/deserializers.

[1422] Embodiment 294: The apparatus of embodiment 288, further comprising a digital application specific integrated circuit mounted on the printed circuit board, in which the third serializers/deserializers are embedded in the application specific integrated circuit, and the serial communication portion of the third serializers/deserializers is electrically coupled to the serial communication portion of the second serializers/deserializers through electrical connections on or in the printed circuit board.

[1423] Embodiment 295: The apparatus of embodiment 288 in which the photonic integrated circuit is configured to receive and process optical signals provided from an external signal source, in which the optical signals are carried by at least one of continuous wave light or pulsed light.

[1424] Embodiment 296: The apparatus of embodiment 288, further comprising a connector module configured to removably attach the substrate to a printed circuit board.

[1425] Embodiment 297: The apparatus of embodiment 288, further comprising control circuitry configured to control the photonic integrated circuit.

[1426] Embodiment 298: A datacenter network switching system that comprises the apparatus or system of any of embodiments 1 to 297, 352a to 358a, 352b to 393.

[1427] Embodiment 299: A supercomputer that comprises the apparatus or system of any of embodiments 1 to 297, 352a to 358a, 352b to 393.

[1428] Embodiment 300: An autonomous vehicle that comprises the apparatus or system of any of embodiments 1 to 297, 352a to 358a, 352b to 393.

[1429] Embodiment 301: The autonomous vehicle of embodiment 300 in which the vehicle comprises at least one of a car, a truck, a train, a boat, a ship, a submarine, a helicopter, a drone, an airplane, a space rover, or a space ship.

[1430] Embodiment 302: A robot that comprises the apparatus or system of any of embodiments 1 to 297, 352a to 358a, 352b to 393.

[1431] Embodiment 303: The robot of embodiment 302 in which the robot comprises at least one of an industrial robot, a helper robot, a medical surgery robot, a merchandise delivery robot, a teaching robot, a cleaning robot, a cooking robot, a construction robot, or an entertainment robot.

[1432] Embodiment 304: A method comprising: [1433] receiving a plurality of channels of first optical signals from a plurality of optical fibers; [1434] generating a plurality of first serial electrical signals based on the received optical signals, in which each first serial electrical signal is generated based on one of the channels of first optical signals; [1435] generating a plurality of sets of first parallel electrical signals based on the plurality of first serial electrical signals, and conditioning the electrical signals, in which each set of first parallel electrical signals is generated based on a corresponding first serial electrical signal; and [1436] generating a plurality of second serial electrical signals based on the plurality of sets of first parallel electrical signals, in which each second serial electrical signal is generated based on a corresponding set of first parallel electrical signals.

[1437] Embodiment 305: The method of embodiment 304, comprising: [1438] generating a plurality of sets of second parallel electrical signals based on the plurality of second serial electrical signals, in which each set of second parallel electrical signals is generated based on a corresponding second serial electrical signal; and [1439] processing the plurality of sets of second

parallel electrical signals, in which the processing comprises at least one of switching data packets, processing graphics data, processing image data, processing video data, processing audio data, performing mathematical computations, performing tensor calculations, performing matrix calculations, performing simulations, performing neural network processing, processing data based on artificial intelligence algorithms, processing digital signals, or processing control signals.

[1440] Embodiment 306: The method of embodiment 304, comprising performing signal conditioning on the electrical signals, the signal conditioning comprising at least one of (i) clock and data recovery, or (ii) signal equalization.

[1441] Embodiment 307: The method of embodiment 304, comprising processing the first parallel electrical signals prior to generating the second serial electrical signals, the processing comprising at least one of switching of data, re-shuffling data, or coding of data.

[1442] Embodiment 308: The method of embodiment 307, in which the first serial electrical signals comprise N lanes of $T \times N / (N - k)$ Gbps electrical signals, N and k are positive integers, and T is a real value; [1443] the second serial electrical signals comprise N lanes of T Gbps electrical signals; and [1444] the method comprises mapping $N - k$ out of the N lanes of $T \times N / (N - k)$ Gbps electrical signals in the first serial signals to the N lanes of T Gbps electrical signals in the second serial signals.

[1445] Embodiment 309: The method of embodiment 307, in which the first serial signals comprise N lanes of $T \times N / (N - k)$ Gbps electrical signals, N and k are positive integers, and T is a real value; [1446] the second serial signals comprise N / M lanes of $M \times T$ Gbps electrical signals, M is different from N ; and [1447] the method comprises mapping $N - k$ out of the N lanes of $T \times N / (N - k)$ Gbps electrical signals in the first serial signals to the N / M lanes of $M \times T$ Gbps electrical signals in the second serial signals.

[1448] Embodiment 310: The method of embodiment 304 in which generating the plurality of first serial electrical signals comprises generating N serial electrical signals, and generating the plurality of second serial electrical signals comprises generating M serial electrical signals based on the plurality of sets of first parallel electrical signals, M and N are positive integers, and M is different from N .

[1449] Embodiment 311: The method of embodiment 310 in which generating the plurality of first serial electrical signals comprises generating N lanes of P Gbps serial electrical signals, and generating the plurality of second serial electrical signals comprises generating N / Q lanes of $P \times Q$ Gbps serial electrical signals, and P and Q are positive numbers.

[1450] Embodiment 312: The method of embodiment 304 in which the first serial electrical signals are modulated according to a first modulation protocol, and the second serial electrical signals are modulated according to a second modulation protocol that is different from the first modulation protocol.

[1451] Embodiment 313: The method of embodiment 304, further comprising: [1452] receiving a plurality of third serial electrical signals; [1453] generating a plurality of sets of third parallel electrical signals based on the plurality of third serial electrical signals, in which each set of third parallel electrical signal is generated based on a corresponding third serial electrical signal; [1454] generating a plurality of fourth serial electrical signals based on the plurality of sets of third parallel signals, in which each fourth serial electrical signal is generated based on a corresponding set of fourth parallel electrical signal; [1455] generating a plurality of channels of second optical signals based on the plurality of fourth serial electrical signals; and [1456] outputting the plurality of channels of second optical signals.

[1457] Embodiment 314: The method of embodiment 313, comprising: [1458] at at least one of a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, or an application specific integrated circuit (ASIC), generating a plurality of sets of fourth parallel electrical signals; and [1459] generating the third serial electrical signals based on

the sets of fourth parallel electrical signals.

[1460] Embodiment 315: The method of embodiment 314 in which generating the third serial electrical signals comprises generating M serial electrical signals based on the sets of fourth parallel electrical signals; and [1461] generating a plurality of fourth serial electrical signals comprises generating N serial electrical signals based on the sets of third parallel signals, M and N are positive integers, and N is different from M.

[1462] Embodiment 316: The method of embodiment 315 in which generating the third serial electrical signals comprises generating N/Q lanes of P*Q Gbps serial electrical signals; and [1463] generating a plurality of fourth serial electrical signals comprises generating N lanes of P Gbps serial electrical signals, and P and Q are positive numbers.

[1464] Embodiment 317: The method of embodiment 313 in which the third serial electrical signals are modulated according to a first modulation protocol, and the fourth serial electrical signals are modulated according to a second modulation protocol that is different from the first modulation protocol.

[1465] Embodiment 318: The method of embodiment 313, comprising performing signal conditioning on the electrical signals, the signal conditioning comprising at least one of (i) clock and data recovery, or (ii) signal equalization.

[1466] Embodiment 319: The method of embodiment 313, comprising aligning multiple serial signals.

[1467] Embodiment 320: The method of embodiment 313, comprising after generating the sets of third parallel electrical signals and prior to generating the fourth serial electrical signals, processing the electrical signals, in which the processing comprises performing at least one of switching of data, re-shuffling data, or coding of data.

[1468] Embodiment 321: The method of embodiment 304 in which receiving a plurality of channels of first optical signals comprises receiving at least 2 channels of optical signals.

[1469] Embodiment 322: The method of embodiment 321 in which generating a plurality of first serial electrical signals comprises processing the at least 2 channels of optical signals and generating at least 2 first serial electrical signals.

[1470] Embodiment 323: The method of embodiment 322 in which generating a plurality of sets of first parallel electrical signals comprises converting the at least 2 first serial electrical signals into at least 2 sets of parallel electrical signals, and each set of parallel electrical signals comprises at least two parallel electrical signals.

[1471] Embodiment 324: The method of embodiment 323 in which each set of parallel electrical signals comprises at least four parallel electrical signals.

[1472] Embodiment 325: The method of embodiment 324 in which each set of parallel electrical signals comprises at least eight parallel electrical signals.

[1473] Embodiment 326: The method of embodiment 325 in which each set of parallel electrical signals comprises at least 32 parallel electrical signals.

[1474] Embodiment 327: The method of embodiment 326 in which each set of parallel electrical signals comprises at least 64 parallel electrical signals.

[1475] Embodiment 328: The method of embodiment 304 in which receiving a plurality of channels of first optical signals comprises receiving at least 4 channels of optical signals.

[1476] Embodiment 329: The method of embodiment 304 in which receiving a plurality of channels of first optical signals comprises receiving at least 8 channels of optical signals.

[1477] Embodiment 330: The method of embodiment 304 in which receiving a plurality of channels of first optical signals comprises receiving at least 16 channels of optical signals.

[1478] Embodiment 331: The method of embodiment 304 in which receiving a plurality of channels of first optical signals comprises receiving at least 32 channels of optical signals.

[1479] Embodiment 332: The method of embodiment 304 in which receiving a plurality of channels of first optical signals comprises receiving at least 64 channels of optical signals.

[1480] Embodiment 333: The method of embodiment 304 in which receiving a plurality of channels of first optical signals comprises receiving at least 100 channels of optical signals.

[1481] Embodiment 334: The method of embodiment 333 in which generating a plurality of first serial electrical signals comprises processing the at least 100 channels of optical signals and generating at least 100 first serial electrical signals.

[1482] Embodiment 335: The method of embodiment 334 in which generating a plurality of sets of first parallel electrical signals comprises converting the at least 100 first serial electrical signals into at least 100 sets of parallel electrical signals, and each set of parallel electrical signals comprises at least two parallel electrical signals.

[1483] Embodiment 336: The method of embodiment 335 in which each set of parallel electrical signals comprises at least four parallel electrical signals.

[1484] Embodiment 337: The method of embodiment 336 in which each set of parallel electrical signals comprises at least eight parallel electrical signals.

[1485] Embodiment 338: The method of embodiment 337 in which each set of parallel electrical signals comprises at least 32 parallel electrical signals.

[1486] Embodiment 339: The method of embodiment 338 in which each set of parallel electrical signals comprises at least 64 parallel electrical signals.

[1487] Embodiment 340: The method of embodiment 333 in which generating a plurality of first serial electrical signals comprises processing at least 500 channels of optical signals and generating at least 500 first serial electrical signals.

[1488] Embodiment 341: The method of embodiment 340 in which generating a plurality of sets of first parallel electrical signals comprises converting the at least 500 first serial electrical signals into at least 500 sets of parallel electrical signals, and each set of parallel electrical signals comprises at least two parallel electrical signals.

[1489] Embodiment 342: The method of embodiment 340 in which generating a plurality of first serial electrical signals comprises processing at least 1000 channels of optical signals and generating at least 1000 first serial electrical signals.

[1490] Embodiment 343: The method of embodiment 342 in which generating a plurality of sets of first parallel electrical signals comprises converting the at least 1000 first serial electrical signals into at least 1000 sets of parallel electrical signals, and each set of parallel electrical signals comprises at least two parallel electrical signals.

[1491] Embodiment 344: The method of embodiment 304 in which the first optical signals have a total bit rate of at least 1 Tbps.

[1492] Embodiment 345: The method of embodiment 304 in which the first optical signals have a total bit rate of at least 10 Tbps.

[1493] Embodiment 346: The method of embodiment 304, comprising receiving a plurality of third serial electrical signals; and [1494] generating a plurality of sets of second parallel electrical signals, [1495] generating a plurality of fourth serial electrical signals based on the plurality of sets of second parallel electrical signals, and [1496] generating second optical signals based on the plurality of fourth serial electrical signals.

[1497] Embodiment 347: The method of embodiment 346, comprising aligning multiple serial output signals.

[1498] Embodiment 348: A method comprising: [1499] operating an autonomous vehicle, comprising performing the method of any of embodiments 304 to 347.

[1500] Embodiment 349: The method of embodiment 348 in which the vehicle comprises at least one of a car, a truck, a train, a boat, a ship, a submarine, a helicopter, a drone, an airplane, a space rover, or a space ship.

[1501] Embodiment 350: A method comprising: [1502] operating a robot, comprising performing the method of any of embodiments 304 to 347.

[1503] Embodiment 351: The method of embodiment 350 in which the robot comprises at least one

of an industrial robot, a helper robot, a medical surgery robot, a merchandise delivery robot, a teaching robot, a cleaning robot, a cooking robot, a construction robot, or an entertainment robot.

[1504] Embodiment 352a: The apparatus of embodiment 1, wherein the optical interconnect module is located on a first side of a printed circuit board.

[1505] Embodiment 353a: The apparatus of embodiment 352a, wherein another optical interconnect module is located on a second side of the printed circuit board

[1506] Embodiment 354a: The apparatus of embodiment 352a, wherein the printed circuit board is located approximate to a front panel of an enclosure.

[1507] Embodiment 355a: The apparatus of embodiment 352a, wherein the printed circuit board is located approximate to a rear panel of an enclosure.

[1508] Embodiment 356a: The apparatus of embodiment 352a, wherein the printed circuit board includes an electrical connector.

[1509] Embodiment 357a: The apparatus of embodiment 356a, wherein the electrical connector connects to another electrical connector of another printed circuit board.

[1510] Embodiment 358a: The apparatus of embodiment 356a, wherein the electrical connector connects to another electrical connector of a line card.

[1511] Embodiment 352b: An apparatus comprising: [1512] at least one of a circuit board or a substrate; and [1513] a first structure attached to the at least one of a circuit board or a substrate, in which the first structure is configured to enable an optical module with connector to be removably coupled to the first structure, and the optical module with connector is configured to enable an optical fiber connector to be removably coupled to the optical module with connector.

[1514] Embodiment 353b: The apparatus of embodiment 352b in which the circuit board or the substrate comprises first electrical contacts, the first structure comprises walls that define a first opening, the walls also define one or more retaining mechanisms such that when the optical module with connector is inserted into the first opening, the one or more retaining mechanisms on the walls of the first structure engage one or more latch mechanisms on the optical module with connector to secure the optical module with connector to the first structure, and second electrical contacts on the optical module with connector are electrically coupled to the first electrical contacts on the circuit board or the substrate.

[1515] Embodiment 354b: The apparatus of embodiment 353b, comprising at least one optical module with connector, in which the optical module with connector comprises an optical module and a mechanical connector structure, the mechanical connector structure is configured to removably couple the optical module to the circuit board or the substrate to enable electrical signals output from the optical module to be transmitted to the first electrical contacts of the circuit board or the substrate.

[1516] Embodiment 355b: The apparatus of embodiment 354b in which the mechanical connector structure is configured to receive the optical fiber connector to enable light signals from the optical fiber connector to be transmitted to the optical module.

[1517] Embodiment 356b: The apparatus of embodiment 355b, comprising the optical fiber connector, in which the optical fiber connector is optically coupled to a fiber cable comprising a plurality of optical fibers, and the optical fiber connector is configured to transmit optical signals carried in the optical fibers to the optical module.

[1518] Embodiment 356c: The apparatus of any of the embodiments 352b to 356b in which the circuit board or the substrate comprises an array of first electrical contacts comprising at least four rows and four columns of electrical contacts, the optical module comprises an array of second electrical contacts comprising at least four rows and four columns of electrical contacts, and the first structure is configured to enable the array of second electrical contacts of the optical module to be electrically coupled to the array of first electrical contacts of the circuit board or the substrate when the optical module with connector is coupled to the first structure.

[1519] Embodiment 357b: The apparatus of any of the embodiments 352b to 356c in which the

first structure comprises a grid structure that defines multiple openings, and each opening is configured to receive a corresponding optical module with connector.

[1520] Embodiment 358b: The apparatus of embodiment 357b in which the grid structure is configured such that when the optical module with connectors are inserted into the openings of the grid structure, the optical module with connectors are oriented such that each optical module with connector is rotated 90 degrees relative to at least one adjacent optical module with connector, and the rotational axis is perpendicular to the circuit board or the substrate.

[1521] Embodiment 359: The apparatus of any of embodiments 352b to 358b in which the first structure is configured to enable the optical module with connectors to be mounted on the first structure in a 90-degree rotate checkerboard fashion.

[1522] Embodiment 360: The apparatus of any of embodiment 352b to 359 in which the first structure is configured to function as a heat spreader.

[1523] Embodiment 361a: The apparatus of any of embodiments 352b to 360 in which each of the at least one of the circuit board or substrate comprises a first side and a second side, the first structure is attached to the first side of the circuit board or the substrate, an application specific integrated circuit is mounted on the second side of the circuit board or the substrate, the first structure has an opening, in which the opening is located opposite the application specific integrated circuit relative to the circuit board or the substrate, discrete circuit components are mounted on the first side of the circuit board or the substrate, and the discrete circuit components extend from the circuit board or the substrate into the opening in the structure.

[1524] Embodiment 361b: The apparatus of embodiment 361a, comprising the circuit board and the substrate, in which the first structure is attached to the first side of the circuit board, an application specific integrated circuit is mounted on the second side of the substrate, the first structure has an opening, the opening is located opposite the application specific integrated circuit relative to the substrate, discrete circuit components are mounted on the first side of the substrate, and the discrete circuit components extend from the substrate into the opening in the structure.

[1525] Embodiment 362: The apparatus of embodiment 361a or 361b in which the discrete circuit components comprise at least one of one or more voltage regulators, one or more filters, or one or more capacitors.

[1526] Embodiment 363: The apparatus of any of embodiments 352b to 362 in which the circuit board or the substrate comprises a first side and a second side, the first structure is attached to the first side of the circuit board or the substrate, a second structure is attached to the second side of the circuit board or the substrate, and the first structure is mechanically and thermally attached to the second structure.

[1527] Embodiment 364: The apparatus of embodiment 363 in which the first structure is attached to the second structure by screws that pass through the printed circuit board or the substrate.

[1528] Embodiment 365: The apparatus of embodiment 363 or 364 in which the first structure is attached to the second structure by thermal vias.

[1529] Embodiment 366: The apparatus of any of embodiments 363 to 365, comprising a heat sink attached to at least one of the first structure or the second structure.

[1530] Embodiment 367: The apparatus of any of embodiments 352b to 366, comprising a snap-in mechanism that is configured to secure the optical module with connector when the optical module with connector is inserted into the first structure.

[1531] Embodiment 368: The apparatus of embodiment 367 in which the snap-in mechanism is configured to enable the optical module with connector to be pulled away from the first structure when a force above a threshold is applied to the optical module with connector.

[1532] Embodiment 369: The apparatus of embodiment 367 or 368 in which the snap-in mechanism comprises one or more grooves formed on walls of the first structure, the optical module with connector comprises one or more elastic wings, each elastic wing comprises a tongue that is configured to engage a corresponding groove when the optical module with connector is

inserted into the first structure.

[1533] Embodiment 370: The apparatus of any of embodiments 367 to 369 in which the snap-in mechanism comprises a lever-based latch mechanism, the latch mechanism is movable between a first position and a second position, the latch mechanism engages a support structure on the first structure when in the first position and disengages from the support structure when in the second position, the optical module with connector is secured to the first structure when the latch mechanism is in the first position and released from the first structure when the latch mechanism is in the second position.

[1534] Embodiment 371: The apparatus of embodiment 370 in which a lever is provided as part of the optical module with connector, the lever is movable between a first position and a second position, the lever is configured such that moving the lever to the first position causes the latch mechanism to move to the first position, and moving the lever to the second position causes the latch mechanism to move to the second position.

[1535] Embodiment 372: The apparatus of embodiment 370 or 371 in which a lever is provided as part of a tool used to insert or remove the optical module with connector into or from the first structure.

[1536] Embodiment 373: The apparatus of any of embodiments 352b to 372 in which the optical module with connector comprises a co-packaged optical module.

[1537] Embodiment 374: The apparatus of any of embodiments 352b to 373, comprising the optical module with connector, in which the optical module with connector comprises a snap-in mechanism configured such that the optical fiber connector locks in place the snap-in mechanism when the optical fiber connector is coupled to the optical module with connector.

[1538] Embodiment 375: The apparatus of embodiment 374 in which the optical module with connector comprises a mechanical connector structure, the optical fiber connector snaps into a part of the mechanical connector structure to hold the optical fiber connector in place when the optical fiber connector is coupled to the optical module with connector.

[1539] Embodiment 376: The apparatus of embodiment 374 or 375, comprising the optical fiber connector, in which the optical fiber connector and the optical module with connector comprise a ball-detent mechanism configured to hold the optical fiber connector in place when the optical fiber connector is coupled to the optical module with connector.

[1540] Embodiment 377: An apparatus comprising: [1541] an optical module with connector configured to be removably coupled to a first structure that is attached to a circuit board or a substrate, in which the optical module comprises a photonic integrated circuit, the optical module with connector is configured to hold the photonic integrated circuit in place when the optical module with connector is coupled to the first structure and to enable electronic signals from the photonic integrated circuit to be transmitted to the circuit board or the substrate; [1542] wherein the optical module with connector is configured to enable an optical fiber connector to be removably coupled to the optical module with connector, in which the optical module with connector is configured to enable optical signals from the optical fiber connector to be transmitted to the photonic integrated circuit.

[1543] Embodiment 378: The apparatus of embodiment 377 in which the optical module with connector comprises the photonic integrated circuit of any of embodiments 1 to 297.

[1544] Embodiment 379: The apparatus of embodiment 378 in which the optical module with connector comprises at least one of the serializers/deserializers module, the serializer unit, or the deserializer unit of any of embodiments 1 to 297.

[1545] Embodiment 380a: The apparatus of embodiment 379, comprising at least one of a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, or an application specific integrated circuit (ASIC) that is configured to process signals provided by the at least one of the serializers/deserializers module, the serializer unit, or the

deserializer unit.

[1546] Embodiment 380b: The apparatus of embodiment 379, comprising at least one of a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, a storage device, or an application specific integrated circuit (ASIC) that is coupled to the circuit board or the substrate and configured to process electrical signal received from or transmitted to the photonic integrated circuit.

[1547] Embodiment 381: The apparatus of any of embodiments 379 to 380b, comprising the bus processing module of any of embodiments 6 to 287.

[1548] Embodiment 382: The apparatus of any of embodiments 377 to 381, comprising the first structure and at least one of the circuit board or the substrate, in which the circuit board or the substrate is attached to a first side of the first structure, and the optical module with connector is configured to be removably coupled to the first structure from the second side of the first structure, and the second side of the first structure is opposite to the first side of the first structure.

[1549] Embodiment 383a: The apparatus of embodiment 382, comprising two or more optical module with connectors that are removably coupled to the first structure in a 90-degree rotate checkerboard fashion.

[1550] Embodiment 383b: The apparatus of embodiment 382 or 383a, comprising the circuit board and the substrate, in which the circuit board is attached to a first side of the first structure, the substrate is attached to the circuit board, and the optical module with connector is configured to hold the photonic integrated circuit in place when the optical module with connector is coupled to the first structure and to enable electronic signals from the photonic integrated circuit to be transmitted to the substrate.

[1551] Embodiment 384: The apparatus of any of embodiments 377 to 383 in which the first structure comprises a grid structure that enables two or more optical module with connectors to be removably coupled to the first structure in an array defined by the grid structure.

[1552] Embodiment 385: A system comprising: [1553] a housing; and [1554] the apparatus of any of embodiments 352b to 384, in which the optical module with connector is configured to be removably coupled to, partially through, or through a front panel of the housing.

[1555] Embodiment 386: The system of embodiment 385 in which the first structure is part of the front panel of the housing.

[1556] Embodiment 387: The system of embodiment 385 or 386 in which the circuit board is part of the front panel of the housing.

[1557] Embodiment 388: The system of embodiment 385 in which the first structure is positioned near and spaced apart from the front panel of the housing.

[1558] Embodiment 389: The system of embodiment 388 in which the first structure has an overall structure that extends along a plane that is substantially parallel to the front panel of the housing.

[1559] Embodiment 390: The system of embodiment 388 or 389 in which the circuit board is substantially parallel to the front panel of the housing.

[1560] Embodiment 391: The system of any of embodiments 385 to 390 in which the optical module with connector is configured to enable the second optical connector of any of embodiments 33 to 59, 64 to 68, 72 to 74, 118 to 144, 149 to 153, 157 to 159, 201 to 205, 236 to 245, 274, 280, or 290 to be removably coupled to the optical module with connector.

[1561] Embodiment 392: The system of any of embodiments 206 to 278, comprising a first structure coupled to the circuit board or a substrate, the first structure configured to enable an optical module comprising the photonic integrated circuit to be removably coupled to the first structure, the first structure is configured such that when the optical module is coupled to the first structure, the photonic integrated circuit is held in place to enable electrical signals from the photonic integrated circuit to be transmitted to the circuit board or the substrate.

[1562] Embodiment 393: The system of embodiment 392 in which the first structure is configured

such that when the optical module is coupled to the first structure, the photonic integrated circuit is held in place to enable electrical signals from the photonic integrated circuit to be transmitted to at least one of a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, a storage device, or an application specific integrated circuit (ASIC) mounted on the circuit board or the substrate.

[1563] Embodiment 394: A method comprising: [1564] removably coupling an optical fiber connector to an optical module with connector; [1565] removably coupling the optical module with connector to a first structure attached to at least one of a circuit board or a substrate; and [1566] electrically coupling the optical module with connector to the circuit board or the substrate.

[1567] Embodiment 395: The method of embodiment 394, comprising: [1568] inserting the optical module with connector through a first opening defined by walls of the first structure; [1569] using one or more retaining mechanisms on the walls of the first structure to engage one or more latch mechanisms on the optical module with connector to secure the optical module with connector to the first structure; and [1570] electrically coupling first electrical contacts on the circuit board or the substrate to second electrical contacts on the optical module with connector.

[1571] Embodiment 396: The method of embodiment 394 or 395, comprising: [1572] transmitting optical signals from the optical fiber connector to the optical module with connector; [1573] at the optical module with connector, generating electrical signals based on the optical signals; and [1574] transmitting the electrical signals from the optical module with connector to the circuit board or the substrate.

[1575] Embodiment 397: The method of any of embodiments 394 to 396 in which the circuit board or the substrate comprises an array of first electrical contacts comprising at least four rows and four columns of electrical contacts, the optical module with connector comprises an array of second electrical contacts comprising at least four rows and four columns of electrical contacts, and [1576] electrically coupling the optical module with connector to the circuit board or the substrate comprises electrically coupling the array of second electrical contacts of the optical module with connector to the array of first electrical contacts of the circuit board or the substrate.

[1577] Embodiment 398: The method of any of embodiments 394 to 397 in which the first structure comprises a grid structure that defines multiple first openings, and the method comprises: [1578] removably coupling a plurality of optical modules with connectors to the grid structure, including inserting each optical module with connector through a corresponding first opening defined by the walls of the grid structure; [1579] for each of the plurality of optical modules with connectors, using one or more retaining mechanisms on the walls of the grid structure to engage one or more latch mechanisms on the optical module with connector to secure the optical module with connector to the grid structure; and [1580] for each of the plurality of optical modules with connectors, electrically coupling second electrical contacts on the optical module with connector to corresponding first electrical contacts on the circuit board or the substrate.

[1581] Embodiment 399: The method of embodiment 398, comprising orienting the optical modules with connectors such that each optical module with connector is rotated 90 degrees relative to at least one adjacent optical module with connector, and the rotational axis is perpendicular to the circuit board or the substrate.

[1582] Embodiment 400: The method of any of embodiments 394 to 399, comprising removably coupling a plurality of optical modules with connectors to the first structure; and [1583] orienting the optical modules with connectors coupled to the first structure in a 90-degree rotate checkerboard fashion.

[1584] Embodiment 401: The method of any of embodiments 394 to 400 in which removably coupling the optical module with connector to the first structure comprises securing, using a snap-in mechanism, the optical module with connector to the first structure.

[1585] Embodiment 402: The method of embodiment 401, comprising applying a force above a

threshold to the optical module with connector to disengage the snap-in mechanism, and pulling the optical module with connector away from the first structure.

[1586] Embodiment 403: The method of embodiment 401 or 402 in which the snap-in mechanism comprises one or more grooves formed on walls of the first structure, the optical module with connector comprises one or more elastic wings, and each elastic wing comprises a tongue; [1587] wherein the method comprises engaging the tongue of the elastic wing with a corresponding groove formed on walls of the first structure when inserting the optical module with connector into an opening defined by the walls of the first structure.

[1588] Embodiment 404: The method of embodiment 403 in which the snap-in mechanism comprises a lever-based latch mechanism, the latch mechanism is movable between a first position and a second position, the latch mechanism engages a support structure on the first structure when in the first position and disengages from the support structure when in the second position, [1589] wherein the method comprises placing the latch mechanism in the first position to secure the optical module with connector to the first structure; and placing the latch mechanism in the second position to release the optical module with connector from the first structure.

[1590] Embodiment 405: The method of embodiment 404 in which a lever is provided as part of the optical module with connector, wherein the method comprises: [1591] moving the lever to a first position to cause the latch mechanism to move to the first position, and [1592] moving the lever to a second position causes the latch mechanism to move to the second position.

[1593] Embodiment 406: The method of any of embodiments 394 to 405 in which the optical module with connector comprises a co-packaged optical module.

[1594] Embodiment 407: A method comprising: [1595] attaching a first structure to a first side of a circuit board or a substrate; [1596] mounting an application specific integrated circuit on a second side of the circuit board or the substrate, in which the first structure has an opening located opposite the application specific integrated circuit relative to the circuit board or the substrate; and [1597] mounting discrete circuit components on the first side of the circuit board or the substrate, in which the discrete circuit components extend from the circuit board or the substrate into the opening in the structure.

[1598] Embodiment 408: The method of embodiment 407, comprising: [1599] attaching the first structure to the first side of the circuit board; [1600] mounting the application specific integrated circuit on the second side of the substrate; and [1601] mounting discrete circuit components on the first side of the substrate, in which the discrete circuit components extend from the substrate into the opening in the structure.

[1602] Embodiment 409: The method of embodiment 407 or 408 in which the discrete circuit components comprise at least one of one or more voltage regulators, one or more filters, or one or more capacitors.

[1603] Embodiment 410: The method of any of embodiments 407 to 409, comprising attaching a second structure to a second side of the circuit board or the substrate, and [1604] mechanically and thermally coupling the first structure to the second structure.

[1605] Embodiment 411: The method of embodiment 410, comprising attaching the first structure to the second structure using screws that pass through the circuit board or the substrate.

[1606] Embodiment 412: The method of embodiment 410 or 411, comprising attaching the first structure to the second structure using thermal vias.

[1607] Embodiment 413: The method of any of embodiments 410 to 412, comprising attaching a heat sink to at least one of the first structure or the second structure.

[1608] Embodiment 414: An apparatus comprising: [1609] at least one of a circuit board or a substrate; [1610] a data processor unit mounted on a first side of the circuit board or the substrate; and [1611] a first structure attached to a second side of the circuit board or the substrate, in which the second side of the circuit board or the substrate comprises a two-dimensional arrangement of second electrical contacts, the first structure is configured to enable an optical module with a two-

dimensional arrangement of first electrical contacts to be removably coupled to the first structure and enable the optical module to be electrically coupled to the circuit board or the substrate through the two-dimensional arrangement of first electrical contacts and the two-dimensional arrangement of second electrical contacts.

[1612] Embodiment 415: The apparatus of embodiment 414 wherein the optical module comprises a connector configured to enable an optical fiber connector to be removably coupled to the optical module.

[1613] Embodiment 416: The apparatus of embodiment 414 or 415 wherein the data processor unit comprises at least one of a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, a storage device, or an application specific integrated circuit (ASIC) that is configured to process electrical signals received from or transmitted to the optical module.

[1614] Embodiment 417: The apparatus of any of embodiments 414 to 416 in which the data processor unit comprises at least one of (i) a bare die processor that is mounted on the first side of the circuit board or the substrate, or (ii) a packaged processor that includes a package substrate that is mounted on the first side of the circuit board or the substrate.

[1615] Embodiment 418: The apparatus of any of embodiments 414 to 417 in which the first structure comprises walls that define a first opening, the walls also define one or more retaining mechanisms such that when the optical module is inserted into the first opening, the one or more retaining mechanisms on the walls of the first structure engage one or more latch mechanisms on the optical module to secure the optical module to the first structure, and the first electrical contacts on the optical module are electrically coupled to the second electrical contacts on the circuit board or the substrate.

[1616] Embodiment 419: The apparatus of embodiment 418, comprising at least one optical module, in which the optical module comprises a mechanical connector structure configured to removably couple the optical module to the circuit board or the substrate to enable electrical signals to be transmitted between the optical module and the second electrical contacts of the circuit board or the substrate.

[1617] Embodiment 420: The apparatus of embodiment 419 in which the optical module comprises a connector configured to enable an optical fiber connector to be removably coupled to the optical module, and the mechanical connector structure is configured to receive the optical fiber connector to enable light signals from the optical fiber connector to be transmitted to the optical module.

[1618] Embodiment 421: The apparatus of embodiment 420, comprising the optical fiber connector, in which the optical fiber connector is optically coupled to a fiber cable comprising a plurality of optical fibers, and the optical fiber connector is configured to transmit optical signals carried in the optical fibers to the optical module.

[1619] Embodiment 422: The apparatus of embodiment 421 wherein the optical module comprises a photonic integrated circuit comprising a two-dimensional array of optical coupling elements, and the optical fiber connector is configured to couple a two-dimensional array of optical fibers to the two-dimensional array of optical coupling elements on the photonic integrated circuit.

[1620] Embodiment 423: The apparatus of embodiment 422 wherein the two-dimension array of optical fibers comprise at least a 3×12 array of optical fibers.

[1621] Embodiment 424: The apparatus of any of embodiments 414 to 423 in which the arrangement of second electrical contacts of the circuit board or the substrate comprises at least four rows and four columns of second electrical contacts, the arrangement of first electrical contacts of the optical module comprises at least four rows and four columns of first electrical contacts, and the first structure is configured to enable the at least four rows and four columns of first electrical contacts of the optical module to be electrically coupled to the at least four rows and four columns of second electrical contacts of the circuit board or the substrate when the optical

module is coupled to the first structure.

[1622] Embodiment 425: The apparatus of any of embodiments 414 to 424 in which the first structure comprises a grid structure that defines multiple openings, and each opening is configured to receive a corresponding optical module.

[1623] Embodiment 426: The apparatus of embodiment 425 in which the grid structure is configured such that when the optical modules are inserted into the openings of the grid structure, the optical modules are oriented such that each optical module is rotated 90 degrees relative to at least one adjacent optical module, and the rotational axis is perpendicular to the circuit board or the substrate.

[1624] Embodiment 427: The apparatus of any of embodiments 414 to 426 in which the first structure is configured to enable the optical modules to be mounted on the first structure in a 90-degree rotate checkerboard fashion.

[1625] Embodiment 428: The apparatus of any of embodiments 414 to 427 in which the first structure is configured to function as a heat spreader.

[1626] Embodiment 429: The apparatus of any of embodiments 414 to 428 in which the first structure has an opening located opposite the data processor unit relative to the circuit board or the substrate, discrete circuit components are mounted on the second side of the circuit board or the substrate, and the discrete circuit components extend from the circuit board or the substrate into the opening in the first structure.

[1627] Embodiment 430: The apparatus of embodiment 429, comprising the circuit board and the substrate, in which the first structure is attached to the second side of the circuit board, the data processor unit is mounted on the first side of the substrate, the opening in the first structure is located opposite the data processor unit relative to the substrate, discrete circuit components are mounted on the second side of the substrate, and the discrete circuit components extend from the substrate into the opening in the structure.

[1628] Embodiment 431: The apparatus of embodiment 429 or 430 in which the discrete circuit components comprise at least one of one or more voltage regulators, one or more filters, or one or more capacitors.

[1629] Embodiment 432: The apparatus of any of embodiments 414 to 431 in which a second structure is attached to a first side of the circuit board or the substrate, and the first structure is mechanically and thermally attached to the second structure.

[1630] Embodiment 433: The apparatus of embodiment 432 in which the first structure is attached to the second structure by screws that pass through the printed circuit board or the substrate.

[1631] Embodiment 434: The apparatus of embodiment 432 or 433 in which the first structure is attached to the second structure by thermal vias.

[1632] Embodiment 435: The apparatus of any of embodiments 432 to 434, comprising a heat sink attached to at least one of the first structure or the second structure.

[1633] Embodiment 436: The apparatus of any of embodiments 414 to 435, comprising a snap-in mechanism that is configured to secure the optical module when the optical module is inserted into the first structure.

[1634] Embodiment 437: The apparatus of embodiment 436 in which the snap-in mechanism is configured to enable the optical module to be pulled away from the first structure when a force above a threshold is applied to the optical module.

[1635] Embodiment 438: The apparatus of embodiment 436 or 437 in which the snap-in mechanism comprises one or more grooves formed on walls of the first structure, the optical module comprises one or more elastic wings, and each elastic wing comprises a tongue that is configured to engage a corresponding groove when the optical module is inserted into the first structure.

[1636] Embodiment 439: The apparatus of any of embodiments 436 to 438 in which the snap-in mechanism comprises a lever-based latch mechanism, the latch mechanism is movable between a

first position and a second position, the latch mechanism engages a support structure on the first structure when in the first position and disengages from the support structure when in the second position, the optical module is secured to the first structure when the latch mechanism is in the first position and released from the first structure when the latch mechanism is in the second position.

[1637] Embodiment 440: The apparatus of any of embodiments 436 to 439 in which a lever is provided as part of the optical module, the lever is movable between a first position and a second position, the lever is configured such that moving the lever to the first position causes the latch mechanism to move to the first position, and moving the lever to the second position causes the latch mechanism to move to the second position.

[1638] Embodiment 441: The apparatus of any of embodiments 436 to 440 in which a lever is provided as part of a tool used to insert or remove the optical module into or from the first structure.

[1639] Embodiment 442: The apparatus of any of embodiments 414 to 441 in which the optical module comprises a co-packaged optical module.

[1640] Embodiment 443: The apparatus of embodiment 373 or 442 in which the co-packaged optical module comprises: [1641] a photonic integrated circuit configured to perform at least one of converting first optical signals to first electrical signals or converting second electrical signals to second optical signals; [1642] the two-dimensional arrangement of first electrical contacts, wherein the two-dimensional arrangement of first electrical contacts are configured to at least one of (i) enable the first electrical signals to be transmitted from the photonic integrated circuit to the circuit board or the substrate, or (ii) enable the second electrical signals to be transmitted from the circuit board or the substrate to the photonic integrated circuit; and [1643] an optical fiber connector part configured to be optically coupled to an array of optical fibers, wherein the optical fiber connector part is configured to at least one of (i) receive the first optical signals from the array of optical fibers and transmit the first optical signals to the photonic integrated circuit, or (ii) receive the second optical signals from the photonic integrated circuit and transmit the second optical signals to the array of optical fibers.

[1644] Embodiment 444: The apparatus of any of embodiments 373, 442, and 443 in which the co-packaged optical module comprises at least one of: [1645] micro-optics assembly facilitating fiber-to-photonic integrated circuit coupling, [1646] one or more driver amplifiers, [1647] one or more trans-impedance amplifiers, [1648] one or more SerDes integrated circuits, [1649] one or more digital signal processors, [1650] one or more lasers or light emitting diodes, or [1651] one or more microcontrollers.

[1652] Embodiment 445: The apparatus of any of embodiments 414 to 444, comprising the optical module, in which the optical module comprises a snap-in mechanism configured such that the optical fiber connector locks in place the snap-in mechanism when the optical fiber connector is coupled to the optical module.

[1653] Embodiment 446: The apparatus of embodiment 445 in which the optical module comprises a mechanical connector structure, the optical fiber connector snaps into a part of the mechanical connector structure to hold the optical fiber connector in place when the optical fiber connector is coupled to the optical module.

[1654] Embodiment 447: The apparatus of embodiment 445 or 446, comprising the optical fiber connector, in which the optical fiber connector and the optical module comprise a ball-detent mechanism configured to hold the optical fiber connector in place when the optical fiber connector is coupled to the optical module.

[1655] Embodiment 448: An apparatus comprising: [1656] an optical module configured to be removably coupled to a first structure that is attached to a circuit board or a substrate, in which the optical module comprises a two-dimensional arrangement of at least 16 electrical contacts distributed in a two-dimensional surface area of the optical module, the optical module comprises a photonic integrated circuit, the optical module is configured to hold the photonic integrated circuit

in place when the optical module is coupled to the first structure and to enable electronic signals from the photonic integrated circuit to be transmitted to the circuit board or the substrate through the two-dimensional arrangement of at least 16 electrical contacts; [1657] wherein the optical module comprises a connector part that is configured to enable an optical fiber connector to be removably coupled to the optical module, in which the optical module is configured to enable optical signals from the optical fiber connector to be transmitted to the photonic integrated circuit.

[1658] Embodiment 449: The apparatus of embodiment 448 wherein the optical fiber connector is configured to couple a two-dimensional array of optical fibers to a two-dimensional array of optical coupling elements on the photonic integrated circuit.

[1659] Embodiment 450: The apparatus of embodiment 449 wherein the two-dimension array of optical fibers comprise at least a 3×12 array of optical fibers.

[1660] Embodiment 451: The apparatus of any of embodiments 448 to 450, comprising at least one of a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, a storage device, or an application specific integrated circuit (ASIC) that is coupled to the circuit board or the substrate and configured to process electrical signals received from or transmitted to the photonic integrated circuit.

[1661] Embodiment 452: The apparatus of any of embodiments 448 to 451, comprising the first structure and at least one of the circuit board or the substrate, in which the circuit board or the substrate is attached to a first side of the first structure, and the optical module is configured to be removably coupled to the first structure from the second side of the first structure, and the second side of the first structure is opposite to the first side of the first structure.

[1662] Embodiment 453: The apparatus of embodiment 452, comprising two or more optical modules that are removably coupled to the first structure in a 90-degree rotate checkerboard fashion.

[1663] Embodiment 454: The apparatus of embodiment 452 or 453, comprising the circuit board and the substrate, in which the circuit board is attached to the first side of the first structure, the substrate is attached to the circuit board, and the optical module is configured to hold the photonic integrated circuit in place when the optical module is coupled to the first structure and to enable electronic signals from the photonic integrated circuit to be transmitted to the substrate.

[1664] Embodiment 455: The apparatus of any of embodiments 448 to 454 in which the first structure comprises a grid structure that enables two or more optical modules to be removably coupled to the first structure in an array defined by the grid structure.

[1665] Embodiment 456: A system comprising: [1666] a housing; and [1667] the apparatus of any of embodiments 414 to 455, in which the optical module is configured to be removably coupled to, partially through, or through a front panel of the housing.

[1668] Embodiment 457: The system of embodiment 456 in which the first structure is part of the front panel of the housing.

[1669] Embodiment 458: The system of embodiment 456 or 457 in which the circuit board is part of the front panel of the housing.

[1670] Embodiment 459: The system of embodiment 456 in which the first structure is positioned near and spaced apart from the front panel of the housing.

[1671] Embodiment 460: The system of embodiment 459 in which the first structure has an overall structure that extends along a plane that is substantially parallel to the front panel of the housing.

[1672] Embodiment 461: The system of embodiment 459 or 460 in which the circuit board is substantially parallel to the front panel of the housing.

[1673] Embodiment 462: A method comprising: [1674] providing at least one of a circuit board or a substrate, in which a data processor unit is mounted on a first side of the circuit board or the substrate, a first structure is attached to a second side of the circuit board or the substrate, and the second side of the circuit board or the substrate comprises a two-dimensional arrangement of

second electrical contacts; [1675] removably coupling an optical module to the first structure, in which the optical module comprises a two-dimensional arrangement of first electrical contacts; and [1676] electrically coupling the optical module to the second side of the circuit board or the substrate through the two-dimensional arrangement of first electrical contacts and the two-dimensional arrangement of second electrical contacts.

[1677] Embodiment 463: The method of embodiment 462, comprising removably coupling an optical fiber connector to the optical module.

[1678] Embodiment 464: The method of embodiment 462 or 463, comprising: [1679] inserting the optical module through a first opening defined by walls of the first structure; [1680] using one or more retaining mechanisms on the walls of the first structure to engage one or more latch mechanisms on the optical module to secure the optical module to the first structure; and [1681] electrically coupling the second electrical contacts on the circuit board or the substrate to the first electrical contacts on the optical module.

[1682] Embodiment 465: The method of any of embodiments 462 to 464, comprising: [1683] transmitting optical signals from an optical fiber connector to the optical module; [1684] at the optical module, generating electrical signals based on the optical signals; and [1685] transmitting the electrical signals from the optical module to the circuit board or the substrate.

[1686] Embodiment 466: The method of embodiment 465 wherein the optical module comprises a photonic integrated circuit comprising a two-dimensional array of optical coupling elements, and [1687] the method, comprises using the optical fiber connector to couple a two-dimensional array of optical fibers to the two-dimensional array of optical coupling elements on the photonic integrated circuit.

[1688] Embodiment 467: The method of embodiment 466 wherein the two-dimension array of optical fibers comprise at least a 3×12 array of optical fibers.

[1689] Embodiment 468: The method of any of embodiments 462 to 467 in which the arrangement of second electrical contacts on the circuit board or the substrate comprises an array of at least four rows and four columns of electrical contacts, the arrangement of first electrical contacts of the optical module comprises an array of at least four rows and four columns of electrical contacts, and [1690] electrically coupling the optical module to the second side of the circuit board or the substrate comprises electrically coupling the array of at least four rows and four columns of electrical contacts of the optical module to the array of at least four rows and four columns of electrical contacts of the circuit board or the substrate.

[1691] Embodiment 469: The method of any of embodiments 462 to 468 in which the first structure comprises a grid structure that defines multiple first openings, and the method comprises: [1692] removably coupling a plurality of optical modules to the grid structure, including inserting each optical module through a corresponding first opening defined by the walls of the grid structure; [1693] for each of the plurality of optical modules, using one or more retaining mechanisms on the walls of the grid structure to engage one or more latch mechanisms on the optical module to secure the optical module to the grid structure; and [1694] for each of the plurality of optical modules, electrically coupling the first electrical contacts on the optical module to corresponding second electrical contacts on the circuit board or the substrate.

[1695] Embodiment 470: The method of embodiment 469, comprising orienting the optical modules such that each optical module is rotated 90 degrees relative to at least one adjacent optical module, and the rotational axis is perpendicular to the circuit board or the substrate.

[1696] Embodiment 471: The method of any of embodiments 462 to 468, comprising removably coupling a plurality of optical modules to the first structure; and [1697] orienting the optical modules coupled to the first structure in a 90-degree rotate checkerboard fashion.

[1698] Embodiment 472: The method of any of embodiments 462 to 471 in which removably coupling the optical module to the first structure comprises securing, using a snap-in mechanism, the optical module to the first structure.

[1699] Embodiment 473: The method of embodiment 472, comprising applying a force above a threshold to the optical module to disengage the snap-in mechanism, and pulling the optical module away from the first structure.

[1700] Embodiment 474: The method of embodiment 472 or 473 in which the snap-in mechanism comprises one or more grooves formed on walls of the first structure, the optical module comprises one or more elastic wings, and each elastic wing comprises a tongue; [1701] wherein the method comprises engaging the tongue of the elastic wing with a corresponding groove formed on walls of the first structure when inserting the optical module into an opening defined by the walls of the first structure.

[1702] Embodiment 475: The method of embodiment 474 in which the snap-in mechanism comprises a lever-based latch mechanism, the latch mechanism is movable between a first position and a second position, the latch mechanism engages a support structure on the first structure when in the first position and disengages from the support structure when in the second position, [1703] wherein the method comprises placing the latch mechanism in the first position to secure the optical module to the first structure; and placing the latch mechanism in the second position to release the optical module from the first structure.

[1704] Embodiment 476: The method of embodiment 475 in which a lever is provided as part of the optical module, wherein the method comprises: [1705] moving the lever to a first position to cause the latch mechanism to move to the first position, and [1706] moving the lever to a second position causes the latch mechanism to move to the second position.

[1707] Embodiment 477: The method of any of embodiments 462 to 476 in which the optical module comprises a co-packaged optical module.

[1708] Embodiment 478: The method of embodiment 406 or 477 in which the co-packaged optical module comprises: [1709] a photonic integrated circuit configured to perform at least one of converting first optical signals to first electrical signals or converting second electrical signals to second optical signals; [1710] the two-dimensional arrangement of first electrical contacts, wherein the two-dimensional arrangement of first electrical contacts are configured to at least one of (i) enable the first electrical signals to be transmitted from the photonic integrated circuit to the circuit board or the substrate, or (ii) enable the second electrical signals to be transmitted from the circuit board or the substrate to the photonic integrated circuit; and [1711] an optical fiber connector part configured to be optically coupled to an array of optical fibers, wherein the optical fiber connector part is configured to at least one of (i) receive the first optical signals from the array of optical fibers and transmit the first optical signals to the photonic integrated circuit, or (ii) receive the second optical signals from the photonic integrated circuit and transmit the second optical signals to the array of optical fibers.

[1712] Embodiment 479: The method of any of embodiments 406, 477, and 478 in which the co-packaged optical module comprises at least one of: [1713] micro-optics assembly facilitating fiber-to-photonic integrated circuit coupling, [1714] one or more driver amplifiers, [1715] one or more trans-impedance amplifiers, [1716] one or more SerDes integrated circuits, [1717] one or more digital signal processors, [1718] one or more lasers or light emitting diodes, or [1719] one or more microcontrollers.

[1720] Embodiment 480: A method comprising: [1721] attaching a first structure to a first side of a circuit board or a substrate, the circuit board or the substrate comprises a plurality of sets of two-dimensional arrangements of electrical contacts on the first side; [1722] mounting an application specific integrated circuit on a second side of the circuit board or the substrate, in which the first structure has a first opening located opposite the application specific integrated circuit relative to the circuit board or the substrate, the first structure defines second openings that are configured to receive optical modules, each second opening corresponding to a set of two-dimensional arrangement of electrical contacts in the plurality of sets of two-dimensional arrangements of electrical contacts, the first structure is configured to enable each optical module to be electrically

coupled to the circuit board or the substrate through the corresponding set of two-dimensional arrangement of electrical contacts when the optical module is inserted through the corresponding second opening defined by the first structure; and [1723] mounting discrete circuit components on the first side of the circuit board or the substrate, in which the discrete circuit components extend from the circuit board or the substrate into the first opening in the first structure.

[1724] Embodiment 481: The method of embodiment 480, comprising: [1725] attaching the first structure to the first side of the circuit board; [1726] mounting the application specific integrated circuit on the second side of the substrate; and [1727] mounting discrete circuit components on the first side of the substrate, in which the discrete circuit components extend from the substrate into the opening in the structure.

[1728] Embodiment 482: The method of embodiment 480 or 481 in which the discrete circuit components comprise at least one of one or more voltage regulators, one or more filters, or one or more capacitors.

[1729] Embodiment 483: The method of any of embodiments 480 to 482, comprising attaching a second structure to a second side of the circuit board or the substrate, and [1730] mechanically and thermally coupling the first structure to the second structure.

[1731] Embodiment 484: The method of embodiment 483, comprising attaching the first structure to the second structure using screws that pass through the circuit board or the substrate.

[1732] Embodiment 485: The method of embodiment 483, comprising attaching the first structure to the second structure using thermal vias.

[1733] Embodiment 486: The method of any of embodiments 483 to 485, comprising attaching a heat sink to at least one of the first structure or the second structure.

[1734] The following is a second set of embodiments. The embodiment numbers below refer to those in the second set of embodiments.

[1735] Embodiment 1: An apparatus comprising: [1736] a first substrate having a first side and a second side; [1737] a first electronic processor mounted on the first side of the first substrate, wherein the first electronic processor is configured to process data; and [1738] a first optical interconnect module mounted on the second side of the first substrate, in which the first optical interconnect module comprises: [1739] an optical port configured to receive optical signals, and [1740] a photonic integrated circuit configured to generate electrical signals based on the received optical signals, and transmit the electrical signals to the first electronic processor.

[1741] Embodiment 2: The apparatus of embodiment 1 in which the first electronic processor comprises at least a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, an application specific integrated circuit (ASIC), or a data storage device.

[1742] Embodiment 3: The apparatus of embodiment 1 or 2 wherein the first optical interconnect module comprises: [1743] a first serializers/deserializers module comprising multiple serializer units and deserializer units, [1744] a second serializers/deserializers module comprising multiple serializer units and deserializer units, [1745] wherein the first photonic integrated circuit is configured to generate first serial electrical signals based on the received optical signals, [1746] wherein the first serializers/deserializers module is configured to generate first parallel electrical signals based on the first serial electrical signals, and condition the electrical signals; [1747] wherein the second serializers/deserializers module is configured to generate second serial electrical signals based on the first parallel electrical signals, and the second serial electrical signals are transmitted toward the first electronic processor.

[1748] Embodiment 4: The apparatus of embodiment 3, comprising a third serializers/deserializers module comprising multiple serializer units and deserializer units, in which the third serializers/deserializers module is configured to generate second parallel electrical signals based on the second serial electrical signals, and transmit the second serial electrical signals to the first

electronic processor.

[1749] Embodiment 5: The apparatus of any of embodiments 1 to 4 in which the first substrate comprises electrical connectors that extend from the first side of the first substrate to the second side of the first substrate, the electrical connectors pass through the first substrate from the first side to the second side in a thickness direction, [1750] wherein the first optical interconnect module is electrically coupled to the first electronic processor by the electrical connectors.

[1751] Embodiment 6: The apparatus of embodiment 5 in which the electrical connectors comprise vias of the first substrate.

[1752] Embodiment 7: The apparatus of any of embodiments 1 to 6 in which the first substrate comprises a first printed circuit board.

[1753] Embodiment 8: The apparatus of any of embodiments 1 to 7, comprising a first structure attached to the second side of the first substrate and configured to enable the first optical interconnect module to be removably coupled to the first structure.

[1754] Embodiment 9: The apparatus of embodiment 8 in which the first substrate comprises a second surface on the second side of the first substrate, and the second surface comprises second electrical contacts that are electrically coupled to the first electronic processor, [1755] wherein the first optical interconnect module comprises electrical contacts that are electrically coupled to the second electrical contacts on the second surface of the first substrate when the first optical interconnect module is coupled to the first structure.

[1756] Embodiment 10: The apparatus of embodiment 9 wherein the first structure is configured to enable an optical fiber connector to be removably coupled to the first optical interconnect module.

[1757] Embodiment 11: The apparatus of any of embodiments 1 to 10, comprising: [1758] a second substrate having a first side and a second side; [1759] a second electronic processor mounted on the first side of the second substrate, wherein the second electronic processor is configured to process data; and [1760] a second optical interconnect module mounted on the second side of the second substrate, in which the second optical interconnect module comprises: [1761] an optical port configured to receive optical signals, and [1762] a photonic integrated circuit configured to generate electrical signals based on the received optical signals, and transmit the electrical signals to the second electronic processor; and [1763] an optical power supply comprising at least one laser, in which the optical power supply is configured to provide a first light source to the photonic integrated circuit of the first optical interconnect module through a first optical link and to provide a second light source to the photonic integrated circuit of the second optical interconnect module through a second optical link.

[1764] Embodiment 12: The apparatus of embodiment 11, in which the first substrate and the second substrate are disposed in a first housing, and the optical power supply is disposed in a second housing that is external to the first housing.

[1765] Embodiment 13: The apparatus of any of embodiments 1 to 10, comprising: [1766] a second substrate having a first side and a second side; [1767] a second electronic processor mounted on the first side of the second substrate, wherein the second electronic processor is configured to process data; and [1768] a second optical interconnect module mounted on the second side of the second substrate, in which the second optical interconnect module comprises: [1769] an optical port configured to receive optical signals, and [1770] a photonic integrated circuit configured to generate electrical signals based on the received optical signals, and transmit the electrical signals to the second electronic processor; [1771] a support structure to support the first and second substrates, in which the second substrate is oriented parallel to the first substrate.

[1772] Embodiment 14: A system comprising: [1773] a plurality of data processing modules, in which each data processing module comprises a substrate having a first side and a second side, an electronic processor mounted on the first side of the substrate, and an optical interconnect module mounted on the second side of the substrate, the optical interconnect module comprises an optical port configured to receive optical signals, and a photonic integrated circuit configured to generate

electrical signals based on the received optical signals and transmit the electrical signals to the electronic processor.

[1774] Embodiment 15: The system of embodiment 14, comprising a structure to support the plurality of data processing modules in a way such that the substrates of the data processing modules are oriented parallel to one another.

[1775] Embodiment 16: The system of embodiment 15 in which the structure supports the data processing modules in a way such that the substrates are oriented vertically to enhance dissipation of heat from at least one of the data processing module or the optical interconnect module of each data processing module.

[1776] Embodiment 17: The system of any of embodiments 14 to 16, comprising an optical power supply comprising at least one laser, in which the optical power supply is configured to provide a plurality of light sources to the plurality of data processing modules, at least one light source is provided to the photonic integrated circuit of each data processing module through an optical link.

[1777] Embodiment 18: The system of any of embodiments 14 to 17 in which the electronic processor of each data processing module comprises at least a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, an application specific integrated circuit (ASIC), or a data storage device.

[1778] Embodiment 19: The system of any of embodiments 14 to 18 in which the plurality of data processing modules comprise a blade pair that comprises a switch blade and a processor blade, the electronic processor of the switch blade comprises a switch, and the electronic processor of the processor blade is configured to process data provided by the switch.

[1779] Embodiment 20: A system comprising: [1780] a plurality of racks of data processing modules, in which multiple racks are stacked vertically, and each rack comprises a plurality of data processing modules; [1781] wherein each data processing module comprises a substrate having a first side and a second side, an electronic processor mounted on the first side of the substrate, and an optical interconnect module mounted on the second side of the substrate, the optical interconnect module comprises an optical port configured to receive optical signals, and a photonic integrated circuit configured to generate electrical signals based on the received optical signals and transmit the electrical signals to the electronic processor.

[1782] Embodiment 21: The system of embodiment 20, comprising a structure to support the plurality of data processing modules in a way such that the substrates of the data processing modules are oriented parallel to one another.

[1783] Embodiment 22: The system of embodiment 21 in which the structure supports the data processing modules in a way such that the substrates are oriented vertically to enhance dissipation of heat from at least one of the data processing module or the optical interconnect module of each data processing module.

[1784] Embodiment 23: The system of any of embodiments 20 to 22, comprising an optical power supply comprising at least one laser, in which the optical power supply is configured to provide a plurality of light sources to the plurality of data processing modules, at least one light source is provided to the photonic integrated circuit of each data processing module through an optical link.

[1785] Embodiment 24: The system of any of embodiments 20 to 23 in which the electronic processor of each data processing module comprises at least a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, an application specific integrated circuit (ASIC), or a data storage device.

[1786] Embodiment 25: The system of any of embodiments 20 to 24 in which the plurality of data processing modules comprise a blade pair that comprises a switch blade and a processor blade, the electronic processor of the switch blade comprises a switch, and the electronic processor of the processor blade is configured to process data provided by the switch.

[1787] Embodiment 26: A method comprising: [1788] operating a plurality of data processing modules, in which each data processing module comprises a substrate having a first side and a second side, an electronic processor mounted on the first side of the substrate, and an optical interconnect module mounted on the second side of the substrate, the optical interconnect module comprises an optical port and a photonic integrated circuit; and [1789] for each data processing module, receiving optical signals at the optical port; [1790] using the photonic integrated circuit to generate electrical signals based on the optical signals received at the optical port; and [1791] transmitting the electrical signals from the photonic integrated circuit to the electronic processor through electrical connectors that extend from the first side of the substrate to the second side of the substrate.

[1792] Embodiment 27: An apparatus comprising: [1793] a first substrate having a first side and a second side; [1794] a first electronic processor mounted on the first side of the first substrate, wherein the first electronic processor is configured to process data; and [1795] a first optical interconnect module comprising: [1796] an optical port configured to receive optical signals from a first optical fiber cable, and [1797] a photonic integrated circuit configured to generate electrical signals based on the received optical signals, and transmit the electrical signals to the first electronic processor; [1798] wherein at least one of the first optical interconnect module or the first optical fiber cable extends through or partially through an opening in the first substrate to enable at least a portion of the first optical fiber cable to be positioned on or near the second side of the first substrate.

[1799] Embodiment 28: The apparatus of embodiment 27 in which the first optical interconnect module and the first optical fiber cable define a signal path that extends from the second side of the substrate through the opening to the first electronic processor.

[1800] Embodiment 29: The apparatus of embodiment 27 or 28 in which the first electronic processor comprises at least a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, an application specific integrated circuit (ASIC), or a data storage device.

[1801] Embodiment 30: The apparatus of any of embodiments 27 to 29 wherein the first optical interconnect module comprises: [1802] a first serializers/deserializers module comprising multiple serializer units and deserializer units, [1803] a second serializers/deserializers module comprising multiple serializer units and deserializer units, [1804] wherein the first photonic integrated circuit is configured to generate first serial electrical signals based on the received optical signals, [1805] wherein the first serializers/deserializers module is configured to generate first parallel electrical signals based on the first serial electrical signals, and condition the electrical signals; [1806] wherein the second serializers/deserializers module is configured to generate second serial electrical signals based on the first parallel electrical signals, and the second serial electrical signals are transmitted toward the first electronic processor.

[1807] Embodiment 31: The apparatus of embodiment 30, comprising a third serializers/deserializers module comprising multiple serializer units and deserializer units, in which the third serializers/deserializers module is configured to generate second parallel electrical signals based on the second serial electrical signals, and transmit the second serial electrical signals to the first electronic processor.

[1808] Embodiment 32: The apparatus of any of embodiments 27 to 31 in which the first substrate comprises a first printed circuit board.

[1809] Embodiment 33: The apparatus of any of embodiments 27 to 32, comprising: [1810] a second substrate having a first side and a second side; [1811] a second electronic processor mounted on the first side of the second substrate, wherein the second electronic processor is configured to process data; and [1812] a second optical interconnect module comprising: [1813] an optical port configured to receive optical signals from a second optical fiber cable, and [1814] a

phonic integrated circuit configured to generate electrical signals based on the received optical signals, and transmit the electrical signals to the second electronic processor; [1815] wherein at least one of the second optical interconnect module or the second optical fiber cable extends through or partially through an opening in the second substrate to enable at least a portion of the second optical fiber cable to be positioned on or near the second side of the second substrate.

[1816] Embodiment 34: The apparatus of embodiment 33, comprising an optical power supply comprising at least one laser, in which the optical power supply is configured to provide a first light source to the photonic integrated circuit of the first optical interconnect module through a first optical link and provide a second light source to the photonic integrated circuit of the second optical interconnect module through a second optical link.

[1817] Embodiment 35: The apparatus of embodiment 34, in which the first substrate and the second substrate are disposed in a first housing, and the optical power supply is disposed in a second housing that is external to the first housing.

[1818] Embodiment 36: The apparatus any of embodiments 33 to 35, comprising a support structure to support the first and second substrates, in which the second substrate is oriented parallel to the first substrate.

[1819] Embodiment 37: A system comprising: [1820] a plurality of data processing modules, in which each data processing module comprises a substrate having a first side and a second side, an electronic processor mounted on the first side of the substrate, and an optical interconnect module comprising an optical port configured to receive optical signals from an optical fiber cable, and a photonic integrated circuit configured to generate electrical signals based on the received optical signals and transmit the electrical signals to the electronic processor, [1821] wherein for each data processing module, at least one of the optical interconnect module or the optical fiber cable extends through or partially through an opening in the substrate to enable at least a portion of the optical fiber cable to be positioned on or near the second side of the substrate.

[1822] Embodiment 38: The system of embodiment 37, comprising a structure to support the plurality of data processing modules in a way such that the substrates of the data processing modules are oriented parallel to one another.

[1823] Embodiment 39: The system of embodiment 38 in which the structure supports the data processing modules in a way such that the substrates are oriented vertically to enhance dissipation of heat from at least one of the data processing module or the optical interconnect module of each data processing module.

[1824] Embodiment 40: The system of any of embodiments 37 to 39 in which for each data processing module, the optical interconnect module and the optical fiber cable define a signal path that extends from the second side of the substrate through the opening to the electronic processor.

[1825] Embodiment 41: The system of any of embodiments 37 to 40, comprising an optical power supply comprising at least one laser, in which the optical power supply is configured to provide a plurality of light sources to the plurality of data processing modules, at least one light source is provided to the photonic integrated circuit of each data processing module through an optical link.

[1826] Embodiment 42: The system of any of embodiments 37 to 41 in which the electronic processor of each data processing module comprises at least a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, an application specific integrated circuit (ASIC), or a data storage device.

[1827] Embodiment 43: The system of any of embodiments 37 to 42 in which the plurality of data processing modules comprise a blade pair that comprises a switch blade and a processor blade, the electronic processor of the switch blade comprises a switch, and the electronic processor of the processor blade is configured to process data provided by the switch.

[1828] Embodiment 44: A system comprising: [1829] a plurality of racks of data processing modules, in which multiple racks are stacked vertically, and each rack comprises a plurality of data

processing modules; [1830] wherein each data processing module comprises a substrate having a first side and a second side, an electronic processor mounted on the first side of the substrate, and an optical interconnect module comprising an optical port configured to receive optical signals from an optical fiber cable, and a photonic integrated circuit configured to generate electrical signals based on the received optical signals and transmit the electrical signals to the electronic processor; [1831] wherein for each data processing module, at least one of the optical interconnect module or the optical fiber cable extends through or partially through an opening in the substrate to enable at least a portion of the optical fiber cable to be positioned on or near the second side of the substrate.

[1832] Embodiment 45: The system of embodiment 44, comprising a structure to support the plurality of data processing modules in a way such that the substrates of the data processing modules are oriented parallel to one another.

[1833] Embodiment 46: The system of embodiment 45 in which the structure supports the data processing modules in a way such that the substrates are oriented vertically to enhance dissipation of heat from at least one of the data processing module or the optical interconnect module of each data processing module.

[1834] Embodiment 47: The system of any of embodiments 44 to 46, comprising an optical power supply comprising at least one laser, in which the optical power supply is configured to provide a plurality of light sources to the plurality of data processing modules, at least one light source is provided to the photonic integrated circuit of each data processing module through an optical link.

[1835] Embodiment 48: The system of any of embodiments 44 to 47 in which the electronic processor of each data processing module comprises at least a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, an application specific integrated circuit (ASIC), or a data storage device.

[1836] Embodiment 49: The system of any of embodiments 44 to 48 in which the plurality of data processing modules comprise a blade pair that comprises a switch blade and a processor blade, the electronic processor of the switch blade comprises a switch, and the electronic processor of the processor blade is configured to process data provided by the switch.

[1837] Embodiment 50: A method comprising: [1838] operating a plurality of data processing modules, in which each data processing module comprises a substrate having a first side and a second side, an electronic processor mounted on the first side of the substrate, and an optical interconnect module comprising an optical port and a photonic integrated circuit, the optical port is optically coupled to an optical fiber cable; and [1839] for each data processing module, defining a signal path using the optical fiber cable and the optical interconnect module, in which the signal path extends from the second side of the substrate through an opening in the substrate to the electronic processor; [1840] using the photonic integrated circuit to generate electrical signals based on the optical signals received at the optical port; and [1841] transmitting the electrical signals from the photonic integrated circuit to the electronic processor.

[1842] The following is a third set of embodiments. The embodiment numbers below refer to those in the third set of embodiments.

[1843] Embodiment 1: A system comprising: [1844] a housing comprising a bottom panel and a front panel, wherein the front panel is at an angle relative to the bottom panel in which the angle is in a range from 30 to 150°; [1845] a first circuit board positioned inside the housing, in which the first circuit board has a length, a width, and a thickness, wherein the length is at least twice the thickness, the width is at least twice the thickness, and the first circuit board has a first surface defined by the length and the width, [1846] wherein the first surface of the first circuit board is at a first angle relative to the bottom panel in which the first angle is in a range from 30° to 150°, [1847] wherein the first surface of the first circuit board is substantially parallel to the front panel or at a second angle relative to the front panel when the front panel is closed in which the second

angle is less than 60°; [1848] a first data processing module electrically coupled to the first circuit board; and [1849] a first optical interconnect module electrically coupled to the first circuit board, in which the optical interconnect module is configured to receive first optical signals from a first optical link, convert the first optical signals to first electrical signals, and transmit the first electrical signals to the first data processing module.

[1850] Embodiment 2: The system of embodiment 1, comprising a second circuit board that has a length, a width, and a thickness, in which the length is at least twice the thickness, the width is at least twice the thickness, and the second circuit board has a first surface defined by the length and the width, [1851] wherein the first surface of the second circuit board is substantially parallel to the bottom panel or at an angle relative to the bottom panel in which the angle is less than 20°, and the second circuit board is electrically coupled to the first circuit board.

[1852] Embodiment 3: The system of embodiment 2 in which the second circuit board comprises a motherboard, the first circuit board comprises a daughter card, and the motherboard is configured to provide electrical power to the daughter card.

[1853] Embodiment 4: The system of embodiment 1 in which the front panel is spaced apart from the rear panel at a mean distance of at least 12 inches, and the first circuit board is spaced apart from the front panel at a mean distance of less than 4 inches.

[1854] Embodiment 5: The system of embodiment 1 in which the first data processing module comprises at least a network switch, a central processor unit, a graphics processor unit, a tensor processing unit, a neural network processor, an artificial intelligence accelerator, a digital signal processor, a microcontroller, an application specific integrated circuit (ASIC), or a data storage device.

[1855] Embodiment 6: The system of embodiment 5 in which the first data processing module is capable of processing data from the first optical interconnect module at a rate of at least 25 gigabits per second.

[1856] Embodiment 7: The system of embodiment 5 in which the first data processing module is capable of processing data from one or more optical interconnect modules at a rate of at least 1 gigabits per second.

[1857] Embodiment 8: The system of embodiment 5 in which the first data processing module is capable of processing data from one or more optical interconnect modules at a rate of at least 10 gigabits per second.

[1858] Embodiment 9: The system of embodiment 5 in which the first data processing module is capable of processing data from one or more optical interconnect modules at a rate of at least 100 gigabits per second.

[1859] Embodiment 10: The system of embodiment 5 in which the first data processing module is capable of processing data from one or more optical interconnect modules at a rate of at least 1 terabits per second.

[1860] Embodiment 11: The system of embodiment 5 in which the first data processing module is capable of processing data from one or more optical interconnect modules at a rate of at least 10 terabits per second.

[1861] Embodiment 12: The system of embodiment 5 in which the first data processing module comprises an integrated circuit or a system on a chip (SoC) that includes at least one thousand transistors.

[1862] Embodiment 13: The system of embodiment 5 in which the first data processing module comprises an integrated circuit or a system on a chip (SoC) that includes at least ten thousand transistors.

[1863] Embodiment 14: The system of embodiment 5 in which the first data processing module comprises an integrated circuit or a system on a chip (SoC) that includes at least one hundred thousand transistors.

[1864] Embodiment 15: The system of embodiment 5 in which the first data processing module

comprises an integrated circuit or a system on a chip (SoC) that includes at least one million transistors.

[1865] Embodiment 16: The system of embodiment 5 in which the first data processing module comprises an integrated circuit or a system on a chip (SoC) that includes at least ten million transistors.

[1866] Embodiment 17: The system of embodiment 5 in which the first data processing module comprises an integrated circuit or a system on a chip (SoC) that includes at least one hundred million transistors.

[1867] Embodiment 18: The system of embodiment 5 in which the first data processing module comprises an integrated circuit or a system on a chip (SoC) that includes at least one billion transistors.

[1868] Embodiment 19: The system of embodiment 5 in which the first data processing module comprises an integrated circuit or a system on a chip (SoC) that includes at least ten billion transistors.

[1869] Embodiment 20: The system of embodiment 5 in which the first data processing module comprises circuitry that is capable of operating at a frequency of 1 MHz or more.

[1870] Embodiment 21: The system of embodiment 5 in which the first data processing module comprises circuitry that is capable of operating at a frequency of 10 MHz or more.

[1871] Embodiment 22: The system of embodiment 5 in which the first data processing module comprises circuitry that is capable of operating at a frequency of 100 MHz or more.

[1872] Embodiment 23: The system of embodiment 5 in which the first data processing module comprises circuitry that is capable of operating at a frequency of 1 GHz or more.

[1873] Embodiment 24: The system of embodiment 5 in which the first data processing module comprises circuitry that is capable of operating at a frequency of 3 GHz or more.

[1874] Embodiment 25: The system of embodiment 1 in which the system comprises a rackmount server, the housing comprises an enclosure for the rackmount server, and the rackmount server has an n rack unit form factor, and n is an integer in a range from 1 to 8.

[1875] Embodiment 26: The system of embodiment 1 in which the first data processing module is mounted on a substrate, and the substrate is electrically coupled to the first circuit board.

[1876] Embodiment 27: The system of embodiment 1 in which the first optical interconnect module is releasably coupled to the first circuit board.

[1877] Embodiment 28: The system of embodiment 27 in which a socket is mounted on the first circuit board, and the first optical interconnect module is releasably coupled to the socket.

[1878] Embodiment 29: The system of embodiment 1 in which the first optical interconnect module comprises a photonic integrated circuit mounted on a substrate, and the substrate is electrically coupled to the first circuit board.

[1879] Embodiment 30: The system of embodiment 1 in which the first optical interconnect module comprise a connector part that enables one or more optical fibers to be releasably connected to the first optical interconnect module.

[1880] Embodiment 31: The system of embodiment 1 in which the optical interconnect module is mounted on the first surface of the first circuit board, and the first surface faces the rear panel and away from the front panel.

[1881] Embodiment 32: The system of embodiment 31 in which the first circuit board defines a first opening, the front panel defines a second opening, the system comprises an optical path that passes through the first and second openings and enables the first optical signals from the first optical link to be transmitted to the first optical interconnect module.

[1882] Embodiment 33: The system of embodiment 1 in which the first electrical signals comprise first serial electrical signals, and the system comprises: [1883] a first serializer/deserializer configured to generate a set of first parallel electrical signals based on the first serial electrical signals, and condition the first parallel electrical signals; and [1884] a second serializer/deserializer

configured to generate a second serial electrical signal based on the set of first parallel electrical signals; [1885] wherein the first data processing module is configured to process data carried in the second serial electrical signal.

[1886] Embodiment 34: The system of embodiment 33, comprising a third serializer/deserializer configured to generate a set of second parallel electrical signals based on the second serial electrical signal; [1887] wherein the first data processing module is configured to process data carried by the set of second parallel electrical signals.

[1888] Embodiment 35: The system of embodiment 34 in which the third serializer/deserializer is embedded in the first data processing module.

[1889] Embodiment 36: The system of embodiment 1 in which the first optical interconnect module comprises a photonic integrated circuit and a first optical connector optically coupled to the photonic integrated circuit, the first optical connector is configured to releasably connect with a second optical connector that is coupled to a bundle of at least 100 optical fibers, and the first optical connector is configured to provide at least 100 optical paths to enable optical signals from the bundle of optical fibers to be coupled to the photonic integrated circuit.

[1890] Embodiment 37: The system of any of embodiments 1 to 36 in which the first optical interconnect module comprises at least one grating coupler, at least one optical waveguide coupled to the grating coupler, and at least one photodetector coupled to the at least one optical waveguide.

[1891] Embodiment 38: The system of any of embodiments 1 to 36 in which the first optical interconnect module comprises an array of grating couplers, a plurality of optical waveguides coupled to the array of grating couplers, and a plurality of photodetectors coupled to the plurality of optical waveguides.

[1892] Embodiment 39: The system of embodiment 38 in which the first optical interconnect module comprises a photonic integrated circuit and an optical fiber connector coupled to the photonic integrated circuit, [1893] wherein the photonic integrated circuit comprises the array of grating couplers, the plurality of optical waveguides, and the plurality of photodetectors, [1894] wherein the optical fiber connector comprises an array of lenses configured to focus light to or from the grating couplers.

[1895] Embodiment 40: A system comprising: [1896] a housing comprising a front panel, in which the front panel comprises a first circuit board; [1897] at least one data processing module electrically coupled to the first circuit board; and [1898] at least one optical/electrical communication interface electrically coupled to the first circuit board.

[1899] Embodiment 41: A system comprising: [1900] a housing comprising a front panel; [1901] a first circuit board oriented at a first angle relative to the front panel, in which the first angle is in a range from -60° to 60° ; [1902] at least one data processor electrically coupled to the first circuit board; and [1903] at least one optical/electrical communication interface electrically coupled to the first circuit board.

[1904] Embodiment 42: A system comprising: [1905] a plurality of rack mount systems, each rack mount system comprising: [1906] a housing comprising a front panel, in which the front panel comprises a first circuit board; [1907] at least one data processor electrically coupled to the first circuit board; and [1908] at least one optical/electrical communication interface electrically coupled to the first circuit board.

[1909] Embodiment 43: A system comprising: [1910] a plurality of rack mount systems, each rack mount system comprising: [1911] a housing comprising a front panel; [1912] a first circuit board oriented at a first angle relative to the front panel, in which the first angle is in a range from -60° to 60° ; [1913] at least one data processor electrically coupled to the first circuit board; and [1914] at least one optical/electrical communication interface electrically coupled to the first circuit board.

[1915] Embodiment 44: An apparatus comprising: [1916] a first substrate having a first side and a second side; [1917] a first electronic processing module mounted on the first side of the first substrate, wherein the first electronic processing module is configured to process data; and [1918] a

first optical interconnect module mounted on the second side of the first substrate, in which the first optical interconnect module comprises: [1919] an optical port configured to receive optical signals, and [1920] a photonic integrated circuit configured to generate electrical signals based on the received optical signals, and transmit the electrical signals to the first electronic processor.

[1921] Embodiment 45: A system comprising: [1922] a housing comprising a bottom panel and a front panel; [1923] a first circuit board or a first substrate positioned in the housing, in which the first circuit board or the first substrate is oriented at an angle relative to the bottom panel in which the angle is in a range from 30° to 150°; [1924] wherein the front panel of the housing is configured to be movable between a closed position and an open position, when the front panel is at the closed position the first circuit board or the first substrate is positioned behind the front panel and substantially parallel to the front panel or at an angle relative to the front panel in which the angle is less than 60°; [1925] a first lattice structure attached to the first circuit board or the first substrate, in which the first lattice structure defines a first plurality of openings; [1926] wherein a plurality of sets of electrical contacts are provided on a surface of the first circuit board or the first substrate, and each of the first plurality of openings of the first lattice structure correspond to one of the sets of electrical contacts and enables an optical interconnect module to pass through the opening and electrically couple to the set of electrical contacts.

[1927] Embodiment 46: A system comprising: [1928] a housing comprising a bottom panel and a front panel, the front panel comprising a plurality of optical connector parts, each optical connector part is configured to be optically coupled to an external optical fiber cable and an internal optical fiber cable; [1929] a first circuit board or a first substrate positioned in the housing, in which the first circuit board or the first substrate is oriented at an angle relative to the bottom panel in which the angle is in a range from 30° to 150°; [1930] wherein the first circuit board or the first substrate is substantially parallel to the front panel or at an angle relative to the front panel in which the angle is less than 60°; [1931] a plurality of optical interconnect modules electrically coupled to the first circuit board; and [1932] a plurality of internal optical fiber cables, in which each internal optical fiber cable is optically coupled to one of the optical interconnect modules and a corresponding optical connector part on the front panel.

[1933] Embodiment 47: The system of embodiment 46 wherein the front panel of the housing is configured to be movable between a closed position and an open position, when the front panel is at the closed position the first circuit board or the first substrate is positioned behind the front panel and substantially parallel to the front panel or at an angle relative to the front panel in which the angle is less than 60°.

[1934] Embodiment 48: A rackmount system configured to be placed on a rack during operation, the rackmount system comprising: [1935] a housing comprising a front panel, in which the housing defines a front opening when the front panel is opened; [1936] a first circuit board or a first substrate positioned in the housing; [1937] a data processing module electrically coupled to the first circuit board or the first substrate, in which the data processing module has a throughput of at least 100 gigabits per second; and [1938] a plurality of optical interface modules electrically coupled to a first surface of the first circuit board or the first substrate, in which at least one of the plurality of optical interface modules are configured to receive first optical signals, convert the first optical signals to first electrical signals, and transmit the first electrical signals to the data processing module, and at least one of the plurality of optical interface modules are configured to receive second electrical signals from the data processing module, convert the second electrical signals to second optical signals, and output the second optical signals; [1939] wherein the first surface of the first circuit board or the first substrate is oriented to face towards the front opening to allow the optical interface modules to be accessed after the front panel is opened without removing the rackmount system from the rack, in which accessing the optical interface module includes at least one of attaching the optical interface module to the first circuit board or the first substrate, or removing the optical interface module from the first circuit board or the first substrate.

[1940] Embodiment 49: A method comprising: [1941] electrically coupling a first optical interconnect module to a first surface of a first circuit board of a system, in which the first circuit board is oriented substantially parallel to a front panel of a housing of the system or at an angle relative to the front panel when the front panel is closed in which the angle is less than 60°, and the first surface faces the front panel when the front panel is closed; [1942] transmitting first optical signals from an optical fiber cable to the first optical interconnect module; [1943] converting, using the first optical interconnect module, the first optical signals to first electrical signals; [1944] transmitting the first electrical signals to a data processing module electrically coupled to the first circuit board; and [1945] processing, using the data processing module, the first electrical signals.

[1946] Embodiment 50: A method comprising: [1947] opening a front panel of a housing of a system to expose a first surface of a first circuit board of the system and a first optical interconnect module that is electrically coupled to the first surface of the first circuit board, in which the first circuit board is oriented substantially parallel to the front panel or at an angle relative to the front panel when the front panel is closed in which the angle is less than 60°, and the first surface faces the front panel when the front panel is closed; [1948] disconnecting the first optical interconnect module from the first surface of the first circuit board; [1949] disconnecting the first optical interconnect module from an optical fiber cable that is optically coupled to the first optical interconnect module; [1950] optically coupling a second optical interconnect module to the optical fiber cable; [1951] electrically coupling the second optical interconnect module to the first surface of the first circuit board; and [1952] closing the front panel.

[1953] Embodiment 51: An apparatus comprising: [1954] a co-packaged optical module comprising: [1955] a photonic integrated circuit; [1956] an optical connector coupled to a first surface of the photonic integrated circuit; and [1957] a first set of at least two electrical integrated circuits that are coupled to the first surface of the photonic integrated circuit.

[1958] Embodiment 52: The apparatus of embodiment 51 in which the first set of at least two electrical integrated circuits comprise two electrical integrated circuits that are positioned on opposite sides of the optical connector along a plane parallel to the first surface of the photonic integrated circuit.

[1959] Embodiment 53: The apparatus of embodiment 51 in which the first set of at least one electrical integrated circuit comprises four electrical integrated circuits that surround three sides of the optical connector along a plane parallel to the first surface of the photonic integrated circuit.

[1960] Embodiment 54: The apparatus of any of embodiments 51 to 53 in which the co-packaged optical module comprises: [1961] a substrate, in which the photonic integrated circuit is mounted on the substrate, and [1962] a second set of at least one electrical integrated circuit mounted on the substrate and electrically coupled to the photonic integrated circuit through one or more signal conductors and/or traces.

[1963] Embodiment 55: The apparatus of embodiment 54 in which the photonic integrated circuit comprises at least one of a photodetector or an optical modulator, and the first set of at least one integrated circuit comprises at least one of a transimpedance amplifier configured to amplify a current generated by the photodetector or a driver configured to drive the optical modulator.

[1964] Embodiment 56: The apparatus of any of embodiments 51 to 55 in which the second set of at least one electrical integrated circuit comprises a serializers/deserializers module.

[1965] Embodiment 57: The apparatus of any of embodiments 51 to 56 in which the photonic integrated circuit comprises a silicon substrate and an active layer at a second surface that is opposite to the first surface relative to the photonic integrated circuit, [1966] wherein the active layer comprises grating couplers, and at least one of photodetectors or optical modulators, [1967] wherein the optical connector is optically coupled to the grating couplers using backside illumination; and [1968] wherein the first set of at least one electrical integrated circuits is coupled to the at least one of photodetectors or optical modulators using through silicon vias.

[1969] Embodiment 58: An apparatus comprising: [1970] a co-packaged optical module

comprising: [1971] a photonic integrated circuit; [1972] an optical connector coupled to a first surface of the photonic integrated circuit; and [1973] a first set of at least one electrical integrated circuit that is coupled to a second surface of the photonic integrated circuit, in which the second surface is opposite to the first surface relative to the photonic integrated circuit.

[1974] Embodiment 59: The apparatus of embodiment 58 in which the photonic integrated circuit comprises an active layer at the first surface, the active layer comprises grating couplers, and at least one of photodetectors or optical modulators, [1975] wherein the optical connector has a footprint that overlaps a footprint of the grating couplers; [1976] wherein the at least one of photodetectors or optical modulators are spaced apart from the grating couplers, and [1977] wherein the first set of at least one electrical integrated circuits is coupled to the at least one of photodetectors or optical modulators using through silicon vias.

[1978] Embodiment 60: The apparatus of embodiment 58 or 59 in which the photonic integrated circuit comprises a silicon substrate and an active layer at the second surface, [1979] wherein the active layer comprises grating couplers, and at least one of photodetectors or optical modulators, [1980] wherein the optical connector is optically coupled to the grating couplers using backside illumination; and [1981] wherein the at least one of photodetectors or optical modulators are spaced apart from the grating couplers, and the first set of at least one electrical integrated circuits is electrically coupled to the at least one of photodetectors or optical modulators.

[1982] Embodiment 61: The apparatus of any of embodiments 58 to 60 in which the photonic integrated circuit comprises at least one of a photodetector or an optical modulator, and the first set of at least one integrated circuit comprises at least one of a transimpedance amplifier configured to amplify a current generated by the photodetector or a driver configured to drive the optical modulator.

[1983] Embodiment 62: The apparatus of any of embodiments 58 to 61 in which the co-packaged optical module comprises: [1984] a substrate, in which the photonic integrated circuit is mounted on the substrate, and [1985] a second set of at least one electrical integrated circuit mounted on the substrate and electrically coupled to the photonic integrated circuit through one or more signal conductors and/or traces.

[1986] Embodiment 63: The apparatus of embodiment 62 in which the second set of at least one electrical integrated circuit comprises a serializers/deserializers module.

Claims

1. An apparatus comprising: an optical interconnect module comprising: a first circuit board or a first substrate having a length, a width, and a thickness, in which the length is at least twice the thickness, and the width is at least twice the thickness, the first circuit board or the first substrate has a first surface defined by the length and the width; an optical input port configured to receive a plurality of channels of optical signals; a photonic integrated circuit mounted on the first circuit board or the first substrate and configured to generate a plurality of first serial electrical signals based on the received optical signals; and an array of first electrical terminals arranged on the first surface of the first circuit board or the first substrate, in which the array of first electrical terminals comprises at least two electrical terminals distributed along the length direction and at least two electrical terminals distributed along the width direction, the first electrical terminals are configured to output the first serial electrical signals.

2. The apparatus of claim 1 wherein the first electrical terminals extend along directions substantially perpendicular to the first surface of the first circuit board or the first substrate.

3. The apparatus of claim 1 wherein the first electrical terminals comprise at least one of spring loaded connectors, compression interposers, or land-grid arrays.

4. The apparatus of claim 1 wherein the first circuit board or the first substrate has a second surface defined by the length and the width, the second surface is spaced apart from the first surface by the

thickness; the photonic integrated circuit is mounted on the second surface of the first circuit board or the first substrate; the photonic integrated circuit comprises second electrical terminals that are electrically coupled to the first electrical terminals through electrical connectors that pass through the first circuit board or the first substrate in the thickness direction.

5. The apparatus of claim 4 wherein the electrical connectors comprise vias in the first circuit board that electrically connect the array of first electrical terminals arranged on the first surface of the first circuit board to an array of third electrical terminals arranged on the second surface of the first circuit board, and wire bonds that electrically connect the array of third electrical terminals arranged on the second surface of the first circuit board to the second electrical terminals of the photonic integrated circuit.

6. The apparatus of claim 4 wherein the electrical connectors comprise first vias in the first circuit board that electrically connect the array of first electrical terminals arranged on the first surface of the first circuit board to an array of third electrical terminals arranged on the second surface of the first circuit board, and second vias in the photonic integrated circuit that are part of the electrical paths between the array of third electrical terminals arranged on the second surface of the first circuit board and the second electrical terminals of the photonic integrated circuit.

7. The apparatus of claim 1 wherein the optical interconnect module comprises at least one of a driver or a transimpedance amplifier, the driver is configured to drive an optical modulator, and the transimpedance amplifier is configured to amplify a signal output from a photo detector; the first circuit board or the first substrate has a second surface defined by the length and the width, the second surface is spaced apart from the first surface by the thickness; the photonic integrated circuit and the at least one of the driver or the transimpedance amplifier are mounted on the second surface of the first circuit board or the first substrate; the at least one of the driver or transimpedance amplifier has second electrical terminals that are electrically coupled to the first electrical terminals through electrical connectors that pass through the first circuit board or the first substrate in the thickness direction.

8. The apparatus of claim 1 wherein the photonic integrated circuit has a length, a width, and a thickness, the length is at least twice the thickness, and the width is at least twice the thickness; the photonic integrated circuit has a first surface defined by the length and the width; the photonic integrated circuit comprises second electrical terminals arranged on the first surface, the second electrical terminals are electrically coupled to the first electrical terminals on the first circuit board or the first substrate; and the optical input port is optical coupled to the first surface of the photonic integrated circuit.

9. The apparatus of claim 1 wherein the photonic integrated circuit has a length, a width, and a thickness, the length is at least twice the thickness, and the width is at least twice the thickness; the photonic integrated circuit has a first surface defined by the length and the width, the photonic integrated circuit has a second surface defined by the length and the width, the second surface is spaced apart from the first surface by the thickness; the photonic integrated circuit comprises second electrical terminals arranged on the first surface, the second electrical terminals are electrically coupled to the first electrical terminals on the first circuit board or the first substrate; the optical input port is optical coupled to the second surface of the photonic integrated circuit.

10. The apparatus of claim 1 wherein the optical interconnect module comprises at least one of a driver or a transimpedance amplifier, the driver is configured to drive an optical modulator, and the transimpedance amplifier is configured to amplify an electrical signal from a photo detector; the photonic integrated circuit and at least one of the driver or the transimpedance amplifier are mounted on the first surface of the first circuit board or the first substrate; and the at least one of the driver or transimpedance amplifier has second electrical terminals that are electrically coupled to the first electrical terminals.

11. The apparatus of claim 1 wherein the optical input port comprises a first optical connector configured to mate with a second optically connector coupled to an optical fiber cable that

comprises a plurality of optical fibers.

12. The apparatus of claim 11 wherein the photonic integrated circuit comprises vertical-coupling elements configured to couple light from the optical input port to the photonic integrated circuit.

13. The apparatus of claim 12 wherein the first optical connector comprises one or more lenses configured to project light onto the vertical coupling elements.

14. The apparatus of claim 12 wherein the first optical connector and the second optical connector comprise one or more optical components configured to couple M spatial paths of the optical fibers and an array of N vertical-coupling elements of the photonic integrated circuit, N is a positive integer, M is a positive integer, and N is equal to or different from M.

15. The apparatus of claim 14 wherein the one or more optical components of the first and second optical connectors are configured to implement at least one of (i) magnifying or de-magnifying by a first factor a minimum core-to-core spacing of the optical fibers at a fiber end face plane to match a minimum spacing between the vertical-coupling elements at a coupling plane; (ii) magnifying or de-magnifying by a second factor a maximum core-to-core spacing of optical fibers at a fiber end face plane to match a maximum spacing between the vertical-coupling elements at a coupling plane; (iii) magnifying or de-magnifying by a third factor an effective core diameter of optical fibers at a fiber end face plane to match an effective size of the vertical coupling elements at a coupling plane; (iv) magnifying or de-magnifying by a fourth factor an effective core diameter of optical fibers at a fiber end face plane to achieve a different effective beam diameter at a connector mating plane than at the fiber end face plane; or (v) changing an effective cross-sectional geometrical layout of the plurality of spatial paths at at least one of a fiber end face plane, a connector mating plane, or a coupling plane.

16. An apparatus comprising: a substrate comprising: a first main surface and a second main surface; a first array of electrical contacts arranged on the first main surface and having a first minimum spacing between the contacts; a second array of electrical contacts arranged on the second main surface and having a second minimum spacing between the contacts, in which the first minimum spacing is larger than the second minimum spacing; and electrical connections between the first array of electrical contacts and the second array of electrical contacts; a photonic integrated circuit having a first main surface and a second main surface; a first optical connector part configured to couple light to the first main surface of the photonic integrated circuit; and an electronic integrated circuit having a first main surface that has a first portion and a second portion, in which the first portion of the first main surface is electrically coupled to the second main surface of the photonic integrated circuit, and the second portion of the first main surface is electrically coupled to the second array of electrical contacts arranged on the second main surface of the substrate.

17. The system of claim 16 wherein the substrate is configured to be removably connectable to a printed circuit board.

18. The system of claim 16, further comprising a second optical connector part that is configured to couple light from an array of optical fibers to the first optical connector part.

19. The system of claim 16 wherein the photonic integrated circuit is configured to receive and process optical signals provided from an external signal source, in which the optical signals are carried by at least one of continuous wave light or pulsed light.

20. The system of claim 16, further comprising a connector module configured to removably attach the substrate to a printed circuit board.

21. An apparatus comprising: a co-packaged optical module comprising: a photonic integrated circuit; an optical connector coupled to a first surface of the photonic integrated circuit; and a first set of at least two electrical integrated circuits that are coupled to the first surface of the photonic integrated circuit.

22. The apparatus of claim 21 wherein the first set of at least two electrical integrated circuits comprise two electrical integrated circuits that are positioned on opposite sides of the optical

connector along a plane parallel to the first surface of the photonic integrated circuit.

23. The apparatus of claim 21 wherein the first set of at least one electrical integrated circuit comprises three electrical integrated circuits that surround three sides of the optical connector along a plane parallel to the first surface of the photonic integrated circuit.

24. The apparatus of claim 21 wherein the co-packaged optical module comprises: a substrate, in which the photonic integrated circuit is mounted on the substrate, and a second set of at least one electrical integrated circuit mounted on the substrate and electrically coupled to the photonic integrated circuit through one or more signal conductors and/or traces.

25. The apparatus of claim 24 wherein the photonic integrated circuit comprises at least one of a photodetector or an optical modulator, and the first set of at least one integrated circuit comprises at least one of a transimpedance amplifier configured to amplify a current generated by the photodetector or a driver configured to drive the optical modulator.

26. The apparatus of claim 21 wherein the second set of at least one electrical integrated circuit comprises a serializers/deserializers module.

27. The apparatus of claim 21 wherein the photonic integrated circuit comprises a silicon substrate and an active layer at a second surface that is opposite to the first surface relative to the photonic integrated circuit, wherein the active layer comprises grating couplers, and at least one of photodetectors or optical modulators, wherein the optical connector is optically coupled to the grating couplers using backside illumination; and wherein the first set of at least one electrical integrated circuits is coupled to the at least one of photodetectors or optical modulators using through silicon vias.

28. An apparatus comprising: a co-packaged optical module comprising: a photonic integrated circuit; an optical connector coupled to a first surface of the photonic integrated circuit; and a first set of at least one electrical integrated circuit that is coupled to a second surface of the photonic integrated circuit, in which the second surface is opposite to the first surface relative to the photonic integrated circuit.

29. The apparatus of claim 28 wherein the photonic integrated circuit comprises an active layer at the first surface, the active layer comprises grating couplers, and at least one of photodetectors or optical modulators, wherein the optical connector has a footprint that overlaps a footprint of the grating couplers; wherein the at least one of photodetectors or optical modulators are spaced apart from the grating couplers, and wherein the first set of at least one electrical integrated circuits is coupled to the at least one of photodetectors or optical modulators using through silicon vias.

30. The apparatus of claim 28 wherein the photonic integrated circuit comprises a silicon substrate and an active layer at the second surface, wherein the active layer comprises grating couplers, and at least one of photodetectors or optical modulators, wherein the optical connector is optically coupled to the grating couplers using backside illumination; and wherein the at least one of photodetectors or optical modulators are spaced apart from the grating couplers, and the first set of at least one electrical integrated circuits is electrically coupled to the at least one of photodetectors or optical modulators.

31. The apparatus of claim 28 wherein the photonic integrated circuit comprises at least one of a photodetector or an optical modulator, and the first set of at least one integrated circuit comprises at least one of a transimpedance amplifier configured to amplify a current generated by the photodetector or a driver configured to drive the optical modulator.

32. The apparatus claim 28 wherein the co-packaged optical module comprises: a substrate, in which the photonic integrated circuit is mounted on the substrate, and a second set of at least one electrical integrated circuit mounted on the substrate and electrically coupled to the photonic integrated circuit through one or more signal conductors and/or traces.
